

Solutions to Final Step | Part – B | Mathematics

Functions

1.(C) $g(x) = f(x+1) = |x-1| + |x-2| + |x-3|$ Now, $g(-x) = |x+1| + |x+2| + |x+3|$

Clearly, $g(x) \neq \pm g(-x) \Rightarrow g(x)$ is neither even nor odd.

2.(D) $f(x) = x - [x+a] - b = x + a - [x+a] - (a+b) = \{x+a\} - (a+b)$ Thus, period of $f(x)$ is 1.

3.(A) $f(x)$ is defined, if $x^2 - 5x + 6 \neq 0$, $\left[x + \frac{1}{2}\right] > 0$, $\left[x + \frac{1}{2}\right] \neq 1$

$$x^2 - 5x + 6 \neq 0, \Rightarrow (x-2)(x-3) \neq 0 \Rightarrow x \neq 2, 3 \quad \dots(i)$$

$$\left[x + \frac{1}{2}\right] > 0 \Rightarrow x \geq \frac{1}{2} \quad \dots(ii)$$

$$\left[x + \frac{1}{2}\right] \neq 1 \Rightarrow x \notin \left[\frac{1}{2}, \frac{3}{2}\right] \quad \dots(iii)$$

From Eq. (i), (ii) and (iii), we get domain of $f = \left[\frac{3}{2}, 2\right) \cup (2, 3) \cup (3, \infty)$.

4.(A) Since, the function $\sec x$ is an even function and $\log(x + \sqrt{1+x^2})$ is odd function, therefore the function

$\sec \left[\log(x + \sqrt{1+x^2}) \right]$ is an even function.

5.(D) Given, $f(x) = \log\left(\frac{1+x}{1-x}\right)$

Then, $f\left(\frac{3x+x^3}{1+3x^2}\right) - f\left(\frac{2x}{1+x^2}\right)$

$$= \log \left(\frac{1 + \left(\frac{3x+x^3}{1+3x^2} \right)}{1 - \left(\frac{3x+x^3}{1+3x^2} \right)} \right) - \log \left(\frac{1 + \frac{2x}{1+x^2}}{1 - \frac{2x}{1+x^2}} \right) = \log \left(\frac{1+x}{1-x} \right)^3 - \log \left(\frac{1+x}{1-x} \right)^2 = \log \left(\frac{1+x}{1-x} \right) = f(x)$$

6.(A) Given, $f(x) = x^2 + \frac{1}{x^2+1} \therefore f(x) = x^2 + \left(\frac{1}{x^2+1} - 1 \right) + 1 = (x^2+1) - \left(\frac{x^2}{x^2+1} \right) = 1 + x^2 \left(1 - \frac{1}{x^2+1} \right) \geq 1, \forall x \in R$

Hence, range of $f(x)$ is $[1, \infty)$.

7.(B) Since, $f(x) = \frac{4^x}{4^x+2} \therefore f(1-x) = \frac{4^{1-x}}{4^{1-x}+2} = \frac{4}{4+2 \cdot 4^x} = \frac{2}{2+4^x} \Rightarrow f(x) + f(1-x) = 1$

Putting, $x = \frac{1}{97}, \frac{2}{97}, \dots, \frac{48}{97}$, we get: $f\left(\frac{1}{97}\right) + f\left(\frac{2}{97}\right) + \dots + f\left(\frac{96}{97}\right) = 48$

8.(C) Put, $x = y = 0$

$$\Rightarrow f(0) + f(0) = 2(f(0))^2 \Rightarrow f(0) = 0, 1 \Rightarrow k = 0, 1$$

Put, $x = 0$, we get, $f(y) + f(-y) = 2f(0)f(y)$

$$\text{If, } f(0) = 0 \Rightarrow f(y) + f(-y) = 0 \Rightarrow f \text{ is odd}$$

$$\text{If, } f(0) = 1 \Rightarrow f(y) + f(-y) = 2f(y) \Rightarrow f \text{ is even}$$

9.(C) The period of the function is found as follows

$$\text{Given, } f(x) + f(x+4) = f(x+2) + f(x+6) \quad \dots\dots\dots(\text{i})$$

$$\therefore \text{ Replacing } x \text{ by } x+2 \text{ we get: } f(x+2) + f(x+6) = f(x+4) + f(x+8) \quad \dots\dots\dots(\text{ii})$$

From equations (i) and (ii), we get :

$$f(x) + f(x+4) = f(x+4) + f(x+8) \Rightarrow f(x) = f(x+8) \Rightarrow \text{period of } f(x) = 8$$

$$\begin{aligned} 10.(A) \quad x &= \left(\frac{81^{\frac{1}{\log_5 9}} + 3^{\frac{3}{\log_3 \sqrt{6}}}}{409} \right) \cdot \left((\sqrt{7})^{\frac{2}{\log_{25} 7}} - 125^{\log_{25} 6} \right) \\ x &= \left(\frac{81^{\frac{1}{\log_9 5}} + 3^{\frac{3}{\log_3 \sqrt{6}}}}{409} \right) \cdot \left(7^{\frac{2}{\log_7 25}} - 6^{\log_{25} 125} \right) = \frac{25 + 6\sqrt{6}}{409} (25 - 6\sqrt{6}) = 1 \Rightarrow \log_2 x = \log_2 1 = 0 \end{aligned}$$

$$11.(D) \quad (A) \quad \log_{1.5} \log_4 \log_{\sqrt{3}} 81 = \log_{1.5} \log_4 8 = \log_{1.5} 1.5 = 1$$

$$(B) \quad \log_2 \sqrt{6} + \log_2 \sqrt{\frac{2}{3}} = \log_2 2 = 1$$

$$(C) \quad -\frac{1}{6} \log_{\frac{\sqrt{3}}{2}} \left(\frac{64}{27} \right) = \frac{1}{6} \log_{\frac{\sqrt{3}}{2}} \left(\frac{27}{64} \right) = \frac{1}{6} \cdot 6 = 1$$

$$(D) \quad \log_{3.5} (1 + 2 + 3 \div 6) = \log_{3.5} 3.5 = 1$$

$$12.(C) \quad \log_a x = \alpha, \log_b x = \beta, \log_c x = \gamma, \log_d x = \delta \Rightarrow a = x^{\frac{1}{\alpha}}, b = x^{\frac{1}{\beta}}, c = x^{\frac{1}{\gamma}}, d = x^{\frac{1}{\delta}}$$

$$\log_{abcd} x = \frac{1}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}}$$

$$13.(C) \quad 4\log_{x/2} \sqrt{x} + 2\log_{4x} x^2 = 3\log_{2x} x^3 \quad \left(x > 0, x \neq \frac{1}{2}, \frac{1}{4}, 2 \right)$$

$$\Rightarrow 2\log_{x/2} x + 4\log_{4x} x = 9\log_{2x} x \Rightarrow \frac{2}{\log_x (x/2)} + \frac{4}{\log_x 4x} = \frac{9}{\log_x 2x}$$

$$\Rightarrow \frac{2}{1 - \log_x 2} + \frac{4}{\log_x 4x} = \frac{9}{\log_x 2x}$$

$$\Rightarrow \frac{2}{1-t} + \frac{4}{2t+1} = \frac{9}{t+1} \quad (\text{Let, } \log_x 2 = t; (x \neq 1))$$

$$\Rightarrow 6(t+1) = 9(t-2t^2+1) \Rightarrow 18t^2 - 3t - 3 = 0 \Rightarrow 6t^2 - t - 1 = 0$$

$$\Rightarrow t = \frac{1}{2}, -\frac{1}{3} \Rightarrow x = 4, \frac{1}{8}$$

Now, checking for $x = 1$

$x = 1$ satisfies the original equation $\therefore$ Integral solution are $\{4, 1\}$

$$14.(B) \quad y = f(x) = \begin{cases} \frac{x^2}{1+x^2}, & x < 0 \\ 0, & x = 0 \\ \frac{-x^2}{1+x^2}, & x > 0 \end{cases}$$

$$\text{Case I : } x < 0 \Rightarrow y = \frac{x^2}{1+x^2} \Rightarrow x^2(1-y) = y \Rightarrow x = -\sqrt{\frac{y}{1-y}} \quad \because (x < 0)$$

$$\text{Case II : } x > 0 \Rightarrow y = \frac{-x^2}{1+x^2} \Rightarrow x^2 = \frac{-y}{1+y} \Rightarrow x = +\sqrt{\frac{-y}{1+y}} \Rightarrow f^{-1}(y) = x = \operatorname{sgn}(-y) \sqrt{\frac{|y|}{1-|y|}}$$

$$15.(A) \quad y = f(x) = \sin\left\{[x+5] + \left\{x - \left\{x - \{x\}\right\}\right\}\right\} = \sin\left\{x - \{[x]\}\right\} = \sin\{x-0\} = \sin x$$

$$\because 0 < x < \frac{\pi}{4} \quad \therefore x = \sin^{-1} y \text{ or } f^{-1}(x) = \sin^{-1} x$$

$$16.(B) \quad G(x) = \left(\frac{1}{a^x-1} + \frac{1}{2}\right)F(x)$$

$$\begin{aligned} G(-x) &= \left(\frac{1}{a^{-x}-1} + \frac{1}{2}\right)F(-x) = -\left(\frac{a^x}{1-a^x} + \frac{1}{2}\right)F(x) = \left(\frac{a^x}{a^x-1} - \frac{1}{2}\right)F(x) \\ &= \left(\frac{a^x-1+1}{a^x-1} - \frac{1}{2}\right)F(x) = \left(1 + \frac{1}{a^x-1} - \frac{1}{2}\right)F(x) = \left(\frac{1}{a^x-1} + \frac{1}{2}\right)F(x) = G(x) \end{aligned}$$

$\therefore G(x)$ is an even function.

$$17.(C) \quad f\left(2x + \frac{y}{8}, 2x - \frac{y}{8}\right) = xy$$

$$\text{Let } m = 2x + \frac{y}{8}, n = 2x - \frac{y}{8} \Rightarrow x = \frac{m+n}{4}, m-n = \frac{y}{4} \Rightarrow y = 4(m-n)$$

$$f(m, n) = \left(\frac{m+n}{4}\right) \cdot 4(m-n) = m^2 - n^2$$

$$\text{Similarly } f(n, m) = n^2 - m^2 \Rightarrow f(m, n) + f(n, m) = 0$$

$$18.(C) \quad f(2x) + f\left(\frac{1}{11} + 2x\right) + f\left(\frac{2}{11} + 2x\right) + f\left(\frac{3}{11} + 2x\right) + \dots + f\left(\frac{21}{11} + 2x\right) = k$$

$$\text{Now, } 2x \rightarrow 2x + \frac{1}{11}$$

$$f\left(2x + \frac{1}{11}\right) + f\left(2x + \frac{2}{11}\right) + f\left(2x + \frac{3}{11}\right) + f\left(2x + \frac{4}{11}\right) + \dots + f\left(2x + \frac{22}{11}\right) = k$$

$$\text{On subtracting } f(2x) = f(2x+2)$$

19.(C) $f(x + f(x)) = 4f(x)$

Put $x = 1$, $f(1 + f(1)) = 4f(1)$

$\Rightarrow f(1 + 4) = 4(4) \Rightarrow f(5) = 16$

Again put $x = 5$

$f(5 + 16) = 4f(5) \Rightarrow f(21) = 4(16) = 64$

20.(A) $f(15 + x) = f(15 - x)$ (i)

And $f(30 + x) = -f(30 - x)$ (ii)

Replace $x \rightarrow 15 - x$ in (i)

$\Rightarrow f(30 - x) = f(x)$

By (ii) $f(x) = -f(30 + x)$ (iii)

Replace $x \rightarrow x + 30$

Then $f(x + 30) = -f(x + 60)$ (iv)

From (iii) & (iv) $f(x) = f(x + 60) \Rightarrow f(x)$ is periodic with period 60

Again $f(30 - x) = f(x)$

$x \rightarrow x + 30 \Rightarrow f(30 + x) = f(-x)$

From (ii) $f(-x) = -f(x)$

$f(x)$ is odd

21.(B) For $x \in \mathbb{R}$, $x \neq 0$, $x \neq 1$, $f_0(x) = \frac{1}{1-x}$ and $f_{n+1} = f_0(f_n(x))$, $n = 0, 1, 2, \dots$

$$f_1(x) = f_0(f_0(x)) = \frac{1}{1 - f_0(x)} = \frac{1}{1 - \frac{1}{1-x}} = \frac{x-1}{x}$$

$$f_2(x) = f_0(f_1(x)) = \frac{1}{1 - f_1(x)} = \frac{1}{1 - \frac{x-1}{x}} = x$$

$$f_3(x) = f_0(f_2(x)) = \frac{1}{1 - f_2(x)} = \frac{1}{1-x}$$

$$f_4(x) = f_0(f_3(x)) = \frac{1}{1 - f_3(x)} = \frac{x-1}{x}$$

$\therefore f_0(x) = f_3(x) = f_6(x) = \dots$ and $f_1(x) = f_4(x) = \dots = f_{100}(x) = \frac{x-1}{x}$

$\therefore f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right) = \frac{3-1}{3} + \frac{\frac{2}{3}-1}{\frac{2}{3}} + \frac{3}{2} = \frac{5}{3}$

22.(D) $f(2) = f(2)$

But $2 \neq -2$, so not one-one.

For onto: $f(x) = y \Rightarrow \frac{|x|-1}{|x|+1} = y \Rightarrow |x| = \frac{y+1}{1-y} \therefore -1 \leq y < 1$. So range of $f(x) = [-1, 1)$

So, other elements of $\mathbb{R}$ (co-domain) has no pre-image in $\mathbb{R}$ (domain). Thus f is not onto.

23.(D) $f(n) = \left[\frac{1}{3} + \frac{3n}{100} \right] n$

$$\frac{1}{3} + \frac{3n}{100} = 1 \Rightarrow n = \frac{200}{9} = 22.22 ; \quad \frac{1}{3} + \frac{3n}{100} = 2 \Rightarrow n = \frac{500}{9} = 55.55$$

$$\therefore \left[\frac{1}{3} + \frac{3n}{100} \right] = 0, \quad 1 \leq n \leq 22$$

$$\therefore f(1) = f(2) = \dots = f(22) = 0$$

$$\left[\frac{1}{3} + \frac{3n}{100} \right] = 1, \quad 23 \leq n \leq 55$$

$$\left[\frac{1}{3} + \frac{3n}{100} \right] = 2, \quad n = 56$$

$$\therefore \sum_{n=1}^{56} f(n) = f(23) + f(24) + \dots + f(55) + f(56)$$

$$= 1.23 + 1.24 + \dots + 1.55 + 2.56 = \frac{33}{2}(23+55) + 2 \times 56 = 1287 + 112 = 1399$$

24.(AB) 25.(A)

26.(1) $f(x) + 2f\left(\frac{2002}{x}\right) = 3x$

$$\Rightarrow f\left(\frac{2002}{x}\right) + 2f(x) = \frac{6006}{x} \Rightarrow 2f\left(\frac{2002}{x}\right) + 4f(x) = \frac{2 \times 6006}{x}$$

$$\Rightarrow 3f(x) = 3\left[\frac{4004}{x} - x\right] \Rightarrow f(2) = 2000$$

27.(1) $f(2) = 2\sin 2 + 2b \cdot \cos 2 + 8$

$$f(-2) = -a \sin 2 - 2b \cos 2 + 8$$

$$f(2) + f(-2) = 16$$

$$\therefore f(-2) = 16 - f(2), \text{ given } f(2) = 15 \Rightarrow f(-2) = 1$$

28.(2) We have, $f(x) = \log_{\sqrt{2}} \left[2 - \log_2 (16 \sin^2 x + 1) \right]$

$$\text{Its domain is } 2 - \log_2 (16 \sin^2 x + 1) > 0$$

$$\Rightarrow -\log_2 (16 \sin^2 x + 1) > -2 \Rightarrow 16 \sin^2 x + 1 < 4 \Rightarrow \sin^2 x < \frac{3}{16}$$

$$\therefore 0 \leq \sin^2 x < \frac{3}{16}$$

$$\text{Now, } 1 \leq 16 \sin^2 x + 1 < 4$$

$$\log_2 1 \leq \log_2 (16 \sin^2 x + 1) < \log_2 4$$

$$\Rightarrow 0 \leq \log_2 (16 \sin^2 x + 1) < 2 \Rightarrow -2 < -\log_2 (16 \sin^2 x + 1) \leq 0 \Rightarrow 0 < 2 - \log_2 (16 \sin^2 x + 1) \leq 2$$

$$\log_{\sqrt{2}} 0 < \log_{\sqrt{2}} \left[2 - \log_2 (16 \sin^2 x + 1) \right] \leq \log_{\sqrt{2}} 2$$

$$-\infty < f(x) \leq 2$$

But it is $(-\infty, a]$

By comparing, we get : $a = 2$

29.(8) We have, $f(x-1) + f(x+3) = f(x+1) + f(x+5) \quad \dots (i)$

Replacing x by $x+2$ in equation (i), we get :

$$f(x+1) + f(x+5) = f(x+3) + f(x+7) \quad \dots (ii)$$

On adding equations (i) and (ii), we get :

$$f(x-1) = f(x+7) \quad \dots (iii)$$

Now, replacing x by $x+1$ in equation (iii), we get : $f(x) = f(x+8)$

Hence, period of $f(x)$ is 8

30.(4) $\sin \frac{\pi}{2}(x) = 0 \quad x \in [4n, 4n+1)$
 $= 1 \quad x \in [4n+1, 4n+2)$
 $= 0 \quad x \in [4n+2, 4n+3)$
 $= -1 \quad x \in [4n+3, 4n+4)$
 $\Rightarrow \sin \frac{\pi}{2}[x]$ is periodic with period = 4

31.(6) We have, $\left(\frac{4a-7}{3}\right)x^3 + (a-3)x^2 + x + 5$

$$\therefore f'(x) = (4a-7)x^2 + 2(a-3)x + 1$$

For one-one polynomial function, $f(x)$ is strictly increasing / decreasing in its entire domain.

$\Rightarrow f'(x)$ doesn't change sign throughout R

$\Rightarrow f'(x)$ has no real roots

$\Rightarrow D < 0$

$$\Rightarrow D = \{2(a-3)\}^2 - 4(4a-7)(1) < 0 \Rightarrow a^2 - 6a + 9 - 4a + 7 < 0$$

$$\Rightarrow a^2 - 10a + 16 < 0 \Rightarrow (a^2 - 8)(a - 2) < 0 \Rightarrow a \in (2, 8)$$

But $a \in [\lambda, \mu] \Rightarrow \lambda = 2, \mu = 8$

$$\therefore |\lambda - \mu| = |2 - 8| = 6$$

32.(5) Here, $f(x) - 10x = (x-1)(x-2)(x-3)(x-\alpha)$

$$\therefore f(12) = 120 + (11 \times 10 \times 9)(12 - \alpha)$$

$$f(-8) = -80 + (-9) + (-10) \times (-11)(-8 - \alpha)$$

$$\Rightarrow f(12) + f(-8) = 40 + (9 \times 10 \times 11) \times 20 = 19840 = (3968)5$$

$$\therefore \lambda = 5$$

33.(4) We have, $f(x) = 2\cos^2 x + x\sqrt{3}\sin 2x + 1$

$$\Rightarrow f(x) = 1 + \cos 2x + \sqrt{3}\sin 2x + 1$$

$$\Rightarrow f(x) = 2 + 2\sin\left(2x + \frac{\pi}{6}\right)$$

$$\therefore \text{Range of } f(x) = [0, 4]$$

Hence, $f^{-1}(x)$ exists, if $B \in [0, 4]$, but $B \in [a, b]$

$$\text{Now, } (a+b) = (0+4) = 4$$

34.(1) We have, $f(x+y) = f(x) + f(y) - xy - 1$ and $f(1) = 1$

On putting $y = 1$, we get :

$$f(x+1) = f(x) + f(1) - x - 1$$

$$\Rightarrow f(n+1) = f(n) - n < f(n)$$

$$\Rightarrow f(n+1) < f(n)$$

$$\text{So, } f(n) < f(n-1) < f(n-2) \dots < f(3) < f(2) \dots < f(1) < 1$$

$$\therefore f(n) = n \text{ holds only for } n = 1$$

35.(2) We have, $f(x) = nx + n - [nx + n] + \tan \frac{\pi x}{2}$

Period of $nx + n - [nx + n]$ or $\{nx + n\}$ is $\frac{1}{n}$ $\{\{nx + n\} = \{nx\}\}$ and $\{nx\}$ has period $\frac{1}{n}$

And period of $\tan \frac{\pi x}{2}$ is $\frac{\pi}{\left(\frac{\pi}{2}\right)} = 2$; Period of $f(x) = \frac{\text{LCM of 1 and 2}}{\text{HCF of } n \text{ and 1}}$

$$\therefore \text{Period of } f(x) = 2$$

36.(18) $f(x) - g(x)$ is quadratic equation with only 1 root $x = -1$

$$\Rightarrow p(x) = k(x+1)^2 \text{ for some constant } k$$

$$p(-2) = 2$$

$$\Rightarrow 2 = k \Rightarrow p(x) = 2(x+1)^2 \Rightarrow p(2) = 18$$

37.(7) Polynomial function satisfying

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = \pm x^n + 1$$

$$\therefore f(3) = \pm 3^n + 1 = -26$$

$$\Rightarrow -3^n + 1 = -26 \Rightarrow -3^n = -27$$

$$\therefore n = 3$$

$$\Rightarrow f(x) = -x^3 + 1 \Rightarrow f(2) = -8 + 1 \Rightarrow f(2) = -7$$

$$\text{Hence, } |f(2)| = 7$$

38.(4) We have, $f(x+y) = f(x)f(y)$, $\forall x, y \in N$ and $f(1) = 2$

$$\therefore \sum_{n=1}^{10} f(n) = f(1) + f(2) + f(3) + f(4) + f(5) + f(6) + f(7) + f(8) + f(9) + f(10)$$

$$= f(1) + f(1+1) + f(1+2) + f(1+3) + f(1+4) + f(1+5) + f(1+6) + f(1+7) + f(1+8) + f(1+9)$$

$$= f(1) + f(1) \cdot f(1) + f(1) \cdot f(1) \cdot f(1) + \dots + \{f(x)\}^{10} = f(1) + \{f(1)\}^2 + \{f(1)\}^3 + \{f(1)\}^4 + \dots + \{f(1)\}^{10}$$

$$= 2 + 2^2 + 2^3 + 2^4 + \dots + 2^{10} = \frac{2(2^{10} - 1)}{2 - 1} = 2^{11} - 2 = 2048 - 2 = 2046$$

$$\therefore \left| \sum_{n=1}^{10} f(n) - 2050 \right| = |2046 - 2050| = 4$$

39.(8) Let coordinates of point A and B be $(a, 0)$ and $(0, b)$ respectively

$$\text{Area of } \Delta(OAB) = \frac{1}{2}|ab| = 16$$

$$\Rightarrow |ab| = 32$$

$$\text{Also, } f(a) = 0 \Rightarrow f^{-1}(0) = a$$

$$f(0) = b \Rightarrow f^{-1}(b) = 0$$

$$\Rightarrow y = f^{-1}(x) \text{ intersects coordinate axes at}$$

$$A' : (b, 0) \text{ and } B' : (0, a)$$

$$OC = \sqrt{a^2 + b^2}$$

$$a^2 + b^2 \geq 2|ab|$$

$$\Rightarrow a^2 + b^2 \geq 64 \Rightarrow \sqrt{a^2 + b^2} \geq 8$$

40.(1) Given, $f(x) = \frac{\alpha x}{x+1}, x \neq -1$

$$\text{and } f\{f(x)\} = x \Rightarrow \frac{\alpha \left(\frac{\alpha x}{x+1} \right)}{\frac{\alpha x}{x+1} + 1} = x$$

$$\Rightarrow \frac{\alpha^2 x}{(\alpha + 1)x + 1} = x$$

$$\Rightarrow (\alpha + 1)x^2 + (1 - \alpha^2)x = 0 \quad \dots\dots(i)$$

$$\therefore \alpha + 1 = 0 \text{ and } 1 - \alpha^2 = 0 \quad [\text{as it is true, } \forall x \neq -1 \text{ Eq.(1) in an identity}]$$

$$\therefore \alpha = -1 \Rightarrow |\alpha| = 1$$

Differential Calculus-1

1.(B) If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$, then $\lim_{x \rightarrow a} (1 + f(x))^{g(x)}$ {From 1^∞ }

$$= e^{\lim_{x \rightarrow a} f(x)g(x)}$$

$$\therefore \lim_{x \rightarrow \infty} \left(1 + \left(\frac{a}{x} - \frac{4}{x^2} \right) \right)^{2x} \quad \text{{From } 1^\infty \text{}}$$

$$= e^{\lim_{x \rightarrow \infty} \left(\frac{a}{x} - \frac{4}{x^2} \right) 2x} = e^{\lim_{x \rightarrow \infty} \left(2a - \frac{8}{x} \right)} = e^{2a}$$

$$\therefore e^{2a} = e^3 \text{ (given)} \Rightarrow a = 3/2$$

2.(C) Given, $f(x)$ is differentiable at $x = 1$

$$\Rightarrow f(x) \text{ is also continuous at } x = 1 \Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$\Rightarrow \lim_{x \rightarrow 1^-} (-x) = \lim_{x \rightarrow 1^+} a + \cos^{-1}(x+b) = a + \cos^{-1}(1+b)$$

$$\Rightarrow -1 = a + \cos^{-1}(1+b) \quad \dots\dots (i)$$

Also, $f(x)$ is differentiable at $x = 1$

$$\therefore L f'(1) = R f', \text{ where } L f'(1) \text{ is left hand derivative at } x = 1$$

$$\Rightarrow \lim_{x \rightarrow 1^-} (-1) = \lim_{x \rightarrow 1^+} \frac{-1}{\sqrt{1-(x+b)^2}} \Rightarrow -1 = \frac{-1}{\sqrt{1-(1+b)^2}} \Rightarrow \sqrt{1-(1+b)^2} = 1$$

$$\Rightarrow 1-(1+b)^2 = 1 \Rightarrow 1+b = 0 \Rightarrow b = -1$$

$$\text{From (i), } -1 = a + \frac{\pi}{2} \Rightarrow a = -\left(\frac{\pi}{2} + 1\right) \therefore \frac{a}{b} = \frac{\pi+2}{2}.$$

3.(D) $\lim_{t \rightarrow x} \frac{t^2 g(x) - x^2 f(t)}{t - x} = 1 \quad \left[\text{Form } \frac{0}{0} \right]$

Different w.r.t t numerator and denominator

$$\lim_{t \rightarrow x} \frac{2t f(x) - x^2 f'(t)}{1} = 1$$

$$2x f(x) - x^2 f'(x) = 1 \Rightarrow 2xy - x^2 \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} - \frac{2}{x} y = -\frac{1}{x^2}$$

$$\text{I.F.} = e^{-\int \frac{2}{x} dx} = e^{-2 \log x} = e^{\log x^{-2}} = \frac{1}{x^2}$$

$$\therefore \frac{y}{x^2} = -\int \frac{1}{x^4} dx = \frac{1}{3x^3} + C$$

$$y(1) = 1 \Rightarrow x = 1, y = 1 \text{ gives } C = \frac{2}{3}$$

$$\therefore \frac{y}{x^2} = \frac{1}{3x^3} + \frac{2}{3} \Rightarrow y = \frac{1}{3x} + \frac{2x^2}{3}$$

$$\therefore f(x) = \frac{1}{3x} + \frac{2x^2}{3} \Rightarrow f\left(\frac{3}{2}\right) = \frac{31}{18}$$

4.(A) $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)^2}{2x \tan x - x \tan 2x}$

$$\lim_{x \rightarrow 0} \frac{(2 \sin^2 x)^2}{2x \tan x - \frac{2x \tan x}{1 - \tan^2 x}}$$

$$\lim_{x \rightarrow 0} \frac{-4 \sin^4 x (1 - \tan^2 x)}{2x \tan^3 x}$$

$$\lim_{x \rightarrow 0} -2 \frac{\sin x}{x} \cos^3 x (1 - \tan^2 x) = -2$$

5.(A) Since $f(x)$ is continuous in $[0, \infty)$,

$\therefore f(x)$ is continuous at $x = 1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$\lim_{x \rightarrow 1^-} \frac{2x^2}{a} = \lim_{x \rightarrow 1^+} a = a$$

$$\frac{2}{a} = a \Rightarrow a^2 = 2 \Rightarrow a = \pm \sqrt{2}$$

Also, $f(x)$ is continuous at $x = \sqrt{2}$

$$\lim_{x \rightarrow \sqrt{2}^-} g(x) = \lim_{x \rightarrow \sqrt{2}^+} f(x) = f(\sqrt{2})$$

$$\lim_{x \rightarrow \sqrt{2}^-} a = \lim_{x \rightarrow \sqrt{2}^+} \frac{2b^2 - 4b}{x^3} = \frac{2b^2 - 4b}{(\sqrt{2})^3} \Rightarrow a = \frac{2b^2 - 4b}{(\sqrt{2})^3} = \frac{2b^2 - 4b}{(\sqrt{2})^3}$$

$$2b^2 - 4b = 2\sqrt{2}a \Rightarrow b^2 - 2b = \sqrt{2}a$$

(i) When $a = -\sqrt{2}$, $b^2 - 2b + 2 = 0$

No real root of b as $D < 0$

(ii) When $a = \sqrt{2}$, $b^2 - 2b - 2 = 0$

$$b = 1 \pm \sqrt{3}$$

$$\therefore (a, b) = (\sqrt{2}, 1 + \sqrt{3}) \text{ or } (\sqrt{2}, 1 - \sqrt{3})$$

6.(B) $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{(e^x - 1) + (1 + \cos x)}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{\sin^2 x} + \frac{1 - \cos x}{\sin^2 x} \times \frac{1 + \cos x}{1 + \cos x}$

$$= \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1)/x^2}{\sin^2 x / x^2} + \frac{\sin^2 x}{\sin^2 x} \times \frac{1}{1 + \cos x} = \frac{1}{1} + \frac{1}{1 + 1} = \frac{3}{2} \quad \left\{ \text{Use } \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right\}$$

$$7.(C) \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Here, $f(x)$ is continuous at $x = 0$

$$\therefore \quad \lim_{x \rightarrow 0} f(x) = f(0) \Rightarrow \lim_{x \rightarrow 0} \frac{(e^x - 1)^2}{\sin\left(\frac{x}{k}\right) \log\left(1 + \frac{x}{4}\right)} = 12$$

$$\Rightarrow \quad \lim_{x \rightarrow 0} \frac{\left(\frac{e^x - 1}{x}\right)^2 \cdot x^2}{\left(\frac{\sin \frac{x}{k}}{\frac{x}{k}}\right) \cdot \frac{x}{k} \cdot \left(\frac{\log\left(1 + \frac{x}{4}\right)}{\frac{x}{4}}\right) \cdot \frac{x}{4}} = 12 \Rightarrow \frac{1^2 \cdot 4k}{1 \times 1} = 12 \Rightarrow k = 3$$

$$8.(B) \quad \lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right) = \lim_{x \rightarrow 0} f\left(\frac{(1 - \cos 3x)(1 + \cos 3x)}{x^2(1 + \cos 3x)}\right) = \lim_{x \rightarrow 0} f\left(\frac{\sin^2 3x}{x^2(1 + \cos 3x)}\right) = f\left(\frac{9}{2}\right) = \frac{2}{9}$$

$$9.(D) \quad y = e^{nx} \Rightarrow \frac{dy}{dx} = ne^{nx} \Rightarrow \frac{d^2y}{dx^2} = n^2 e^{nx}$$

$$\log y = \log e^{nx} = nx$$

$$\therefore \quad \frac{dx}{dy} = \frac{1}{ny} \Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{n} \cdot \frac{1}{y^2}$$

$$\therefore \quad \left(\frac{d^2x}{dx^2}\right) \left(\frac{dx}{dy}\right)^2 = n^2 e^{nx} \left(-\frac{1}{n} \cdot \frac{1}{y^2}\right) = -ne^{-nx}.$$

$$10.(D) \quad \lim_{x \rightarrow 2} \frac{\tan(x-2) \cdot [(x-2)(x+k)]}{(x-2)^2} = 5 \Rightarrow 1 \cdot (2+k) = 5 \Rightarrow k = 3$$

$$11.(C) \quad f(x) = x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$$

$$g(x) = \sin x$$

$$h(x) = (g \circ f)(x) = g(g(f(x))) = \begin{cases} \sin x^2, & x \geq 0 \\ \sin(-x^2) = -\sin x^2 & x < 0 \end{cases}$$

Obviously, $h(x)$ is continuous at $x = 0$

$$\therefore \quad h'(x) = \begin{cases} 2x \cos x^2, & x \geq 0 \\ -2x \cos x^2, & x < 0 \end{cases}$$

$$\lim_{x \rightarrow 0^+} h'(x) = \lim_{x \rightarrow 0^-} h'(x) = h'(0) = 0$$

$$h'(x) = \lim_{x \rightarrow 0} h'(x) = h'(0) = 0$$

$h'(x)$ is continuous at $x = 0$

$$\text{Again, } h''(x) = \begin{cases} 2 \cos x^2 - 4x^2 \sin x^2, & x \geq 0 \\ -2 \cos x^2 + 4x^2 \sin x^2, & x < 0 \end{cases}$$

$$Rh''(0) = 2 \text{ and } Lh''(0) = -2$$

$$\therefore \text{R.H.D} \neq \text{L.H.D. at } x = 0$$

$$\therefore h'(x) \text{ is not differentiable at } x = 0$$

12.(C) $g(x)$ is inverse of $f(x)$

$$\therefore f(g(x)) = x \Rightarrow f'(g(x)) \cdot g'(x) = 1$$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))} \Rightarrow g'(7) = \frac{1}{f'(g(7))} \dots\dots (i)$$

$$f'(x) = 2x - 1 \dots\dots (ii)$$

To find pre-image of 7 in f , put $f(x) = 7$

$$\Rightarrow x^2 - x + 5 = 7 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x = 2 \text{ as } x > 1/2$$

$$\therefore f(2) = 7 \Rightarrow f^{-1}(7) = 2 \Rightarrow g(7) = 2 \quad (\because f^{-1} = g).$$

$$\text{From (i), } g'(7) = \frac{1}{f'(2)} = \frac{1}{3} \quad [\text{using (ii)}]$$

13.(B) Statement I : $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 = f(0)$

$$\therefore f \text{ is cont. at } x = 0$$

$$\text{Statement II : } g(x) = x f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$f(x)$ is diff. at $x = a$ if $Rf'(a) = Lf'(a)$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}$$

$$\therefore Rg'(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{1}{h}\right) - 0}{h} = 0$$

$$Lg'(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin\left(-\frac{1}{h}\right) - 0}{-h} = 0$$

$$\therefore Rg'(0) = Lg'(0) \Rightarrow g \text{ is diff. at } x = 0$$

14.(D) $f(x)$ is continuous at $x = \pi$

$$\therefore \lim_{x \rightarrow \pi^+} f(x) = \lim_{x \rightarrow \pi^-} f(x) = f(\pi)$$

$$\begin{aligned} \lim_{x \rightarrow \pi^+} f(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{2 + \cos(\pi + h)} - 1}{(\pi - (\pi + h))^2} = \lim_{h \rightarrow 0} \frac{\sqrt{2 - \cos h} - 1}{h^2} \times \frac{\sqrt{2 - \cos h} + 1}{\sqrt{2 - \cos h} + 1} = \lim_{h \rightarrow 0} \frac{(2 - \cos h) - 1}{h^2 (\sqrt{2 - \cos h} + 1)} \\ &= \lim_{h \rightarrow 0} \frac{1 - \cos h}{(\sqrt{2 - \cos h} + 1)} \times \frac{1 + \cos h}{1 + \cos h} = \lim_{h \rightarrow 0} \frac{\sin^2 h}{h^2} \cdot \frac{1}{(\sqrt{2 - \cos h} + 1)} \cdot \frac{1}{1 + \cos h} = \frac{1}{4} \end{aligned}$$

15.(B) $f(x) \leq x^2$, for all $x \in R$

Obviously, f is continuous as well as differentiable at $x = 0$.

16.(B) $\lim_{x \rightarrow 0} \frac{1}{x} \left[\tan^{-1} \left(\frac{x+1}{2x+1} \right) - \tan^{-1} 1 \right]$

$$\lim_{x \rightarrow 0} \frac{1}{x} \tan^{-1} \left(\frac{\frac{x+1}{2x+1} - 1}{1 + \frac{x+1}{2x+1} \cdot 1} \right) = \lim_{x \rightarrow 0} \frac{1}{x} \tan^{-1} \left(\frac{-x}{3x+2} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x} \cdot \frac{\tan^{-1} \left(\frac{-x}{3x+2} \right)}{\frac{-x}{3x+2}} \times \frac{-x}{3x+2} = 1 \cdot \left(\frac{-1}{2} \right).$$

$$\left[\because \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1 \right]$$

17.(D) First find $\frac{dy}{dx} = -\frac{y}{x} \quad \therefore \quad 1 + \left(\frac{dy}{dx} \right)^2 = \frac{x^2 + y^2}{x^2}$

18.(D) $g(x) = \begin{cases} 1-x, & x \geq 0 \\ 1+x, & x < 0 \end{cases}$

$$f(x) = \begin{cases} -1+x-2 = x-3, & x \geq 2 \\ -1-x+2 = 1-x, & x < 2 \end{cases}$$

$$fog(x) = \begin{cases} f(1-x), & x \geq 0 \\ f(1+x), & x < 0 \end{cases} = \begin{cases} 1-(1-x) = x, & x \geq 0 \\ 1-(1+x) = -x, & x < 0 \end{cases}$$

fog is continuous everywhere. So, fog is not discontinuous at any point.

19.(D) $f(x) = \sin(\sin x) \Rightarrow f'(x) = \cos(\sin x) \cos x$

$$f''(x) = -\sin(\sin x) \cos^2 x - \cos(\sin x) \sin x$$

Now $f''(x) + \tan x f'(x) + g(x) = 0$

$$\Rightarrow -\sin(\sin x) \cos^2 x - \cos(\sin x) \sin x + \tan x \cos(\sin x) \cos x + g(x) = 0$$

$$\Rightarrow -\sin(\sin x) \cos^2 x - \cos(\sin x) \sin x + \sin x \cos(\sin x) + g(x) = 0$$

$$\Rightarrow g(x) = \sin(\sin x) \cos^2 x$$

20.(B) A composite function is continuous if both the functions are continuous.

$$u(x) = \frac{1}{x-1}, x \neq 1$$

$$\therefore f(u) = \frac{1}{u^2 + u - 2} = \frac{1}{(u+2)(u-1)} = \frac{1}{\left(\frac{1}{x-1} + 2\right)\left(\frac{1}{x-1} - 1\right)} = \frac{(x-1)^2}{(2x-1)(2-x)}, x \neq 2, x \neq \frac{1}{2}$$

So, f is discontinuous at $x = 1/2, 1, 2$.

21.(C) $\lim_{x \rightarrow 1^+} \left(\{x\} \right)^{\frac{1}{\ln x}} \Rightarrow \lim_{h \rightarrow 0} \{1+h\}^{\frac{1}{\ln(1+h)}}$

$$\lim_{h \rightarrow 0} \{h\}^{\frac{1}{\ln(1+h)}} = 0 \quad (0^\infty \text{ form which is not an indeterminate form})$$

$$\begin{aligned}
 22.(C) \quad \lim_{x \rightarrow \infty} \left\{ \cos(\sqrt{x+1}) - \cos(\sqrt{x}) \right\} &= \lim_{x \rightarrow \infty} 2 \sin\left(\frac{\sqrt{x} + \sqrt{x+1}}{2}\right) \cdot \sin\left(\frac{\sqrt{x} - \sqrt{x+1}}{2}\right) \\
 &= \lim_{x \rightarrow \infty} 2 \sin\left(\frac{\sqrt{x} + \sqrt{x+1}}{2}\right) \cdot \sin\left(\frac{x - x - 1}{2 \cdot (\sqrt{x} + \sqrt{x+1})}\right) \\
 &= 2 \lim_{x \rightarrow \infty} \sin\left(\frac{\sqrt{x} + \sqrt{x+1}}{2}\right) \cdot \lim_{x \rightarrow \infty} \sin\left[\frac{-1}{2 \cdot (\sqrt{x} + \sqrt{x+1})}\right] = 2 \times (\text{oscillating } -1 \text{ to } 1) \times 0 = 0
 \end{aligned}$$

$$\begin{aligned}
 23.(C) \quad \lim_{x \rightarrow \infty} \left((x+1)^{2/3} - (x-1)^{2/3} \right) &= \lim_{x \rightarrow \infty} \frac{\left[(x+1)^{2/3} - (x-1)^{2/3} \right] \left[(x+1)^{4/3} + (x+1)^{2/3}(x-1)^{2/3} + (x-1)^{4/3} \right]}{\left[(x+1)^{4/3} + (x+1)^{2/3}(x-1)^{2/3} + (x-1)^{4/3} \right]} \\
 &= \lim_{x \rightarrow \infty} \frac{(x+1)^2 - (x-1)^2}{\left[(x+1)^{4/3} + (x+1)^{2/3}(x-1)^{2/3} + (x-1)^{4/3} \right]} \\
 &= \lim_{x \rightarrow \infty} \frac{4x}{(x+1)^{4/3} \left[1 + \left(\frac{x-1}{x+1} \right)^{2/3} + \left(\frac{x-1}{x+1} \right)^{4/3} \right]} \\
 &= \lim_{x \rightarrow \infty} \frac{4x}{(x+1)(x+1)^{1/3} \left[1 + \left(\frac{x-1}{x+1} \right)^{2/3} + \left(\frac{x-1}{x+1} \right)^{4/3} \right]} \quad [\text{Divide } N^r \text{ and } D^r \text{ by } x] \\
 &= \frac{4}{(1+0) \times (\infty) \times [1+1+1]} = 0
 \end{aligned}$$

$$\begin{aligned}
 24.(A) \quad \lim_{n \rightarrow \infty} \frac{\sqrt{n^3 - 2n^2 + 1} + \sqrt[3]{n^4 + 1}}{\sqrt[4]{n^6 + 6n^5 + 2} - \sqrt[5]{n^7 + 3n^3 + 1}} &= \lim_{n \rightarrow \infty} \frac{n^{3/2} \sqrt{1 - \frac{2}{n} + \frac{1}{n^3}} + n^{4/3} \sqrt[3]{1 + \frac{1}{n^4}}}{n^{3/2} \sqrt{1 + \frac{6}{n} + \frac{2}{n^6}} - n^{7/5} \sqrt[5]{1 + \frac{3}{n^4} + \frac{1}{n^7}}} \quad [\text{Divide } N^r \text{ and } D^r \text{ by } n^{3/2}] \\
 &= \lim_{n \rightarrow \infty} \frac{\sqrt{1 - \frac{2}{n} + \frac{1}{n^3}} + n^{-1/6} \sqrt[3]{1 + \frac{1}{n^4}}}{\sqrt{1 + \frac{6}{n} + \frac{2}{n^6}} - n^{-1/10} \sqrt[5]{1 + \frac{3}{n^4} + \frac{1}{n^7}}} = \frac{1+0}{1-0} = 1
 \end{aligned}$$

$$\begin{aligned}
 25.(B) \quad \lim_{x \rightarrow \infty} \left(((x+1)(x+2)(x+3)(x+4))^{1/4} - x \right) &= \lim_{x \rightarrow \infty} x \left(\left(1 + \frac{1}{x} \right) \left(1 + \frac{2}{x} \right) \left(1 + \frac{3}{x} \right) \left(1 + \frac{4}{x} \right) \right)^{1/4} - x \\
 &= \lim_{x \rightarrow \infty} x \left[1 + \frac{10}{x} + \frac{\sum 1.2}{x^2} + \frac{\sum 1.2.3}{x^3} + \frac{24}{x^4} \right]^{1/4} - x \\
 &= \lim_{x \rightarrow \infty} x \left[1 + \frac{\left(\frac{10}{x} + \frac{\sum 1.2}{x^2} + \frac{\sum 1.2.3}{x^3} + \frac{24}{x^4} \right)}{4} + \dots \right] - x \quad \left[\text{Using the series: } (1+x)^n = \left[1 + nx + \frac{n(n-1)}{2} x^2 + \dots \right] \right] \\
 &= \frac{10}{4} = \frac{5}{2}
 \end{aligned}$$

$$26.(A) \quad \lim_{x \rightarrow 1^-} ([x])^{1-x} = \lim_{h \rightarrow 0^+} [1-h]^{1-(1-h)} = \lim_{h \rightarrow 0^+} (0)^h$$

27.(C) $\lim_{x \rightarrow 2} \frac{(x+2)^{1/2} - (15x+2)^{1/5}}{(7x+2)^{1/4} - x}$ Let $x = 2 + h$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{(4+h)^{1/2} - (32+15h)^{1/5}}{(16+7h)^{1/4} - (2+h)} = \lim_{h \rightarrow 0} \frac{2 \left[1 + \frac{h}{4} \right]^{1/2} - 2 \left[1 + \frac{15h}{32} \right]^{1/5}}{2 \left[1 + \frac{7h}{16} \right]^{1/4} - (2+h)} \\
 &= \lim_{h \rightarrow 0} \frac{2 \left[1 + \frac{h}{8} + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2} \left(\frac{h^2}{16}\right) + \dots \right] - 2 \left[1 + \frac{3h}{32} + \frac{\left(\frac{1}{5}\right)\left(-\frac{4}{5}\right)}{2} \left(\frac{15h}{32}\right)^2 + \dots \right]}{2 \left[1 + \frac{7h}{64} + \dots \right] - (2+h)} \\
 &= \lim_{h \rightarrow 0} \frac{h \left(\frac{1}{4} - \frac{3}{16} \right) + h^2 \left(-\frac{1}{64} - \frac{9}{256} \right) + \dots}{h \left(\frac{7}{32} - 1 \right) + \dots} = \frac{\frac{1}{4} - \frac{3}{16}}{\frac{7}{32} - 1} = -\frac{2}{25}
 \end{aligned}$$

28.(A) $f(x) = (x+2)(x-2)(x-3)$

$$h(x) = \begin{cases} (x+2)(x-2) & , x \neq 3 \\ k & , x = 3 \end{cases}$$
 for continuity $k = \lim_{x \rightarrow 3} h(x) = 5$
 $h(x) = (x+2)(x-2) = x^2 - 4$ which is even $\forall x \in R$.

29.(A) $f(x)$ is continuous at $x = 1 \Rightarrow a - b = -1$

L.H.D. = $\lim_{h \rightarrow 0} \frac{a(1-h)^2 - b + 1}{-h} = 2a$ R.H.D. = $\lim_{h \rightarrow 0} \frac{\frac{-1}{1+h} + 1}{h} = 1$ It gives $\Rightarrow a = \frac{1}{2}, b = \frac{3}{2}$

30.(A) $f: R \rightarrow R$ and $f(x+y) = f(x)f(y) \forall x, y \in R$

Put $x = y = 0$ $f(0) = f^2(0)$ Since $f(0) \neq 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+y) - f(x)}{h} = \lim_{h \rightarrow 0} f(x) \frac{[f(h) - 1]}{h} = f(x) f'(0)$$

Let $f(x) = y \Rightarrow \frac{dy}{dx} = y \cdot f'(0)$

On solving $\ln y = x f'(0) + c$

$y = f(x) = e^c \cdot e^{x f'(0)} \because f(0) = 1 \Rightarrow c = 0$. Thus $f(x) = e^{x \cdot f'(0)} \forall x \in R$

31.(B) $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin\left(\frac{x}{p}\right) \ln\left(1 + \frac{x^2}{3}\right)} = \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{4^x - 1}{x}\right)^3}{\left(\frac{x}{p}\right) \left[\frac{\sin\left(\frac{x}{p}\right)}{\frac{x}{p}}\right] \cdot \ln\left(1 + \frac{x^2}{3}\right)} = 3p \cdot \lim_{x \rightarrow 0} \frac{\left(\frac{4^x - 1}{x}\right)}{\left[\frac{\sin\left(\frac{x}{p}\right)}{\frac{x}{p}}\right]} \cdot \frac{\left(\frac{x^2}{3}\right)}{\ln\left(1 + \frac{x^2}{3}\right)} = 3p(\ln 4)^3$

32.(B)

$$\begin{aligned}
 33.(C) \quad \lim_{x \rightarrow \pi/2} \tan^2 x \left(\sqrt{2 \sin^2 x + 3 \sin x + 4} - \sqrt{\sin^2 x + 6 \sin x + 2} \right) \\
 = \lim_{x \rightarrow \pi/2} \tan^2 x \left(\frac{2 \sin^2 x + 3 \sin x + 4 - (\sin^2 x + 6 \sin x + 2)}{\sqrt{2 \sin^2 x + 3 \sin x + 4} + \sqrt{\sin^2 x + 6 \sin x + 2}} \right) = \lim_{x \rightarrow \pi/2} \tan^2 x \cdot \frac{(\sin^2 x - 3 \sin x + 2)}{\sqrt{2+3+4} + \sqrt{1+6+2}} \\
 = \lim_{x \rightarrow \pi/2} \frac{1}{6} \left[\frac{\sin^2 x - 3 \sin x + 2}{\cos^2 x} \right] \quad \left(\frac{0}{0} \text{ form} \right) \quad (\text{Use L'Hospital rule}) \\
 = \frac{1}{6} \lim_{x \rightarrow \pi/2} \frac{2 \sin x \cos x - 3 \cos x}{2 \cos x (-\sin x)} = \frac{1}{6} \lim_{x \rightarrow \pi/2} \frac{2 \sin x - 3}{-2 \sin x} = \left(\frac{1}{6} \right) \left(\frac{1}{2} \right) = \frac{1}{12}
 \end{aligned}$$

$$34.(D) \quad \lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} = \lim_{n \rightarrow \infty} \frac{5 \cdot 5^n + 3^n - 4^n}{5^n + 2^n + 27 \cdot 9^n} = \lim_{n \rightarrow \infty} \frac{5 \cdot \left(\frac{5}{9} \right)^n + \left(\frac{3}{9} \right)^n - \left(\frac{4}{9} \right)^n}{\left(\frac{5}{9} \right)^n + \left(\frac{2}{9} \right)^n + 27} = \frac{0+0-0}{0+0+27} = 0$$

$$35.(AB) (A) \quad \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{-3x-6} = \lim_{x \rightarrow -\infty} \frac{\sqrt{1+\frac{2}{x^2}}}{-3-\frac{6}{x}} = -\frac{1}{3} \quad (B) \quad \lim_{x \rightarrow \infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x \rightarrow \infty} \frac{\sqrt{1+\frac{2}{x^2}}}{3-\frac{6}{x}} = \frac{1}{3}$$

36.(A)

$$37.(C) \quad \lim_{x \rightarrow 0} \frac{e^{-x^2/2} - \cos x}{x^3 \sin x} = \lim_{x \rightarrow 0} \frac{\left[1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots \right] - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right]}{x^3 \left[x - \frac{x^3}{3!} + \dots \right]} = \lim_{x \rightarrow 0} \frac{x^4 \left[\frac{1}{8} - \frac{1}{4!} \right] - x^6 \left[\frac{1}{8 \cdot 3!} - \frac{1}{6!} \right] + \dots}{x^4 \left[1 - \frac{x^2}{3!} + \dots \right]} = \frac{1}{8} - \frac{1}{24} = \frac{1}{12}$$

$$38.(A) \quad S_3 : \quad \lim_{x \rightarrow \infty} \sqrt{\frac{x - \sin x}{x + \cos^2 x}} = \lim_{y \rightarrow 0^+} \sqrt{\frac{\frac{1}{y} - \sin \frac{1}{y}}{\frac{1}{y} + \cos^2 \frac{1}{y}}} = \lim_{y \rightarrow 0^+} \sqrt{\frac{1 - y \sin \frac{1}{y}}{1 + y \cos^2 \frac{1}{y}}} = \sqrt{\frac{1-0}{1+0}} = 1$$

$$S_4 : \quad \lim_{n \rightarrow \infty} \frac{(n+2)(n+1)! + (n+1)!}{(n+3)(n+2)(n+1)!} = \lim_{n \rightarrow \infty} \frac{(n+2)+1}{(n+3)(n+2)} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} + \frac{3}{n^2}}{\left(1 + \frac{3}{n} \right) \left(1 + \frac{2}{n} \right)} = 0$$

$$39.(AD) \quad \text{For } \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2}$$

$$\Rightarrow 1 + a = 0 \Rightarrow a = -1$$

$$\text{And for } \lim_{x \rightarrow 0} \frac{b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{b}{x^2} \Rightarrow b = 0$$

$$\text{Now } \ell = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} = \lim_{x \rightarrow 0} \frac{2}{4} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2 = \frac{1}{2} \cdot (1)^2 = \frac{1}{2}$$

$$\therefore (a, b) \equiv (-1, 0) \text{ and } \ell = \frac{1}{2}$$

40.(B)

$$41.(D) \quad \text{Let } \ell = \lim_{n \rightarrow \infty} \frac{1}{n^4} \left(\left[1^3 x \right] + \left[2^3 x \right] + \dots + \left[n^3 x \right] \right) \\
 1^3 x - 1 < \left[1^3 x \right] \leq 1^3 x$$

$$2^3 x - 1 < \left[2^3 x \right] \leq 2^3 x$$

$$\begin{array}{ccc} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ n^3 x - 1 < \left[n^3 x \right] \leq n^3 x \end{array}$$

Adding all these inequalities

$$(1^3 + 2^3 + 3^3 + \dots + n^3)x - n < \left[1^3 x \right] + \left[2^3 x \right] + \dots + \left[n^3 x \right] \leq (1^3 + 2^3 + \dots + n^3)x$$

$$\frac{\frac{n^2(n+1)^2}{4}x - n}{n^4} < \frac{\left[1^3 x \right] + \left[2^3 x \right] + \dots + \left[n^3 x \right]}{n^4} \leq \frac{\left(\frac{n(n+1)}{2} \right)^2 x}{n^4}$$

$$\text{Now } \ell = \lim_{n \rightarrow \infty} \left(\frac{n(n+1)}{2} \right)^2 \cdot \frac{x}{n^4} = \frac{x}{4}$$

42.(C)

43.(D)

44.(C)

45.(B)

46.(D)

47.(B)

$$48.(\text{ABC}) \quad f(x) = \begin{cases} |x-3| & , x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4} & , x < 1 \end{cases}$$

Clearly it is continuous at $x = 1$ as $f(1^-) = f(1^+) = f(1)$

At $x = 3$ $f(3^-) = f(3^+) = f(3) = 0$ it is continuous at $x = 3$

$$\text{Now RHD at } x = 1 = f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = -1$$

$$\begin{aligned} \text{Now LHD at } x = 1 = f'(1^-) &= \lim_{h \rightarrow 0} \frac{(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{-h^2 - 2h + 1 - 6 + 6h + 13 - 8}{-4h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 + 4h}{-4h} = \lim_{h \rightarrow 0} \frac{h+4}{-4} = -1 \end{aligned} \quad \text{Hence diff. at } x = 1$$

$$49.(\text{B}) \quad \lim_{x \rightarrow 2^-} f(x) = \frac{3}{5} = f(2) \neq \lim_{x \rightarrow 2^+} f(x) = 1 \quad f(x) \text{ is not continuous at } x = 2$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3) = \frac{9}{2}$$

$$\text{Now } f'(3^-) = \lim_{h \rightarrow 0} \frac{\frac{1}{4}((3-h)^3 - (3-h)^2) - \frac{9}{2}}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - 8h + 21}{4} = \frac{21}{4}$$

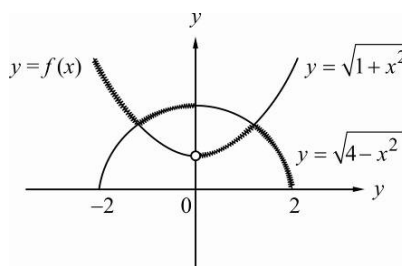
$$\text{and } f'(3^+) = \lim_{h \rightarrow 0} \frac{\frac{9}{4}(|h-1| + |-1-h|) - \frac{9}{2}}{h} = 0$$

$f(x)$ is not differentiable at $x = 2$ and $x = 3$.

50.(B)

51.(BD)

- 52.(D) It is clear from the graph that it is not differentiable at more than one point.



- 53.(B) 54.(C)

55.(D) $f(x+2y) = f(x) + f(2y) + 4xy \quad \forall x, y \in \mathbb{R} \xrightarrow{2y \rightarrow y} f(x+y) = f(x) + f(y) + 2xy \quad \forall x, y \in \mathbb{R}$

Partially Diff. w.r.t. x we get :

$$f'(x+y) = f'(x) + 2y \xrightarrow{x=1, y=-1} f'(0) = f'(1) - 2$$

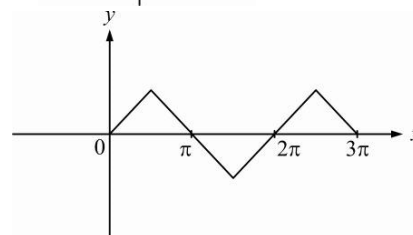
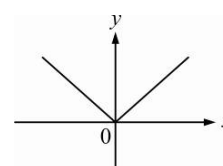
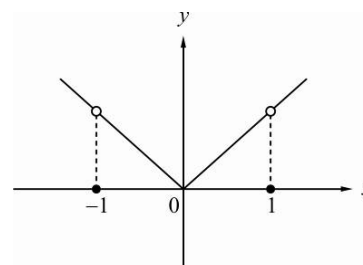
56.(A) S₁ : $f(x) = \begin{cases} -x & , x \in (-\infty, -1) \cup (-1, \infty) \\ 0 & , x = -1, 0, 1 \\ x & , x \in (0, 1) \cup (1, \infty) \end{cases}$

Function is discontinuous at $x = -1, 1$ and non-differentiable at $x = -1, 0, 1$

S₃ : $f(x) = (x^2 |x|)^{1/3} = \begin{cases} (x^3)^{1/3} x & , x \geq 0 \\ (-x^2)^{1/3} = (-1)^{1/3} x = -x & , x < 0 \end{cases}$
 $= -x$

$f(x)$ is differentiable every where except at $x = 0$

- S₄ : $f(x)$ will be non differentiable if $\sin^{-1}(\sin x) = 0$ or graph of $f(x)$ has a sharp point. Hence number of points of non-differentiable will be 5.



- 57.(B) 58.(C) 59.(A)

60.(A) $\lim_{x \rightarrow \infty} (\sqrt{x^4 + ax^3 + 3x^2 + bx + 2} - \sqrt{x^4 + 2x^3 - cx^2 + 3x - d}) = 4$

$$\Rightarrow \lim_{y \rightarrow 0^+} \left\{ \frac{\sqrt{1+ay+3y^2+by^3+2y^4} - \sqrt{1+2y-cy^2+3y^3-dy^4}}{y^2} \right\} = 4 \quad [\text{Put } x = \frac{1}{y}, x \rightarrow \infty, y \rightarrow 0^+]$$

$$\Rightarrow \lim_{y \rightarrow 0^+} \frac{(a-2)y + (3+c)y^2 + (b-3)y^3 + (2+d)y^4}{2y^2} = 4$$

$$\Rightarrow a-2=0 \quad \text{and} \quad \frac{3+c}{2} = 4 \Rightarrow c=5 \Rightarrow a=2, b \in \mathbb{R}, c=5, d \in \mathbb{R}$$

- 61.(B)

62.(C) $\lim_{x \rightarrow 0} \left[\sin^2 \left(\frac{\pi}{2-ax} \right) \right]^{\sec^2 \left(\frac{\pi}{2-bx} \right)} \quad (1^\infty \text{ form})$

$$= e^{\lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{\pi}{2-ax} \right) - 1 \right) \sec^2 \left(\frac{\pi}{2-bx} \right)} = e^{\lim_{x \rightarrow 0} \frac{-\cos^2 \left(\frac{\pi}{2-ax} \right)}{\cos^2 \left(\frac{\pi}{2-bx} \right)}}$$

$$\text{Applying L'Hospital Rule} = e^{\lim_{x \rightarrow 0} \frac{-\sin \left(\frac{2\pi}{2-ax} \right)}{\sin \left(\frac{2\pi}{2-bx} \right)} \cdot \frac{a\pi(2-bx)^2}{b\pi(2-ax)^2}} = e^{\lim_{x \rightarrow 0} \frac{-\sin \left(\frac{2\pi}{2-ax} \right)}{\sin \left(\frac{2\pi}{2-bx} \right)} \cdot \frac{a\pi}{b\pi}} = e^{\lim_{x \rightarrow 0} \frac{-\cos \left(\frac{2\pi}{2-ax} \right)}{\cos \left(\frac{2\pi}{2-bx} \right)} \cdot \frac{\frac{2a^2\pi^2}{(2-ax)^2}}{\frac{2b^2\pi^2}{(2-bx)^2}}} = e^{-\frac{a^2}{b^2}}$$

63.(A)

64.(B) R.H.L. $= \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\left(\sin^{-1}(1-\{x\}) \cdot \cos^{-1}(1-\{x\}) \right)}{\sqrt{2\{x\}}(1-\{x\})} = \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h) \cdot \cos^{-1}(1-h)}{\sqrt{2h}(1-h)}$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(1-h) \sin^{-1} \sqrt{h(2-h)}}{(1-h)\sqrt{2h}} = \frac{\pi}{2} \lim_{h \rightarrow 0} \frac{\sin^{-1} \sqrt{h(2-h)}}{\sqrt{2h-h^2} \sqrt{2h}} = \lim_{h \rightarrow 0} \frac{\pi}{2} \cdot 1 \cdot \sqrt{1-\frac{h}{2}} = \frac{\pi}{2}$$

L.H.L. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\sin^{-1}(1-\{x\}) \cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}(1-\{x\})} = \lim_{h \rightarrow 0} \frac{\sin^{-1} h \cdot \cos^{-1} h}{\sqrt{2(1-h)h}} = \frac{\pi}{2\sqrt{2}}$

65.(B) Differentiability at $x = 1$

L.H.D. $= f'(1^-) = \lim_{h \rightarrow 0} \frac{\frac{\sin[(1-h)^2] \pi}{(1-h)^2 - 3(1-h) + 8} + a(1-h)^3 + b - (a+b)}{-h}$

L.H.D. $= \lim_{h \rightarrow 0} \frac{a(1-h)^3 - a}{-h} = \lim_{h \rightarrow 0} \frac{3a(1-h)^2}{1} = 3a$

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{2\cos(1+h)\pi + \tan^{-1}(1+h) - a - b}{h} = \lim_{h \rightarrow 0} \frac{(-2\cos \pi h + \tan^{-1}(1+h) - a - b)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-2\cos \pi h + \tan^{-1}(1+h) + 2 - \frac{\pi}{4}}{h} = \lim_{h \rightarrow 0} \frac{2\pi \sin \pi h + \frac{1}{1+(1+h)^2}}{1} = \frac{1}{2}$$

Now $f'(1^-) = f'(1^+) \Rightarrow 3a = \frac{1}{2} \Rightarrow a = \frac{1}{6} \dots \text{(i)}$

Also it is continuous at $x = 1 \Rightarrow -2 + \frac{\pi}{4} = a + b \dots \text{(ii)}$ From (i) and (ii) $b = \frac{\pi}{4} - \frac{13}{6}$

66.(A) $y = \sec(\tan^{-1} x)$

Let $\tan^{-1} x = \theta$

$x = \tan \theta$

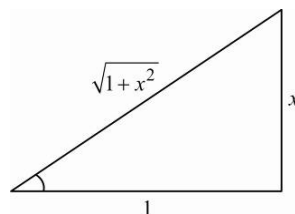
$y = \sec \theta$

$y = \sqrt{1+x^2}$

$\frac{dy}{dx} = \frac{1}{2\sqrt{1+x^2}} \cdot 2x$

At $x = 1$

$\frac{dy}{dx} = \frac{1}{\sqrt{2}}$



67.(B) 68.(C) $\lim_{x \rightarrow \infty} \left(x \sin\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x^2}\right) \right) \Rightarrow \lim_{x \rightarrow \infty} \left(\frac{\sin(1/x)}{\frac{1}{x}} + \sin\left(\frac{1}{x^2}\right) \right) = 1 + 0 = 1$

69.(C) $\lim_{x \rightarrow a^-} \left(\frac{|x|^3}{a} - \left[\frac{x}{a} \right]^3 \right) (a > 0)$

$$x = a - h \Rightarrow \lim_{h \rightarrow 0} \left(\frac{|a-h|^3}{a} - \left[\frac{a-h}{a} \right]^3 \right) = \lim_{h \rightarrow 0} \left(\frac{|a|^3}{a} - \left[1 - \frac{h}{a} \right]^3 \right) = a^2 - 0 = a^2$$

70.(A) RHL = $\lim_{x \rightarrow 0^+} \left[(1 - e^x) \frac{\sin x}{x} \right]$

When $x \in (0, h)$ and $h \rightarrow 0$ then $(1 - e^x) \in (-1, 0)$ & $\frac{\sin x}{x} < 1 \Rightarrow -1 < (1 - e^x) \frac{\sin x}{x} < 0 \Rightarrow \lim_{x \rightarrow 0^+} \left[(1 - e^x) \frac{\sin x}{x} \right] = -1$

LHL = $\lim_{x \rightarrow 0^-} \left[(1 - e^x) \frac{\sin x}{-x} \right] = \lim_{x \rightarrow 0^-} \left[(e^x - 1) \frac{\sin x}{x} \right]$

When $x \in (-h, 0)$ and $h \rightarrow 0$, then $e^x - 1 \in (-1, 0)$ and $\frac{\sin x}{x} < 1 \Rightarrow -1 < (e^x - 1) \frac{\sin x}{x} < 0 \Rightarrow \lim_{x \rightarrow 0^-} \left[(e^x - 1) \frac{\sin x}{x} \right] = -1$

LHL = RHL = -1

71.(B) (i) $\ell = \lim_{x \rightarrow \infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$

$$\Rightarrow \ell = \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x+1} + \sqrt{x}}{2} \right) \cdot \sin \left(\frac{\sqrt{x+1} - \sqrt{x}}{2} \right) \Rightarrow \ell = 2 \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x} + \sqrt{x+1}}{2} \right) \cdot \sin \left(\frac{x+1-x}{2(\sqrt{x+1} + \sqrt{x})} \right)$$

$$\Rightarrow \ell = 2 \lim_{x \rightarrow \infty} 2 \cos \left(\frac{\sqrt{x} + \sqrt{x+1}}{2} \right) \cdot \sin \left(\frac{1}{2(\sqrt{x} + \sqrt{x+1})} \right) \Rightarrow \ell = (\text{Oscillating value } -1 \text{ to } 1) \times 0 = 0$$

(ii) $m = \lim_{x \rightarrow -\infty} [\sin \sqrt{x+1} - \sin \sqrt{x}]$ [When $x \rightarrow -\infty$, then $\sqrt{x}$ and $\sqrt{x+1}$ undefined, m is undefined]

72.(A)

73.(C) LHL = $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \sqrt{\{x\} \cot \{x\}} = \lim_{h \rightarrow 0} \sqrt{(1-h) \cot(1-h)} = \sqrt{\cot 1}$

RHL = $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\tan^2 [x]}{x^2 - [x]^2} = \lim_{h \rightarrow 0} \frac{\tan^2 [0+h]}{(0+h)^2 - [0+h]^2} = \lim_{h \rightarrow 0} \frac{\tan^2 0}{h^2} = 0$

$\therefore$ LHL $\neq$ RHL so $\lim_{x \rightarrow 0} f(x)$ does not exist. $f(x)$ is not continuous at $x = 0$.

Now, $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = \cot^{-1} (\sqrt{\cot 1})^2 = \cot^{-1} (\cot 1) = 1$

74.(C)

75.(A) $\lim_{x \rightarrow \infty} f(x)$ exist and is finite and non zero $\lim_{x \rightarrow \infty} \left[f(x) + \frac{3f(x)-1}{f^2(x)} \right] = 3$ let $\lim_{x \rightarrow \infty} f(x) = A$

$$A + \frac{3A-1}{A^2} = 3 \Rightarrow A = 1 \quad \text{So} \quad \lim_{x \rightarrow \infty} f(x) = 1$$

76.(B)

$$\begin{aligned}
 77.(B) \quad \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\exp\left(x \ln\left(1 + \frac{ay}{x}\right)\right) - \exp\left(x \ln\left(1 + \frac{by}{x}\right)\right)}{y} \right) &= \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\left(1 + \frac{ay}{x}\right)^x - \left(1 + \frac{by}{x}\right)^x}{y} \right) \text{ by expansion} \\
 &= \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\left(1 + ay + \frac{x(x-1)}{2!} \cdot \frac{a^2 y^2}{x^2} + \dots\right) - \left(1 + by + \frac{x(x-1)}{2!} \cdot \frac{b^2 y^2}{x^2} + \dots\right)}{y} \right) \\
 &= \lim_{y \rightarrow 0} \left[\frac{y(a-b) + \frac{y^2}{2}(a^2 - b^2) + \dots}{y} \right] = a - b
 \end{aligned}$$

78.(D)

$$79.(B) \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -\sqrt{h} \left(1 - h \sin\left(-\frac{1}{h}\right) \right) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \sqrt{h} \left(1 + h \sin \frac{1}{h} \right) = 0$$

$$f(0) = 0$$

$f(x)$ is continuous at $x = 0$

$$\text{LHD at } x = 0 \text{ is } \lim_{h \rightarrow 0} \frac{-\sqrt{h} \left(1 - h \sin\left(-\frac{1}{h}\right) \right) - 0}{-h} = \lim_{h \rightarrow 0} \frac{1 - h \sin\left(-\frac{1}{h}\right)}{\sqrt{h}} = \infty$$

LHD is not define so $f(x)$ is not differentiable at $x = 0$

$$80.(A) \quad \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 e^{2(x-1)} = 1$$

$$f(1) = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} a \operatorname{sgn}(x+1) \cos 2(x-1) + bx^2 = a \cdot 1 \cdot 1 + b$$

For continuity $a + b = 1$

$$\text{LHD } (x=1) \text{ is } \lim_{h \rightarrow 0} \frac{(1-h)^2 e^{-2h} - 1}{-h} = \lim_{h \rightarrow 0} 2e^{-2h} + he^{-2h} + \left(\frac{e^{-2h} - 1}{-h} \right) = 4$$

$$\text{RHD } (x=1) \text{ is } \lim_{h \rightarrow 0} \frac{a \operatorname{sgn}(2+h) \cos 2h + b(1+h)^2 - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a \cos 2h + b + bh^2 + 2bh - (a+b)}{h} = \lim_{h \rightarrow 0} a \left(\frac{\cos 2h - 1}{h} \right) + bh + 2b = 2b$$

$f(x)$ is differentiable at $x = 1$ if $2b = 4 \Rightarrow b = 2 \quad a = -1$

81.(ABCD) $f(x) = \frac{|x + \pi|}{\sin x}$

(A) $f(-\pi^+) = \lim_{h \rightarrow 0} \frac{|-\pi + h + \pi|}{\sin(-\pi + h)} = \lim_{h \rightarrow 0} \frac{|h|}{-\sin h} = -1$

(B) $f(-\pi^-) = \lim_{h \rightarrow 0} \frac{|-\pi - h + \pi|}{\sin(-\pi - h)} = \lim_{h \rightarrow 0} \frac{|h|}{\sin h} = 1$

(C) $f(-\pi^+) \neq f(-\pi^-)$ so $\lim_{x \rightarrow -\pi} f(x)$ does not exist

(D) For $\lim_{x \rightarrow \pi} f(x)$

$$\text{LHL} = \lim_{x \rightarrow \pi^-} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi - h}{\sin h} = \frac{2\pi}{0} = \infty$$

$$\text{RHL} = \lim_{x \rightarrow \pi^+} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi + h}{-\sin h} = -\frac{2\pi}{0} = -\infty$$

LHL $\neq$ RHL

So $\lim_{x \rightarrow \pi} f(x)$ does not exist

82.(ABCD) $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A}$

(A) If $n \in \mathbb{N}$ then $\lim_{x \rightarrow \infty} \frac{\left(a + \frac{1}{x}\right)^n}{1 + \frac{A}{x^n}} = \frac{(a+0)^n}{1+0} = a^n$

(B) If $n \in \mathbb{Z}^-$ and $a = A = 0$ then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{x^n} = \infty$

(C) If $n = 0$ then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{1 + A} = \frac{1}{1 + A}$

(D) If $n \in \mathbb{Z}^-$, $A = 0$ and $a \neq 0$

then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n} = \lim_{x \rightarrow \infty} \left(a + \frac{1}{x}\right)^n = (a+0)^n$

83.(C)
$$L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} = \lim_{x \rightarrow 0} \frac{a - a \cdot \left[1 - \frac{1}{2} \cdot \frac{x^2}{a^2} + \frac{1}{2} \cdot \frac{\left(\frac{1}{2} - 1\right)}{2} \cdot \frac{x^4}{a^4} - \dots\right] - \frac{x^2}{4}}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x^2}{2a} + \frac{1}{8} \cdot \frac{x^4}{a^3} + \dots - \frac{x^2}{4}}{x^4}$$

Since L is finite $\Rightarrow 2a = 4 \Rightarrow a = 2 \quad \therefore L = \lim_{x \rightarrow 0} \frac{1}{8 \cdot a^3} = \frac{1}{64}$

$$84.(D) \quad \lim_{x \rightarrow 0} e^{\frac{x \ln(1+b^2)}{x}} = 1+b^2 = 2b \sin^2 \theta \quad \Rightarrow \quad \sin^2 \theta = \frac{1}{2} \left(b + \frac{1}{b} \right)$$

We know $b + \frac{1}{b} \geq 2$

$$\Rightarrow \quad \sin^2 \theta \geq 1 \quad \text{but} \quad \sin^2 \theta \leq 1 \quad \Rightarrow \quad \sin^2 \theta = 1 \quad \Rightarrow \quad \theta = \pm \frac{\pi}{2}$$

$$85.(B) \quad \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1}{x + 1} - ax - b \right) = 4 \quad \Rightarrow \quad \lim_{x \rightarrow \infty} \left(\frac{x^2(1-a) + x(1-a-b) + (1-b)}{x+1} \right) = 4$$

Limit is finite

$$\text{It exists when } 1-a=0 \quad \Rightarrow \quad a=1 \quad \text{then} \quad \lim_{x \rightarrow \infty} \left(\frac{1-a-b + \frac{1-b}{x}}{1 + \frac{1}{x}} \right) = 4$$

$$\therefore \quad 1-a-b=4 \quad \Rightarrow \quad b=-4$$

$$86.(B) \quad \left((1+a)^{1/3} - 1 \right) x^2 + \left((a+1)^{1/2} - 1 \right) x + \left((a+1)^{1/6} - 1 \right) = 0$$

Let $a+1=t^6$

$$\Rightarrow \quad (t^2-1)x^2 + (t^3-1)x + (t-1) = 0 \quad \Rightarrow \quad (t+1)x^2 + (t^2+1+1)x + 1 = 0$$

$$\text{As } a \rightarrow 0, t \rightarrow 1 \quad \Rightarrow \quad 2x^2 + 3x + 1 = 0 \quad \Rightarrow \quad x = -1 \text{ and } x = -\frac{1}{2}$$

$$87.(A) \quad \lim_{x \rightarrow a} \frac{f(a)g(x) - f(a) - g(a)f(x) + g(a)}{g(x) - f(x)} = 4$$

Apply L'Hospital rule

$$\Rightarrow \quad \lim_{x \rightarrow a} \left(\frac{f(a)g'(x) - g(a)f'(x)}{g'(x) - f'(x)} \right) = 4 \quad \Rightarrow \quad K \lim_{x \rightarrow a} \left(\frac{g'(x) - f'(x)}{g'(x) - f'(x)} \right) = 4 \quad \Rightarrow \quad k = 4$$

88.(B) Continuity at $x = 0$

$$\text{LHL} = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} (0-h)e^{-\left(\frac{1}{|h|} + \frac{1}{(-h)}\right)} = \lim_{h \rightarrow 0} (-h)e^{-\left(\frac{1}{h} - \frac{1}{h}\right)} = \lim_{h \rightarrow 0} (-h) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} (0+h)e^{-\left(\frac{1}{|h|} + \frac{1}{h}\right)} = \lim_{h \rightarrow 0} he^{-\left(\frac{1}{h} + \frac{1}{h}\right)} = \lim_{h \rightarrow 0} \frac{h}{e^{2/h}} = 0 \Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

Therefore, $f(x)$ is continuous for all x .

Differentiability at $x = 0$

$$Lf'(0) = \lim_{h \rightarrow 0} \frac{(-h)e^{-\left(\frac{1}{h} - \frac{1}{h}\right)} - 0}{(-h)} = \lim_{h \rightarrow 0} e^0 = 1$$

$$Rf'(0) = \lim_{h \rightarrow 0} \frac{he^{-\left(\frac{1}{h} + \frac{1}{h}\right)} - 0}{h} = \lim_{h \rightarrow 0} \frac{1}{e^{2/h}} = 0 \quad \Rightarrow \quad Rf'(0) \neq Lf'(0)$$

Therefore, $f(x)$ is not differentiable at $x = 0$

89.(D) Now $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{2}{e^{2x} - 1} \right)$

$$= \lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x(e^{2x} - 1)} = \lim_{x \rightarrow 0} \frac{2e^{2x} - 2}{(e^{2x} - 1) + 2xe^{2x}} \quad (\text{Using L'Hospital rule})$$

$$= \lim_{x \rightarrow 0} \frac{4e^{2x}}{4e^{2x} + 4xe^{2x}} = 1 \quad (\text{Using L'Hospital's rule})$$

$\therefore f(x)$ is continuous at $x = 0$, then $\lim_{x \rightarrow 0} f(x) = f(0) \Rightarrow 1 = f(0)$

90.(B) Now, $f'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{(1-h-1)\sin\left(\frac{1}{1-h-1}\right) - 0}{-h} = \lim_{h \rightarrow 0} \sin\left(-\frac{1}{h}\right) = -\lim_{h \rightarrow 0} \sin \frac{1}{h}$

and $f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h-1)\sin\left(\frac{1}{1+h-1}\right) - 0}{h} = \lim_{h \rightarrow 0} \sin \frac{1}{h} \quad \therefore f'(1^-) \neq f'(1^+)$
 f is not differentiable at $x = 1$.

Now $f'(0^-) = \frac{(h+1)\sin \frac{1}{1+h} - \sin 1}{-h} = \frac{(h+1)\cos\left(\frac{1}{1+h}\right)\left(-\frac{1}{(1+h)^2}\right) + \sin \frac{1}{1+h}}{-1}$

$$= \lim_{h \rightarrow 0} \frac{\left[-\left\{ (h+1)\cos\left(\frac{1}{h+1}\right) \times \left(\frac{1}{(h+1)^2}\right) \right\} + \sin\left(\frac{1}{h+1}\right) \right]}{-1} \quad (\text{Using L'Hospital's rule})$$

 $= \cos 1 - \sin 1$

and $f'(0^+) = \lim_{h \rightarrow 0} \frac{(0+h-1)\sin\left(\frac{1}{0+h-1}\right) - \sin 1}{h}$

$$= \lim_{h \rightarrow 0} \frac{(h-1)\cos\left(\frac{1}{h-1}\right) \times \left(\frac{-1}{(h-1)^2}\right) + \sin\left(\frac{1}{h-1}\right)}{1} \quad (\text{Using L'Hospital's rule})$$

 $= \cos 1 - \sin 1 \Rightarrow f'(0^-) = f'(0^+)$

$\therefore f$ is differentiable at $x = 0$.

91.(D) $\lim_{x \rightarrow 0} \frac{f(3x)}{f(x)} = 1$

$$f(x) < f(2x) < f(3x) \text{ Divide by } f(x) \Rightarrow 1 < \frac{f(2x)}{f(x)} < \frac{f(3x)}{f(x)}$$

Using sandwich theorem $\Rightarrow \lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1$

92.(D)

93.(C) $f(0) = q$

$$f(0^+) = \lim_{x \rightarrow 0^+} \frac{(1+x)^{1/2} - 1}{x} = \lim_{x \rightarrow 0^+} \frac{1 + \frac{1}{2}x + \dots - 1}{x} = \frac{1}{2}$$

$$f(0^-) = \lim_{x \rightarrow 0^-} \frac{\sin(p+1)x + \sin x}{x} = \lim_{x \rightarrow 0^-} \frac{(\cos(p+1)x)(p+1) + (\cos x)}{1} = (p+1) + 1 = p+2$$

$$\therefore p+2 = q = \frac{1}{2} \Rightarrow p = -\frac{3}{2}, q = \frac{1}{2}$$

94.(C) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\left(1 - \tan \frac{x}{2}\right)(1 - \sin x)}{\left(1 + \tan \frac{x}{2}\right)(\pi - 2x)^3} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)(1 - \sin x)}{(\pi - 2x)^3}$

Put $x = \frac{\pi}{2} + y$ $y \rightarrow 0^+$

$$= \lim_{y \rightarrow 0} \frac{\tan\left(-\frac{y}{2}\right)(1 - \cos y)}{(-2y)^3} = \lim_{y \rightarrow 0} \frac{\left(\tan \frac{y}{2}\right)(1 - \cos y)}{8y^3} = \lim_{y \rightarrow 0} \left(\frac{\tan \frac{y}{2}}{\frac{y}{2}}\right) \left(\frac{\sin^2 \frac{y}{2}}{\frac{y^2}{4}}\right) \cdot \frac{2}{64} = (1)(1) \cdot \frac{1}{32} = \frac{1}{32}$$

95.(C) $\lim_{x \rightarrow 0} \frac{\ln(3+x) - \ln(3-x)}{x} = k \Rightarrow \lim_{x \rightarrow 0} \frac{\ln\left(1 + \frac{x}{3}\right) - \ln\left(1 - \frac{x}{3}\right)}{x} = k$

$$\Rightarrow \lim_{x \rightarrow 0} \left[\frac{\ln\left(1 + \frac{x}{3}\right)}{\frac{x}{3}} \cdot \frac{1}{3} - \frac{\ln\left(1 - \frac{x}{3}\right)}{\left(-\frac{x}{3}\right)} \left(-\frac{1}{3}\right) \right] = k \Rightarrow \frac{1}{3} + \frac{1}{3} = k \Rightarrow k = \frac{2}{3}$$

96.(B)

97.(D)

98.(B)

99.(C) $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a} = \lim_{x \rightarrow a} \frac{2xf(a) - a^2 f'(x)}{1} = 2af(a) - a^2 f'(a)$

Alter $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$

$$= \lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(a) + a^2 f(a) - a^2 f(x)}{x - a} = \lim_{x \rightarrow a} \frac{(x^2 - a^2) f(a) - a^2 (f(a) - f(x))}{x - a}$$

$$= \lim_{x \rightarrow a} (x+a) f(a) - a^2 \left\{ \frac{f(x) - f(a)}{(x-a)} \right\} = 2af(a) - a^2 f'(a)$$

100.(A)

101.(A)

102.(D)

103.(A)

104.(A)

105.(C)

106.(D) $f(x) = |x|^{\sin x}$

At $x = \frac{\pi}{4}$, $|x| = x$ and $|\sin x| = \sin x$

$$\therefore f(x) = x^{\sin x} \Rightarrow \ln(f(x)) = \sin x \cdot \ln x$$

$$\Rightarrow \frac{1}{f(x)} f'(x) = \cos x \ln x + \frac{\sin x}{x} \Rightarrow f'\left(\frac{\pi}{4}\right)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{4} \ln \frac{\pi}{4} + \frac{2\sqrt{2}}{4}\right)$$

107.(D)

$$108.(B) \quad x\sqrt{1+y} + y\sqrt{1+x} = 0 \Rightarrow x^2(1+y) = y^2(1+x) \Rightarrow x^2 - y^2 + x^2y - y^2x = 0$$

$$(x+y)(x-y) + xy(x-y) = 0 ; \quad (x-y)(x+y+xy) = 0$$

$$\therefore x \neq y \Rightarrow y = -\frac{x}{1+x} \Rightarrow \frac{dy}{dx} = \frac{-(1+x) + x}{(1+x)^2} = -\frac{1}{(1+x)^2}$$

$$109.(B) \quad f'(x) = \frac{x^{10}}{1+x^2}, g(2) = a$$

$$g \text{ is inverse of } f \Rightarrow f(g(x)) = x$$

Differentiating w.r.t. x

$$f'(g(x)) \cdot g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))} \Rightarrow g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(a)} \Rightarrow g'(2) = \frac{1+a^2}{a^{10}}$$

110.(D)

$$111.(C) \quad \frac{dy}{dx} = ax + b \quad \text{where } y = \frac{(1-x+x^2)(1+x+x^2)}{(1+x+x^2)} = (1-x+x^2)$$

$$\Rightarrow \frac{d}{dx}(1-x+x^2) = ax + b \Rightarrow 2x-1 = ax + b \Rightarrow a = 2, b = -1$$

$$112.(B) \quad y = f\left(\frac{2x-1}{x^2+1}\right) \text{ and } f'(x) = \sin x$$

$$\Rightarrow \frac{dy}{dx} = f'\left(\frac{2x-1}{x^2+1}\right) \cdot \left\{ \frac{(x^2+1) \cdot 2 - (2x-1) \cdot 2x}{(x^2+1)^2} \right\} = \sin\left(\frac{2x-1}{x^2+1}\right) \cdot \frac{2(1-x-x^2)}{(1+x^2)^2}$$

$$113.(BC) \quad \therefore t = \frac{1}{2} \ln(x^2 + y^2) \Rightarrow \frac{1}{2} \ln(x^2 + y^2) = \sin^{-1}\left(\frac{y}{\sqrt{x^2 + y^2}}\right)$$

Case 1 : When $x \geq 0$

$$\Rightarrow \ln(x^2 + y^2) = 2 \tan^{-1}\left(\frac{y}{x}\right) \Rightarrow \frac{1}{x^2 + y^2} (2x + 2yy') = \frac{2}{1 + \frac{y^2}{x^2}} \left(\frac{xy' - y}{x^2}\right)$$

$$\Rightarrow xy' - yy' = x + y \Rightarrow y' = \frac{x+y}{x-y}$$

$$\text{Let } y = x \tan \theta ; \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow \sin^{-1}\left(\frac{y}{\sqrt{x^2 + y^2}}\right) = \sin^{-1}\left(\frac{x \tan \theta}{x \sec \theta}\right) = \sin^{-1}\left(\frac{x}{|x|} \sin \theta\right) = \begin{cases} \theta = \tan^{-1}\left(\frac{y}{x}\right) & , x \geq 0 \\ -\theta = -\tan^{-1}\left(\frac{y}{x}\right) & , x < 0 \end{cases}$$

$$\text{Case 2 : When } x < 0 ; \quad \ln(x^2 + y^2) = -2 \tan^{-1}\left(\frac{y}{x}\right) \Rightarrow \frac{(2x + 2yy')}{x^2 + y^2} = \frac{-2}{1 + \frac{y^2}{x^2}} \left(\frac{xy' - y}{x^2}\right)$$

$$y'x + y = y - x ; \quad h' = \frac{y-x}{y+x}$$

114.(CD)

115.(BC)

$$y = \cos^{-1} \sqrt{\frac{\sqrt{1+x^2} + 1}{2\sqrt{1+x^2}}}$$

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - \left(\frac{1}{2} + \frac{1}{2\sqrt{1+x^2}}\right)^2}} \cdot \frac{1}{2\sqrt{\frac{1}{2} + \frac{1}{2\sqrt{1+x^2}}}} \cdot \frac{1}{2} \cdot \frac{2x}{(-2)(1+x^2)^{3/2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{\frac{1}{4} - \frac{1}{4(1+x^2)}}} \cdot \frac{x}{4(1+x^2)^{3/2}} \Rightarrow \frac{dy}{dx} = \frac{2\sqrt{1+x^2}}{\sqrt{1+x^2}-1} \cdot \frac{x}{4\sqrt{1+x^2}(1+x^2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{2|x|(1+x^2)}$$

When $x < 0$ and $x > 0$

$$\frac{dy}{dx} = \frac{-1}{2(1+x^2)} \quad \frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

Alternate :

Put $x = \tan \theta$

$$\tan^{-1} x = \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$y = \cos^{-1} \sqrt{\frac{1 + |\sec \theta|}{2|\sec \theta|}} = \cos^{-1} \sqrt{\frac{\cos \theta + 1}{2}} \Rightarrow y = \cos^{-1}(\cos \theta / 2)$$

$$y = \begin{cases} -\theta/2 & , -\pi/2 < \theta \leq 0 \\ \theta/2 & , 0 < \theta < \pi/2 \end{cases} \Rightarrow \frac{dy}{dx} = \begin{cases} -\frac{1}{2(1+x^2)} & , x \leq 0 \\ \frac{1}{2(1+x^2)} & , x > 0 \end{cases}$$

116.(A) Let $f(x) = a_1x^2 + a_2x + a_3$

$$\therefore f(1) = f(-1)$$

$$\Rightarrow a_1 + a_2 + a_3 = a_1 - a_2 + a_3 \Rightarrow a_2 = 0 \Rightarrow f(x) = a_1x^2 + a_3$$

$$f'(x) = 2a_1x \Rightarrow f'(a) = 2a_1a \quad \text{and} \quad f'(b) = 2a_1b \quad \text{also} \quad f'(c) = 2a_1c$$

$\therefore a, b, c$ are in A.P.

$\Rightarrow f'(a), f'(b), f'(c)$ are also in AP.

117.(1) Say I is an integer

$$f(I) = 0 \quad \lim_{x \rightarrow I^+} f(x) = 0$$

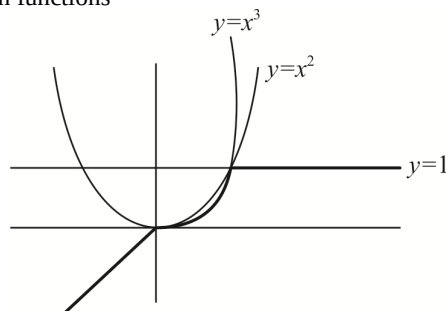
$$\lim_{x \rightarrow I^-} f(x) = (I-1)^2 - (I^2 - 1) = I^2 - 2I + 1 - I^2 + 1 = 2 - 2I$$

$$2 - 2I = 0 \Rightarrow I = 1$$

$$118.(2) \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\int_{\frac{\pi}{4}}^{\sec^2 x} f(t) dt}{x^2 - \frac{\pi^2}{16}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{f(\sec^2 x) \cdot 2 \sec^2 x \tan x}{2x} = \frac{8}{\pi} f(2)$$

$$119.(1) \quad \lim_{x \rightarrow 0^+} (\sin x)^{1/x} = 0 ; \quad \lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\sin x} = 1$$

120.(0) Plot then functions



Zero point of discontinue by

121.(0) The only point of vulnerability is 0, where the function is differentiable.

$$122.(1) \quad \text{Since } |f(x) - f(y)|^2 \leq |x - y|^2$$

$$\lim_{y \rightarrow x} \frac{|f(x) - f(y)|}{|x - y|} \leq \lim_{y \rightarrow x} |x - y|$$

$$|f'(x)| \leq 0 \quad \Rightarrow \quad f'(x) = 0 \quad f(x) \text{ is constant}$$

123.(0) The only points of contention are integers. The function is continuous at integral points

$$124.(2) \quad \lim_{x \rightarrow 1} (1 - x) \tan\left(\frac{\pi x}{2}\right) \cdot \pi$$

$$\text{Say } 1 - x = t$$

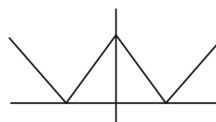
$$\lim_{t \rightarrow 0} t \tan\left(\frac{\pi}{2}(1 - t)\right) \cdot \pi = \lim_{t \rightarrow 0} t \tan\left(\frac{\pi}{2} - \frac{\pi}{2}t\right) \cdot \pi = \lim_{t \rightarrow 0} \frac{t}{\tan\left(\frac{\pi}{2}t\right)} \cdot \pi = 2.$$

$$125.(1) \quad \lim_{x \rightarrow \infty} \sqrt{\frac{x - \sin^2 x}{x + \cos^2 x}} = \lim_{x \rightarrow \infty} \sqrt{\frac{1 - \frac{\sin^2 x}{x}}{1 + \frac{\cos^2 x}{x}}} = 1$$

$$126.(1) \quad \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \cos^2 t dt}{x \sin x} = \lim_{x \rightarrow 0} \frac{2x \cos^2(x^2)}{\sin x + x \cos x} = 1$$

127.(1) The only point of non-differentiability is 2

128.(3)



$$129.(5) \quad F(x) = (f(x))^2 + (g(x))^2 ; \quad F'(x) = 2f(x)f'(x) + 2g(x)g'(x) = 0 ; \quad F \text{ is constant}$$

$$130.(1) \quad x^2 + y^2 = 1 ; \quad 2x + 2yy' = 0 ; \quad yy' + x = 0$$

131.(0) The function is identically zero everywhere.

Differential Calculus-2

1.(C) Let $A(x_1, y_1)$ be a point on the curve at the minimum distance from origin

$$\therefore OA^2 = x_1^2 + y_1^2$$

$$z = x_1^2 + (x_1^2 - 4)^2 \quad \{ \because y_1 = x_1^2 - 4 \}$$

$$\frac{dz}{dx_1} = 2x_1 + 2(x_1^2 - 4)2x_1 = 4x_1^3 - 14x_1$$

$$\frac{dz}{dx_1} = 0 \text{ gives } x_1 = 0, x_1^2 = \frac{7}{2}$$

$$\Rightarrow x_1 = \pm \sqrt{\frac{7}{2}}; \quad \frac{d^2z}{dx_1^2} = 12x_1^2 - 14$$

$$\text{At } x_1 = \pm \sqrt{\frac{7}{2}}, \frac{d^2z}{dx_1^2} = 12\left(\frac{7}{2}\right) - 14 = 28 > 0$$

$$\therefore OA \text{ is minimum when } x_1 = \pm \sqrt{\frac{7}{2}}$$

$$\therefore \text{Minimum distance} = \sqrt{\frac{7}{2} + \left(\frac{7}{2} - 4\right)^2} = \frac{\sqrt{15}}{2}$$

2.(C) Given curve is $y(x) = 1 + \sqrt{4x+3}$

$$\frac{dy}{dx} = \frac{2}{\sqrt{4x+3}} = \frac{2}{3} \text{ (given)} \quad \Rightarrow \quad \sqrt{4x+3} = 3 \quad \Rightarrow \quad x = 3$$

When $x = 3$, then $y = 1 + 3 = 4$

$$\therefore \text{At } P(3, 4), \text{ slope of tangent} = \frac{2}{3}$$

$$\therefore \text{Equation of normal at } P(3, 4) \text{ is } y - 4 = -\frac{3}{2}(x - 3) \quad \Rightarrow \quad 3x + 2y = 17$$

Clearly, this normal passes through $(1, 7)$

3.(C) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \tan t$

$$\text{At } t = \frac{\pi}{4}, \frac{dy}{dx} = 1 = \text{slope of tangent}$$

$$\text{At } t = \frac{\pi}{4}, x = 2\cos\frac{\pi}{4} + 2\frac{\pi}{4}\sin\frac{\pi}{4} = \frac{2}{\sqrt{2}} + \frac{\pi}{2\sqrt{2}}$$

Equation of normal at $t = \frac{\pi}{4}$ is

$$y - \left(\frac{2}{\sqrt{2}} + \frac{\pi}{2\sqrt{2}}\right) = -1 \left[x - \left(\frac{2}{\sqrt{2}} + \frac{\pi}{2\sqrt{2}}\right) \right] \quad \Rightarrow \quad x + y - 2\sqrt{2} = 0$$

$$\therefore \text{Distance of normal from origin} = \frac{|0 + 0 - 2\sqrt{2}|}{\sqrt{1^2 + 1^2}} = 2$$

- 4.(B) Rolle's theorem states that if $f(x)$ is continuous in $[a, b]$, $f(x)$ is differentiable in (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$

$$f(x) = 2x^3 + bx^2 + cx, x \in [-1, 1]$$

$$f(-1) = f(1)$$

$$\Rightarrow -2 + b - c = 2 + b + c \Rightarrow c = -2$$

$$f'(x) = 6x^2 + 2bx + c$$

$$\therefore \text{Rolle's theorem holds at } x = \frac{1}{2}$$

$$\therefore f'\left(\frac{1}{2}\right) = 0 \Rightarrow \frac{6}{4} + \frac{2b}{2} - 2 = 0 \Rightarrow b = \frac{1}{2} \quad \therefore 2b + c = 1 - 2 = -1.$$

- 5.(D) Minimum value of $\left|a + \frac{1}{a}\right|$ is 2 when $a = 1$ or $a = -1$. $x^2 + y^2 = 4^2$

Equation of line PA is

$$y - h = m(x - 0)$$

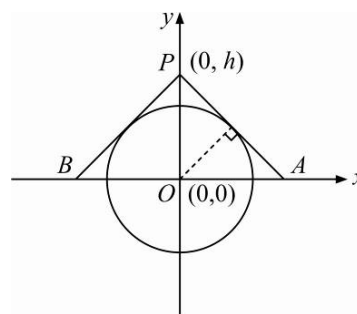
$$mx - y + h = 0$$

PA to be tangent, perpendicular from $(0, 0) = \text{radius}$

$$\left| \frac{h}{\sqrt{m^2 + 1}} \right| = 4$$

$$h^2 = 16(m^2 + 1) \quad \dots\dots (i)$$

PA intersects x-axis at $A\left(-\frac{h}{m}, 0\right)$



$$\text{Area of } \triangle APB = 2 \left(\text{area of } \triangle POA \right) = 2 \times \frac{1}{2} \times OA \times OP = \left| \frac{-h}{m} \times h \right| = \left| \frac{h^2}{m} \right| = \left| \frac{16(m^2 + 1)}{m} \right| = 16 \left| m + \frac{1}{m} \right|$$

Area is minimum when $m = 1$ or $m = -1$

$$\text{From (i), } h^2 = 16(1+1) = 32 \Rightarrow h = 4\sqrt{2}.$$

- 6.(A) Given $\sin y = x \sin\left(\frac{\pi}{3} + y\right)$

$$\text{When } x = 0, \text{ then } \sin y = 0 \Rightarrow y = 0$$

$$\text{Now, } \cos y \frac{dy}{dx} = \sin\left(\frac{\pi}{3} + y\right) + x \cos\left(\frac{\pi}{3} + y\right) \frac{dy}{dx}$$

$$\text{When, } x = 0 \text{ and } y = 0, \text{ then } \frac{dy}{dx} = \frac{\sqrt{3}}{2} = \text{Slope of tangent}$$

$$\therefore \text{Slope of normal} = -\frac{2}{\sqrt{3}}$$

$$\text{Therefore, the equation of normal at } (0, 0) \text{ is } y - 0 = -\frac{2}{\sqrt{3}}(x - 0)$$

$$2x + \sqrt{3}y = 0$$

- 7.(D) For max./min. value of $f(x)$ in $[a, b]$, find $f'(x) = 0$ which gives critical points. Then find $f(x)$ at critical points, $f(a)$ and $f(b)$ which give max./min. values of $f(x)$.

$$f(x) = \frac{(1+x)^{0.6}}{1+x^{0.6}} \text{ in } [0, 1]$$

$$f'(x) = \frac{(1+x^{0.6})(0.6)(1+x)^{-0.4} - (1+x)^{0.6}(0.6)x^{-0.4}}{(1+x^{0.6})^2} = \frac{0.6}{(1+x^{0.6})^2} \left[\frac{1+x^{0.6}}{(1+x)^{0.4}} - \frac{(1+x)^{0.6}}{x^{0.4}} \right]$$

$$= \frac{0.6}{(1+x^{0.6})^2} \left[\frac{x^{0.4} + x - (1+x)}{(1+x)^{0.4} x^{0.4}} \right] = \frac{0.6}{(1+x^{0.6})^2} \left[\frac{x^{0.4} - 1}{(1+x)^{0.4} x^{0.4}} \right]$$

$$f'(x) = 0 \text{ gives } x = 1$$

So, critical points are $x = 1$ and $x = 0$

$$f(1) = \frac{2^{0.6}}{2} = 2^{-0.4} \text{ and } f(0) = 1$$

$$\therefore k = \text{min. value} = 2^{-0.4} \text{ and } K = \text{max. value} = 1$$

$$\therefore (k, K) = (2^{-0.4}, 1)$$

- 8.(B) Rolle's theorem holds then

$$f(-1) = f(1) \text{ and } f'(c) = 0$$

$$\therefore -2 + a - b = 2 + a + b \Rightarrow -2$$

$$f'(x) = 6x^2 + 2ax + b$$

$$f'(c) = 0 \Rightarrow 6c^2 + 2ac + b = 0$$

$$c = \frac{1}{2} \therefore 6\left(\frac{1}{4}\right) + a + b = 0 \Rightarrow a = \frac{1}{2}$$

$$\therefore 2b + a = 1 - 2 = -1$$

- 9.(C) $y = \frac{3}{2} \sin 2\theta$, $x = e^\theta \sin \theta$

$$\frac{dy}{d\theta} = 3 \cos 2\theta, \frac{dx}{d\theta} = e^\theta (\sin \theta + \cos \theta)$$

$$\frac{dy}{dx} = \frac{3 \cos 2\theta}{e^\theta (\sin \theta + \cos \theta)}$$

Since tangent is parallel to x-axis.

$$\therefore \frac{dy}{dx} = 0 \Rightarrow \cos 2\theta = 0 \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\text{But at } \theta = \frac{3\pi}{4}, \frac{dy}{dx} = \frac{0}{0} \text{ (not defined)}$$

$$\therefore \theta = \frac{3\pi}{4} \text{ (rejected). Hence } \theta = \frac{\pi}{4}$$

10.(A) In ΔABC , $\cos C = \frac{a^2 + b^2 + c^2}{2ab}$

$$\frac{dx}{dt} = 20 \text{ Km/hr},$$

$$\frac{dy}{dt} = 30 \text{ Km/hr}$$

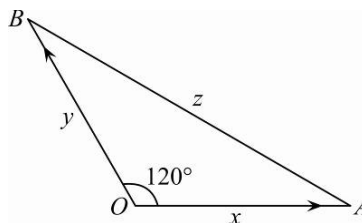
$$z^2 = x^2 + y^2 - 2xy \cos 120^\circ$$

$$z = \sqrt{x^2 + y^2 + xy}$$

$$\frac{dz}{dt} = \frac{1}{2\sqrt{x^2 + y^2 + xy}} \cdot \left(2x \frac{dx}{dt} + 2y \frac{dy}{dt} + x \frac{dy}{dt} + y \frac{dx}{dt} \right)$$

When $x = 8 \text{ Km}$, $y = 6 \text{ Km}$

$$\frac{dz}{dt} = \frac{1}{2\sqrt{64 + 36 + 48}} (2.8.20 + 2.6.30 + 8.30 + 6.20) = \frac{320 + 360 + 240 + 120}{2\sqrt{148}} = \frac{260}{\sqrt{37}} \text{ Km/hr}$$



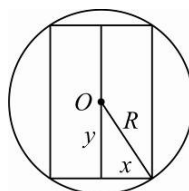
11.(C) $x^2 + y^2 = R^2$

$$V = \pi x^2 y = \pi (R^2 - y^2) y = 2\pi (R^2 y - y^3)$$

$$\frac{dV}{dy} = 2\pi (R^2 - 3y^2) = 0 \Rightarrow R = \sqrt{3}y$$

$$\frac{d^2V}{dy^2} < 0 \Rightarrow V \text{ is max. at } R = \sqrt{3}y$$

$$\therefore \text{Max volume} = 2\pi \left(R^2 - \frac{R^2}{3} \right) \frac{R}{\sqrt{3}} = \frac{4\pi R^3}{3\sqrt{3}} = 4\pi \quad (\because R = \sqrt{3})$$



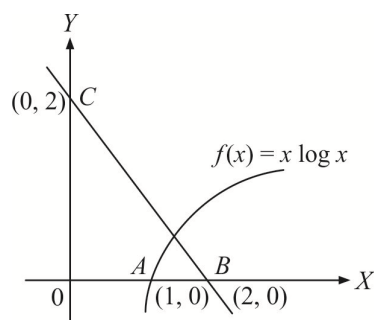
12.(B)

13.(C) $V = 288\pi \Rightarrow \frac{4}{3}\pi r^3 = 288\pi \Rightarrow r = 6$

$$\frac{dV}{dt} = 4\pi \text{ cc/sec}$$

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \Rightarrow 4\pi = 4\pi r^2 \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{36} \quad (\because r = 6)$$

14.(A) **Statement I:** To solve $x \log x = 2 - x$, draw graphs of $f(x) = x \log x$, $g(x) = 2 - x$. They intersect between 1 and 2.



Statement II : $f'(x) = 1 + \log x > 0$, $g'(x) = -1 < 0$.

So, $f(x)$ is increasing and $g(x)$ is decreasing in $[1, 2]$. Also, they intersect at a point in $[1, 2]$.

15.(D) $\frac{dA}{dt} = 8\text{cm}^2/\text{s}$

$$A(\text{S. Area}) = 4\pi r^2 \Rightarrow \frac{dA}{dt} = 8\pi r \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi r}$$

$$V(\text{volume}) = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi r^2 \frac{1}{\pi r} = 4r \quad \therefore \frac{dV}{dt} \propto r.$$

16.(C) **Statement I :** Let $f(x) = x^2(e^x + e^{-x})$

$$f'(x) = 2x(e^x + e^{-x}) + x^2(e^x - e^{-x})$$

Here $2x(e^x + e^{-x}) > 0 \forall x > 0$; $x^2(e^x - e^{-x}) = x^2\left(e^x - \frac{1}{e^x}\right) > 0 \forall x > 0$

$\therefore f'(x) > 0 \forall 0$, so $f(x)$ is increasing Statement I is true.

Statement II : Let $f(x) = x^2e^x$

$\Rightarrow f'(x) = xe^x(x+2) > 0 \forall x > 0$, so increasing.

Let $g(x) = x^2e^{-x} \Rightarrow g'(x) = xe^{-x}(2-x)$

$g'(x) > 0$ for $0 < x < 2$, so increasing.

and $g'(x) \leq 0$ for $x \geq 2$, so decreasing

$\therefore g(x)$ is not increasing $\forall x > 0$. Statement I is false.

17.(D) $x^2 + y^2 = h^2$

$$A = \frac{1}{2}xy \Rightarrow A^2 = \frac{1}{4}x^2y^2$$

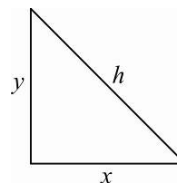
$$\Rightarrow A^2 = \frac{1}{4}x^2(h^2 - x^2) \Rightarrow f(x) = \frac{1}{4}(h^2x^2 - x^4)$$

$$f'(x) = \frac{1}{4}(2h^2x - 4x^3) = 0$$

$$h^2 = 2x^2$$

$$f''(x) < 0. A \text{ is maximum when } x = y = \frac{h}{\sqrt{2}}$$

$$\text{Maximum area } \frac{1}{2} \cdot \frac{h}{\sqrt{2}} \cdot \frac{h}{\sqrt{2}} = \frac{h^2}{4}$$



18.(C) Here Cost (v) = $av + \frac{b}{v}$ (i)

When $v = 30$, $C = 75$

$\therefore 30a + \frac{b}{30} = 75$ (ii)

When $v = 40$, $C = 65$

$\therefore 40a + \frac{b}{40} = 65$ (iii)

Solve (ii) and (iii), $a = \frac{1}{2}$, $b = 1800$

From (i), $C(v) = \frac{v}{2} + \frac{1800}{v}$

$$\Rightarrow \frac{dC}{dv} = \frac{1}{2} - \frac{1800}{v^2} = 0 \Rightarrow v = 60$$

$$\frac{d^2C}{dv^2} = \frac{3600}{v^3} > 0$$

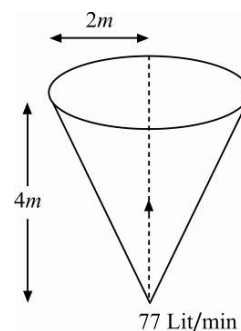
$\therefore C(v)$ is min. when $v = 60 \text{ km/h}$

19.(B) $V = \frac{1}{3}\pi r^2 h$ $\left(\because \frac{r}{h} = \frac{2}{4} = \frac{1}{2} \right)$

$$V = \frac{1}{3}\pi \frac{h^3}{4} = \frac{\pi}{12} h^3 \Rightarrow \frac{dV}{dt} = \frac{\pi}{4} h^2 \frac{dh}{dt}$$

$$77 \times 10^3 = \frac{22}{7} \times \frac{1}{4} \times 70 \times 70 \times \frac{dh}{dt} \quad (\because 1 \text{ litre} = 10^3 \text{ c.c.})$$

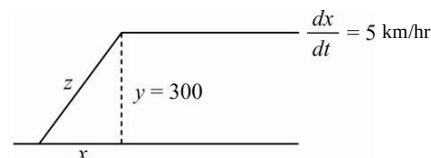
$$\therefore \frac{dh}{dt} = 20 \text{ cm/min.}$$



20.(A) From figure $z^2 = x^2 + y^2$

$$z \frac{dz}{dt} = x \frac{dx}{dt}$$

$$\text{If } z = 500 \text{ then } x = 400 \Rightarrow 500 \frac{dz}{dt} = (400) \times 5 \Rightarrow \frac{dz}{dt} = 4$$



21.(B)

22.(C) $2y^3 = ax^2 + x^3 \Rightarrow 6y^2 \frac{dy}{dx} = 2ax + 3x^2 \Rightarrow \frac{dy}{dx} \Big|_{(a, a)} = \frac{5a^2}{6a^2} = \frac{5}{6}$

Tangent at (a, a) is $5x - 6y = -a$ now put $(\alpha, 0)$ to get $\alpha = \frac{-a}{5}$ and put $(0, \beta)$ to get $\beta = \frac{a}{6}$

$$\text{Now } \alpha^2 + \beta^2 = 61 \Rightarrow \frac{a^2}{25} + \frac{a^2}{36} = 61 \Rightarrow a^2 = 25 \times 36 \Rightarrow a = \pm 30$$

23.(AB)

24.(AD) $2y = x^2 \Rightarrow 2y' = 2x \Rightarrow y' = x$ (y' at $(h, k) = h$)

Equation of normal at (h, k) is : $(y - k) = -\frac{1}{h}(x - h)$. As it passes through $(0, 3)$

$$\Rightarrow (3 - k)h = -(-h) \Rightarrow (3 - k)h = h \Rightarrow h(3 - k - 1) = 0$$

$$\Rightarrow h(2 - k) = 0 \Rightarrow 2h - \frac{h^2}{2} = 0 \quad \left(\text{As } k = \frac{h^2}{2} \right)$$

$$\Rightarrow 4h - h^2 = 0 \Rightarrow h = 0, \pm 2$$

$\therefore$ Required point are $(2, 2)$ and $(-2, 2)$ (Rejecting $(0, 0)$ since, its distance from point $(0, 3)$ is 3 which is not shortest)

25.(B) $f'(x) = \left(\frac{\sqrt{p+4}}{1-p} \right) 5x^4 - 3$

It is sufficient to solve for p, the condition $f'(x) \leq 0 \quad \forall x \in R$

$$\left(\frac{\sqrt{p+4}}{1-p} - 1 \right) 5x^4 - 3 \leq 0 \quad \forall x \in R$$

Case 1: $1-p < 0 \quad p > 1 \Rightarrow$ Inequality holds true.

Case 2: $1-p > 0 \quad p < 1 \Rightarrow$ Inequality holds if $\frac{\sqrt{p+4}}{1-p} - 1 \leq 0$

$$\Rightarrow p \geq -4, p+4 \leq (1-p)^2 \Rightarrow p \geq -4, p^2 - 3p - 3 \geq 0$$

$$\Rightarrow -4 \leq p \leq \frac{3-\sqrt{21}}{2} \quad \text{Hence } p \in \left[-4, \frac{3-\sqrt{21}}{2} \right] \cup (1, \infty)$$

26.(AD) $f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{x^2+1}} = 1 - \frac{1}{1+x^2} + 1 - \frac{1}{\sqrt{x^2+1}} = \frac{x^2}{1+x^2} + \left(1 - \frac{1}{\sqrt{x^2+1}} \right) \geq 0$

27.(BC) $g(x) = 2f\left(\frac{x}{2}\right) + f(1-x)$ and $g'(x) = f'\left(\frac{x}{2}\right) - f'(1-x)$

Now $g(x)$ is increasing if $g'(x) \geq 0$

$$f'\left(\frac{x}{2}\right) \geq f'(1-x) \quad [\because f''(x) < 0 \text{ i.e. } f'(x) \text{ is decreasing}]$$

$$\Rightarrow \frac{x}{2} \leq 1-x \Rightarrow x \leq \frac{2}{3} \Rightarrow 0 \leq x \leq \frac{2}{3} \Rightarrow g(x) \text{ increases in } 0 \leq x \leq \frac{2}{3}$$

If $g(x)$ is decreasing if $g'(x) \leq 0$

$$\Rightarrow f'\left(\frac{x}{2}\right) \leq f'(1-x) \Rightarrow \frac{x}{2} \geq 1-x \Rightarrow \frac{2}{3} \leq x \leq 1$$

28.(AC)

29.(ABC) $f'(x) = \frac{1}{1+x^2} - \frac{1}{2x}, x > 0$; $f'(x) = -\frac{(x-1)^2}{2x(1+x^2)} \leq 0 \quad \forall x > 0$

$f(x)$ is decreasing $\forall x > 0$

For $x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3} \right]$, greatest value is $f\left(\frac{1}{\sqrt{3}}\right) = \left(\frac{\pi}{6}\right) - \frac{1}{2} \ln\left(\frac{1}{\sqrt{3}}\right)$ and least value is $f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{2} \ln\sqrt{3}$

30.(D) Let R and H be the radius and height of cone and r and h be the radius and height of the cylinder.

As $\frac{H}{R} = \frac{H-h}{r}$

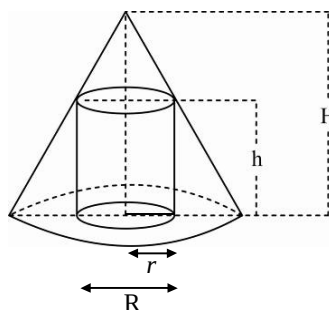
Now $S = 2\pi rh$
 $= 2\pi H \left(r - \frac{r^2}{R} \right)$

$$\frac{dS}{dr} = 2\pi H \left(1 - \frac{2r}{R} \right)$$

$$\Rightarrow \text{Maximum at } r = \frac{R}{2}$$

31.(D)

32.(D)



33.(ACD) $f(x) = \ln(x-2) - \frac{1}{x}$

$$f'(x) = \frac{1}{x-2} + \frac{1}{x^2} = \frac{x^2 + x - 2}{x^2(x-2)} = \frac{x^2 + 2x - x - 2}{x^2(x-2)} \Rightarrow \frac{(x+2)(x-1)}{x^2(x-2)}$$

As $\ln(x-2)$ is defined when $x > 2$

$$\Rightarrow f(x) \text{ is M.I. for } x \in (2, \infty) \Rightarrow f^{-1}(x) \text{ is M.I. wherever defined}$$

$$\text{Also } f''(x) = \frac{-1}{(x-2)^2} - \frac{2}{x^3} < 0 \Rightarrow f(x) \text{ is always concave downward}$$

34.(C)

35.(C) $f'(x) = \frac{x^{1/x}}{x^2} (1 - \ln x)$

$$\text{Now } f'(x) \leq 0, \text{ when } x \geq e \Rightarrow f(x) \text{ is decreasing function, when } x \geq e$$

$$\text{If } \pi > e \Rightarrow f(\pi) < f(e) \Rightarrow \pi^{1/\pi} < e^{1/e} \Rightarrow e^\pi > \pi^e$$

36.(B) $f'(x) = 50x^{49} - 20x^{19} = 10x^{19}(5x^{30} - 2)$

$x = 0$ is stationary point. Hence statement 2 is true.

$$\text{Now } f(0) = 0 \text{ and } f\left(\left(\frac{2}{5}\right)^{1/30}\right) = \left(\frac{2}{5}\right)^{5/3} - \left(\frac{2}{5}\right)^{2/3} < 0 \text{ Also } f(1) = 0$$

$\Rightarrow$ Global maximum is 0.

37.(B)

38.(A)

39.(B)

40.(B)

41.(D)

42.(D) $f(x) = a^{\left\{a^{|x|} \operatorname{sgn} x\right\}}; g(x) = a^{\left[a^{|x|} \operatorname{sgn} x\right]}$

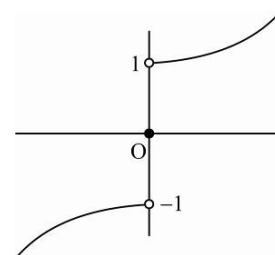
Consider $\ln a \cdot h(x) = \ln f(x) + \ln g(x)$ [For $a > 1, a \neq 1$ and $x \in \mathbb{R}$]

$$\Rightarrow (\ln a) h(x) = \left\{a^{|x|} \operatorname{sgn} x\right\} \ln a + \left[a^{|x|} \operatorname{sgn} x\right] \ln a$$

$$\Rightarrow h(x) = \left\{a^{|x|} \operatorname{sgn} x\right\} + \left[a^{|x|} \operatorname{sgn} x\right] \Rightarrow h(x) = a^{|x|} \operatorname{sgn} x$$

$$\text{Now } h(-x) = a^{|-x|} \operatorname{sgn}(-x) = -h(x)$$

Also from graph of $h(x)$ is clear that $h(x)$ is an increasing function



43.(D)

44.(B)

45.(C)

46.(AB) $f'(x) = \frac{m}{n} x^{\left(\frac{m-n}{n}\right)}$

If $m-n$ is odd.

$$f'(x) < 0 \quad \forall x \in (-\infty, 0) \quad \text{and} \quad f'(x) > 0 \quad \forall x \in (0, \infty)$$

47.(AD) $\phi'(x) = \left(3(f(x))^2 - 6(f(x)) + 4\right) f'(x) + 5 + 3\cos x - 4\sin x$

$$\text{Now } 5 - \sqrt{9+16} \leq 5 + 3\cos x - 4\sin x \leq 5 + \sqrt{9+16}$$

Adding $\left(3(f(x))^2 - 6(f(x)) + 4\right)f'(x)$ both sides

$$\left(3(f(x))^2 - 6(f(x)) + 4\right)f'(x) \leq \phi'(x) \leq \left((30f(x))^2 - 6(f(x)) + 4\right)f'(x) + 10$$

$[3(f(x))^2 - 6f(x) + 4 = 3(f(x) - 1)^2 + 1 > 0 \Rightarrow (3(f(x))^2 - 6(f(x)) + 4)f'(x) \geq 0$ when ever $f(x)$ is increasing]

$\Rightarrow \phi'(x) \geq 0 \Rightarrow \phi(x)$ is increasing, when $f(x)$ is increasing.

If $f'(x) = -11$ then $\left(3(f(x))^2 - 6f(x) + 4\right)f'(x) + 10 - 33(f(x) - 1)^2 - 1 < 0$

$\Rightarrow \phi'(x) < 0 \Rightarrow \phi(x)$ is decreasing.

48.(ABCD)

49.(ABD)

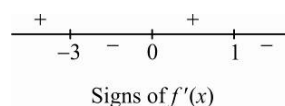
50.(C) Let $f(x) = x^7 + 14x^5 + 16x^3 + 30x - 560$

$$\Rightarrow f'(x) = 7x^6 + 70x^4 + 48x^2 + 30 > 0, \forall x \in R$$

So, $f(x)$ is increasing.

Hence, $f(x) = 0$ has only one solution.

51.(ACD)
$$f'(x) = \frac{-40 \times 12x(x+3)(x-1)}{(3x^4 + 8x^3 - 18x^2 + 60)^2}$$



Put $f'(x) = 0 \Rightarrow x = 0, x = -3, x = 1$

So, at $x = 0$, $f(x)$ has local minima and at $x = -3, x = 1$; $f(x)$ has local maxima

$$f(1) = \frac{40}{53}, f(-3) = 0, f(1) > 0 \text{ and } f(x) \neq 0$$

$\Rightarrow f(x)$ is undefined at point(s) in $(-3, 1)$. Hence $f(x)$ has no absolute maxima.

52.(B) 53.(D) 54.(C)

$$\frac{da}{dt} = 2 \Rightarrow a = 2t + c$$

$$\Rightarrow c = 0 \quad \{ \text{As } a = 0, \text{ when } t = 0 \}$$

$$a = 2t$$

$\therefore$ The curve $y = x^2 - 2ax + a^2 + a$ becomes $y = x^2 - 4tx + 4t^2 + 2t$

Now if $x = 0$, then $y = 4t^2 + 2t$

$$\frac{dy}{dx} = 2x - 4t \quad \therefore \quad \frac{dy}{dx} \Big|_{at \ x=0} = c - 4t$$

$$\therefore \text{Equation of the tangent } y - (4t^2 + 2t) = -4t(x - 0) \Rightarrow y = -4tx + 4t^2 + 2t$$

Vertex of $y = x^2 - 4tx + 4t^2 + 2t$ is $(2t - 2t)$ $\therefore$ Distance of vertex from the origin $= 2\sqrt{2}t$

$\therefore$ Rate of change of distance of vertex from origin with respect to $t = 2\sqrt{2} \Rightarrow k = 2\sqrt{2}$

$$\text{If } c(t) = 4t^2 + 2t \Rightarrow \frac{dc}{dt} = 8t + 2 \Rightarrow \frac{dc}{dt} \Big|_{at \ t=2\sqrt{2}} = 16\sqrt{2} + 2 \Rightarrow \ell = 16\sqrt{2} + 2$$

$$\text{Again } m(t) = -4t \Rightarrow \frac{dm}{dt} = -4 \Rightarrow \frac{dm}{dt} \Big|_{at \ t=\ell} = -4$$

55-57. 55.(C) 56.(B) 57.(A)

Let $g(x) = \frac{x + \sin x}{2}$, $x \in [0, \pi]$, $g(x)$ is increasing function of x .

$$\therefore \text{Range of } g(x) \text{ is } \left[0, \frac{\pi}{2}\right] \quad \therefore f(x) = \frac{x \sin x}{2}, x \in [0, \pi]$$

Now if $\pi \leq t \leq 2\pi$, then $f(t) + f(2\pi - t) = \pi \Rightarrow f(t) + \frac{2\pi - t + \sin(2\pi - t)}{2} = \pi$

$$\Rightarrow f(t) + \pi - \frac{t}{2} - \frac{\sin t}{2} = \pi \Rightarrow f(t) = \frac{t + \sin t}{2}$$

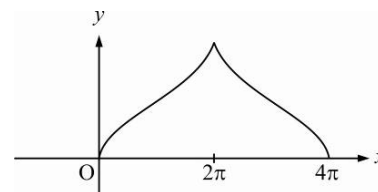
$$\therefore f(x) = \frac{x + \sin x}{2} \text{ for } \pi \leq x \leq 2\pi$$

Also $f(x) = f(4\pi - x)$ for all $x \in [2\pi, 4\pi]$

$\Rightarrow f(x)$ is symmetric about $x = 2\pi$

$$\therefore \text{From graph of } f(x) \Rightarrow \alpha = 2\pi - 0 = 2\pi \Rightarrow \beta = \alpha$$

and maximum value is $f(2\pi) = \pi = \frac{\beta}{2}$



58.(D) Clearly $f(x) = e^{x^2} + e^{-x^2}$

$$\Rightarrow f'(x) = 2x(e^{x^2} - e^{-x^2}) \geq 0 \Rightarrow f(x) \text{ increasing} \Rightarrow f_{\max} = f(1) = e + \frac{1}{e}$$

$$g(x) = xe^{x^2} + e^{-x^2}$$

$$\Rightarrow g'(x) = e^{x^2} + 2xe^{x^2} - 2xe^{-x^2} > 0 \Rightarrow g(x) \text{ is increasing} \Rightarrow g_{\max} = g(1) = e + \frac{1}{e}$$

$$h(x) = x^2e^{x^2} + e^{-x^2}$$

$$\Rightarrow h'(x) = 2xe^{x^2} + 2x^3e^{x^2} - 2xe^{-x^2} = 2x(e^{x^2} + x^2e^{x^2} - e^{-x^2}) > 0 \Rightarrow h_{\max} = h(1) = e + \frac{1}{e}$$

$$\Rightarrow a = b = c$$

59.(C) $x^2 = x \sin x + \cos x$

Let $f(x) = x^2$ and $g(x) = x \sin x + \cos x$

Draw the graph of $f(x)$ and $g(x)$ to get the answer as two points of intersection.

60.(AC) Let $\ell = 8x, b = 15x$

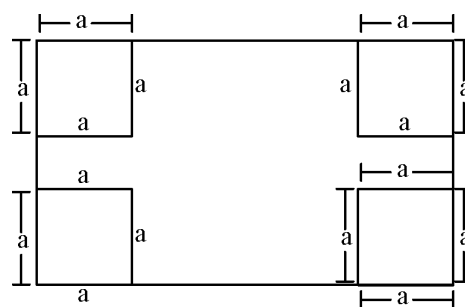
$$\therefore \text{Volume} = (8x - 2a)(15x - 2a)(a) = 4a^3 - 46a^2x + 120ax^2$$

$$\frac{dV}{da} = 6a^2 - 46ax + 60x^2 \Rightarrow \left(\frac{dV}{da}\right)_{at a=5} = 0$$

$$\Rightarrow x = 3 \text{ and } \frac{5}{6}$$

$$\text{Now } \frac{d^2V}{da^2} = 6a - 23x \Rightarrow \left(\frac{d^2V}{da^2}\right)_{at a=5 \text{ and } x=3} < 0 \Rightarrow \text{At } x = 3 \text{ gives maxima}$$

$$\text{Again } \left(\frac{d^2V}{da^2}\right)_{at a=5 \text{ and } x=\frac{5}{6}} > 0 \Rightarrow \text{At } x = \frac{5}{6} \text{ gives minima.}$$



Now $\frac{dV}{da} = 0$ when $a = 5$ given (As $4a^2 = 100$ given for maximum volume)

At $a = 5$ and $\frac{dV}{da} = 0$ we get : $6x^2 - 23x + 15 = 0 \Rightarrow x = 3$ or $\frac{5}{8}$

So at $x = 3$ (for max volume) $\Rightarrow 8x = 24, 15x = 45$

$$61.(B) \quad f(x) = 2|x| + |x+2| - ||x+2| - 2||x|| = \begin{cases} -2x-4 & x \leq -2 \\ 2+4x & -2 < x \leq -\frac{2}{3} \\ -4x & -\frac{2}{3} < x \leq 0 \\ 4x & 0 < x \leq 2 \\ 2x+4 & x > 2 \end{cases}$$

Draw the graph of $y = f(x)$ is minima at $x = -2, 0$; maxima at $x = -\frac{2}{3}$

62.(AB)

63.(A) $y = x^2 - 5x + 6$

$$\frac{dy}{dx} = 2x - 5 \Rightarrow \left. \frac{dy}{dx} \right|_{(2,0)} = -1, \left. \frac{dy}{dx} \right|_{(3,0)} = 1$$

Product of slopes $= -1 \Rightarrow$ angle between tangents is $\frac{\pi}{2}$

64.(B) 65.(A)

66.(AC) By LMVT we get : $\frac{f(6) - f(1)}{6 - 1} = f'(x) \quad x \in (1, 6)$

$$\text{And } f'(x) \geq 2 \Rightarrow \frac{f(6) - f(1)}{6 - 1} \geq 2 \Rightarrow \frac{f(6) + 2}{5} \geq 2 \Rightarrow f(6) \geq 8$$

67.(AC) $\frac{dy}{dx} = \frac{a(\cos \theta - \cos \theta + \theta \sin \theta)}{a(-\sin \theta + \sin \theta + \theta \cos \theta)} \Rightarrow \frac{dy}{dx} = \frac{\sin \theta}{\cos \theta}$

$$\text{Equation of normal is } y - a(\sin \theta - \theta \cos \theta) = \frac{-\cos \theta}{\sin \theta} (x - a(\cos \theta + \theta \sin \theta))$$

$$\Rightarrow x \cos \theta + y \sin \theta = a \Rightarrow \text{Distance from origin} = |a| (\text{constant})$$

$$\text{Slope of normal} = -\frac{\cos \theta}{\sin \theta} = \tan\left(\frac{\pi}{2} + \theta\right) \Rightarrow \text{angle made of } x\text{-axis is } \frac{\pi}{2} + \theta$$

68.(A) By LMVT we get : $\frac{f(3) - f(1)}{3 - 1} = f'(c), \quad 1 < c < 3$

$$\Rightarrow \frac{\log_e^3 - 0}{2} = \frac{1}{c} \Rightarrow c = \frac{2}{\log_e^3} = \log_3 e$$

69.(B)

70.(B) $P'(x) = 4x^3 + 3ax^2 + 2bx + c$ and $P'(0) = 0 \Rightarrow c = 0$

$$P'(x) = x(4x^2 + 3ax + 2b)$$

$$\text{Now } D = 9a^2 - 32b < 0 \Rightarrow b > \frac{9a^2}{32} > 0 \quad (\because P'(x) = 0 \text{ has only one root } x = 0)$$

$$\begin{aligned}
 P(-1) < P(1) &\Rightarrow a > 0 \\
 P'(x) \text{ has only one changed sign.} &\Rightarrow x = 0 \text{ is a point of minima.} \\
 P(-1) = 1 - a + b + d, P(0) = d \\
 \Rightarrow P(-1) < P(1), P(0) < P(1), P(-1) > P(0) &\Rightarrow P(-1) \text{ is not minimum but } P(1) \text{ is maximum.}
 \end{aligned}$$

71.(C)

$$72.(C) \quad V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \left(\frac{dr}{dt} \right) \Rightarrow 45 \times 25 \times 3 = r^3 \left(\text{As } 4500\pi = \frac{4\pi r^3}{3} \right)$$

$$\Rightarrow r = 15m$$

$$\text{Final volume at the end of 49 min is : } (4500 - 49 \times 72) \pi = 972\pi \text{ m}^3$$

$$\text{If } r \text{ is the radius at the end of 49 min. then } r^3 = 3 \times 243 = 3 \times 3^6 \Rightarrow r = 9$$

$$\Rightarrow \text{Radius of the balloon at the end of 49 min is } 9 \text{ m}$$

$$\Rightarrow 72\pi = 4\pi \times 9 \times 9 \left(\frac{dr}{dt} \right) \Rightarrow \frac{dr}{dt} = \left(\frac{2}{9} \right)$$

$$73.(D) \quad f(x) = 2x^3 + 3x + k$$

$$f'(x) = 6x^2 + 3 > 0 \quad \forall x \in R$$

$$\Rightarrow f(x) \text{ is strictly increasing function} \Rightarrow f(x) = 0 \text{ has only one real root, so two roots are not possible.}$$

$$74.(C) \quad \text{Given that, } \frac{dV}{dt} = 50 \text{ cm}^3 / \text{min} \Rightarrow \frac{d}{dt} \left(\frac{4}{3}\pi r^3 \right) = 50$$

$$\Rightarrow 3r^2 \frac{dr}{dt} = \frac{50 \times 3}{4\pi} \Rightarrow \frac{dr}{dt} = \frac{50}{4\pi r^2} \Rightarrow \left(\frac{dr}{dt} \right)_{r=15} = \frac{50}{4\pi \times 225} = \frac{1}{18\pi} \text{ cm/min}$$

$$75.(D) \quad y^2 = 18x \quad \dots (i)$$

Differentiating w.r.t. t

$$2y \frac{dy}{dt} = 18 \frac{dx}{dt} \Rightarrow 4y = 18 \quad \left(\because \frac{dy}{dt} = \frac{2dx}{dt} \right)$$

$$\Rightarrow y = \frac{9}{2}$$

$$\text{From equation (i), we get : } x = \frac{9}{8} \Rightarrow \text{Required points is } \left(\frac{9}{8}, \frac{9}{2} \right)$$

$$76.(D) \quad e^x + 2e^{-x} \geq 2\sqrt{2} \quad (AM \geq GM)$$

$$\frac{1}{e^x + 2e^{-x}} \leq \frac{1}{2\sqrt{2}} \Rightarrow \frac{1}{2\sqrt{2}} \geq f(x) > 0 \quad \text{so statement - 2 is correct}$$

$$\text{As } f(x) \text{ is continuous and } \frac{1}{3} \text{ belongs to range } \left[0, \frac{1}{2\sqrt{2}} \right] \text{ of } f(x) \Rightarrow f(c) = \frac{1}{3} \text{ for some } c \in R.$$

$$77.(D) \quad \text{Here, } f(0) = 2 = g(1), g(0) = 0 \text{ and } f(1) = 6$$

$$\because f \text{ and } g \text{ are differentiable in } (1, 0).$$

$$\text{Let } h(x) = f(x) - 2g(x)$$

$$h(0) = f(0) - 2g(0) \Rightarrow h(0) = 2 - 0 \Rightarrow h(0) = 2$$

$$\text{Now, } h(1) = f(1) - 2g(1) = 6 - 2(2)$$

$$h(1) = 2, h(0) = h(1) = 2$$

Hence, using Rolle's theorem,

There exists $c \in (0, 1)$, such that $h'(c) = 0$

$$\Rightarrow f'(c) - 2g'(c) = 0, \text{ for some } c \in (0, 1) \quad \Rightarrow f'(c) = 2g'(c)$$

78.(C) Here, $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$, then

$$f'(x) = \frac{\alpha}{x} + 2\beta x + 1$$

$$\therefore f'(-1) = -\alpha - 2\beta + 1 = 0 \quad [\text{At extreme point, } f'(x) = 0] \quad \dots(\text{i})$$

$$\text{and } f'(2) = \frac{\alpha}{2} + 4\beta + 1 = 0 \quad \dots(\text{ii})$$

On solving equation (i) and (ii), we get :

$$\Rightarrow \alpha = 2, \beta = -\frac{1}{2}$$

79.(C) Statement (i) A function f , such that

$$f(x) = \log|x| + bx^2 + ax, x \neq 0$$

Statement (ii) The function f has extrema at $x = -1$ and $x = 2$,

$$\text{i.e., } f'(-1) = f'(2) = 0 \text{ and } f''(-1) \neq 0 \neq f''(2)$$

Now, given function f is given by $f(x) = \log|x| + bx^2 + ax$

$$\Rightarrow f'(x) = \frac{1}{x} + 2bx + a \quad \Rightarrow f''(x) = \frac{-1}{x^2} + 2b$$

Since, f has extrema at $x = -1$ and $x = 2$

$$\text{Hence } f'(-1) = 0 = f'(2) \quad \Rightarrow a = \frac{1}{2} \text{ and } b = \frac{-1}{4}$$

$$\Rightarrow f''(x) = \frac{-1}{x^2} + \frac{-1}{2} = -\left(\frac{x^2 + 2}{2x^2}\right) \quad \Rightarrow f''(-1) < 0 \text{ and } f''(2) < 0$$

Hence, f has local maxima at both $x = -1$ and $x = 2$

Hence, Statement I is correct.

Also, while solving for Statement I, we found the value of a and b , which justify that Statement II is also correct.

However, Statement II does not explain Statement I in any way.

80.(0) PLAN Any function have extreme values (maximum or minimum) at its critical points, where $f'(x) = 0$.

Since, the function have extreme values at $x = 1$ and $x = 2$

$$f'(x) = 0 \text{ at } x = 1 \text{ and } x = 2$$

$$\Rightarrow f'(1) = 0 \text{ and } f'(2) = 0$$

$$\text{Also, it is given that, } \lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$$

$$\Rightarrow 1 + \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 3 \quad \Rightarrow \quad \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2$$

$$\Rightarrow f(x) \text{ will be of the form } ax^4 + bx^3 + 2x^2. \quad [\because f(x) \text{ is of four degree polynomial}]$$

$$\text{Let } f(x) = ax^4 + bx^3 + 2x^2$$

$$\Rightarrow f'(x) = 4ax^3 + 3bx^2 + 4x$$

$$\Rightarrow f'(1) = 4a + 3b + 4 = 0 \quad \dots (i)$$

$$\text{and } f'(2) = 32a + 12b + 8 = 0$$

$$\Rightarrow 8a + 3b + 2 = 0 \quad \dots (ii)$$

On solving Eqs. (i) and (ii),

$$\text{We get } a = \frac{1}{2}, b = -2 \quad \therefore f(x) = \frac{x^4}{2} - 2x^3 + 2x^2 \Rightarrow f(2) = 8 - 16 + 8 = 0$$

81.(2) PLAN The given equation contains algebraic and trigonometric functions called transcendental equation. To solve transcendental equations we should always plot the graph for LHS and RHS.

$$\text{Here, } x^2 = x \sin x + \cos x$$

$$\text{Let } f(x) = x^2 \text{ and } g(x) = x \sin x + \cos x$$

$$\text{We know that, the graph for } f(x) = x^2$$

$$\text{To plot, } g(x) = x \sin x + \cos x$$

$$g'(x) = x \cos x + \sin x - \sin x$$

$$g'(x) = x \cos x \quad \dots (i)$$

$$g''(x) = -x \sin x \cos x \quad \dots (ii)$$

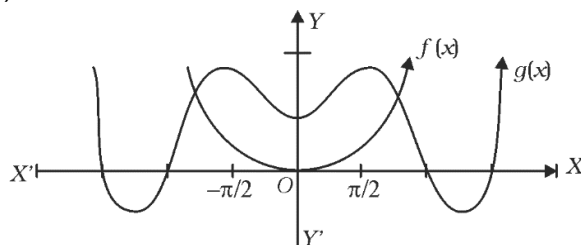
$$\text{Put } g'(x) = 0 \Rightarrow x \cos x = 0$$

$$x = 0, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\text{At } x = 0, \frac{3\pi}{2}, \frac{7\pi}{2}, \dots, f''(x) > 0, \text{ so minimum}$$

$$\text{At } x = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \dots, f''(x) < 0, \text{ so maximum}$$

So, graph of $f(x)$ and $g(x)$ are shown as



So, number of solutions are 2.

82.(1) The maximum value of $f(x) = \cos x + \cos(\sqrt{2}x)$ is 2 which occurs at $x = 0$. Also, there is no other value of x for which this value will be attained again.

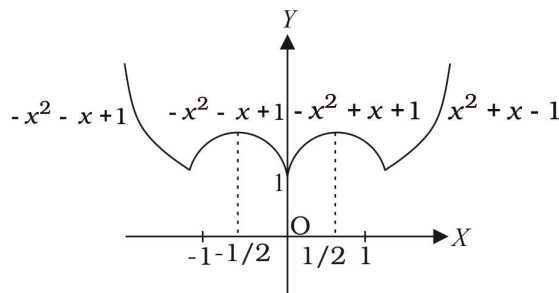
- 83.(5)** (i) Local maximum and local minimum are those points at which $f'(x) = 0$, when defined for all real numbers.
 (ii) Local maximum and local minimum for piecewise functions are also been checked a sharp edges.

Description of Situation $y = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$

Also, $y = |x^2 - 1| = \begin{cases} (x^2 - 1), & \text{if } x \leq -1 \text{ or } x \geq 1 \\ (1 - x^2), & \text{if } -1 \leq x \leq 1 \end{cases}$

$$y = |x| + |x^2 - 1| = \begin{cases} -x+1-x^2, & \text{if } x \leq -1 \\ -x+1-x^2, & \text{if } -1 \leq x \leq 1 \\ x+1-x^2, & \text{if } 0 \leq x \leq 1 \\ x+x^2-1, & \text{if } x \geq 1 \end{cases}$$

$$= \begin{cases} -x^2-x+1, & \text{if } x \leq -1 \\ -x^2-x+1, & \text{if } -1 \leq x \leq 1 \\ -x^2+x+1, & \text{if } 0 \leq x \leq 1 \\ x^2+x-1, & \text{if } x \geq 1 \end{cases}$$



which could be graphically shown as

Thus $f(x)$ attains maximum at $x = \frac{1}{2}, \frac{-1}{2}$ and $f(x)$ attains minimum at $x = -1, 0, 1$.

$\Rightarrow$ Total number of points = 5

84.(9) If $f(x)$ is least degree polynomial having local maximum and local minimum at α and β .

Then, $f'(x) = \lambda(x - \alpha)(x - \beta)$

Here, $p'(x) = \lambda(x - 1)(x - 3) = \lambda(x^2 - 4x + 3)$

On integrating both sides between 1 to 3, we get

$$\int_1^3 p'(x) dx = \int_1^3 \lambda(x^2 - 4x + 3) dx$$

$$\Rightarrow (p(x))_1^3 = \lambda \left(\frac{x^3}{3} - 2x^2 + 3x \right)_1^3 \Rightarrow p(3) - p(1) = \lambda \left((9 - 18 + 9) - \left(\frac{1}{3} - 2 + 3 \right) \right)$$

$$\Rightarrow 2 - 6 = \lambda \left(\frac{-4}{3} \right) \Rightarrow \lambda = 3 \Rightarrow p'(x) = 3(x - 1)(x - 3)$$

$$\therefore p'(0) = 9$$

85.(2) $f(x) = x^4 - 4x^3 + 12x^2 + x - 1$

$$f'(x) = 4x^3 - 12x^2 + 24x + 1$$

$$f''(x) = 12x^2 - 24x + 24 = 12(x^2 - 2x + 2) = 12\{(x - 1)^2 + 1\} > 0 \forall x$$

$\Rightarrow f'(x)$ is increasing.

Since, $f'(x)$ is cubic and increasing.

$\Rightarrow f'(x)$ has only one real root and two imaginary roots.

$\therefore f(x)$ cannot have all distinct roots.

$\Rightarrow$ Atmost 2 real roots.

Now, $f(-1) = 15, f(0) = -1, f(1) = 9$

$\therefore f(x)$ must have one root in $(-1, 0)$ and other in $(0, 1)$.

$\Rightarrow$ 2 real roots.

- 86.(1) Let $g(x) = e^{f(x)}, \forall x \in R \Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$
 $\Rightarrow f'(x)$ changes its sign from positive to negative in the neighbourhood of $x = 2009$.
 $\Rightarrow f(x)$ has local maxima at $x = 2009$.
 So, the number of local maximum is one.

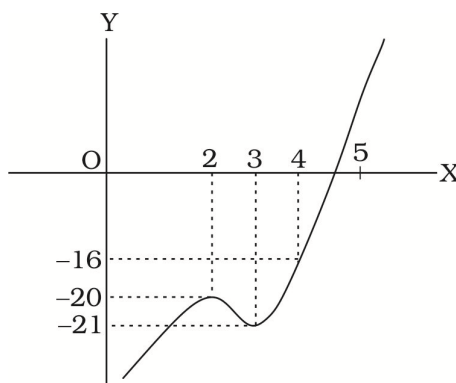
87.(2) Let
$$f(\theta) = \frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$$

Again let, $g(\theta) = \sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta$

$$= \frac{1 - \cos 2\theta}{2} + 5 \left(\frac{1 + \cos^2 \theta}{2} \right) + \frac{3}{2} \sin 2\theta = 3 + 2 \cos^2 \theta + \frac{3}{2} \sin 2\theta$$

$\therefore$ Maximum value of $f(\theta) = \frac{1}{1/2} = 2$

88.(7) Given, $A = \{x | x^2 + 20 \leq 9x\} = \{x | x \in [4, 5]\}$



Now, $f'(x) = 6(x^2 - 5x + 6)$

Put $f'(x) = 0 \Rightarrow x = 2, 3$

$f'(2) = -20, f(3) = -21, f(4) = -16, f(5) = -7$

From graph, maximum value of $f(x)$ on set A is $f(5) = -7$.

- 89.(3) The equation $x^3 - 3x + c = 0$ will have its roots real and distinct if the curves $y = x^3 - 3x$ and $y = -c$ intersect in distinct points.
- The function $f(x) = x^3 - 3x$ attains its local minima and maxima at $x = -1$ and $x = 1$

$$f(x)|_{x=-1} = 2 \quad f(x)|_{x=1} = -2$$

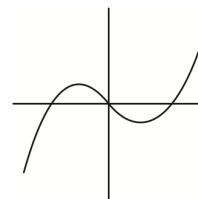
Now for three distinct solutions

$$-2 < -c < 2 \Rightarrow -2 < c < 2$$

Three integral values

90.(1)
$$f(x) = \begin{cases} 4x^3 - x(x-2); & 2 \leq x \leq 3 \\ 4x^3 - x(2-x); & 0 \leq x < 2 \end{cases}$$

$$f'(x) = \begin{cases} 12x^2 - 2x + 2; & 2 \leq x \leq 3 \\ 12x^2 + 2x - 2; & 0 \leq x < 2 \end{cases}$$



Clearly there's only one point of minima lying in $(0, 1)$

91.(1) $f(x) = 3 \tan x + x^3 - 2$

$f(0) < 0$

$f\left(\frac{\pi}{4}\right) > 0$

$f'(x) = 3 \sec^2 x + 3x^2 > 0$

Therefore only one solution

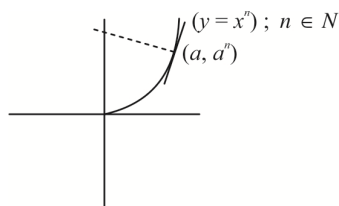
92.(2) Equation of normal at (a, a^n)

$y - a^n = -\frac{1}{na^{n-1}}(x - a)$

$b = \frac{a}{na^{n-1}} + a^n$

$\lim_{a \rightarrow 0} \left(\frac{a}{na^{n-1}} + a^n \right) = \frac{1}{2}$

$\lim_{a \rightarrow 0} \left(\frac{1}{na^{n-2}} \right) = \frac{1}{2} \Rightarrow n = 2$



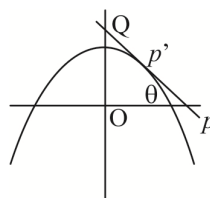
93.(4) Take a parametric point $p'(t, 3 - t^2)$

$\frac{dy}{dx} = -2t$

Area of triangle $OPQ = \frac{1}{2} OQ \cdot \frac{OQ}{\tan \theta}$

$= \frac{1}{2} \frac{(t^2 + 3)^2}{2t} = \frac{1}{4} \frac{(t^2 + 3)^2}{t}$

Whose minimum value for $t > 0$ is 4

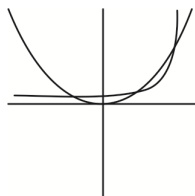


94.(3) The given equation is

$4x^2 e^{-x} - 1 = 0$

$4x^2 e^{-x} = 1$

$4x^2 = e^x$



Matrices & Determinants

$$1.(C) \quad \begin{vmatrix} 1+x & 1 & 1 \\ 1+y & 1+2y & 1 \\ 1+z & 1+z & 1+3z \end{vmatrix} = xyz \begin{vmatrix} 1+\frac{1}{x} & \frac{1}{x} & \frac{1}{x} \\ 1+\frac{1}{y} & 2+\frac{1}{y} & \frac{1}{y} \\ 1+\frac{1}{z} & 1+\frac{1}{z} & 3+\frac{1}{z} \end{vmatrix}$$

Use, $R_1 \rightarrow R_1 + R_2 + R_3$

$$= xyz \begin{vmatrix} 1 & 1 & 1 \\ 1+\frac{1}{y} & 2+\frac{1}{y} & \frac{1}{y} \\ 1+\frac{1}{z} & 1+\frac{1}{z} & 3+\frac{1}{z} \end{vmatrix}$$

Use, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$= xyz \begin{vmatrix} 1 & 0 & 0 \\ 1+\frac{1}{y} & 1 & -1 \\ 1+\frac{1}{z} & 0 & 2 \end{vmatrix} = 2xyz \left(3 + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) = 0 \Rightarrow x^{-1} + y^{-1} + z^{-1} = -3$$

$$2.(A) \quad f = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix} \Rightarrow f = \begin{vmatrix} x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \\ 1 & 1 & 1 \end{vmatrix}$$

$$f = (xy + yz + zx)(x - y)(y - z)(z - x)$$

$$3.(A) \quad g(x) = \begin{vmatrix} a^{-x} & e^{\log_e a^x} & x^2 \\ a^{-3x} & e^{\log_e a^{3x}} & x^4 \\ a^{-5x} & e^{\log_e a^{5x}} & 1 \end{vmatrix} = \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix} \quad \left(\because e^{\log a^x} = a^x \right)$$

$$g(-x) = \begin{vmatrix} a^x & a^{-x} & x^2 \\ a^{3x} & a^{-3x} & x^4 \\ a^{5x} & a^{-5x} & 1 \end{vmatrix} = - \begin{vmatrix} a^{-x} & a^x & x^2 \\ a^{-3x} & a^{3x} & x^4 \\ a^{-5x} & a^{5x} & 1 \end{vmatrix} \left(\text{Interchanging I}^{\text{st}} \text{ and II}^{\text{nd}} \text{ columns} \right) = -g(x) \Rightarrow g(x) + g(-x) = 0$$

4.(C) Since $3 < \pi < 3.2$ and $2.7 < e < 2.8$, $[3\pi] = 9$ and $[2e] = 5$, so that the three equations are consistent provided $[4a] = 6$. This gives, $6 \leq 4a < 7$

$$\Rightarrow \frac{3}{2} \leq a < \frac{7}{4} \quad \text{i.e. } a \text{ lies in } \left[\frac{3}{2}, \frac{7}{4} \right)$$

$$\begin{aligned}
 5.(D) \quad \text{We have } \phi'(x) &= \begin{vmatrix} f''(x) & g''(x) & h''(x) \\ f''(x) & g''(x) & h''(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f'''(x) & g'''(x) & h'''(x) \\ f'''(x) & g'''(x) & h'''(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \\ f^{iv}(x) & g^{iv}(x) & h^{iv}(x) \end{vmatrix} \\
 &= 0 + 0 + \begin{vmatrix} f'(x) & g'(x) & h(x) \\ f''(x) & g''(x) & h''(x) \\ 0 & 0 & 0 \end{vmatrix} = 0 \quad [\because f, g, h \text{ are polynomials of degree 3, } f^{iv}(x) = g^{iv}(x) = h^{iv}(x) = 0]
 \end{aligned}$$

$\Rightarrow \phi(x)$ must be a constant.

$$6.(A) \quad \begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix}$$

$$\text{Use, } R_2 \rightarrow R_2 - (R_1 + R_3) = \text{gives } \begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ -4 & 0 & 0 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = 12(2x - 1) = Ax + B$$

$$\Rightarrow A = 24, B = -12 \quad \Rightarrow A + 2B = 0$$

$$7.(C) \quad \text{Let } A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}. \text{ Now } A^T = \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix}$$

As A is orthogonal $\Rightarrow AA^T = I$

$$\Rightarrow \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix} \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 4b^2 + c^2 & 2b^2 - c^2 & -2b^2 + c^2 \\ 2b^2 - c^2 & a^2 + b^2 + c^2 & a^2 - b^2 - c^2 \\ -2b^2 + c^2 & a^2 - b^2 - c^2 & a^2 + b^2 + c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we get :

$$4b^2 + c^2 = 1 \dots (i) \quad , \quad 2b^2 - c^2 = 0 \dots (ii) \quad \text{and} \quad a^2 + b^2 + c^2 = 1 \dots (iii)$$

$$\text{On solving (i), (ii) and (iii), we get } a = \pm \frac{1}{\sqrt{2}}; b = \pm \frac{1}{\sqrt{6}}; c = \pm \frac{1}{\sqrt{3}}$$

$$8.(A) \quad A^2 = AA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix}$$

$$\begin{aligned}
 \text{Again } A^3 &= A^2A = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} = \begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+r^2p & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix} \\
 &= \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix} + \begin{bmatrix} 0 & q & 0 \\ 0 & 0 & q \\ pq & q^2 & qr \end{bmatrix} + \begin{bmatrix} 0 & 0 & r \\ pr & qr & r^2 \\ pr^2 & pr+qr^2 & qr+r^3 \end{bmatrix}
 \end{aligned}$$

$$= p \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + q \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} + r \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix}$$

$$= pI + qA + rA^2 \Rightarrow A^3 - rA^2 - qA = pI$$

9.(D) We have, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ $A^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ $A^3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$

Similarly, $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$. Now, go option by option

Clearly A and C are ruled out.

(B) $\begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} \neq \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} + \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$ (D) Now, $nA - (n-1)I = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

10.(C) Now, $P^3 = P(I - P) = PI - P^2 = PI - (I - P) = P - I + P = 2P - I$

$$P^4 = P \cdot P^3 \Rightarrow P^4 = P(2P - I) \Rightarrow P^4 = 2P^2 - P$$

$$\Rightarrow P^4 = 2I - 2P - P \Rightarrow P^4 = 2I - 3P \Rightarrow P^5 = P(2I - 3P)$$

$$\Rightarrow P^5 = 2P - 3(I - P) \Rightarrow P^5 = 5P - 3I \Rightarrow P^6 = P(5P - 3I)$$

$$\Rightarrow P^6 = 5P^2 - 3P \Rightarrow P^6 = 5(I - P) - 3P \Rightarrow P^6 = 5I - 8P, \text{ So, } n = 6.$$

11.(A) $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ x/3 & 2/3 & y/3 \end{bmatrix} \Rightarrow AA^T = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ x/3 & 2/3 & y/3 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & x/3 \\ 2/3 & 1/3 & 2/3 \\ 2/3 & -2/3 & y/3 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & \frac{x+2y+4}{9} \\ 0 & 1 & \frac{2x-2y+2}{9} \\ \frac{x+2y+4}{9} & \frac{2x-2y+2}{9} & \frac{x^2+4+y^2}{9} \end{bmatrix}$$

$$\Rightarrow x+2y+4=0 \quad \dots (i) \quad 2x-2y+2=0 \quad \dots (ii)$$

Solve (i) and (ii) to get : $x = -2$ and $y = -1 \Rightarrow x+y = -3$

12.(A) $P^T P = PP^T = I \Rightarrow Q^2 = PAP^T PAP^T = PA^2 P^T \Rightarrow Q^n = PA^n P^T$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \Rightarrow A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \Rightarrow P^T Q^{2005} P = P^T PA^{2005} P^T P = A^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

13.(B) $2x-y = \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & 2 \end{bmatrix} \dots (i) \text{ and } x+2y = \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix} \dots (ii)$

Multiply (i) by 2 and add to (ii) to get :

$$5x = \begin{bmatrix} 6 & -6 & 0 \\ 6 & 6 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 1 & 5 \\ -1 & 4 & -4 \end{bmatrix} = \begin{bmatrix} 10 & -5 & 5 \\ 5 & 10 & 0 \end{bmatrix} \Rightarrow x = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

Multiplying (ii) by 2 and subtract (i) from it to get :

$$5y = \begin{bmatrix} 8 & 2 & 10 \\ -2 & 8 & -8 \end{bmatrix} - \begin{bmatrix} 3 & -3 & 0 \\ 3 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 5 & 10 \\ -5 & 5 & -10 \end{bmatrix} \Rightarrow y = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 1 & -2 \end{bmatrix}$$

$$\Rightarrow x + y = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & -2 \end{bmatrix} \text{ and } x - y = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 1 & 2 \end{bmatrix}$$

- 14.(B)** Solve $2x + y = 5$ and $x + 3y = 5$ to get $x = 2$ and $y = 1$. As $(x, y) \equiv (2, 1)$ satisfies the equation $x - 2y = 0$, we have unique solution for given system of equations.

$$\mathbf{15.(C)} \quad f(x+y) = \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(x) \cdot f(y)$$

$$\mathbf{16.(B)} \quad A = \frac{\sin 2\theta}{2} \begin{bmatrix} \cot \theta & 1 \\ 1 & \tan \theta \end{bmatrix} \text{ assuming } \begin{bmatrix} \cos \theta \neq 0 \\ \sin \theta \neq 0 \end{bmatrix}; \quad B = \frac{\sin 2\phi}{2} \begin{bmatrix} \cot \phi & 1 \\ 1 & \tan \phi \end{bmatrix} \text{ assuming } \begin{bmatrix} \cos \phi \neq 0 \\ \sin \phi \neq 0 \end{bmatrix}$$

$$\Rightarrow AB = \frac{\sin 2\theta \sin 2\phi}{4} \begin{bmatrix} \cot \theta \cot \phi + 1 & \cot \theta + \tan \phi \\ \cot \phi + \tan \theta & 1 + \tan \theta \tan \phi \end{bmatrix}$$

As $\sin 2\theta \sin 2\phi \neq 0$

$$\Rightarrow \cot \theta \cot \phi + 1 = 0 \Rightarrow |\theta - \phi| = \frac{(2n+1)\pi}{2}$$

$$\text{and } \cot \theta \tan \phi = 0 \Rightarrow |\theta - \phi| = \frac{(2n+1)\pi}{2} \quad \text{and } \cot \phi \tan \theta = 0 \Rightarrow |\theta - \phi| = \frac{(2n+1)\pi}{2}$$

$$\text{and } 1 + \tan \theta \tan \phi = 0 \Rightarrow |\theta - \phi| = \frac{(2n+1)\pi}{2} \Rightarrow |\theta - \phi| = \frac{\pi}{2}$$

$$\mathbf{17.(A)} \quad (A+B) = \begin{bmatrix} 1+\omega & \omega+\omega^2 & 1+\omega^2 \\ \omega+\omega^2 & \omega^2+1 & 1+\omega \\ \omega^2+\omega & 1+\omega^2 & \omega+1 \end{bmatrix} = \begin{bmatrix} -\omega^2 & -1 & -\omega \\ -1 & -\omega & -\omega^2 \\ -1 & -\omega & -\omega^2 \end{bmatrix}$$

$$(A+B)C = \begin{bmatrix} -\omega^2 & -1 & -\omega \\ -1 & -\omega & -\omega^2 \\ -1 & -\omega & -\omega^2 \end{bmatrix} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix} = \begin{bmatrix} -\omega^2 - \omega - \omega^3 \\ -1 - \omega^2 - \omega^4 \\ -1 - \omega^2 - \omega^4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{18.(A)} \quad \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} a^2 & ab & ac \\ ba & b^2 & bc \\ ca & cb & c^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O.$$

- 19. (A)** p, q, r are the entries of first row and C_{21}, C_{22}, C_{23} are cofactors of second row

$$\therefore p \cdot C_{21} + q \cdot C_{22} + r \cdot C_{23} = 0$$

- 20. (C)** x, y, z are the entries of second row and C_{21}, C_{22}, C_{23} are cofactors second row

$$\therefore x \cdot C_{21} + y \cdot C_{22} + z \cdot C_{23} = \Delta$$

- 21. (B)** $q \cdot M_{12} - y \cdot M_{22} + m \cdot M_{32} = -q \cdot C_{12} - y \cdot C_{22} - m \cdot C_{32} = -(q \cdot C_{12} + y \cdot C_{22} + m \cdot C_{32}) = -\Delta$

[$\because q, y, m$ are the entries of second column and C_{12}, C_{22}, C_{32} are cofactors second column]

$$22.(A) f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix} \quad (\text{Differentiating column by column})$$

$$f'(x) = \begin{vmatrix} 1 & 2x & 0 \\ 1 & 4x & 0 \\ 1 & 6x & 0 \end{vmatrix} + \begin{vmatrix} x+a & 2x & 1 \\ x+b & 4x & 1 \\ x+c & 6x & 1 \end{vmatrix} + \begin{vmatrix} x+a & x^2+1 & 0 \\ x+b & 2x^2-1 & 0 \\ x+c & 3x^2-2 & 0 \end{vmatrix}$$

$$= 0 + 2x \begin{vmatrix} x+a & 1 & 1 \\ b-a & 1 & 0 \\ c-b & 1 & 0 \end{vmatrix} + 0 \quad \begin{pmatrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{pmatrix}$$

$$= 2x(2b - a - c) = 0 \quad (\because a, b, c \text{ are in A.P.})$$

$$23.(D) \Delta = 0$$

$$\Rightarrow f(x) f\left(\frac{1}{x}\right) - f(x) - f\left(\frac{1}{x}\right) = 0$$

$$\Rightarrow f(x) f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow (a + bx^n) \left(a + \frac{b}{x^n}\right) = (a + bx^n) + \left(a + \frac{b}{x^n}\right)$$

Comparing coeffs. of x^n on both sides,

$$ab = b \Rightarrow a = 1$$

Equating constant terms on both sides.

$$a^2 + b^2 = 2a$$

$$\Rightarrow 1 + b^2 = 2 \Rightarrow b^2 = 1 \Rightarrow b = \pm 1$$

$$\therefore f(x) = 1 \pm x^n$$

$$f(2) = 17 = 1 + (2)^4 \Rightarrow n = 4$$

$$\therefore f(5) = 1 + (5)^4 = 626$$

$$24.(A) \text{ Determinant of coefficients}$$

$$= \begin{vmatrix} \lambda & -1 & \cos \theta \\ 3 & 1 & 2 \\ \cos \theta & 1 & 2 \end{vmatrix} = \cos \theta - \cos^2 \theta + 6$$

and this is positive for all θ since $|\cos \theta| \leq 1$.

The only solution is therefore the trivial solution.

$$25.(B) \text{ We have } [1 \ x \ 1] \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$$

$$\Rightarrow [1 \ 5x+6 \ x+4] \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$$

$$\Rightarrow [1 + 5x + 6 + x^2 + 4x] = 0$$

$$\Rightarrow x^2 + 9x + 7 = 0 \Rightarrow x = \frac{-9 \pm \sqrt{53}}{2}$$

$$26.(A) \begin{vmatrix} 1 & 1 & 1 \\ (2^x + 2^{-x})^2 & (3^x + 3^{-x})^2 & (5^x + 5^{-x})^2 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix}$$

Operate $R_2 \rightarrow R_2 - R_3 - 4R_1$ to get:

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ (2^x - 2^{-x})^2 & (3^x - 3^{-x})^2 & (5^x - 5^{-x})^2 \end{vmatrix} = 0$$

$$27.(D) \text{ Since, } A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$$

$$\therefore B = \text{adj } A = \begin{bmatrix} 3 & 1 & 1 \\ -6 & -2 & 3 \\ -4 & -3 & 2 \end{bmatrix}$$

$$\Rightarrow \text{adj } B = \begin{bmatrix} 5 & -5 & 5 \\ 0 & 10 & -15 \\ 10 & 5 & 0 \end{bmatrix}$$

$$\Rightarrow |\text{adj } B| = \begin{vmatrix} 5 & -5 & 5 \\ 0 & 10 & -15 \\ 10 & 5 & 0 \end{vmatrix} = 625$$

Given that $C = 5A$

$$\Rightarrow |C| = 5^3 |A| = 5^3 \begin{vmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{vmatrix} = 625$$

$$\Rightarrow \frac{|\text{adj } B|}{|C|} = 1$$

28.(C) $D = \text{diagonal } [d_1, d_2, d_3, \dots, d_n]$

$$= \begin{bmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ 0 & 0 & d_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n \end{bmatrix}$$

As $DD^{-1} = I_n$

$$\Rightarrow D^{-1} = \begin{bmatrix} \frac{1}{d_1} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{d_2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{d_n} \end{bmatrix}$$

30.(C) For $n = 2$

$$\begin{aligned} \Rightarrow (A^{-1}BA)(A^{-1}BA) &= A^{-1}B^2A \\ (A^{-1}BA)^3 &= (A^{-1}BA)^2(A^{-1}BA) \\ &= (A^{-1}B^2A)(A^{-1}BA) = A^{-1}B^3A \text{ and so on } \dots \\ \text{thus } (A^{-1}BA)^n &= A^{-1}B^nA \end{aligned}$$

31.(D) Each element has 2 options. No. of matrices = $2^9 = 512$.

32.(D) $AB' = A_{m \times n} B'_{n \times p}$ $B'A = B'_{n \times m} A_{m \times n}$ Hence B has order $m \times n$

33.(A) $C = AB' - BA'$

$$\begin{aligned} \Rightarrow C' &= (AB' - BA')' = (AB')' - (BA')' \\ &= BA' - AB' = -C \end{aligned}$$

Hence, C is skew symmetric matrix.

$$34.(D) \begin{vmatrix} a-b & b+c & a \\ b-a & c+a & b \\ c-a & a+b & c \end{vmatrix} = \begin{vmatrix} 2(a-b) & -(a-b) & (a-b) \\ (b-c) & -(b-c) & (b-c) \\ (c-a) & a+b & c \end{vmatrix} = (a-b)(b-c) \begin{vmatrix} 2 & -1 & 1 \\ 1 & -1 & 1 \\ c-a & a+b & c \end{vmatrix}$$

$$29.(D) \begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0 \quad \left(\begin{array}{l} \because R_1 \rightarrow R_1 - R_2 \\ R_2 \rightarrow R_2 - R_3 \end{array} \right)$$

$[\because \text{system has non-zero solution also, } D_1 = D_2 = D_3 = 0.]$

So, D must be equal to zero

$$\begin{vmatrix} p-a & b-q & 0 \\ 0 & q-b & c-r \\ a & b & r \end{vmatrix} = 0$$

$$\Rightarrow (p-a)(q-b)(r-c) \begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ \frac{a}{p-a} & \frac{b}{q-b} & \frac{r}{r-c} \end{vmatrix} = 0$$

As $p \neq a, q \neq b, r \neq c$, we have

$$\Rightarrow \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ \frac{a}{p-a} + \frac{b}{q-b} + \frac{r}{r-c} & \frac{b}{q-b} & \frac{r}{r-c} \end{vmatrix} = 0$$

$(\because C_1 \rightarrow C_1 + C_2 + C_3)$

$$\Rightarrow \frac{a}{p-a} + \frac{b}{q-b} + \frac{r}{r-c} = 0$$

$$\Rightarrow \left(1 + \frac{a}{p-a}\right) + \left(\frac{b}{q-b} + 1\right) + \left(\frac{r}{r-c}\right) = 2$$

$$\Rightarrow \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2$$

$$(a-b)(b-c) \begin{vmatrix} 2 & -1 & 0 \\ 1 & -1 & 0 \\ c-a & a+b & a+b+c \end{vmatrix} = -(a-b)(b-c)(a+b+c)$$

$$35.(D) \begin{vmatrix} b(b-a) & b-c & c(b-a) \\ a(b-a) & a-b & b(b-a) \\ c(b-a) & c-a & a(b-a) \end{vmatrix} = (b-a)^2 \begin{vmatrix} b & b-c & c \\ a & a-b & b \\ c & c-a & a \end{vmatrix} = (b-a)^2 \begin{vmatrix} b-c & b-c & c \\ a-b & a-b & b \\ c-a & c-a & a \end{vmatrix} = 0 \quad [C_1 \rightarrow C_1 - C_3]$$

$$36.(A) f(t) = (\cos t)[t^2 - 2t^2] - 2 \sin t[t^2 - t] + \sin t[2t^2 - t] = -t^2 \cos t + t \sin t$$

$$\frac{f(t)}{t^2} = -\cos t + \frac{\sin t}{t} \Rightarrow \lim_{t \rightarrow 0} \frac{f(t)}{t} = 0$$

$$37.(B) \begin{vmatrix} x & x+y & x+2y \\ x+2y & x & x+y \\ x+y & x+2y & x \end{vmatrix} = \begin{vmatrix} -2y & y & y \\ y & -2y & y \\ x+y & x+2y & x \end{vmatrix}$$

$$= y^2 \begin{vmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ x+y & x+2y & x \end{vmatrix} = y^2 \{(-2)[-2x - x - 2y] - (x - x - y) + (x + 2y + 2x + 2y)\}$$

$$= y^2 \{6x + 4y + y + 3x + 4y\} = 9y^2(x + y)$$

$$38.(A) A^2 = A.A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix}$$

$$A^2 + 4A - 5I = \begin{bmatrix} 8 & 4 \\ 8 & 0 \end{bmatrix} = 4 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$$

$$39.(A) B = -A^{-1}BA \Rightarrow AB = -AA^{-1}BA \Rightarrow AB = -BA$$

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2 = A^2 + B^2$$

$$40.(C) A^2 = A, B^2 = B, AB = BA = 0$$

$$A^2 + B^2 = A + B \Rightarrow (A+B)^2 = A + B \quad [\because AB = BA = 0]$$

$$(A-B)^2 = (A-B)(A-B) = A^2 + B^2 - AB - BA = A^2 + B^2 = A + B$$

$$41.(C) D = \begin{vmatrix} t & t+1 & t-1 \\ t+1 & t & t+2 \\ t-1 & t+2 & t \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -1 & 1 & -3 \\ 2 & -2 & 2 \\ t-1 & t+2 & t \end{vmatrix} = 0 \quad \begin{bmatrix} R_1 \rightarrow R_1 - R_2 \\ R_2 \rightarrow R_2 - R_3 \end{bmatrix}$$

$$\Rightarrow (-1)[-t - t - 2] - 1[t - t + 1] - 3(t + 2 + t - 1) = 0 \Rightarrow 2t + 2 - 6t - 3 = 0 \Rightarrow t = -\frac{1}{2}$$

$$42.(A) a_{ij} = a_{ji} \forall i, j \text{ then } A = [a_{ij}] \text{ is symmetric matrix.}$$

$$43.(D) x - a = x - b = x - c = 0 \Rightarrow x = a, b, c \quad \text{i.e., } a = b = c$$

$$y - a = y - b = y - c = 0 \Rightarrow y = a, b, c \quad \text{i.e., } a = b = c$$

$$z - a = z - b = z - c = 0 \Rightarrow z = a, b, c \quad \text{i.e., } a = b = c$$

$$\text{Hence, } x = y = z = a = b = c$$

$$44.(D) \quad A+B = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} \Rightarrow (A+B)^{-1} = -\frac{1}{6} \begin{bmatrix} 2 & -2 \\ -4 & 1 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} -2 & 2 \\ 4 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{-1}{4} \begin{bmatrix} 4 & -4 \\ -4 & 3 \end{bmatrix}, \quad B^{-1} = \frac{1}{4} \begin{bmatrix} -2 & 2 \\ 0 & -2 \end{bmatrix} \quad (A+B)^{-1} \neq A^{-1} + B^{-1}$$

$$45.(D) \quad |A| = 1(2) - 1(0) + (2) = 4 \Rightarrow |5A| = 5^3 |A| = 500$$

$$46.(D) \quad \det(A+B) = 0 \text{ does not imply } |A| = |B| = 0$$

$$\text{For example, } A = \begin{bmatrix} 6 & 1 \\ 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$$

$$|A| = 3, \quad |B| = -1 \text{ whereas } A+B = \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} \text{ such that } \det(A+B) = 0$$

$$47.(A) \quad \det(\text{adj}(A)) = |A|^2 \quad \left[\because A \text{adj}(A) = |A|I \right]$$

$$48.(A) \quad D = \begin{vmatrix} 4 & k & 2 \\ k & 4 & 1 \\ 2 & 2 & 1 \end{vmatrix} = 0 \Rightarrow 8 - k(k-2) + 2(2k-8) = 0 \Rightarrow k^2 - 6k + 8 = 0 \Rightarrow k = 2, 4$$

$$49.(D) \quad \text{Let } u_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$Au_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ 2x_1 + y_1 \\ 3x_1 + 2y_1 + z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Hence, } \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\text{Similarly, } Au_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_2 \\ 2x_2 + y_2 \\ 3x_2 + 2y_2 + z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$u_1 + u_2 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$50.(C) \quad P^3 = Q^3 \text{ and } P^2Q = Q^2P$$

$$P^2(P-Q) = Q^2(Q-P) \Rightarrow P^2(P-Q) + Q^2(P-Q) = 0 \Rightarrow (P^2 + Q^2)(P-Q) = 0$$

$$\text{If } \det(P^2 + Q^2) \neq 0 \text{ then inverse of } P^2 + Q^2 \text{ exists}$$

$$\Rightarrow (P^2 + Q^2)^{-1} (P^2 + Q^2)(P-Q) = 0 \Rightarrow P-Q = 0$$

$$\text{But } P = Q \text{ is false (Given) Hence, } \det(P^2 + Q^2) = 0$$

$$51.(D) \quad BB' = (A^{-1}A') (A^{-1}A')' = A^{-1}A'(A')' (A^{-1})' = A^{-1}A'A(A^{-1})' = A^{-1}AA'(A')^{-1} = I$$

52.(B) $P = \text{adj}(A) \Rightarrow |P| = |\text{adj}(A)| = |A|^2 = 16$

$1(12-12) - \alpha(4-6) + 3(4-6) = 16 \Rightarrow \alpha = 11$

53.(D) If $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ have infinite number of solutions, then by Cramer's rule $D = D_1 = D_2 = D_3 = 0$, where

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$\therefore \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{vmatrix} = 0 \Rightarrow a = 8$$

$$\begin{vmatrix} 1 & 2 & 6 \\ 1 & 3 & 9 \\ 2 & 5 & b \end{vmatrix} = 0 \Rightarrow b = 15$$

54.(C) Apply $R_1 \rightarrow R_1 + R_2 + R_3$ and take $2\sin x + \cos x$ common

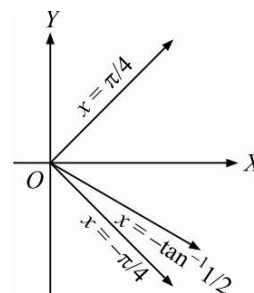
$$(2\sin x + \cos x) \begin{vmatrix} 1 & 1 & 1 \\ \sin x & \cos x & \sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = 0$$

$C_1 \rightarrow C_1 - C_3$ and expand we get $(2\sin x + \cos x)(\sin x - \cos x)^2 = 0$

$\Rightarrow \sin x - \cos x = 0$ or $2\sin x + \cos x = 0$

$\Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}$

$\tan x = -\frac{1}{2} \Rightarrow x = -\tan^{-1} \frac{1}{2} \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$



Hence, two solutions

55.(C) $A^2 - 5A + 7I = 0 \Rightarrow 7I = A(5I - A)$

$|7I| = A(5I - A) \Rightarrow 7 = |A||5I - A|$

$A \neq 0 \Rightarrow A^{-1}$ exists.

Statement I : $7I = 5A - A^2$

$7A^{-1} = 5I - A \Rightarrow A^{-1} = \frac{1}{7}(5I - A)$. True

Statement II : $A^2 = 5A - 7I$ given

$$\begin{aligned} \therefore A^3 - 2A^2 - 3A + I &= A(5A - 7I) - 2A^2 - 3A + I = 3A^2 - 10A + I \\ &= 3(5A - 7I) - 10A + I = 5(A - 4I). \text{ True} \end{aligned}$$

$\therefore$ Both the statements are true.

56.(D) $A^{2016} - 2A^{2015} - A^{2014} = A^{2014}(A^2 - 2A - I)$

$$A = \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix}, A^2 = \begin{bmatrix} 13 & 3 \\ -9 & -2 \end{bmatrix}, |A| = -1$$

$$\Rightarrow A^2 - 2A - I = A^2 - 2 \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 20 & 5 \\ -15 & -5 \end{bmatrix}$$

$$\therefore |A^{2016} - 2A^{2015} - A^{2014}| = |A|^{2014} |A^2 - 2A - I| = (-1)^{2014} (-100 + 75) = -25$$

57.(A) Given equation $\begin{bmatrix} 3p+3q+2r & 4p+2q & p+3q+2r \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$

$$\Rightarrow 3p+3q+2r = 3 \quad \dots(i); \quad 4p+2q = 0 \quad \dots(ii) \quad \text{and} \quad p+3q+2r = 1 \quad \dots(iii)$$

From (i), (ii) and (iii) we get: $p = 1, q = -2, r = 3$

58.(C) $A^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -I$

$$A^3 = -A \text{ and } A^4 = I$$

(A) $A^4 - I = O = A^2 + I$ Correct

(B) $A^3 - I = -A - I$

$$A(A - I) = A^2 - A = -I - A$$

$$\therefore A^3 - I = -A(A - I) \quad \text{Correct}$$

(C) $A^2 + I = O$

$$A(A^2 - I) = A^3 - A = -2A$$

$$A^2 + I \neq A(A^2 - I) \quad \text{Not Correct}$$

(D) $A^3 + I = A + I$

$$A(A^3 - I) = A^4 - A = I - A$$

$$\therefore A^3 + I = A(A^3 - I) \quad \text{Correct}$$

59.(A) We know that

$$|kA| = k^n |A|, \text{ where } n \text{ is the order of } A \text{ and } |\text{adj } A| = |A|^{n-1}.$$

$$\text{Given } |5 \cdot \text{adj } A| = 5$$

$$5^3 |\text{adj } A| = 5 \quad \because \text{order of } A \text{ is } 3 \times 3$$

$$5^2 |A|^{3-1} = 1 \Rightarrow |A|^2 = \frac{1}{5^2} \Rightarrow |A| = \pm \frac{1}{5}$$

60.(B) Apply $R_2 \rightarrow R_2 - (R_1 + R_3)$

$$\begin{vmatrix} x^2+x & x+1 & x-2 \\ -4 & 0 & 0 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = ax-12$$

Expand along R_2 , we get

$$24x - 12 = ax - 12 \Rightarrow a = 24$$

61.(B) $(a-1)x - y - z = 0, x - (b-1)y + z = 0$

$$x + y - (c-1)z = 0$$

For non-trivial solution

$$\begin{vmatrix} a-1 & -1 & -1 \\ 1 & -(b-1) & 1 \\ 1 & 1 & -(c-1) \end{vmatrix} = 0 \Rightarrow ab + bc + ca = abc$$

62.(A) $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

$$B^2 = 0 \Rightarrow B^3 = B^4 = \dots = B^{50} = 0$$

$$(I - B)^{50} - 50B = I + {}^{50}C_1 B + {}^{50}C_2 B^2 + {}^{50}C_3 B^3 + \dots + {}^{50}C_{50} B^{50} - 50B$$

$$= I + 50 + 0 - 50B = I$$

$$\therefore \det[(I + B)^{50} - 50B] = 1$$

63.(A) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} 3 & 1 & 2 \\ 3 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

64.(D) Let $p_1(x) = a_1x^2 + b_1x + c_1$

$$p_2(x) = a_2x^2 + b_2x + c_2$$

$$p_3(x) = a_3x^2 + b_3x + c_3$$

$$A = \begin{vmatrix} a_1x^2 + b_1x + c_1 & 2a_1x + b_1 & 2a_1 \\ a_2x^2 + b_2x + c_2 & 2a_2 + b_2 & 2a_2 \\ a_3x^2 + b_3x + c_3 & 2a_3 + b_3 & 2a_3 \end{vmatrix} = \begin{vmatrix} a_1x^2 + b_1x + c_1 & b_1 & a_1 \\ a_2x^2 + b_2x + c_2 & b_2 & a_2 \\ a_3x^2 + b_3x + c_3 & b_3 & a_3 \end{vmatrix} \quad C_2 \rightarrow C_2 - C_3 \quad x$$

$$= 2 \begin{vmatrix} c_1 & b_1 & a_1 \\ c_2 & b_2 & a_2 \\ c_3 & b_3 & a_3 \end{vmatrix} \quad C_1 \rightarrow C_1 - C_3 x^2 - C_2 x$$

= constant

Now, $|A| = |A^{-1}|$ and $|AB| = |A||B|$

$$\therefore |B(x)| = \left[A(x) \right]^T A(x) = |A(x)|^2 = \text{constant. From (i)}$$

Hence determine of $B(x)$ does not depends on x .

65.(D) $AB = \begin{bmatrix} y+3x \\ 3y-x-2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$

$$y+3x=6, 3y-x+2=8 \Rightarrow 3y-x=6$$

$$\therefore y+3x=3y-3 \Rightarrow y=2x$$

66.(A) Apply (i) $R_2 \rightarrow R_2 - R_3$ (ii) $R_3 \rightarrow R_3 - R_1 + \frac{1}{2}R_2$

$$4\lambda^3 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = k\lambda \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} \quad \therefore \quad k = 4\lambda^2$$

67.(B) $A^T = A$ and $B^T = -B$

$$(AB - BA)^T = (AB)^T - (BA)^T = B^T A^T - A^T B^T = -BA - A(-B) = AB - BA \quad \therefore \quad \text{Hence symmetric}$$

68.(B) Given $\Delta = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = -\frac{1}{2}(a+b+c)(2a^2+2b^2+2c^2-2ab-2bc-2ca)$
 $= -\frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2] < 0$

As all sides are different

69.(B) **Statement I :** For non-trivial solution $\begin{vmatrix} 1 & \sin \alpha & \cos \alpha \\ 1 & \cos \alpha & \sin \alpha \\ 1 & -\sin \alpha & -\cos \alpha \end{vmatrix} = 0$

Apply $R_3 \rightarrow R_3 + R_1$ and expand, we get $\cos 2\alpha = 0 \Rightarrow \alpha = \frac{\pi}{4}, \frac{3\pi}{4} \therefore$ One value of α lies in $(0, \pi/2)$

Statement II : Apply $R_3 \rightarrow R_3 + R_1$ and expand $2\cos \alpha (\sin^2 \alpha - \cos^2 \alpha) = 0$

$\Rightarrow -2\cos \alpha \cos 2\alpha = 0 \Rightarrow \alpha = \frac{\pi}{2}, \alpha = \frac{\pi}{4}$. So one value of α lies in $(0, \pi/2)$.

70.(B) Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\begin{bmatrix} a^2+bc & b(a+d) \\ c(a+d) & bc+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow b(a+d) = 0, c(a+d) = 0 \Rightarrow a+d = 0$ or $b = c = 0$

Case (i) $b = c = 0$

$$a^2 + bc = 1, bc + d^2 = 1 \Rightarrow a^2 = d^2 = 1 \Rightarrow a = \pm 1, b = \pm 1 \therefore A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \text{ as } A \neq I, -I$$

Case (ii) $a + d = 0 \Rightarrow d = -a$

$$a^2 + bc = 1, bc + d^2 = 1$$

$$\therefore A = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$$

$$\text{Tr}(A) = 0 \text{ and } |A| = -1 \text{ (Both cases)}$$

71.(225) $|A| = a$ $b = a^2$ $a = 3, b = 9$

$$(a^5 + a^4 + 1)S$$

$$\frac{S}{2} = \frac{\frac{a}{b}}{1 - \frac{a}{b^2}} = \frac{ab}{b^2 - a}$$

$$\frac{S}{2} = \frac{27}{81-3} \Rightarrow S = \frac{54}{78} \Rightarrow S = \frac{27}{39}$$

$$(3^5 + 3^4 + 1) \times \frac{27}{39} = 225$$

72.(8) $|A| = 2$ $|B| = 1$

$$|A^{-1}| = \frac{1}{2} \quad |B^{-1}| = 1$$

$$|2A^{-1}| = 4 \quad \text{adj}(B^{-1}) = (B^{-1})^2 = 1$$

$$|\text{adj}(2A^{-1})| = 16$$

73.(1) $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\lambda^3 = 5\lambda^2 - 5\lambda + 1$$

$$A^3 = 5A^2 - 5A + I$$

$$A^2 = 5A - 5I + A^{-1}$$

$$\Rightarrow A^{-1} = A^2 - 5A + 5I$$

$$x + y + z = 1$$

74.(100) If $AB = (AB)^{-1} \Rightarrow (AB)^2 = I$

$$AB = \begin{bmatrix} 5x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2 & 5x \end{bmatrix}$$

$$(AB)^2 = \begin{bmatrix} 5x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2 & 5x \end{bmatrix} \begin{bmatrix} 5x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2 & 5x \end{bmatrix} = \begin{bmatrix} 25x^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2+50x^2 & 25x^2 \end{bmatrix}$$

$$25x^2 = 1$$

$$x = \frac{1}{5} \text{ or } -\frac{1}{5}$$

$$AB \neq I \Rightarrow x = -\frac{1}{5}$$

$$AB = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & -1 \end{bmatrix}$$

$$\text{Tr}(AB) = -1$$

$$\text{Tr}(AB)^2 = 3$$

$$\text{Tr}(AB)^3 = \text{Tr}(AB) = -1$$

$$\text{Tr}(AB)^4 = \text{Tr}(AB)^2 = 3$$

$$75.(1) \quad A^{-1} = \begin{bmatrix} \sin^2 \alpha & 0 & 0 \\ 0 & \sin^2 \beta & 0 \\ 0 & 0 & \sin^2 \gamma \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} \cos^2 \alpha & 0 & 0 \\ 0 & \cos^2 \beta & 0 \\ 0 & 0 & \cos^2 \gamma \end{bmatrix}$$

$$C = (A^{-1})^3 + (B^{-1})^3 + 3A^{-1}B^{-1}(A^{-1}B^{-1})$$

$$C = \begin{bmatrix} (\sin^2 \alpha + \cos^2 \alpha)^3 & 0 & 0 \\ 0 & (\sin^2 \beta + \cos^2 \beta)^3 & 0 \\ 0 & 0 & (\sin^2 \gamma + \cos^2 \gamma)^3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \det(C) = 1$$

$$76.(9) \quad \text{Given } A^2 = A$$

$$\Rightarrow I = (I - 0.4A)(I - \alpha A) = I - I\alpha A - 0.4AI + 0.4\alpha A^2 = I - A\alpha - 0.4A + 0.4\alpha A = I - A(0.4 + \alpha) + 0.4\alpha A$$

$$\Rightarrow 0.4\alpha = 0.4 + \alpha \Rightarrow \alpha = -\frac{2}{3} \Rightarrow |9\alpha| = 6$$

$$77.(52) \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\text{We have } P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 4(1+2) & 4 & 1 \end{bmatrix}$$

$$\text{Again, } P^3 = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 4(1+2+3) & 6 & 1 \end{bmatrix}$$

$$\text{Then by induction, } P^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 100 & 1 & 0 \\ \frac{4 \cdot 50 \cdot 51}{2} & 100 & 1 \end{bmatrix}$$

As $P^{50} - Q = I$. We have $q_{31} = \frac{4 \cdot 50 \cdot 51}{2}$

Again, $q_{32} = 100, q_{21} = 100$

$$\therefore \frac{q_{31} + q_{32}}{q_{21}} = \frac{4 \cdot 50 \cdot 51}{2 \cdot 100} + 1 = 51 + 1 = 52$$

$$\begin{aligned} 78.(2) \quad \begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^2 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10 & \Rightarrow x^3 \begin{vmatrix} 1 & 1 & 1+x^3 \\ 2 & 4 & 1+8x^3 \\ 3 & 9 & 1+27x^3 \end{vmatrix} = 10 \Rightarrow x^3 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 3 & 9 & 1 \end{vmatrix} + x^6 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ 3 & 9 & 27 \end{vmatrix} = 10 \\ \Rightarrow x^3 \cdot 2 + x^6 \cdot 12 = 10 & \Rightarrow 6x^6 + x^3 - 5 = 0 \Rightarrow (x^3 + 1)(6x^3 - 5) = 0 \end{aligned}$$

Therefore two real solutions.

$$79.(9) \quad \text{Let } M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \text{ from the system of equations we get } \left. \begin{matrix} a = 0 \\ e = 2 \\ i = 7 \end{matrix} \right\} \text{ by solving simultaneous equation.}$$

Sum of diagonal entries = $0 + 2 + 7 = 9$

80.(1) According to question, $(-1)x + cy + bz = 0, cx + (-1)y + az = 0$ and $bx + ay + (-1)z = 0$

Above is a system of three linear homogeneous equations and since x, y, z are not all zero, it has a nontrivial solution. So we must have determinant must be zero.

$$\therefore \begin{vmatrix} -1 & c & b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

Expanding $(-1)(1 - a^2) - c(-c - ab) + b(ca + b) = 0$

$$\Rightarrow a^2 + b^2 + c^2 + 2abc - 1 = 0 \Rightarrow a^2 + b^2 + c^2 + 2abc = 1$$

$$81.(4) \quad \Delta = \begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$$

Applying $R_3 \rightarrow R_3 - R_2$ and $R_2 \rightarrow R_2 - R_1$

$$\Delta = \begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ x+1 & x+1 & x+1 \\ x+2 & 2(x+2) & 6(x+2) \end{vmatrix} = 0$$

$$\therefore \Delta = (x+1)(x+2) \begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 1 & 1 & 1 \\ 1 & 2 & 6 \end{vmatrix} = 0$$

$$\therefore \Delta = (x+1)(x+2)[(x+2) \cdot 4 - (2x+3) \cdot 5 + (3x+4) \cdot 1] = 0$$

$$\Delta = (x+1)(x+2)(-3x-3) = 0$$

$$\text{or } (x+1)^2(x+2) = 0$$

$$\therefore x = -1, -1, -2$$

$$82.(4) \quad \Delta = \begin{vmatrix} 1 & x+y & x+y+z \\ 2 & 3x+2y & 4x+3y+2z \\ 3 & 6x+3y & 10x+6y+3z \end{vmatrix} = x^2 \begin{vmatrix} 1 & 1 & x+y \\ 2 & 3 & 4x+3y \\ 3 & 6 & 10+6y \end{vmatrix} \begin{matrix} C_3 \rightarrow C_3 - zC_1 \\ C_2 \rightarrow C_2 - yC_1 \end{matrix}$$

$$= x^3 \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 6 & 10 \end{vmatrix} [C_3 \rightarrow C_3 - yC_2] = x^3(6-8+3) = 64 \Rightarrow x^3 = 64 \text{ or } x = 4$$

$$83.(0) \quad A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = A \cdot A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow A^4 = A^2 \cdot A^2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 3^2 & 0 \\ 0 & 3^2 \end{bmatrix}$$

$$\Rightarrow A^8 = \begin{bmatrix} 3^4 & 0 \\ 0 & 3^4 \end{bmatrix} \text{ and } A^6 = A^4 \cdot A^2 = \begin{bmatrix} 3^2 & 0 \\ 0 & 3^2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 3^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

$$\text{Let } V = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$A^8 + A^6 + A^4 + A^2 + I$$

$$\begin{bmatrix} 81 & 0 \\ 0 & 81 \end{bmatrix} + \begin{bmatrix} 27 & 0 \\ 0 & 27 \end{bmatrix} + \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 121 & 0 \\ 0 & 121 \end{bmatrix}$$

$$(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} 121 & 0 \\ 0 & 121 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix} \text{ or } \begin{bmatrix} 121x \\ 121y \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix} \Rightarrow x = 0 \text{ and } y = \frac{1}{11} \Rightarrow V = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{11} \end{bmatrix}$$

$$84.(4) \quad \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} \text{ is an idempotent matrix.}$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} = \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$$

$$\text{or } \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} = \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} \text{ or } \begin{bmatrix} a^2+bc & ab+b-ab \\ ac+c-ac & bc+(1-a)^2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$$

$$\text{or } \begin{bmatrix} a^2+bc & b \\ c & bc+(1-a)^2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix} \text{ or } a^2+bc = a$$

$$a-a^2 = bc = \frac{1}{4} \text{ (given)} \quad f(a) = \frac{1}{4}$$

$$85.(8) \quad \text{Given, } (A-3I)(A-5I) = O$$

$$\therefore A^2 - 8A + 15I = O$$

Post multiplying by A^{-1} on both sides, we have $A \cdot AA^{-1} - 8A \cdot A^{-1} + 15I \cdot A^{-1} = O$

$$\Rightarrow A - 8I + 15A^{-1} = O \Rightarrow A + 15A^{-1} = 8I$$

$$\Rightarrow \frac{1}{2}A + \frac{15}{2}A^{-1} = 4I \quad \dots\dots\dots (i)$$

$$\text{Comparing (i) with } \alpha A + \beta A^{-1} = 4I, \text{ we get } \alpha = \frac{1}{2} \text{ and } \beta = \frac{15}{2} \therefore \alpha + \beta = \frac{1}{2} + \frac{15}{2} = 8$$

Integral Calculus-1

$$1.(B) \quad I = \int \frac{dx}{\cos^3 x \sqrt{2 \sin 2x}} = \int \frac{dx}{\cos^3 x \sqrt{4 \sin x \cos x}} = \frac{1}{2} \int \frac{dx}{\cos^3 x \sqrt{\frac{\sin x}{\cos x} \cos^2 x}}$$

$$= \frac{1}{2} \int \frac{dx}{\cos^4 x + \sqrt{\tan x}} = \frac{1}{2} \int \frac{(1 + \tan^2 x) \sec^2 x dx}{\sqrt{\tan x}}$$

Put $\tan x = t \Rightarrow \sec^2 x dx = dt$

$$\therefore I = \frac{1}{2} \int \frac{1+t^2 \cdot dt}{\sqrt{t}} = \sqrt{t} + \frac{1}{5} t^{5/2} + k = \sqrt{\tan x} + \frac{1}{5} (\tan x)^{5/2} + k$$

On equating, $A = \frac{1}{2}$, $B = \frac{5}{2}$ and $C = \frac{1}{5}$

$$\therefore A + B + C = \frac{1}{2} + \frac{5}{2} + \frac{1}{5} = \frac{16}{5}$$

2.(B) Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$

$$I = \int \frac{2 \sin \theta \cos \theta d\theta}{(1 + \sin \theta) \sqrt{\sin^2 \theta - \sin^4 \theta}} = \int \frac{2 \sin \theta \cos \theta d\theta}{(1 + \sin \theta) \sin \theta \cos \theta} = 2 \int \frac{1 - \sin \theta}{\cos^2 \theta} d\theta$$

$$= 2 [\tan \theta - \sec \theta] + C = 2 \left[\frac{\sin \theta}{\cos \theta} - \frac{1}{\sin \theta} \right] + C = 2 \int \left(\frac{1 - \sin \theta}{\cos \theta} \right) + C$$

$$= -2 \left(\frac{1 - \sin \theta}{\sqrt{1 - \sin^2 \theta}} \right) + C = -2 \left(\frac{1 - \sqrt{x}}{\sqrt{1 - x}} \right) + C = -2 \frac{1 - \sqrt{x}}{\sqrt{(1 + \sqrt{x})(1 - \sqrt{x})}} + C = -2 \sqrt{\frac{1 - \sqrt{x}}{1 + \sqrt{x}}} + C$$

3.(C) If $m + n = 2$, put $\frac{x-a}{x-b} = t$ or $\frac{x+b}{x+a} = t$

$$I = \int \frac{dx}{(x+1)^{3/4} (x-2)^{5/4}} = \int \frac{dx}{\left(\frac{x+1}{x-2} \right)^{3/4} (x-2)^{3/4} (x-2)^{5/4}} = \int \frac{dx}{\left(\frac{x+1}{x-2} \right)^{3/4} (x-2)^2}$$

Put $\frac{x+1}{x-2} = t \Rightarrow \frac{-3}{(x-2)^2} dx = dt$

$$\therefore I = -\frac{1}{3} \int \frac{dt}{t^{3/4}} = -\frac{4}{3} t^{1/4} + C = \frac{4}{3} \left(\frac{x+1}{x-2} \right)^{1/4} + C$$

4.(B) Put $\log(t + \sqrt{1+t^2}) = x$

$$\Rightarrow \frac{1}{t + \sqrt{1+t^2}} \left(1 + \frac{2t}{2\sqrt{1+t^2}} \right) dt = dx \Rightarrow \frac{1}{t + \sqrt{1+t^2}} \left(\frac{\sqrt{1+t^2} + t}{\sqrt{1+t^2}} \right) dt = dx \Rightarrow \frac{1}{\sqrt{1+t^2}} dt = dx$$

$$I = \int t dt = \frac{1}{2} t^2 + C = \frac{1}{2} \left(\log(t + \sqrt{1+t^2}) \right)^2 + C = \frac{1}{2} (g(t))^2 + C$$

$$\therefore g(t) = \log(t + \sqrt{1+t^2}) \quad \therefore g(2) = \log(2 + \sqrt{5})$$

$$\begin{aligned}
 5.(B) \quad I &= \int \frac{(\sin^4 x + \cos^4 x)(\sin^2 x - \cos^2 x)(\sin^2 x + \cos^2 x)}{(1 - 2\sin^2 x \cos^2 x)} dx \\
 &= \int \frac{[(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x](-\cos 2x) dx}{(1 - 2\sin^2 x \cos^2 x)} = - \int \cos 2x dx = -\frac{1}{2} \sin x + C
 \end{aligned}$$

$$\begin{aligned}
 6.(A) \quad I &= 2 \int (\tan^{-1} x) x dx \quad (\text{By Parts}) \\
 &= 2 \left[(\tan^{-1} x) \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2} dx \right] = x^2 \tan^{-1} x - \int \frac{x^2 + 1 - 1}{x^2 + 1} dx \\
 &= x^2 \tan^{-1} x - x + \tan^{-1} x + c = -x + (1+x^2) \tan^{-1} x + c
 \end{aligned}$$

7.(C) Divide numerator and denominator by $\cos^6 x$

$$I = \int \frac{\tan^2 x \sec^2 x}{(1 + \tan^3 x)} dx$$

$$\text{Put } 1 + \tan^3 x = t \Rightarrow 3 \tan^2 x \sec^2 x dx = dt$$

$$\therefore I = \frac{1}{3} \int \frac{dt}{t^2} = \frac{-1}{3} \frac{1}{t} + C$$

$$8.(B) \quad \text{Let } \frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2} = t$$

$$\text{Differentiating, we get } \frac{x^{5m-1} + 2x^{4m-1}}{(x^{2m} + x^m + 1)^3} dx = dt$$

$$\therefore I = \int dt = t + c = \frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2} + c$$

$$\therefore f(x) = \frac{x^{4m}}{2m(x^{2m} + x^m + 1)^2}$$

$$9.(A) \quad \int \frac{x^6}{x+x^7} dx = \int \frac{(x^7+x)^{-x}}{x(x+x^7)} dx = \int \frac{1}{x} dx - \int \frac{1}{x+x^7} dx = \ln|x| - p(x) + c$$

$$10.(B) \quad \text{Put } \cot^{-1} x = t \Rightarrow x = \cot t \Rightarrow dx = -\operatorname{cosec}^2 t dt$$

$$I = - \int \frac{\cot^2 t - \cot t + 1}{\cot^2 t + 1} \cdot e^t \cdot \operatorname{cosec}^2 t dt = \int e^t (\cot t - \operatorname{cosec}^2 t) dt = e^t \cdot \cot t + c = x e^{\cot^{-1} x} + C.$$

$$\text{So } A(x) = x$$

$$11.(B) \quad \text{Put } 2-x^2 = t^2 \Rightarrow x dx = -t dt$$

$$I = - \int \frac{t dt}{t^2 + t} = - \int \frac{1}{t+1} dt = -\log|t+1| + c$$

12.(B) Let $I = \int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$

$\therefore I = 2 \int \frac{dt}{t} = 2 \log |t| + c$ [Put : $1 + \sqrt{x} = t \Rightarrow \frac{1}{2} \cdot \frac{1}{\sqrt{x}} dx = dt \Rightarrow \frac{1}{\sqrt{x}} dx = 2 dt$]
 $= 2 \log |1 + \sqrt{x}| + c$

13.(A) Let $I = \int \frac{1 + \cot x}{x + \log \sin x} dx$

$\therefore I = \int \frac{dt}{t} = \log |t| + c$ [Put $x + \log \sin x = t \Rightarrow \left(1 + \frac{1}{\sin x} \cdot \cos x\right) dx = dt \Rightarrow (1 + \cot x) dx = dt$]
 $= \log |x + \log \sin x| + c$

14.(A) Let $I = \int \frac{\log(\sin x)}{\tan x} dx = \int \log(\sin x) \cdot \cot x dx$ [Put $\log(\sin x) = t$]

$\frac{1}{\sin x} \cdot \cos x dx = dt \Rightarrow \cot x dx = dt$

$I = \int t \cdot dt = \frac{t^2}{2} + C = \frac{[\log(\sin x)]^2}{2} + C$

15.(A) Let $I = \int x\sqrt{x^2 + x} dx$

$= \int \sqrt{x^2 + x} \left[\frac{1}{2}(2x+1) - \frac{1}{2} \right] dx = \frac{1}{2} \int \sqrt{x^2 + x} (2x+1) dx - \frac{1}{2} \int \sqrt{x^2 + x} dx$

$= \frac{1}{2} I_1 - \frac{1}{2} I_2 \quad \dots \text{(i)}$

Now $I_1 = \int \sqrt{x^2 + x} (2x+1) dx$

$\therefore I_1 = \int \sqrt{t} dt = \int \frac{t^{3/2}}{3/2} = \frac{2}{3} (x^2 + x)^{3/2}$ [Put : $x^2 + x = t \Rightarrow (2x+1) dx = dt$] $\dots \text{(ii)}$

and $I_2 = \int \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} dx$

$= \frac{\left(x + \frac{1}{2}\right) \sqrt{x^2 + x}}{2} - \frac{1}{8} \log \left(x + \frac{1}{2} + \sqrt{x^2 + x}\right) + c \quad \dots \text{(iii)}$

From (i), (ii) and (iii), we get : $I = \frac{1}{3} (x^2 + x)^{3/2} - \frac{1}{4} \left[\left(x + \frac{1}{2}\right) \sqrt{x^2 + x} - \frac{1}{4} \log \left(x + \frac{1}{2} + \sqrt{x^2 + x}\right) \right] + c$

16.(A) $\int \frac{dx}{\sin x \cdot \sin(x+\alpha)} = \frac{1}{\sin \alpha} \int \frac{\sin(\alpha + x - x)}{\sin x \sin(x+\alpha)} dx = \csc \alpha \int \frac{\sin(x+\alpha) \cos x - \cos(x+\alpha) \sin x}{\sin x \sin(x+\alpha)} dx$

$= \csc \alpha \left(\int \cot x dx - \int \cot(x+\alpha) dx \right) + C = \csc \alpha \left[\log |\sin x| - \log |\sin(x+\alpha)| \right] + C$

$= \csc \alpha \log \left| \frac{\sin x}{\sin(x+\alpha)} \right| + C$

17.(B)

18.(A) $I = \int \frac{\cos 2x}{\cos x} dx = \int \frac{2\cos^2 x - 1}{\cos x} dx = 2 \sin x - \int \sec x dx = 2 \sin x - \ln |\sec x + \tan x| + C$

19.(C) Let $5^{5^x} = t \Rightarrow 5^{5^x} \cdot \ln 5 \cdot 5^x \cdot \ln 5 \, dx = dt \quad \therefore \quad I = \int \frac{5^t dt}{(\ln 5)^2} = \frac{5^t}{(\ln 5)^3} + C = \frac{5^{5^x}}{(\ln 5)^3} + C$

20.(A)

21.(C) $\int \tan^3 2x \sec 2x \, dx$

$I = \int (\sec^2 2x - 1) \sec 2x \cdot \tan 2x \, dx \quad [\text{Put } \sec 2x = t \Rightarrow 2 \sec 2x \tan 2x \, dx = dt]$

$\Rightarrow \quad I = \frac{1}{2} \int (t^2 - 1) dt = \frac{t^3}{6} - \frac{t}{2} + C = \frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$

22..(B)

23.(A) Let $I = \int x^{13/2} (1+x^{5/2})^{1/2} dx = \int x^5 \cdot x^{3/2} (1+x^{5/2})^{1/2} dx \quad [\text{Put } 1+x^{5/2} = t \Rightarrow \frac{5}{2} x^{3/2} dx = dt]$

$\Rightarrow \quad I = \frac{2}{5} \int (t-1)^2 t^{1/2} dt$
 $= \frac{2}{5} \int (t^{5/2} + t^{1/2} - 2t^{3/2}) dt = \frac{2}{5} \left[\frac{2}{7} t^{7/2} + \frac{2}{3} t^{3/2} - 2 \cdot \frac{2}{5} t^{5/2} \right] + C$
 $= \frac{4}{35} (1+x^{5/2})^{7/2} + \frac{4}{15} (1+x^{5/2})^{3/2} - \frac{8}{25} (1+x^{5/2})^{5/2} + C$

On comparing with given, we have : $A = \frac{4}{35}, B = -\frac{8}{25}$ and $C = \frac{4}{15}$

24.(CD) $I = \int e^{3x} \cos 4x \, dx = e^{3x} (A \sin 4x + B \cos 4x) + C \quad \dots \dots (i)$

Apply by parts in LHS

$I = \frac{1}{4} e^{3x} \sin 4x - \int \frac{3}{4} e^{3x} \sin 4x \, dx = \frac{1}{4} e^{3x} \sin 4x + \frac{3}{16} e^{3x} \cos 4x - \int \frac{9}{16} e^{3x} \cos 4x \, dx$

$\frac{25}{16} I = \frac{1}{16} (4e^{3x} \sin 4x + 3e^{3x} \cos 4x)$

Comparing with equation (i)

$\Rightarrow \quad A = \frac{4}{25}, B = \frac{3}{25} \Rightarrow \frac{A}{B} = \frac{4}{3} \Rightarrow 3A = 4B \Rightarrow 4A + 3B = 1$

25.(D)

26.(C) $I = \int \frac{dx}{x^3(x+1)} = \frac{A}{x^2} + \frac{B}{x} + \ln \left(\frac{x}{x+1} \right) + C$

Let $\frac{1}{x^3(x+1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x^3} + \frac{d}{(x+1)}$

$1 = a(x+1)x^2 + bx(x+1) + c(x+1) + dx^3 = a(x^3+x^2) + b(x^2+x) + c(x+1) + dx^3$

$0 = a + d, \quad 0 = a + b, \quad 0 = b + c \Rightarrow c = 1, b = -1, a = 1, d = -1$

$\Rightarrow \quad \int \frac{dx}{x^3(x+1)} = \int \frac{dx}{x} - \int \frac{dx}{x^2} + \int \frac{dx}{x^3} - \int \frac{1}{x+1} dx = \ln |x| + \frac{1}{x} - \frac{1}{2x^2} - \ln |x+1| + C$

Comparing with given, we get : $B = 1, A = -\frac{1}{2}$

27.(B)

28.(C) $I = \int \sqrt{\sec x - 1} = \int \frac{\sqrt{1 - \cos x}}{\sqrt{\cos x + 1} - 1} dx = \int \frac{\sqrt{2} \sin \frac{x}{2}}{\sqrt{\left(\sqrt{2} \cos \frac{x}{2} \right)^2 - 1}} dx \quad [\text{Put } \sqrt{2} \cos \frac{x}{2} = t \Rightarrow \sin \frac{x}{2} dx = -\sqrt{2} dt]$

$\Rightarrow \quad I = -2 \int \frac{dt}{\sqrt{t^2 - 1}} = -2 \ln \left(t + \sqrt{t^2 - 1} \right)$

$$-2\ell n\left(\sqrt{2}\cos\frac{x}{2}+\sqrt{2\cos^2\frac{x}{2}-1}\right)+C_1=-2\times\frac{1}{2}\ell n2-2\ell n\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+C$$

$$=-2\ell n\left(\cos\frac{x}{2}+\sqrt{\cos^2\frac{x}{2}-\frac{1}{2}}\right)+C$$

29.(ACD) $f'(x) = 3x^2 \sin\left(\frac{1}{x}\right) - x \cos\left(\frac{1}{x}\right)$

Now $\lim_{x \rightarrow 0} f'(x) = 0 - 0 = 0$ $\therefore$ At $x = 0$, $f(x)$ is derivable so continuous also, and $f'(x)$ is continuous at $x = 0$

Now $f''(x) = 6x \sin\left(\frac{1}{x}\right) + 3x^2 \cos\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) - \frac{1}{x} \sin\frac{1}{x} - \cos\left(\frac{1}{x}\right)$

So $f''(x) = 6x \sin\left(\frac{1}{x}\right) - \frac{1}{x} \sin\left(\frac{1}{x}\right) - 4 \cos\left(\frac{1}{x}\right)$

At $x = 0$, $\cos\left(\frac{1}{x}\right)$ is not define so at $x = 0$, $f''(x)$ is not continuous or $f'(x)$ is not differentiable at $x = 0$.

30.(A) $I = \int \left((\log_x e) - (\log_x e)^2 \right) dx = \int \left(\frac{1}{\ell n x} - \frac{1}{(\ell n x)^2} \right) dx$ (Put $\ell n x = t \Rightarrow x = e^t \Rightarrow dx = e^t dt$)

$$\Rightarrow I = \int \left(\frac{1}{t} - \frac{1}{t^2} \right) e^t dt = \frac{e^t}{t} + C = \frac{x}{\ell n x} + C = x \log_x e + C$$

31.(A) $\int \tan x \tan 2x \tan 3x dx$

$$= \int (\tan 3x - \tan x - \tan 2x) dx \quad [\text{As } \tan 3x = \frac{\tan 2x + \tan x}{1 - \tan 2x \cdot \tan x}]$$

$$= -\frac{1}{3} \ell n |\cos 3x| + \ell n |\cos x| + \frac{1}{2} \ell n |\cos 2x| + C = -\ell n |\sec x| + \frac{1}{3} \ell n |\sec 3x| - \frac{1}{2} \ell n |\sec 2x| + C$$

32.(C) $\int e^{\sin x} \cdot \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$

$$= \int e^{\sin x} (x \cos x - \tan x \sec x) dx = \int e^{\sin x} \cdot x \cos x dx - \int e^{\sin x} \tan x \sec x dx \quad (\text{By parts integration})$$

$$= x e^{\sin x} - \int 1 \cdot e^{\sin x} dx - \left[e^{\sin x} \sec x - \int e^{\sin x} \cos x \cdot \sec x dx \right] + C = e^{\sin x} (x - \sec x) + C$$

33.(A) Put $x = 2 \tan \theta$

34.(B) $\int \frac{1 + x \cos x}{x(1 - x^2 e^{2 \sin x})} dx$

Put $x e^{\sin x} = t \Rightarrow (x e^{\sin x} \cdot \cos x + e^{\sin x}) dx = dt \Rightarrow e^{\sin x} (x \cos x + 1) dx = dt$

$$\Rightarrow I = \int \frac{dt}{t(1-t^2)} = \int \frac{1}{t(1-t)(1+t)} dt$$

Let $\frac{1}{t(1-t)(1+t)} = \frac{A}{t} + \frac{B}{(1-t)} + \frac{C}{1+t}$

Solve to get : $A = 1, B = \frac{1}{2}, C = -\frac{1}{2}$

$$\Rightarrow I = \int \left\{ \frac{1}{t} + \frac{1}{2(1-t)} - \frac{1}{2(1+t)} \right\} dt = \ell n |t| - \frac{1}{2} \ell n |1-t| - \frac{1}{2} \ell n |1+t| + K = \ell n |x e^{\sin x}| - \frac{1}{2} \log |1 - x^2 e^{2 \sin x}| + K$$

35.(B) $f(x) = \int \frac{2 \sin x - \sin 2x}{x^3} dx$

$$f'(x) = \frac{2 \sin x - \sin 2x}{x^3} = \frac{2 \sin x}{x^3} \cdot \frac{1 - \cos x}{x^2} = 2 \left(\frac{\sin x}{x} \right) \cdot \frac{2 \sin^2 \frac{x}{2}}{\frac{x^2}{4} \times 4} = \frac{2 \times 2}{4} \left(\frac{\sin}{x} \right) \cdot \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2$$

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2 = 1$$

36.(C)

37.(D) $I = \int e^{\tan \theta} (\sec \theta - \sin \theta) d\theta = \int e^{\tan \theta} \left(\frac{\sec \theta - \sin \theta}{\sec^2 \theta} \right) \sec^2 \theta d\theta$

$$I = \int e^{\tan \theta} \left(\frac{1}{\sec \theta} - \frac{\sin \theta}{\sec^2 \theta} \right) \sec^2 \theta d\theta = \int e^{\tan \theta} \left(\frac{1}{\sqrt{1 + \tan^2 \theta}} - \frac{\tan \theta}{(1 + \tan^2 \theta)^{3/2}} \right) \sec^2 \theta d\theta$$

$$= e^{\tan \theta} \cdot \frac{1}{\sqrt{1 + \tan^2 \theta}} = e^{\tan \theta} \cdot \cos \theta + C$$

38.(C)

39.(C)

40.(A) $\int \left\{ \ln(1 + \sin x) + x \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right\} dx = \int \ln(1 + \sin x) dx + \int x \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) dx$

$$= x \cdot \ln(1 + \sin x) - \int \frac{1}{1 + \sin x} \cos x \cdot x dx + \int x \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) dx$$

$$= x \cdot \ln(1 + \sin x) - \int \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)^2} \cdot x dx + \int x \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) dx = x \ln(1 + \sin x) + C$$

41.(C) $I = \int \sqrt{\frac{x-1}{x+1}} \times \frac{1}{x^2} dx$ [Put $\frac{1}{x} = \cos 2\theta \Rightarrow -\frac{dx}{x^2} = -2 \sin 2\theta d\theta$]

$$I = \int \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} 2 \sin 2\theta d\theta = \int 4 \sin^2 \theta d\theta = 2 \int (1 - \cos 2\theta) d\theta = 2\theta - \sin 2\theta + C = \cos^{-1} \left(\frac{1}{x} \right) - \sqrt{1 - \frac{1}{x^2}} + C$$

42.(A) S₂: If primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ is $\frac{x}{x^4 + x + 1} + C$, then derivative of $\frac{x}{x^4 + x + 1} + C$ must be $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$

$$\text{Now let } f(x) = \frac{x}{x^4 + x + 1} + C \Rightarrow f'(x) = \frac{(x^4 + x + 1) - x(4x^3 + 1)}{(x^4 + x + 1)^2} = \frac{-3x^4 + 1}{(x^4 + x + 1)^2}$$

Hence it is false.

S₃: Let $I = \int \frac{dx}{\sqrt{\sin^3 x \cos x}} = \int \sin^{-3/2} x \cos^{-1/2} x dx = \int \tan^{-3/2} \cdot \sec^2 x dx$ [Put $\tan x = t \Rightarrow \sec^2 x dx = dt$]

$$\Rightarrow I = \int t^{-3/2} dt = \frac{-2}{\sqrt{\tan x}} + C$$

S₄: Let $I = \int \left(\sqrt{\frac{a+x}{a-x}} - \sqrt{\frac{a-x}{a+x}} \right) dx = \int \frac{2x}{\sqrt{a^2 - x^2}} dx = -2\sqrt{a^2 - x^2} + C$

$$\begin{aligned}
 43.(ABCD) \quad I &= \int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx = \int \frac{2 \tan x \sec^2 x}{\tan^4 x + 1} dx \quad [\text{Put } \tan^2 x = t \Rightarrow 2 \tan x \sec^2 x dx = dt] \\
 &= \int \frac{dt}{t^2 + 1} = \tan^{-1}(t) + C = \tan^{-1}(\tan^2 x) + C = \frac{\pi}{2} - \cot^{-1}(\tan^2 x) + C \\
 &= -\cot^{-1}(\tan^2 x) + C_1 = -\cot^{-1}\left(\frac{1}{\cot^2 x}\right) + C_1 = \cot^{-1}(\cot^2 x) + C_1
 \end{aligned}$$

Also $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x} \Rightarrow \left(\frac{1 - \cos 2x}{1 + \cos 2x}\right) = \tan^2 x$, using these values in given integral

$$I = \int \frac{\sin 2x}{(\cos^2 x - \sin^2 x)^2 + 2 \sin^2 x \cos^2 x} dx = \int \frac{\sin 2x}{(\cos 2x)^2 + \left(\frac{1 - \cos^2 2x}{2}\right)} dx = \int \frac{2 \sin 2x}{(\cos 2x)^2 + 1} dx$$

[Put $\cos 2x = t \Rightarrow -2 \sin 2x dx = dt$]

$$I = \int \frac{-dt}{t^2 + 1} = -\tan^{-1} t + C_2 = -\tan^{-1}(\cos 2x) + C_2$$

$$44.(ABD) \quad I = \int \frac{dx}{\sqrt{x-x^2}} = \int \frac{dx}{\sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2}} = \sin^{-1} \frac{\left(x - \frac{1}{2}\right)}{1/2} = \sin^{-1}(2x-1) + C \quad \dots (i)$$

Also $I = \int \frac{dx}{\sqrt{x}\sqrt{1-x}} \quad [\text{Put } \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt]$

$$\Rightarrow I = \int \frac{2dt}{\sqrt{1-t^2}} = 2 \sin^{-1} \sqrt{x} + C \quad \dots (ii) \quad (\text{Now let } \theta = \sin^{-1}(2x-1))$$

$$\Rightarrow \sin \theta = 2x-1 \Rightarrow \cos \theta = \sqrt{1-(2x-1)^2} = \sqrt{1-4x^2+4x-1} = 2\sqrt{x-x^2}$$

$$\Rightarrow \theta = \cos^{-1} 2\sqrt{x-x^2}$$

$$45.(AB) \quad I_n = \int \cot^n x dx = \int \cot^{n-2} x (\operatorname{cosec}^2 x - 1) dx = \frac{-\cot^{n-1} x}{n-1} - I_{n-2} + C_1$$

$$\Rightarrow I_n + I_{n-2} = \frac{-\cot^{n-1} x}{n-1} + C_1$$

Now, $I_0 + I_1 + 2(I_2 + \dots + I_8) + I_9 + I_{10}$
 $= (I_2 + I_0) + (I_3 + I_1) + (I_4 + I_2) + (I_5 + I_3) + (I_6 + I_4) + (I_7 + I_5) + (I_8 + I_6) + (I_9 + I_7) + (I_{10} + I_8)$
 $= -\left(\frac{\cot x}{1} + \frac{\cot^2 x}{2} + \dots + \frac{\cot^9 x}{9}\right) + C \quad \therefore A = -1$

$$46-48. \quad 46.(A) \quad I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx = 2 \int \frac{(\sqrt{\tan x} + \sqrt{\cot x})(\tan x + \cot x)}{2(\tan x + \cot x)} dx$$

$$I = 2 \int \frac{d(\sqrt{\tan x} - \sqrt{\cot x})}{(\sqrt{\tan x} - \sqrt{\cot x})^2 + 2} = \sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$$

$$\begin{aligned}
 47.(B) \quad \text{If } x \in \left(0, \frac{\pi}{2}\right), \text{ then } I &= \int (\sqrt{\tan x} + \sqrt{\cot x}) dx = \int \left(\frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}} \right) dx \\
 &= \int \frac{\sin x + \cos x}{\sqrt{\sin x \cdot \cos x}} dx = \int \frac{\sqrt{2} d(\sin x - \cos x)}{\sqrt{1 - (\sin x - \cos x)^2}} = \sqrt{2} \sin^{-1}(\sin x - \cos x) + C
 \end{aligned}$$

48.(A) If $x \in \left(\pi, \frac{3\pi}{2}\right)$, then $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx = \int \left(\frac{\sqrt{-\sin x}}{\sqrt{-\cos x}} + \frac{\sqrt{-\cos x}}{\sqrt{-\sin x}} \right) dx$

$$= -\int \frac{\sin x + \cos x}{\sqrt{\sin x \cdot \cos x}} dx = -\int \frac{\sqrt{2}d(\sin x - \cos x)}{\sqrt{1 - (\sin x - \cos x)^2}} = \sqrt{2} \sin^{-1}(\cos x - \sin x) + C$$

49.(A) For any natural number m , the given integral can be written as,

$$I = \int (x^{3m} + x^{2m} + x^m) \frac{(2x^{3m} + 3x^{2m} + 6x^m)^{1/m}}{x} dx = \int (2x^{3m} + 3x^{2m} + 6x^m)^{1/m} (x^{3m-1} + x^{2m-1} + x^{m-1}) dx$$

[Put $2x^{3m} + 3x^{2m} + 6x^m = t \Rightarrow 6m(x^{3m-1} + x^{2m-1} + x^{m-1})dx = dt$]

$$\Rightarrow I = \int t^{1/m} \cdot \frac{dt}{6m} = \frac{1}{6m} \frac{t^{1/m+1}}{\left(\frac{1}{m}+1\right)} + C = \frac{1}{6(m+1)} \cdot (2x^{3m} + 3x^{2m} + 6x^m)^{(m+1)/m} + C$$

50.(A) $f(f(x)) = \frac{f(x)}{(1+f(x)^n)^{1/n}} = \frac{\frac{x}{(1+x^n)^{1/n}}}{\left(1 + \frac{x^n}{1+x^n}\right)^{1/n}} = \frac{x}{(1+2x^n)^{1/n}}$

and $f(f(f(x))) = \frac{\frac{x}{(1+2x^n)^{1/n}}}{\left(1 + \frac{x^n}{1+2x^n}\right)^{1/n}} = \frac{x}{(1+3x^n)^{1/n}} \Rightarrow g(x) = \frac{x}{(1+nx^n)^{1/n}}$

Hence $I = \int x^{n-2} \frac{x}{(1+nx^n)^{1/n}} dx = \int \frac{x^{n-1}}{(1+nx^n)^{1/n}} dx$ [Put $1+nx^n = t \Rightarrow x^{n-1}dx = \frac{1}{n^2}dt$]

$$\Rightarrow I = \frac{1}{n^2} \int \frac{dt}{t^{1/n}} = \frac{1}{n^2} \frac{t^{-1/n+1}}{-1/n+1} = \frac{1}{n(n-1)} (1+nx^n)^{1-1/n} + K$$

51.(C) Put $\sec x + \tan x = t \Rightarrow (\sec x \tan x + \sec^2 x) dx = dt \Rightarrow \sec x \cdot t dx = dt \Rightarrow \sec x - \tan x = \frac{1}{t} \Rightarrow \sec x = \frac{t + \frac{1}{t}}{2}$

$$\int \sec x \cdot dt = \int \frac{1}{2} \left(\frac{t + \frac{1}{t}}{t \cdot t^{9/2}} \right) dt = \frac{1}{2} \int \left(\frac{1}{t^{9/2}} + \frac{1}{t^{13/2}} \right) dt = -\frac{1}{2} \left[\frac{2}{7t^{7/2}} + \frac{2}{11t^{11/2}} \right] + k = -\frac{1}{t^{11/2}} \left[\frac{t^2}{7} + \frac{1}{11} \right] + k$$

52.(A) 53.(B) 54.(D) 55.(A) 56.(A) 57.(C) 58.(D) 59.(C)

60.(B) $\int \left(1 + x - \frac{1}{x}\right) e^{x + \frac{1}{x}} dx$

$$= \int e^{x + \frac{1}{x}} dx + \int x \left(1 - \frac{1}{x^2}\right) e^{x + \frac{1}{x}} dx = \int e^{x + \frac{1}{x}} dx + x e^{x + \frac{1}{x}} - \int \frac{d}{dx}(x) e^{x + \frac{1}{x}}$$

$$= \int e^{x + \frac{1}{x}} dx + x e^{x + \frac{1}{x}} - \int e^{x + \frac{1}{x}}$$

$$\left[\because \int \left(x - \frac{1}{x^2}\right) e^{x + \frac{1}{x}} = e^{x + \frac{1}{x}} \right]$$

$$= x e^{x + \frac{1}{x}} + C$$

61.(1) Let $f(x) = ax^3 + bx^2 + cx + d$

Since $f(0) = -1$ and $f(1) = 0$

$\Rightarrow d = -1$ and $a + b + c + d = 0 \Rightarrow d = -1$ and $a + b + c = 1$... (i)

Since 0 is stationary point of $f(x)$,

$\therefore f'(0) = 0 \Rightarrow 3a(0)^2 + 2b(0) + c = 0 \Rightarrow c = 0$

Since $f(x)$ does not have an extremum at $x = 0$

$\therefore f''(0) = 0 \Rightarrow b = 0$ and $\therefore$ from (i), $a = 1$

So, $f(x) = x^3 - 1$ $\therefore \int \frac{f(x)}{x^3 - 1} dx = \int 1 dx = x + C$

62. (2) Let $e^x + C_1$ and $e^x + C_2$ be the two antiderivatives.

Then difference between them is $C_1 - C_2$ which is fixed and equal to 2.

63.(1) $\int \frac{1}{x} \log_{ex} e \log_{e^2x} e dx = \int \frac{1}{x} \left(\frac{\log e}{\log e + \log x} \right) \left(\frac{\log e}{2 \log e + \log x} \right) dx = \int \frac{1}{x} \frac{1}{(1 + \log x)} \frac{1}{(2 + \log x)} dx$

Let $\log x = k \Rightarrow \int \frac{1}{e^k} \frac{1}{(1+k)} \frac{1}{(2+k)} e^k dk = \int \left(\frac{1}{1+k} - \frac{1}{2+k} \right) dk = \log \left(\frac{1+k}{2+k} \right)$

$\Rightarrow f(x) = k = \log x \Rightarrow \lim_{x \rightarrow 0} \frac{\log(x+1)}{x} \left(\frac{0}{0} \text{ form} \right) \Rightarrow \text{from L.H. rule } \lim_{x \rightarrow 0} \frac{1}{x+1} = 1$

64. (3) $F(x) = \int \frac{dx}{\cos^2 x \sqrt{1 + \tan x}} = \int \frac{\sec^2 x dx}{\sqrt{1 + \tan x}}$

Put $\tan x = t$ to get: $F(x) = \int \frac{dt}{\sqrt{1+t}} = 2\sqrt{1+t} + C = 2\sqrt{1 + \tan x} + C$

$F(0) = 4 \Rightarrow C = 2$

65.(1) Put $x - 2 = u$ to get $I = \int \frac{x dx}{x^2 - 4x + 8} = \int \frac{u + 2}{u^2 + 4} du$

$= \frac{1}{2} \int \frac{2u}{u^2 + 4} du + 2 \int \frac{du}{u^2 + 4} = \frac{1}{2} \log(u^2 + 4) + \tan^{-1} \left(\frac{u}{2} \right) + C = \frac{1}{2} \log(x^2 - 4x + 8) + \tan^{-1} \left(\frac{x-2}{2} \right) + C$

66.(202) $I = \int \sqrt{1 + \sec x} dx = \int \frac{\sqrt{\sec^2 x - 1}}{\sqrt{\sec x - 1}} dx = \int \frac{\tan x}{\sqrt{\sec x - 1}} dx$

[Put $\sec x - 1 = t^2 \Rightarrow \sec x \tan x dx = 2t dt \Rightarrow \tan x dx = \frac{2t dt}{t^2 + 1}$]

$I = \int \frac{1}{t} \left(\frac{2t}{t^2 + 1} \right) dt = 2 \tan^{-1}(t) + C = 2 \tan^{-1}(\sqrt{\sec x - 1}) + C$

67.(1) $f(x) = \int \frac{2 \sin x - \sin 2x}{x^3}, x \neq 0$ $\lim_{x \rightarrow 0} f'(x) = \frac{2 \sin x}{x} \left(\frac{1 - \cos x}{x^2} \right) = 1$

68.(32) Put $1 + x^{4/3} = z^7 \Rightarrow x^{1/3} dx = \frac{21}{4} z^6 dz$

$\therefore \int \sqrt[3]{x} \sqrt{1 + \sqrt[3]{x^4}} dx = \int z \left(\frac{21}{4} z^6 dz \right) = \frac{21}{4} \cdot \frac{z^8}{8} + C \Rightarrow \frac{21}{32} \left(1 + \sqrt[3]{x^4} \right)^{8/7} + C.$

69.(10) Let $I = \int \frac{(5x^4 + 4x^5)}{(x^5 + x + 1)^2} dx$

Dividing above and below by x^{10} , we get: $I = \frac{\left(\frac{5}{x^6} + \frac{4}{x^5}\right) dx}{\left(1 + \frac{1}{x^4} + \frac{1}{x^5}\right)^2}$

Putting $1 + \frac{1}{x^4} + \frac{1}{x^5} = t$, $\left(-\frac{4}{x^5} - \frac{5}{x^6}\right) dx = dt$ or $\left(\frac{4}{x^5} + \frac{5}{x^6}\right) dx = -dt$, we get:

$$I = -\int \frac{dt}{t^2} = \frac{1}{t} + c = \frac{1}{\left(1 + \frac{1}{x^4} + \frac{1}{x^5}\right)} + c = \frac{x^5}{(x^5 + x + 1)} + c = f(x) \quad f(0) = 0 \Rightarrow c = 0 \Rightarrow f(1) = \frac{1}{3}$$

70.(18) Let $I = \int \frac{\sin^{1/2} x}{\cos^{9/2} x} dx = \int \frac{dx}{\sin^{-1/2} x \cos^{9/2} x}$

Here $m + n = \frac{1}{2} - \frac{9}{2} = -4$ (negative even integer).

Divide numerator and denominator by $\cos^4 x$.

$$I = \int \sqrt{\tan x} \sec^4 x dx = \int \sqrt{\tan x} (1 + \tan^2 x) \sec^2 x dx = \int \sqrt{t} (1 + t^2) dt \quad (\text{Using } \tan x = t)$$

$$= \frac{2}{3} t^{3/2} + \frac{2}{7} t^{7/2} + c = \frac{2}{3} \tan^{3/2} x + \frac{2}{7} \tan^{7/2} x + c$$

71.(1024) $\int \sin x \cos x \cos 2x \cos 4x \cos 8x \cos 16x dx = \int \frac{\sin 32x}{32} dx = -\frac{\cos 32x}{1024} + c$

72.(0) $\int \frac{x + (\cos^{-1} 3x)^2}{\sqrt{1-9x^2}} dx = \int \frac{x}{\sqrt{1-9x^2}} dx + \int \frac{(\cos^{-1} 3x)^2}{\sqrt{1-9x^2}} dx = -\frac{1}{18} 2\sqrt{1-9x^2} - \frac{1}{9} (\cos^{-1} 3x)^3 + C$

$\therefore A = -\frac{1}{9}, B = -\frac{1}{9}$

73.(9) Differentiating both sides gives $x^{26} \cdot (x-1)^{17} (5x-3) = \frac{1}{k} [x^{27} \cdot 18(x-1)^{17} + (x-1)^{18} 27x^{26}]$

$$= \frac{x^{26}(x-1)^{17}}{k} \cdot [18x + 27(x-1)] = \frac{x^{26}(x-1)^{17}}{k} (45x - 27) = 9 \frac{x^{26}(x-1)^{17}}{k} (5x - 3) \Rightarrow k = 9$$

74.(3) $\int \frac{f(x) dx}{x^2(x+1)^3} = \int \left(\frac{1}{x^2} - \frac{x}{(x+1)^3} \right) dx$; $\int \frac{f(x) dx}{x^2(x+1)^3} = \int \frac{3x^2 + 3x + 1}{x^2(x+1)^3} dx \Rightarrow f(x) = 3x^2 + 3x + 1 \Rightarrow f'(0) = 3$

75.(2) $I = \int \frac{(x-x^3)^{1/3}}{x^4} dx = \int \frac{\left(\frac{1}{x^2} - 1\right)^{1/3}}{x^3} dx$

Putting $\frac{1}{x^2} - 1 = t$, $\frac{1}{x^3} dx = -\frac{dt}{2}$, we get:

$$I = -\frac{1}{2} \int t^{1/3} dt = -\frac{3}{8} t^{4/3} + C = -\frac{3}{8} \left(\frac{1}{x^2} - 1 \right)^{4/3} + C \Rightarrow a = -\frac{3}{8}, b = \frac{4}{3} \Rightarrow ab = -\frac{1}{2} \Rightarrow \frac{1}{|ab|} = 2$$

Integral Calculus-2

$$1.(C) \quad 2 \int_0^1 \tan^{-1} x \cdot dx = \int_0^1 \cot^{-1}(1-x+x^2) dx \quad \dots\dots (i)$$

$$\begin{aligned} \therefore I &= \int_0^1 \tan^{-1}(1-x+x^2) dx = \int_0^1 \left(\frac{\pi}{2} - \cot^{-1}(1-x+x^2) \right) dx = \int_0^1 \frac{\pi}{2} dx - \int_0^1 \cot^{-1}(1-x+x^2) dx \\ &= \frac{\pi}{2} - 2 \int_0^1 \tan^{-1} x \cdot 1 \cdot dx \quad \{\text{From (i)}\} \\ &= \frac{\pi}{2} - 2 \left[x \tan^{-1} x - \int \frac{1}{1+x^2} x dx \right]_0^1 = \frac{\pi}{2} - 2 \left[x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right]_0^1 \\ &= \frac{\pi}{2} - 2 \left[1 \cdot \frac{\pi}{4} - \frac{1}{2} \log 2 - 0 \right] = \log 2 \end{aligned}$$

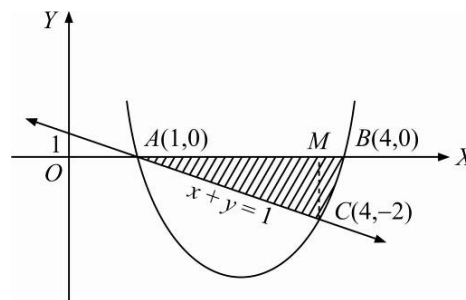
$$2.(B) \quad \text{Here } y \geq x^2 - 5x + 4 \Rightarrow y \geq \left(x - \frac{5}{2}\right)^2 - \frac{9}{4} \Rightarrow y + \frac{9}{4} \geq \left(x - \frac{5}{2}\right)^2. \text{ Parabola upwards.}$$

$$\text{If } y = 0, \text{ then } x^2 - 5x + 4 = 0 \Rightarrow x = 1, 4$$

So, parabola intersect x-axis at $A(1, 0)$ and $B(4, 0)$. $x+y \geq 1$ and $y \leq 0$

$$\text{Req. area} = \text{ar}(AMC) + \text{ar}(BCM)$$

$$\begin{aligned} &= \frac{1}{2} \cdot AM \cdot MC + \left| \int_3^4 (x^2 - 5x + 4) dx \right| \\ &= \frac{1}{2} \cdot 2 \cdot 2 + \left| \left[\frac{x^3}{3} - \frac{5x^2}{2} + 4x \right]_3^4 \right| \\ &= 2 + \left| \frac{64}{3} - \frac{80}{2} + 16 - \frac{27}{3} + \frac{45}{2} - 12 \right| \\ &= 2 + \left| \frac{-7}{6} \right| = 2 + \frac{7}{6} = \frac{19}{6} \end{aligned}$$



$$3.(B) \quad I = \int_4^{10} \frac{[x^2] dx}{[x^2 - 28x + 196] + [x^2]} = \int_4^{10} \frac{[x^2] dx}{[(14-x)^2] + [x^2]} \quad \dots\dots (i)$$

$$= \int_4^{10} \frac{[(14-x)^2] dx}{[14 - (14-x)^2] + [(14-x)^2]} = \int_4^{10} \frac{(14-x)^2 dx}{[x^2] + [(14-x)^2]} \quad \dots\dots (ii)$$

(i) + (ii)

$$2I = \int_4^{10} dx = 6 \Rightarrow I = 3$$

$$4.(C) \quad \text{Given } x \int_1^x y(t) dt = (x+1) \int_1^x ty(t) dt$$

Differentiating w.r.t. x , we get

$$x \int_1^x y(t) dt + x [y(x) \cdot 1 - y(1) \cdot 0] = \int_1^x ty(t) dt + (x+1) [xy(x) \cdot 1 - 1 \cdot y(1) \cdot 0]$$

$$\Rightarrow \int_1^x y(t) dt + xy(x) = \int_1^x ty(t) dt + (x+1)xy(x) \Rightarrow \int_1^x y(t) dt = \int_1^x ty(t) dt + x^2y(x)$$

Again, differentiating w.r.t. x , we get : $y(x) = xy(x) + x^2 \frac{dy}{dx} + 2xy(x)$

$$\Rightarrow x^2 \frac{dy}{dx} = y(x) - 3xy(x) = y - 3xy \Rightarrow \frac{dy}{y} = \left(\frac{1}{x^2} - \frac{3}{x} \right) dx \Rightarrow \int \frac{dy}{y} = \int \left(\frac{1}{x^2} - \frac{3}{x} \right) dx$$

$$\Rightarrow \log y = -\frac{1}{x} - 3 \log x + \log C \Rightarrow \log \frac{yx^3}{C} = -\frac{1}{x} \Rightarrow \frac{yx^3}{C} = e^{-\frac{1}{x}} \Rightarrow y = \frac{C}{x^3} e^{-\frac{1}{x}}$$

5.(B) $f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\log t}{1+t} dt$

Put $t = \frac{1}{u} \Rightarrow dt = -\frac{1}{u^2} du$

When $t = 1, u = 1$; when $t = 1/x, u = x$

$$\therefore f\left(\frac{1}{x}\right) = -\int_1^x \frac{\log \frac{1}{u}}{1+\frac{1}{u}} \cdot \frac{1}{u^2} du = \int_1^x \frac{\log u}{u+1} \cdot \frac{1}{u} du \quad \{\because \log 1 = 0\}$$

$$= \int_1^x \frac{\log t}{t+1} \cdot \frac{1}{t} dt$$

$$\therefore f(x) + f\left(\frac{1}{x}\right) = \int_1^x \frac{\log t}{1+t} dt + \int_1^x \frac{\log t}{t+1} \cdot \frac{1}{t} dt$$

$$= \int_1^x \frac{\log t}{t+1} \left(1 + \frac{1}{t}\right) dt = \int_1^x \frac{\log t}{t} dt = \left[\frac{(\log t)^2}{2} \right]_1^x = \frac{(\log x)^2}{2}$$

6.(D) Required area = $2 \left[\int_{-2}^1 x dy - \int_{-2}^0 x dy \right] = 2 \left[\frac{1}{\sqrt{3}} \int_{-2}^1 \sqrt{1-y} - \frac{1}{\sqrt{2}} \int_{-2}^0 \sqrt{-y} dy \right]$

$$= -\frac{1}{\sqrt{3}} \cdot \frac{2}{3} \left[(1-y)^{3/2} \right]_{-2}^1 + \frac{1}{\sqrt{2}} \cdot \frac{2}{3} \left[(-y)^{3/2} \right]_{-2}^0 = 2 \left[-\frac{2}{3\sqrt{3}} (-3^{3/2}) + \frac{2}{3\sqrt{2}} (-2^{3/2}) \right]$$

$$= 2 \left[2 - \frac{4}{3} \right] = \frac{4}{3}$$

7.(B) $f(2-x) = f(2+x)$

Replace x by $x-2$, we get : $f(2-x+2) = f(2+x-2) \Rightarrow f(4-x) = f(x)$

$$\Rightarrow f(4+x) = f(x) \quad \{\because f(4-x) = f(4+x)\}$$

$\therefore$ Period of $f(x)$ is 4.

$$\int_0^2 f(x) dx = \int_0^2 f(2x) dx = \int_0^2 f(x+2) dx$$

If T is period of $f(x)$, then

$$I = \int_{10}^{50} f(x) dx = \int_{10}^{10+40} f(x) dx = \int_0^{40} f(x) dx = \int_0^{10 \times 4} f(x) dx = 10 \int_0^4 f(x) dx$$

$$= 10 \left[\int_0^2 f(x) dx + \int_2^4 f(x) dx \right] = 10 \times 5 + 10 \int_2^4 f(x) dx \quad \{\text{Put } x = 2+t\}$$

$$= 50 + 10 \int_0^2 f(2+t) dt = 50 + 10 \int_0^2 f(t) dt = 50 + 10 \int_0^2 f(x) dx = 50 + 10 \times 5 = 100$$

8.(B) Given $\int_0^{\sin x} f(t) dt = \frac{\sqrt{3}}{2} x$

Differentiating w.r to x on both sides

$$\frac{d}{dx} \left(\int_0^{\sin x} f(t) dt \right) = \frac{d}{dx} \left(\frac{\sqrt{3}}{2} x \right)$$

$$f(\sin x) \frac{d}{dx}(\sin x) - f(0) \frac{d}{dx}(0) = \frac{\sqrt{3}}{2} \Rightarrow f(\sin x) \cos x = \frac{\sqrt{3}}{2}$$

When $x = \frac{\pi}{3}$, then $f\left(\sin \frac{\pi}{3}\right) \cos \frac{\pi}{3} = \frac{\sqrt{3}}{2} \Rightarrow f\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}$

9.(C) Put $2x = \tan \theta \Rightarrow 2 dx = \sec^2 \theta d\theta$

$$\therefore I = \frac{1}{2} \int_0^{\pi/4} \frac{\log(1 + \tan \theta)}{1 + \tan^2 \theta} \sec^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/4} \log(1 + \tan \theta) d\theta = \frac{1}{2} \cdot \frac{\pi}{8} \log 2.$$

10.(C) $\int (\log x)^n \cdot 1 \cdot dx = (\log x)^n \cdot x - \int \frac{n(\log x)^{n-1}}{x} \cdot x dx = x(\log x)^n - n \int (\log x)^{n-1} \cdot 1 \cdot dx$

$$= x(\log x)^n - n \left[(\log x)^{n-1} \cdot x - (n-1) \int (\log x)^{n-2} dx \right]$$

$$= x(\log x)^n - nx(\log x)^{n-1} + n(n-1) \int (\log x)^{n-2} dx$$

$$\therefore P_n = \int_1^e (\log x)^n dx = \left[x(\log x)^n - nx(\log x)^{n-1} \right]_1^e + n(n+1) \int_1^e (\log x)^{n-2} dx$$

$$\therefore P_{10} = \left[x(\log x)^{10} - 10x(\log x)^9 \right]_1^e + 10 \times 9 \int_1^e (\log x)^8 dx = (e \cdot 1 - 10e \cdot 1) - (0 - 0) + 90P_8$$

$$\{\because \log e = 1, \log 1 = 0\}, P_{10} - 90P_8 = -9e$$

11.(A) $\int_0^\pi [\cos x] dx = \int_0^{\pi/2} [\cos x] dx + \int_{\pi/2}^\pi [\cos x] dx = \int_0^{\pi/2} 0 dx + \int_{\pi/2}^\pi (-1) dx = -\frac{\pi}{2}$

12.(A) $\int_{-\pi}^t (f(x) + x) dx = \pi^2 - t^2$

Differentiating w.r.t. t

$$f(t) + t = 2t \Rightarrow f(t) = -3t$$

$$\therefore f\left(-\frac{\pi}{3}\right) = -3\left(-\frac{\pi}{3}\right) = \pi$$

13.(D) $F(x) = \int_1^x \frac{e^t}{t} dt, \quad x > 0$

$$\int_1^x \frac{e^t}{t} dt, \text{ put } t+a = u$$

$$= \int_{1+a}^{x+a} \frac{e^u}{u} du = e^{-a} \int_{1+a}^{x+a} \frac{e^u}{u} du = e^{-a} \left[\left(\int_{1+a}^{x+a} \frac{e^u}{u} du + \int_1^{1+a} \frac{e^u}{u} du \right) - \int_1^{1+a} \frac{e^u}{u} du \right]$$

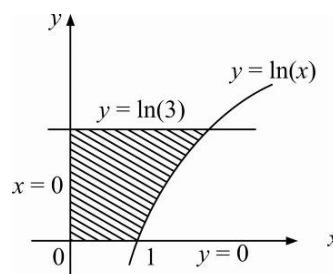
$$= e^{-a} \left[- \int_1^{1+a} \frac{e^u}{u} du - F(1+a) \right] = e^{-a} [F(x+a) - F(1+a)]$$

$$14.(A) \quad \therefore \quad x = \int_0^y \frac{dt}{\sqrt{1+t^2}} \Rightarrow \frac{dx}{dy} = \frac{1}{\sqrt{1+y^2}} \Rightarrow \frac{dy}{dx} = \sqrt{1+y^2}$$

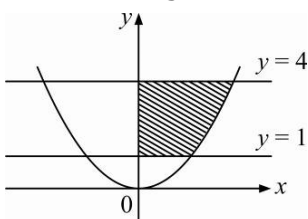
$$\frac{d^2y}{dx^2} = \frac{1}{2\sqrt{1+y^2}} \cdot 2y \frac{dy}{dx} = \frac{y}{\sqrt{1+y^2}} \cdot \sqrt{1+y^2} = y$$

$$15.(D) \quad y = \ln(x) \Rightarrow x = e^y$$

$$\text{Area} = \int_0^{\ln 3} x dy = \int_0^{\ln 3} e^y dy = [e^y]_0^{\ln 3} = [e^{\ln 3} - e^0] = 3 - 1 = 2$$



$$16.(D) \quad \text{Area} \int_1^4 x dy = \frac{1}{3} \int_1^4 \sqrt{4} dy = \frac{14}{9}$$



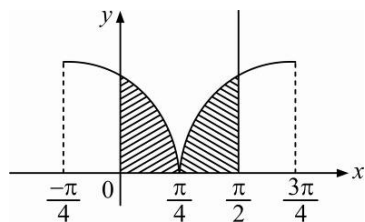
$$17.(A) \quad |\tan x| = \begin{cases} -\tan x, & \frac{7\pi}{4} \leq x < 2\pi \\ \tan x, & 2\pi \leq x \leq \frac{7\pi}{3} \end{cases}$$

$$\begin{aligned} \int_{7\pi/4}^{7\pi/3} \sqrt{\tan^2 x} dx &= \int_{7\pi/4}^{7\pi/2} |\tan x| dx \\ &= \int_{7\pi/4}^{2\pi} -\tan x dx + \int_{2\pi}^{7\pi/3} \tan x dx = -[\log|\sec x|]_{7\pi/4}^{2\pi} + [\log|\sec x|]_{2\pi}^{7\pi/3} \\ &= [\log 1 - \log 1 - \sqrt{2}] + [\log 2 - \log 1] = \log \sqrt{2} + \log 2 = \log 2\sqrt{2} \end{aligned}$$

$$18.(D) \quad \int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_0^a f(-x) dx$$

$$\therefore \quad I = \int_0^{\pi/2} \frac{\sin^2 x}{1+2^x} dx + \int_0^{\pi/2} \frac{\sin^2 x}{1+2^{-x}} dx = \int_0^{\pi/2} \frac{\sin^2 x}{1+2^x} dx (1+2^x) dx = \int_0^{\pi/2} \sin^2 x = \frac{x}{4}$$

$$19.(B) \quad y = |\cos x - \sin x| = \sqrt{2} \left| \cos x \cdot \frac{1}{\sqrt{2}} - \sin x \cdot \frac{1}{\sqrt{2}} \right| = \sqrt{2} \left| \cos \left(x + \frac{\pi}{4} \right) \right|$$



$$\text{Required area} = 2 \cdot \sqrt{2} \int_0^{\pi/4} \cos \left(x + \frac{\pi}{4} \right) dx = 2\sqrt{2} \left[\sin \left(x + \frac{\pi}{4} \right) \right]_0^{\pi/4} = 2\sqrt{2} \left[1 - \frac{1}{\sqrt{2}} \right] = 2\sqrt{2} - 2$$

$$20.(C) \quad \int_{-1}^2 (2x - [2x]) dx = \int_{-1}^2 2x dx - \int_{-1}^2 [2x] dx = \int_{-1}^2 2x dx - \left(\int_{-1}^{-1/2} [2x] dx + \int_{-1/2}^0 [2x] dx + \int_0^{1/2} [2x] dx \right) + \int_{1/2}^1 [2x] dx + \int_1^{3/2} [2x] dx + \int_{3/2}^2 [2x] dx = \frac{3}{2}$$

$$21.(B) \quad I = \int_0^{20\frac{\pi}{2}} (|\sin x| + |\cos x|) dx = 20 \int_0^{\pi/2} (\sin x + \cos x) dx = 40 \quad \left[\text{period of } |\sin x| + |\cos x| = \frac{\pi}{2} \right]$$

$$22.(C) \quad f(x) = \sqrt{3+x^3} \Rightarrow f'(x) = \frac{3x^2}{2\sqrt{3+x^3}} > 0 \forall x \in (1, 3) \Rightarrow f \text{ is strictly increasing in } (1, 3)$$

$$\text{So } m = \text{least value } f(1) = \sqrt{3+1} = 2 \text{ and } M = \text{greatest value } = f(3) = \sqrt{30}$$

$$\text{Now } m(b-a) \leq \int_a^b f(x) dx \leq M(b-a) \Rightarrow 2(2) \leq \int_1^3 \sqrt{3+x^3} dx \leq \sqrt{30} \times 2 \Rightarrow 4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$$

$$23.(C) \quad \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{3}{n} \sqrt{\frac{n}{n+3r}} = \lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{3}{n} \sqrt{\frac{n/r}{3+n/r}} = \int_0^1 3 \sqrt{\frac{x}{x+3}} dx = \frac{\pi}{2}$$

$$24.(B) \quad f(x) = \text{Maximum} \{x^2, (1-x)^2, 2x(1-x)\} \Rightarrow f(x) = \begin{cases} (1-x)^2 & \text{for } 0 \leq x \leq \frac{1}{3} \\ 2x(1-x) & \text{for } \frac{1}{3} \leq x \leq \frac{2}{3} \\ x^2 & \text{for } \frac{2}{3} \leq x \leq 1 \end{cases}$$

$$\text{The required area A is given by } A = \int_0^1 f(x) dx = \int_0^{1/3} (1-x)^2 dx + \int_{1/3}^{2/3} 2x(1-x) dx + \int_{2/3}^1 x^2 dx = \frac{17}{27} \text{ square units}$$

$$25.(A) \quad \text{Area} = \int_0^1 e^y \sin(\pi y) dy = \frac{e^y}{1+\pi^2} (\sin \pi y - \pi \cos \pi y) \Big|_0^1 = \frac{(e+1)\pi}{1+\pi^2}$$

$$26.(A) \quad f(x) = \min\{x+1, \sqrt{1-x}\} = \begin{cases} x+1 & , -1 < x < 0 \\ \sqrt{1-x} & , 0 < x < 1 \end{cases} \Rightarrow \frac{12}{7} \int_{-1}^1 f(x) dx = \frac{12}{7} \left[\int_{-1}^0 (x+1) dx + \int_0^1 \sqrt{1-x} dx \right] = 2$$

$$27.(C) \quad 28.(C)$$

$$29.(C) \quad I = \int_0^1 x f''(2x) dx \text{ [by applying by parts]}$$

$$I = \left[\frac{xf'(2x)}{2} - \frac{f(2x)}{4} \right]_0^1 = \frac{f'(2)}{2} - \frac{f(2)}{4} + \frac{f(0)}{4} = \frac{5}{2} - \frac{3}{4} + \frac{1}{4} = 2$$

$$30.(A) \quad \int_{-1}^3 |x-2| + [x] dx = \int_{-1}^0 (2-x-1) dx + \int_0^1 (2-x) dx + \int_1^2 (2-x+1) dx + \int_2^3 (x-2+2) dx = 7 \quad 31.(A)$$

$$32.(A) \quad I_1 = \int_e^{e^2} \frac{dx}{\ell nx} \quad (\text{Let } \ell nx = t \Rightarrow x = e^t \Rightarrow dx = e^t dt) \Rightarrow I_1 = \int_1^2 \frac{e^t}{t} dt = I_2 \quad 33.(C)$$

$$34.(AC) \quad I = \int_0^\infty \frac{x dx}{(1+x)(1+x^2)} \quad (x = \tan \theta \Rightarrow dx = \sec^2 \theta \cdot d\theta)$$

$$I = \int_0^{\pi/2} \frac{\tan \theta}{(1+\tan \theta) \cdot \sec^2 \theta} \cdot \sec^2 \theta d\theta$$

$$I = \int_0^{\pi/2} \frac{\sin \theta}{\sin \theta + \cos \theta} d\theta \quad \dots \text{(i)} \quad \text{and} \quad I = \int_0^{\pi/2} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta \quad \dots \text{(ii)}$$

Now from (i) + (ii) we get : $2I = \int_0^{\pi/2} 1 \cdot d\theta = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$

Also $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)} \quad (x = \tan \theta \Rightarrow dx = \sec^2 \theta \cdot d\theta)$

$$I = \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(1+\tan \theta) \cdot \sec^2 \theta} = \int_0^{\pi/2} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta = \frac{\pi}{4}$$

35.(ABCD) **Case I :** If $0 < a < b$ then $|x| = x \Rightarrow \int_a^b \frac{|x|}{x} dx = b - a$

Case II : If $a < 0 < b$ then $\int_a^b \frac{|x|}{x} dx = \int_a^0 \frac{|x|}{x} dx + \int_0^b \frac{|x|}{x} dx = a + b$

Case III : If $a < b < 0$ then $\int_a^b \frac{|x|}{x} dx = \int_a^b (-1) dx = a - b$ Also $\int_a^b \frac{|x|}{x} dx = |b| - |a|$

36.(C) $I = \int \frac{-\ell n \lambda f\left(\frac{x^2}{4}\right) [f(x) - f(-x)]}{\ell n \lambda g\left(\frac{x^2}{4}\right) [g(x) - g(-x)]} dx = 0.$ (Since $\frac{f\left(\frac{x^2}{4}\right) [f(x) - f(-x)]}{g\left(\frac{x^2}{4}\right) [g(x) + g(-x)]}$ is an odd function).

37.(D) $I = \int_{2-\ell n 3}^{3+\ell n} \frac{\ell n(4+x)}{\ell n(4+x) + \ell n(9-x)} dx = \int \frac{\ell n(9-x)}{\ell n(9-x) + \ell n(4-x)} dx \Rightarrow 2I = \int_{2-\ell n 3}^{3+\ell n 3} 1 \cdot dx = 3 + \ell n 3 - (2 - \ell n 3) = 1 + 2\ell n 3$

38.(B) $I = \int_0^{\pi} x f(\sin x) dx = \int_0^{\pi} (\pi - x) f(\sin x) dx = \pi \int_0^{\pi/2} f(\sin x) dx$

39.(A) $f(x + \pi) = \int_0^{x+\pi} (\cos^4 t + \sin^4 t) dt = \int_0^x (\cos^4 t + \sin^4 t) dt + \int_x^{x+\pi} (\sin^4 t + \cos^4 t) dt$

$$= f(x) + \int_x^{\pi} (\cos^4 t + \sin^4 t) dt \quad (\text{Put } t = x + y)$$

$$= f(x) + \int_x^{\pi} (\sin^4 y + \cos^4 y) dy = f(x) + f(\pi)$$

40.(B) $I = \lim_{h \rightarrow 0} \frac{\int_x^x \ell n^2 t dt + \int_x^{x+h} \ell n^2 t dt - \int_a^x \ell n^2 t dt}{h} = I = \lim_{h \rightarrow 0} \frac{\int_x^{x+h} \ell n^2 t dt}{h}$

Using L Hospital we get : $I = \lim_{h \rightarrow 0} \ell n^2 (x+h) = \ell n^2 x$

41.(A)

42.(B)

43.(C)

44.(C)
$$S = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right) \right]^{1/n}$$

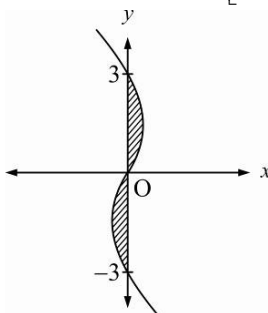
$$\ln S = \lim_{n \rightarrow \infty} \frac{1}{n} \left[\ln \left(1 + \frac{1}{n^2}\right) + \ln \left(1 + \frac{2^2}{n^2}\right) + \dots + \ln \left(1 + \frac{n^2}{n^2}\right) \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \left(1 + \frac{r^2}{n^2}\right) = \int_0^1 \ln(1+x^2) dx \Rightarrow \ln S = \ln \left[2e^{\frac{\pi}{2}-2} \right] = \frac{2}{e^2} e^{\frac{\pi}{2}}$$

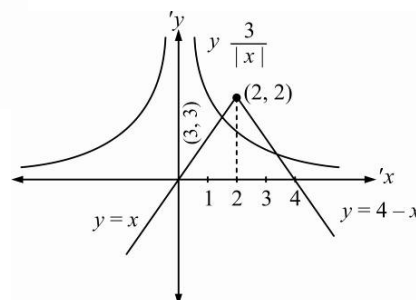
45.(D)

46.(C) Put $x=0 \Rightarrow y=0, -3, 3$

$$\text{Required area} = 2 \int_0^3 (9y - y^3) dy = \frac{81}{2}$$



47.(B)
$$A = \int_{\sqrt{3}}^2 \left(x - \frac{3}{x}\right) dx + \int_2^3 \left(4 - x - \frac{3}{x}\right) dx = \frac{4-3\ln 3}{2}$$

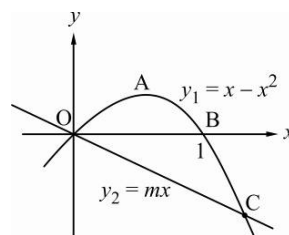


48.(BD) Point of intersection of $y = x - x^2$; $y = mx \Rightarrow x - x^2 = mx$

$$\Rightarrow x^2 + (m-1)x = 0 \Rightarrow x = 0, 1-m$$

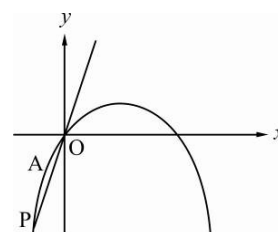
Case I : Area of OABCO

$$A = \int_0^{1-m} (y_1 - y_2) dx = \int_0^{1-m} (x - x^2 - mx) dx = \frac{9}{2} \Rightarrow m = -2$$



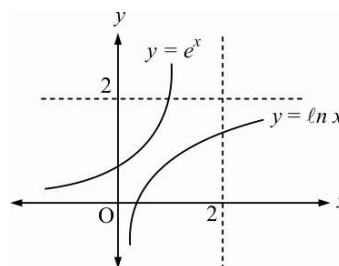
Case II : Area of PAOP

$$\int_{1-m}^0 (x - x^2 - mx) dx = \frac{9}{2} \Rightarrow m = 4$$



49.(A)
$$A = \int_1^2 \ln x dx = 2\ln 2 - 1$$

$$\Rightarrow \text{Required area} = 4 - 2(2\ln 2 - 1) = 6 - 4\ln 2$$



50.(C)

$$51.(A) \quad \int_0^t \{x\} dx = \int_0^{\lfloor t \rfloor} \{x\} dx + \int_{\lfloor t \rfloor}^t \{x\} dx = \lfloor t \rfloor \int_0^1 x dx + \int_0^{\{t\}} x dx = \frac{\lfloor t \rfloor}{2} + \frac{\{t\}^2}{2} \Rightarrow \text{Statement 2 is true.}$$

$$\int_0^{5.5} \{x\} dx = \frac{5}{2} + \frac{(0.5)^2}{2} = \frac{21}{8} \Rightarrow \text{Statement 1 is true and also it is correct explanation of statement 2.}$$

52.(D) 53.(B)

$$54.(C) \quad I = \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt \quad [\text{Put } t = \frac{1}{x}]$$

$$I = \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_e^{\tan x} \frac{1}{x \left(1 + \frac{1}{x^2}\right)} \cdot \left(-\frac{1}{x^2}\right) dx$$

$$I = \int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{\tan x}^e \frac{x}{1+x^2} dx = \int_{1/e}^e \frac{t}{1+t^2} dt = \left(\frac{1}{2} \ln(1+t^2) \right) \Big|_{1/e}^e = 1$$

$$55.(AB) \quad I_n = \int_0^1 \frac{dx}{(1+x^2)^n} = \int_0^1 (1+x^2)^{-n} dx = \left[\frac{x}{(1+x^2)^n} \right]_0^1 - \int_0^1 (-n)(1+x^2)^{-n-1} 2x^2 dx$$

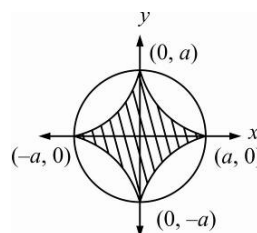
$$= \frac{1}{2^n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx = \frac{1}{2^n} + 2n \int_0^1 \frac{1+x^2-1}{(1+x^2)^{n+1}} dx = \frac{1}{2^n} + 2n I_n - 2n I_{n+1}$$

$$\Rightarrow 2n I_{n+1} = 2^{-n} + (2n-1) I_n \Rightarrow 2I_2 = \frac{1}{2} + I_1 = \frac{1}{2} + \left[\tan^{-1} x \right]_0^1 \Rightarrow I_2 = \frac{1}{4} + \frac{\pi}{8}$$

$$56.(A) \quad x = a \cos^3 t, y = a \sin^3 t \Rightarrow x^{2/3} + y^{2/3} = a^{2/3}$$

$$A = 4 \int_0^{\pi/2} y \frac{dx}{dt} dt = 4 \int_0^{\pi/2} 3a^2 \sin^3 t \cos^2 t (-\sin t) dt$$

$$= \left| -12a^2 \int_0^{\pi/2} \sin^4 t \cos^2 t dt \right| = \frac{3}{8} \pi a^2 \text{ square units}$$



$$57.(C) \quad \text{Required area } (b-1)\sin(3b+4) = \int_1^b f(x) dx$$

Differentiate w.r.t. b

$$3(b-1)\cos(3b+4) + \sin(3b+4) = f(b) \Rightarrow f(x) = 3(x-1)\cos(3x+4) + \sin(3x+4)$$

58.(A)

$$59.(A) \quad \text{We have } I(m, n) = \int_0^1 t^m (1+t)^n dt = \left[\frac{t^{m+1}}{m+1} (1+t)^n \right]_0^1 - \frac{n}{m+1} \int_0^1 t^{m+1} (1+t)^{n-1} dt = \frac{2^n}{m+1} - \frac{n}{m+1} I_{(m+1, n-1)}$$

$$60.(B) \quad f'(x) = 2xe^{-(x^2+1)^2} \left[1 - e^{2x^2+1} \right]$$

$$\text{If } x > 0, e^{2x^2+1} > 1 \Rightarrow \text{Thus, } f'(x) < 0 \text{ for } 0 < x < \infty$$

Hence $f(x)$ decreases in $(0, \infty)$

61.(B) $y(x) = \cos x \cdot \int_{\pi^2/16}^{x^2} \frac{\cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta$

$$\frac{dy}{dx} = -\sin x \int_{\pi^2/16}^{x^2} \frac{\cos \sqrt{\theta}}{1 + \sin^2 \sqrt{\theta}} d\theta + \frac{2x \cos x \cdot \cos x}{1 + \sin^2 x} \Rightarrow \left. \frac{dy}{dx} \right|_{x=\pi} = 0 + \frac{2\pi(-1)(-1)}{1+0} = 2\pi$$

62.(A) Let $I = \int_{-\pi/3}^{\pi/3} \frac{\pi + 4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$

$$I = \int_{-\pi/3}^{\pi/3} \frac{\pi}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx + \int_{-\pi/3}^{\pi/3} \frac{4x^3}{2 - \cos\left(x + \frac{\pi}{3}\right)} dx$$

The second integral becomes zero (As integrand being an odd function of x)

$$I = 2\pi \int_0^{\pi/3} \frac{dx}{2 - \cos\left(x + \frac{\pi}{3}\right)}$$

(Using the prop. Of even function and also $|x| = x$ for $0 \leq x \leq \pi/3$)

$$I = 2\pi \int_{\pi/3}^{2\pi/3} \frac{dy}{2 - \cos y} = 2\pi \int_{\pi/3}^{2\pi/3} \frac{dy}{2 - \frac{1 - \tan^2 y/2}{1 + \tan^2 y/2}} = \frac{4\pi}{\sqrt{3}} \tan^{-1} \left(\frac{1}{2} \right) \quad [\text{Put } x + \pi/3 = y \Rightarrow dx = dy]$$

63.(C) 64.(CD) 65.(C)

66.(ABC)

(A) $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1 + \pi^x) \sin x} dx$

$$I_n = \int_{-\pi}^{\pi} \frac{\pi^x \sin nx}{(1 + \pi^x) \sin x} dx \quad (\text{By property } \int_a^b f(x) dy = \int_a^b f(a+b-x) dx)$$

$$2I_n = 2 \int_0^{\pi} \frac{\sin nx}{\sin x} dx \Rightarrow I_n = \int_0^{\pi} \frac{\sin nx}{\sin x} dx$$

$$I_{n+2} - I_n = \int_0^{\pi} \frac{\sin(n+2)x - \sin nx}{\sin x} dx = \int_0^{\pi} \frac{2 \cos(n+1)x \sin x}{\sin x} dx = 2 \left[\frac{\sin(n+1)x}{(n+1)} \right]_0^{\pi} = 0$$

$$\Rightarrow I_{n+2} = I_n$$

(B) $I_3 = I_5 = \dots = I_{21} \quad \therefore \sum_{m=1}^{10} I_{2m+1} = 10I_3 = 10 \int_0^{\pi} \frac{\sin 3x}{\sin x} dx = 10 \int_0^{\pi} (3 - 4 \sin^2 x) dx = 10\pi$

(C) $I_2 = I_4 = \dots = I_{20} \quad \sum_{m=1}^{10} I_{2m} = 10 \int_0^{\pi} \frac{\sin 2x}{\sin x} dx = 0$

67.(A) Put $x^2 = t$

$$x dx = \frac{dt}{2} \quad \therefore \quad I = \int_{\ell n 2}^{\ell n 3} \frac{\sin t}{\sin t + \sin(\ell n 6 - t)} \cdot \frac{dt}{2} \quad \dots(i)$$

$$I = \frac{1}{2} \int_{\ell n 2}^{\ell n 3} \frac{\sin(\ell n 6 - t)}{\sin(\ell n 6 - t) + \sin t} dt \quad \dots(ii) \quad \left[\text{Applying } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$\text{Adding (i) and (ii)} \quad 2I = \frac{1}{2} \int_{\ell n 2}^{\ell n 3} 1 \cdot dt \Rightarrow I = \frac{1}{4} \ell n \frac{3}{2}$$

$$68.(B) \quad \int \left(x^2 + \ell n \left(\frac{\pi+x}{\pi-x} \right) \right) \cos x dx = 2 \int_0^{\pi/2} x^2 \cos x dx + 0 \quad \left(\because \ell n \left(\frac{\pi+x}{\pi-x} \right) \text{ is an odd function} \right)$$

$$= \frac{\pi^2}{2} - 4$$

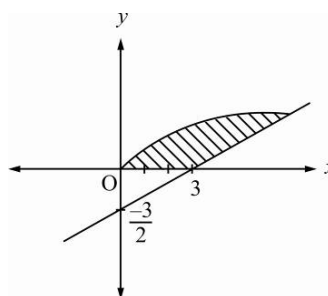
69.(A) $y = \sqrt{x}$

...(i)

And $2y - x + 3 = 0$

...(ii)

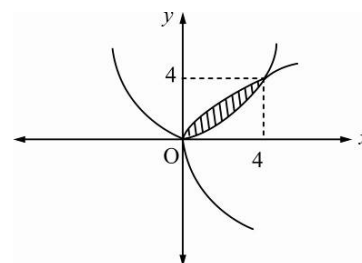
$$\text{Required area} = \int_0^3 \left\{ (2y+3) - y^2 \right\} dy = 9$$



$$70.(D) \quad I = \int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}} = \int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan \left(\frac{\pi}{2} - x \right)}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\tan x} dx}{1 + \sqrt{\tan x}}$$

$$\Rightarrow \quad 2I = \int_{\pi/6}^{\pi/3} dx \Rightarrow I = \frac{1}{2} \left[\frac{\pi}{3} - \frac{\pi}{6} \right] = \frac{\pi}{12} \quad \text{statement-1 is false}$$

$$\text{and } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \text{ is a property.}$$



71.(BC)

72.(C)

$$73.(B) \quad \text{Area} = \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4} \right) dx = \frac{16}{3}$$

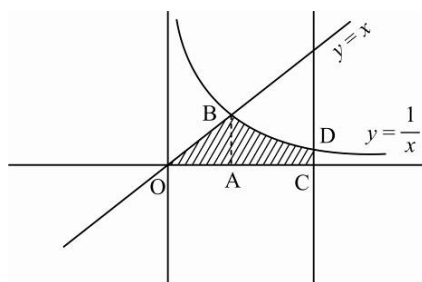
74.(C) Required area

$$= OAB + ACDB$$

$$= \frac{1}{2} \times 1 \times 1 + \int_1^e \frac{1}{x} dx$$

$$= \frac{1}{2} + (\ell n x)_1^e$$

$$= \frac{3}{2} \text{ square unit}$$

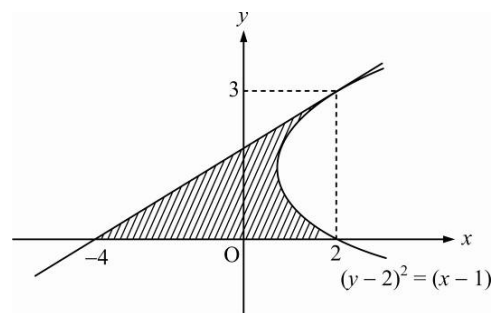


$$75.(C) \quad I = \int_0^1 x [x^2] dx + \int_1^{\sqrt{2}} x [x^2] dx + \int_{\sqrt{2}}^{1.5} x [x^2] dx \Rightarrow \int_0^1 x \cdot 0 dx + \int_1^{\sqrt{2}} x dx + \int_{\sqrt{2}}^{1.5} 2x dx = \frac{3}{4}$$

76.(D)

77.(B) The equation of tangent at (2, 3) to the given parabola is $x = 2y - 4$

$$\begin{aligned} \therefore \text{Required area} &= \int_0^3 \left\{ (y-2)^2 + 1 - 2y + 4 \right\} dy \\ &= \left(\frac{(y-2)^3}{3} - y^2 + 5y \right)_0^3 = \frac{1}{3} - 9 + 15 + \frac{8}{3} = 9 \text{ square unit} \end{aligned}$$



78.(A) $P'(x) = p'(1-x) \Rightarrow p(x) = -p(1-x) + c$

[Put $x = 0$ we get $c = 42$]

$$\Rightarrow I = \int_0^1 p(x) dx \Rightarrow I = \int_0^1 p(1-x) dx$$

[Applying $(a-x)$ property]

$$\Rightarrow 2I = \int_0^1 (p(x) + p(1-x)) dx = \int_0^1 c dx = \int_0^1 42 dx \Rightarrow 2I = 42 \Rightarrow I = 21$$

79.(C) Let $I = \int_0^\pi [\cot x] dx$... (i)

$$\Rightarrow I = \int_0^\pi [\cot(\pi - x)] dx = \int_0^\pi [-\cot x] dx \quad \dots \text{(ii)}$$

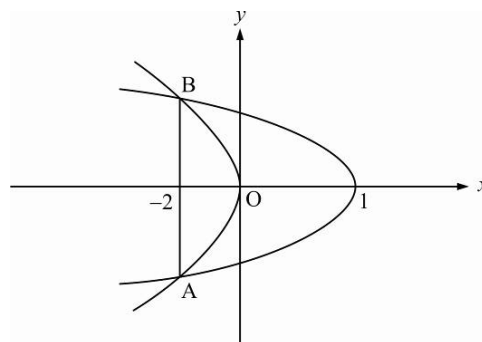
From (i) + (ii) we get :

$$\Rightarrow 2I = \int_0^\pi [\cot x] dx + \int_0^\pi [-\cot x] dx = \int_0^\pi (-1) dx = -\pi \Rightarrow I = -\frac{\pi}{2}$$

80.(D) $x + 2y^2 = 0 \Rightarrow y^2 = -\frac{1}{2}(x-1)$

The required area is AOBDA, given by

$$= \left| \int_{-1}^1 (1 - y^2) dy \right| = \left| \left[y - \frac{y^3}{3} \right]_{-1}^1 \right| = 2 \times \frac{2}{3} = \frac{4}{3} \text{ square units}$$



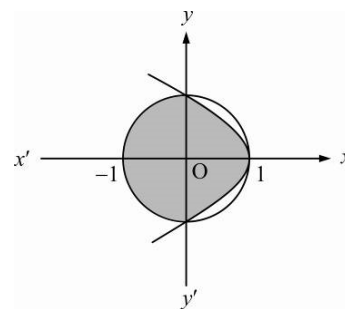
81.(D) $\int_0^{\pi/3} \sqrt{1 - 2 \sin \frac{x}{2}} dx = \int_0^\pi \left| 1 - 2 \sin \frac{x}{2} \right| dx$

$$= \int_0^{\pi/3} \left(1 - 2 \sin \frac{x}{2} \right) dx - \int_{\pi/3}^\pi \left(1 - 2 \sin \frac{x}{2} \right) dx \quad \left[\because \left| x - a \right| = \begin{cases} x - a & , x \geq a \\ -(x - a) & , x < a \end{cases} \right]$$

$$= \left[x + 4 \cos \frac{x}{2} \right]_0^{\pi/3} - \left[x + 4 \cos \frac{x}{2} \right]_{\pi/3}^\pi = 4\sqrt{3} - 4 - \frac{\pi}{3}$$

82.(A) $A = \{(x, y) : x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$

$$\begin{aligned} \text{Required area} &= \frac{1}{2} \pi r^2 + 2 \int_0^1 (1 - y^2) dy \\ &= \frac{1}{2} \pi (1)^2 + 2 \left[y - \frac{y^3}{3} \right]_0^1 = \frac{\pi}{2} + \frac{4}{3} \end{aligned}$$



83.(D) Let $I = \int_0^1 \frac{8 \log(1+x)}{(1+x^2)} dx$ [Put $x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$]

When $x = 0 \Rightarrow \tan \theta = 0$

$\therefore \theta = 0$

When $x = 1 = \tan \theta \Rightarrow \theta = \frac{\pi}{4}$

$\therefore I = \int_0^{\pi/4} \frac{8 \log(1 + \tan \theta)}{1 + \tan^2 \theta} \cdot \sec^2 \theta d\theta = 8 \int_0^{\pi/4} \log(1 + \tan \theta) d\theta$

$I = 8 \int_0^{\pi/4} \log(1 + \tan \theta) d\theta \quad \dots \text{(i)}$

Using $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$\therefore I = 8 \int_0^{\pi/4} \log \left\{ 1 + \tan \left(\frac{\pi}{4} - \theta \right) \right\} d\theta = 8 \int_0^{\pi/4} \log \left\{ 1 + \frac{1 - \tan \theta}{1 + \tan \theta} \right\} d\theta = 8 \int_0^{\pi/4} \log \left\{ \frac{2}{1 + \tan \theta} \right\} d\theta \quad \dots \text{(ii)}$

On adding equations (i) and (ii), we get :

$$2I = 8 \int_0^{\pi/4} \left[\log \{1 + \tan \theta\} + \log \left\{ \frac{2}{1 + \tan \theta} \right\} \right] d\theta$$

$\Rightarrow I = 4 \int_0^{\pi/4} \log 2 d\theta = 4 \cdot \log 2 (\theta)_0^{\pi/4} = 4 \log 2 \cdot \left(\frac{\pi}{4} - 0 \right) = \pi \log 2$

84.(A) $\int_1^a [x] f'(x) dx = \int_1^2 f'(x) dx + \int_2^3 2 f'(x) dx + \dots + \int_{[a]}^a [a] f'(x) dx$

$$= [f(x)]_1^2 + 2[f(x)]_2^3 + \dots + [a][f(x)]_{[a]}^a$$

$$= f(2) - f(1) + 2f(3) - 2f(2) + \dots + [a]f(a) - [a]f([a])$$

$$= [a]f(a) - \{f(1) + f(2) + \dots + f([a])\}$$

$$85.(19) \quad I_n = \int_n^{n+1} f(x) dx = n^2 \Rightarrow I_{\text{Given}} = I_{-2} + I_{-1} + \dots + I_3 = 2^2 + 1^2 + 0^2 + 1^2 + 2^2 + 3^2 = 19$$

$$86.(4) \quad f(x) = Ax^2 + Bx + C$$

$$f'(x) = 2Ax + B \text{ \& } f'(1) = 8 \Rightarrow 2A + B = 8$$

$$f''(x) = 2A \text{ \& } f(2) + f''(2) = 33 \Rightarrow 6A + 2B + C = 33$$

$$F(x) = \int f(x) dx = \frac{Ax^3}{3} + \frac{Bx^2}{2} + Cx$$

$$\text{Again } F(1) - F(0) = \frac{7}{3} \Rightarrow 2A + 3B + 6C = 14$$

$$\text{Solve to get: } \begin{cases} A = 7 \\ B + 3C = 3 \\ C - B = 9 \end{cases} \Rightarrow \begin{cases} A = 7 \\ B = -6 \\ C = 3 \end{cases}$$

$$87.(4) \quad \lim_{x \rightarrow 1} \frac{(f(x^2) - x^2)2x}{2(x-1)} = \lim_{x \rightarrow 1} \frac{(f'(x^2)2x - 2x)2x + (f(x^2) - x^2)2}{2} \\ = \frac{(2f'(1) - 2)2 + 2(f(1) - 1)}{2} = (2(3) - 2) + (1 - 1) = 4$$

$$88.(10) \quad \lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$$

$$T_r = \frac{1}{\sqrt{\frac{r}{n}} \left(3\sqrt{\frac{r}{n}} + 4 \right)^2} \quad \therefore \quad S = \lim_{n \rightarrow \infty} \frac{1}{n} \sum \frac{1}{\left(3\sqrt{\frac{r}{n}} + 4 \right)^2 \sqrt{\frac{r}{n}}} = \int_0^4 \frac{dx}{\sqrt{x}(3\sqrt{x} + 4)^2}$$

$$\text{Put } 3\sqrt{x} + 4 = t \quad \text{or} \quad \frac{3}{2} \frac{1}{\sqrt{x}} dx = dt \quad \therefore \quad S = \frac{2}{3} \int_4^{10} \frac{dt}{t^2} = \frac{2}{3} \left[\frac{1}{t} \right]_4^{10} = \frac{1}{10}$$

89.(0)

$$90.(36) \quad I = \int_{-10}^{20} [\cot^{-1} x] dx, \text{ we know } \cot^{-1} x \in (0, \pi) \forall x \in \mathbb{R}$$

$$\text{Thus } [\cot^{-1} x] = \begin{cases} 3, & x \in (-\infty, \cot 3) \\ 2, & x \in (\cot 3, \cot 2) \\ 1, & x \in (\cot 2, \cot 1) \\ 0, & x \in (\cot 1, \infty) \end{cases}$$

$$\text{Hence } I = \int_{-10}^{\cot 3} 3 dx + \int_{\cot 3}^{\cot 2} 2 dx + \int_{\cot 2}^{\cot 1} 1 dx + \int_{\cot 1}^{20} 0 dx = 30 + \cot 1 + \cot 2 + \cot 3$$

$$91.(21) \quad \text{Integrating by parts } I_{10} = \int_0^{\pi/2} x^{10} \sin x dx$$

$$= \left[x^{10} (-\cos x) \right]_0^{\pi/2} + 10 \int_0^{\pi/2} x^9 \cos x dx = 0 + 10 \left[x^9 \sin x \right]_0^{\pi/2} - 9 \int_0^{\pi/2} x^8 \sin x dx$$

$$I_{10} = 10 \left(\frac{\pi}{2} \right)^9 - 90I_8 \quad \therefore \quad I_{10} + 90I_8 = 10 \left(\frac{\pi}{2} \right)^9$$

$$92.(2) \quad I_1 = \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})} = \int_{-100}^{101} \frac{dx}{(5+2(1-x)-2(1-x)^2)(1+e^{2-4(1-x)})}$$

$$\text{or} \quad 2I_1 = \int_{-100}^{101} \frac{dx}{5+2x-2x^2} = I_2 \quad \text{or} \quad \frac{I_1}{I_2} = \frac{1}{2}$$

$$93.(4) \quad \text{Let } I = \int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)] dx$$

$$= \int_{-2}^0 [(x+1)^3 + 2 + (x+1)\cos(x+1)] dx$$

Put $x+1=t$

$$\Rightarrow \quad dx = dt$$

$$\therefore \quad I = \int_{-1}^1 (t^3 + 2 + t \cos t) dt$$

$$= \int_{-1}^1 t^3 dt + 2 \int_{-1}^1 dt + \int_{-1}^1 t \cos t dt$$

$$= 0 + 2 \cdot 2[x]_0^1 + 0 = 4$$

[since, t^3 and $t \cos t$ are odd functions]

$$94.(4) \quad \text{Since } \int_{\sin x}^1 t^2 f(t) dt = 1 - \sin x, \text{ thus to find } f(x).$$

On differentiating both sides using Newton Leibnitz formula

$$\text{i.e.} \quad \frac{d}{dx} \int_{\sin x}^1 t^2 f(t) dt = \frac{d}{dx} (1 - \sin x)$$

$$\Rightarrow \quad \{1^2 f(1)\} \cdot (0) - (\sin^2 x) \cdot f(\sin x) \cdot \cos x = -\cos x$$

$$\Rightarrow \quad f(\sin x) = \frac{1}{\sin^2 x}$$

$$\text{For } f\left(\frac{1}{\sqrt{3}}\right) \text{ is obtained when } \sin x = \frac{1}{\sqrt{3}}$$

$$\text{i.e.} \quad f\left(\frac{1}{\sqrt{3}}\right) = (\sqrt{3})^2 = 3$$

$$95.(2) \quad \int_0^x \phi(x) dx = x + \int_x^2 x \phi(x) dx$$

Differentiating both sides with respect to x ,

$$\phi(x) = 1 + (0 - x \phi(x)).$$

$$\Rightarrow \quad \phi(x) = \frac{1}{1+x} \Rightarrow \phi(4) = \frac{1}{1+4} = \frac{1}{5}$$

$$96.(4) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \left(\left(\frac{1}{n} \right)^2 + \left(\frac{2}{n} \right)^2 + \dots + \left(\frac{n}{n} \right)^2 \right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^a$$

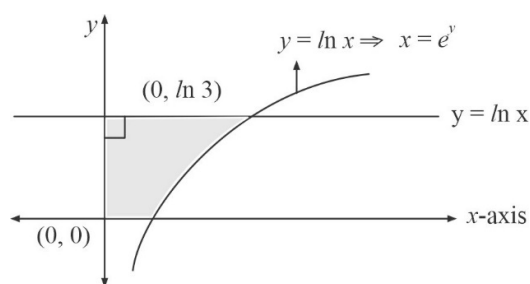
$$\int_0^1 x^a dx = \frac{1}{a+1} = \frac{1}{5} \Rightarrow a = 4$$

$$97.(1) \quad I_n = \int_0^1 (x \ln x)^n dx$$

$$I_n = \left((\ln x)^n \frac{x^{n+1}}{n+1} \right) \Big|_0^1 - \frac{n}{n+1} \int_0^1 x^n (\ln x)^{n-1} dx$$

$$\therefore I_4 = -\frac{4}{5} \int_0^1 x^4 \ln x^3 dx$$

98.(2)



$$\therefore \text{Area} = \int_0^{\ln 3} e^y dy = (e^y)_0^{\ln 3} = 3 - 1 = 2$$

$$99.(0) \quad \int_1^3 f'(x) dx = f(x) \Big|_1^3 = f(3) - f(1) = 0$$

Differential Equations

1.(C) Given $2y \frac{dy}{dx} + y^2 \sec x = \tan x$

Put $y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx}$

$$\frac{dt}{dx} + (\sec x)t = \tan x$$

I.F. = $e^{\int \sec x dx} = e^{\log(\sec x + \tan x)} = \sec x + \tan x$

$$\therefore t(\sec x + \tan x) = \int \tan x (\sec x + \tan x) dx = \sec x + \int (\sec^2 x - 1) dx = \sec x + \tan x - x + C$$

$$\therefore y^2 (\sec x + \tan x) = \sec x + \tan x - x + C$$

Given $y(0) = 1 \Rightarrow x = 0, y = 1 \Rightarrow C = 0$

$$\therefore y^2 (\sec x + \tan x) = \sec x + \tan x - x \Rightarrow y^2 = 1 - \frac{x}{\sec x + \tan x}$$

2.(A) $dy = \frac{x^2 + 4x - 9}{x + 2} = \frac{(x + 2)^2 - 13}{x + 2}$

$$\int dy = \int \left[(x + 2) - \frac{13}{x + 2} \right] dx$$

$$y = \frac{(x + 2)^2}{2} - 13 \log|x + 2| + C$$

Given $y(0) = 0$ means $x = 0, y = 0$

$$\therefore 0 = 2 - 13 \log 2 + C \Rightarrow C = 13 \log 2 - 2$$

When $x = -4, y = 2 - 13 \log 2 + 13 \log 2 - 2 = 0$

$$\therefore y(-4) = 0$$

3.(B) $x = f(y)$ is the solution of the given differential equation. So, $f(-1) = 1$ means when $y = -1$, then $x = 1$.

Given $y dx - (x + 2y^2) dy = 0$

$$\frac{dx}{dy} - \frac{x}{y} = 2y$$

I.F. = $e^{-\int \frac{1}{y} dy} = e^{\log \frac{1}{y}} = \frac{1}{y} \therefore \frac{x}{y} = \int \frac{2y}{y} dy = 2y + C$

$$x = 2y^2 + Cy$$

When, $y = -1$, then $x = 1 \quad \{ \because f(-1) = 1 \}$

$$1 = 2 - C \Rightarrow C = 1 \therefore x = 2y^2 + y$$

When $y = 1$, then $x = 2 + 1 = 3$

So, $x = f(y) \Rightarrow f(1) = 3$

4.(C) Let the circle with centre $(0, a)$ and radius a touches the x -axis at origin.

$$\therefore x^2 + (y-a)^2 = a^2 \quad \dots\dots (i)$$

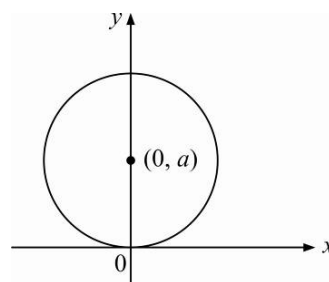
$$\Rightarrow 2x + 2(y-a)y_1 = 0 \Rightarrow y-a = -\frac{x}{y_1} \Rightarrow a = \frac{yy_1 + x}{y_1}$$

$$\text{From (i), } x^2 + \frac{x^2}{y_1^2} = \frac{(yy_1 + x)^2}{y_1^2}$$

$$\Rightarrow x^2(y_1^2 + 1) = (yy_1 + x)^2 \Rightarrow (x^2 - y^2)y_1^2 = 2xyy_1$$

$$\Rightarrow (x^2 - y^2)y_1 = 2xy \Rightarrow (x^2 - y^2)\frac{dy}{dx} = 2xy$$

Comparing with $(x^2 - y^2)\frac{dy}{dx} = g(x)y$, we get $g(x) = 2x$



5.(D) Given D.E. is $y' = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$ (i)

Its solution is $y \ln|cx| = x$ (ii)

$$\Rightarrow \frac{dy}{dx} \ln|cx| + y \cdot \frac{1}{cx} \cdot c = 1 \Rightarrow \frac{dy}{dx} \cdot \frac{x}{y} + \frac{y}{x} = 1 \quad \text{From (ii)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{y^2}{x^2} \quad \dots\dots (iii)$$

$\Rightarrow$ This is the differential equation of (ii)

$$\therefore \text{ (i) and (iii) are identical } \phi\left(\frac{x}{y}\right) = -\frac{y^2}{x^2} = -\frac{1}{\left(\frac{x}{y}\right)^2}$$

$$\therefore \phi(t) = -\frac{1}{t^2} \Rightarrow \phi(2) = -\frac{1}{2^2} = -\frac{1}{4}$$

6.(B) $\frac{dy}{dx} - y \operatorname{cosec} 2x = \sqrt{\tan x}$

$$\text{I.F.} = e^{-\int \operatorname{cosec} 2x dx} = e^{\frac{1}{2} \log|\tan x|} = \frac{1}{\sqrt{\tan x}}$$

$$\therefore \frac{y}{\sqrt{\tan x}} = x + c \Rightarrow y\sqrt{\cot x} = x + c$$

7.(C) I.F. = $e^{\int \tan x dx} = e^{\log|\sec x|} = \sec x$

$$y \sec x = \int \sin 2x \sec x dx = 2 \int \sin x dx = -2 \cos x + c$$

$$\text{When } x = 0, y = 1, \text{ then } c = 1 + 2 = 3 \therefore y \sec x = -2 \cos x + 3$$

$$\text{When } x = \pi, y(-1) = -2(-1) + 3 \Rightarrow y = -5 \therefore y(\pi) = -5$$

8.(C)

9.(B)

10.(B) $ax^2 + by^2 = 1$... (i)
 $2ax + 2byy' = 0$... (ii) (Differentiate w.r.t. x)
 $2a + 2b(y'^2 + yy'') = 0$... (iii) (Again differentiate w.r.t. x)

Solving (i), (ii) and (iii) we get : $xyy'' + xy'^2 - yy' = 0$

11.(C) $xy = ae^{-x} + be^x \Rightarrow (y + y'x) = -ae^{-x} + be^x$ (Differentiate w.r.t. x)
 $\Rightarrow y' + y''x + y' = ae^{-x} + be^x$ (Again differentiate w.r.t. x)
 $\Rightarrow x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = xy$

12.(A) 13.(B) 14.(A) 15.(D) 16.(A) 17.(B)

18.(B) $\phi(x) = \phi'(x)$ and $\phi(1) = 2$
 $\Rightarrow \frac{d\phi}{dx} = \phi(x) \Rightarrow \ln \phi(x) = x + c \Rightarrow \phi(3) = 2e^2$ ($\ln 2 = 1 + c \Rightarrow c = \ln 2 - 1$)

19.(ACD) $x^2 y_1^2 + xy y_1 - 6y^2 = 0$

It is quadratic equation in y_1

$$y_1 = \frac{-xy \pm \sqrt{x^2 y^2 + 24y^2 x^2}}{2x^2} = \frac{-xy \pm 5xy}{2x^2}$$

Case 1 : $y_1 = -\frac{3y}{x} \Rightarrow \frac{dy}{dx} = -\frac{3y}{x} \Rightarrow -\frac{dy}{y} = 3 \frac{dx}{x} \Rightarrow -\ln y = 3 \ln x + \ln c$

Case 2 : $y_1 = \frac{2y}{x} \Rightarrow \frac{dy}{dx} = \frac{2y}{x} \Rightarrow \frac{dy}{y} = \frac{2y}{x} \Rightarrow \ln y = 2 \ln x + \ln c$

Now check Option (C) also we get : $\frac{1}{2} \log y = c + \log x \Rightarrow \log y = \ln c_1 + \log x^2 \Rightarrow y = C_1 x^2$

20.(A) Given differential equation is $\frac{dv}{dt} + \frac{K}{m}v = -g$ [Which is LDE]

Integrating factor (I.F.) = $e^{\int \frac{K}{m} dt} = e^{\frac{K}{m}t}$

$\Rightarrow$ Solution is $V e^{\frac{K}{m}t} = -\int g e^{\frac{K}{m}t} dt \Rightarrow V e^{\frac{K}{m}t} = \frac{-gm}{K} e^{\frac{K}{m}t} + c \Rightarrow V = C \cdot e^{-\frac{K}{m}t} - \frac{mg}{K}$

21.(C) Let equation of Straight Line $Y - y = m(X - x)$

Distance from origin $\Rightarrow \left| \frac{mx - y}{\sqrt{1 + m^2}} \right| = 1$

$\therefore (mx - y)^2 = 1 + m^2 \Rightarrow \left(y - \frac{dy}{dx}x \right)^2 = 1 + \left(\frac{dy}{dx} \right)^2$

22.(D)

23.(B) Statement 1 : $x - \frac{y^2}{2} + \frac{1}{3}(x^2 + y^2)^{3/2} = c \Rightarrow 1 - y \frac{dy}{dx} + \frac{1}{2}\sqrt{x^2 + y^2} \left(2x + 2y \frac{dy}{dx} \right) = 0$
 $\Rightarrow dx - y dy + x \sqrt{x^2 + y^2} dx + y \sqrt{x^2 + y^2} dy = 0 \Rightarrow (1 + x\sqrt{x^2 + y^2})dx + y(-1 + \sqrt{x^2 + y^2})dy = 0$

Statement 2 : $\ln \frac{x}{y} - \frac{1}{xy} = c \Rightarrow \ln x - \ln y - \frac{1}{xy} = c$

Differentiate w.r.t. x, we get : $\frac{1}{x} - \frac{1}{y} \frac{dy}{dx} + \frac{y + x \frac{dy}{dx}}{(xy)^2} = 0 \Rightarrow xy^2 dx - x^2 y dy + y dx + x dy = 0$
 $\Rightarrow (1 + xy)y dx + x(1 - xy) dy = 0$

24.(A) Let the equation of tangent be : $Y - y = m(X - x)$

$$X_{int} = x - \frac{y}{m}$$

and Point of intersection with y-axis be : $\frac{x - \frac{y}{m} + x}{2} = 0 \Rightarrow 2x = \frac{y}{m} \Rightarrow \frac{dy}{dx} = \frac{y}{2x}$

$$\Rightarrow \ell n y = \frac{1}{2} \ell n x + c \Rightarrow \frac{y^2}{x} = k \Rightarrow y^2 = kx$$

25.(B)

26.(C)

27.(B)

28.(A)

29.(B) $S_2 : (2x \ell n y) dx + \left(\frac{x^2}{y} + 3y^2 \right) dy = 0 \Rightarrow d((\ell n y)x^2) + 3y^2 dy = 0 \Rightarrow (\ell n y)x^2 + y^3 = c$

$S_3 : x^2 = e^{\left(\frac{x}{y}\right)^{-1} \left(\frac{dy}{dx}\right)} \Rightarrow x^2 = e^{\left(\frac{y}{x}\right) \left(\frac{dy}{dx}\right)} \Rightarrow \ell n x^2 = \frac{y}{x} \frac{dy}{dx} \Rightarrow \int x \ell n x^2 dx = \int y dy$
 $\Rightarrow y^2 = x^2 \ell n x^2 - x^2 + c$

30.(A) $\frac{dy}{1+y^2} + \frac{dx}{\sqrt{1-x^2}} = 0 \Rightarrow \tan^{-1} y + \sin^{-1} x + c = 0$

31.(C) Given DE can be written as $\frac{dy}{dx} - \left(1 + \frac{f'(x)}{f(x)}\right)y = f(x)$ [Which is L.D.E.] I.F. = $e^{-x - \ell n f(x)} = \frac{e^{-x}}{f(x)}$

So the general solution $y \frac{e^{-x}}{f(x)} = \int f(x) \frac{e^{-x}}{f(x)} dx + c = -e^{-x} + c \Rightarrow y = -f(x) + ce^x f(x)$

32.(A) Given differential equation is :

$$\sin x dy + y \cos x dx = -\cos x dx - 2dy$$

$$\Rightarrow d(y \sin x) = d(-\sin x) - 2d(y)$$

$$\Rightarrow y \sin x = -\sin x - 2y + c$$

Since $y(0) = 1 \Rightarrow c = 2$

Thus, $y = \frac{2 - \sin x}{2 + \sin x}$. Now $y\left(\frac{\pi}{2}\right) = \frac{1}{3}$

33.(BCD) Equation of the tangent is

$$Y - y = \frac{dy}{dx}(X - x)$$

Given that $\frac{BP}{AP} = \frac{3}{1}$ so that

$$\Rightarrow \frac{dx}{x} = -\frac{dy}{3y} \Rightarrow x \frac{dy}{dx} + 3y = 0 \Rightarrow y = \frac{1}{x^3}$$

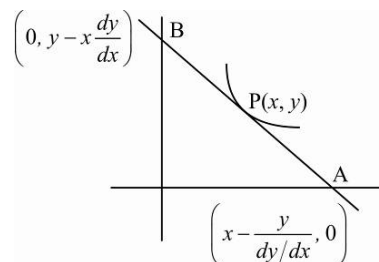
34.(A) $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ [Using L'Hospital]

$$\lim_{t \rightarrow x} \frac{2tf(x) - x^2 f'(t)}{1} = 1 \quad (\text{Differentiating with respect to } t)$$

So $2xf(x) - x^2 f'(x) = 1 \quad \therefore y = f(x)$

So $2xy - x^2 \frac{dy}{dx} = 1 \Rightarrow x^2 \frac{dy}{dx} - 2xy = -1$

$$\frac{dy}{dx} - \frac{2y}{x} = \frac{-1}{x^2} \quad [\text{Which is LDE}]$$



$$\text{Integrating factor} = e^{\int -\frac{2}{x} dx} = \frac{1}{x^2}$$

$$\text{So the solution is } \frac{y}{x^2} = \int \frac{-1}{x^4} dx \Rightarrow \frac{y}{x^2} = \frac{1}{3x^3} + c \quad \because \quad f(1) = 1 \Rightarrow c = \frac{2}{3}$$

$$y = \frac{1}{3x} + \frac{2}{3}x^2$$

35.(A) Given slope at (x, y) is : $\frac{dy}{dx} = \frac{y}{x} + \sec\left(\frac{y}{x}\right)$

$$\begin{aligned} \text{Let } \frac{y}{x} = t \Rightarrow y = xt &\Rightarrow \frac{dy}{dx} = t + x \frac{dt}{dx} \\ &\Rightarrow t + x \frac{dt}{dx} = t + \sec(t) \Rightarrow \int \cos t \, dt \int \frac{1}{x} dx \\ &\Rightarrow \sin t = \ln x + c \Rightarrow \sin\left(\frac{y}{x}\right) = \ln x + c \end{aligned}$$

$$\text{This curve passes through } \left(1, \frac{\pi}{6}\right) \Rightarrow \sin\left(\frac{\pi}{6}\right) = \ln(1) + c \Rightarrow c = \frac{1}{2}$$

$$\sin\left(\frac{y}{x}\right) = \ln x + \frac{1}{2}$$

36.(B) $y dx + (x + x^2 y) dy = 0$

$$\begin{aligned} \Rightarrow y dx + x dy &= -x^2 y dy \Rightarrow \frac{y dx + x dy}{x^2 y^2} = -\frac{1}{y} dy \Rightarrow d\left(-\frac{1}{xy}\right) = -\frac{1}{y} dy \\ -\frac{1}{xy} &= -\log y + c \Rightarrow -\frac{1}{xy} + \log y = c \end{aligned}$$

37.(B) The differential equation of a family of curves of n parameters is a differential equation of n order.

Equation of family of curves is

$$y^2 = 2c(x + \sqrt{c}) \quad \dots \text{.(i)}$$

On differentiating equation (i) w.r.t. x , then :

$$2yy_1 = 2c \Rightarrow c = yy_1$$

On putting the value of c in equation (i), we get :

$$y^2 = 2yy_1(x + \sqrt{yy_1}) \Rightarrow (y^2 - 2yy_1 x)^2 = (yy_1)^3$$

Hence, the degree and order of above equation are 3 and 1 respectively.

38.(D)

39.(A) $\frac{dp}{dt} = \frac{1}{2}p(t) - 200$

$$\text{IF} = e^{\int -\left(\frac{1}{2}\right) dt} = e^{-t/2}$$

$$\text{Hence, } p(t) \cdot e^{-t/2} = \int -200 \cdot e^{-t/2} dt$$

$$p(t) \cdot e^{-t/2} = 400e^{-t/2} + K$$

$$\because p(0) = 100 \Rightarrow K = -300$$

$$\therefore p(t)e^{-t/2} = 400e^{-t/2} - 300 \Rightarrow p(t) = 400 - 300e^{t/2}$$

40.(B)

41.(A)

42.(B)

43.(C)

44.(3) $(x-h)^2 + (y-k)^2 = r^2$

$$2(x-h) + 2(y-k)y_1 = 0$$

$$1 + y_1^2 + (y-k)y_2 = 0$$

$$\Rightarrow 2y_1y_2 + y_1y_2 + (y-k)y_3 = 0$$

$$\Rightarrow (1 + y_1^2)y_3 = 3y_1y_2^2 \Rightarrow \lambda = 3$$

45.(3) Order of D.E. = Number of arbitrary constants.

46.(5) $\left(4 + \left(\frac{dy}{dx}\right)^2\right)^2 = \left(\frac{d^2y}{dx^2}\right)^3$

Order = 2 ; Degree = 3

47.(2) $e^y dy = (x^2 + e^x) dx$

$$e^y = \frac{x^3}{3} + e^x$$

$$y = \ln\left(\frac{x^3}{3} + e^x\right) = f(x) \Rightarrow [f(2)] = 2$$

48.(10) $\frac{dy}{dx} = \frac{3y-7x-3}{3x-7y+7}$

$$y = vx + 1$$

$$x \frac{dv}{dx} = \frac{3v-7}{3-7v} - v = \frac{7v^2-7}{3-7v}$$

$$\frac{7v-3}{7v^2-7} dv + \frac{dx}{x} = 0$$

$$\frac{1}{2} \ln(v^2-1) - \frac{3}{7} \cdot \frac{1}{2} \ln\left(\frac{v-1}{v+1}\right) + \ln x = c$$

$$7 \ln(v^2-1) - 3 \ln\left(\frac{v-1}{v+1}\right) + 14 \ln x = \ln k$$

$$(v^2-1)^7 \left(\frac{v+1}{v-1}\right)^3 x^{14} = k$$

$$(v+1)^{10} (v-1)^4 x^{14} = k$$

$$\left(\frac{y-1}{x} + 1\right)^{10} \left(\frac{y-1}{x} - 1\right)^4 x^{14} = k$$

$$(x+y-1)^{10} (y-x-1)^4 = k$$

$$(x+y-1)^5 (y-x-1)^2 = \text{constant}$$

49.(3) $\frac{dx}{dy} = \frac{\tan^{-1} y - x}{1 + y^2}$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1} y}{1+y^2}$$

$$I.F. = e^{\tan^{-1} y}$$

$$\text{solution } x \cdot e^{\tan^{-1} y} = (\tan^{-1} y - 1) e^{\tan^{-1} y} + c$$

$$x = \tan^{-1} y - 1 + 2e^{-\tan^{-1} y}$$

$$\text{As passes through } (1, 0) \Rightarrow c = 2$$

$$50.(16) (2x - 10y^3) \frac{dy}{dx} + y = 0$$

$$\frac{dx}{dy} + \frac{2x}{y} = 10y^2$$

$$I.F. = y^2$$

$$\text{solution } x \cdot y^2 = 2y^5 + c$$

$$x = 2y^3 + \frac{1}{y^2} \text{ as } f(1) = 3$$

$$51.(5) \text{ Order } = m = 3$$

$$\text{Degree } = n = 2$$

$$52.(1) y^2 = k(x - h), h, k \text{ are arbitrary}$$

$$2y y' = k$$

$$y y'' + (y')^2 = 0$$

$$53.(1) \frac{dy}{dx} = e^y (e^x + e^{-x})$$

$$e^{-y} dy = (e^x + e^{-x}) dx$$

$$-e^{-y} = e^x - e^{-x} - e \text{ as passing through } (1, 1)$$

$$e^{-y} = e^{-x} + e - e^x$$

$$y = -\ln(e^{-x} + e - e^x)$$

$$54.(0) y dy - x dx + (x dy + y dx) = 0$$

$$\frac{y^2}{2} - \frac{x^2}{2} + xy = \frac{c}{2}$$

$$y^2 - x^2 + 2xy = c$$

$$55.(4) \frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} y' = 0$$

$$\frac{dy}{dx} = \frac{-x^{-1/3}}{y^{-1/3}} = \frac{-y^{1/3}}{x^{1/3}}$$

Differential equation of orthogonal trajectory is

$$\frac{dy}{dx} = \frac{x^{1/3}}{y^{1/3}}$$

$$x^{1/3}dx = y^{1/3}dy$$

$$\frac{3}{4}x^{4/3} = \frac{3}{4}y^{4/3} + \frac{3}{4}c^{4/3}$$

$$x^{4/3} - y^{4/3} = c^{4/3}$$

$$56.(3) \quad \frac{dy}{dx} - \frac{y}{x+1} = \frac{(x+1)^2 - 3}{x+1}$$

$$I.F. = \frac{1}{x+1}$$

$$\text{solution is } \frac{y}{x+1} = x + \frac{3}{x+1} - 3 \text{ as passing through } (2, 0) \text{ then } y = x^2 - 2x$$

$$57.(5) \quad \frac{dy}{dx} = -\left(\frac{x-2y+5}{2x-y+4}\right)$$

$$x = X - 1$$

$$y = Y + 2$$

$$\frac{dY}{dX} = \frac{2Y - X}{2X - Y}$$

$$58.(2) \quad x \frac{dy}{dx} + y = x \frac{f(xy)}{f'(xy)}$$

$$\frac{f'(xy)}{f(xy)} d(xy) = x dx$$

$$\ln(f(xy)) = \frac{x^2}{2} + c$$

$$f(xy) = k \cdot e^{x^2/2}$$

$$59.(2) \quad y = \frac{x+c_1}{x+c_2}$$

$$y' = \frac{c_2 - c_1}{(x+c_2)^2}$$

$$y'' = \frac{-2(c_2 - c_1)}{(x+c_2)^3}$$

$$\frac{y''}{y'} = \frac{-2}{x+c_2}$$

$$x+c_2 = \frac{-2y'}{y''}$$

$$x+c_1 = \frac{-2yy'}{y''}$$

$$y' \left(\frac{-2y'}{y''} \right)^2 = \frac{2yy' - 2y'}{y''}$$

$$2(y')^2 = (y-1)y'' \Rightarrow y'' + \frac{2}{1-y}(y')^2 = 0$$

Vectors

1.(A) $\vec{r} \times \vec{a} = \vec{b}$

Take cross with $\vec{a} \Rightarrow \vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$

$$\Rightarrow (\vec{a} \cdot \vec{a}) \vec{r} - (\vec{a} \cdot \vec{r}) \vec{a} = \vec{a} \times \vec{b} \Rightarrow \vec{r} = \frac{(\vec{a} \cdot \vec{r})}{|\vec{a}|^2} \vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$$

Let $\frac{\vec{a} \cdot \vec{r}}{|\vec{a}|^2} = x \Rightarrow \vec{r} = x\vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$

If we take cross with $\vec{b} : \vec{b} \times (\vec{r} \times \vec{a}) \equiv \vec{b} \times \vec{b} = 0 \Rightarrow (\vec{a} \cdot \vec{b}) \vec{r} - (\vec{b} \cdot \vec{r}) \vec{a} = 0$

This will give no condition as $\vec{a} \perp_r \vec{b}$ and $\vec{b} \perp_r \vec{r}$. You will only prove $\wedge - \wedge$

Alternative approach :

Hence, $\vec{r}$, $\vec{a}$ and $\vec{a} \times \vec{b}$ are coplanar.

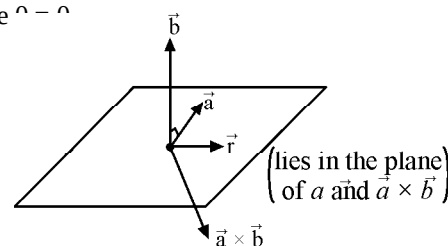
$$\Rightarrow \vec{r} = \vec{a} \cdot l + m(\vec{a} \times \vec{b})$$

Take cross with $\vec{a}$

$$\Rightarrow \vec{r} \times \vec{a} = 0 + m(\vec{a} \times \vec{b}) \times \vec{a} \Rightarrow \vec{b} = m [(\vec{a} \cdot \vec{a}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a}]$$

$$\Rightarrow \vec{b} = m |\vec{a}|^2 \vec{b} \Rightarrow m = \frac{1}{|\vec{a}|^2} \Rightarrow \vec{r} = l\vec{a} + \frac{1}{|\vec{a}|^2} (\vec{a} \times \vec{b})$$

Take dot with $\vec{a} : \vec{a} \cdot \vec{r} = l |\vec{a}|^2 + 0 \Rightarrow l = \frac{\vec{a} \cdot \vec{r}}{|\vec{a}|^2} \Rightarrow \vec{r} = \left(\frac{\vec{a} \cdot \vec{r}}{|\vec{a}|^2} \right) \vec{a} + \frac{1}{|\vec{a}|^2} (\vec{a} \times \vec{b})$



2.(C) $\vec{AB} = -2\hat{i} + 3\hat{j} - 3\hat{k} ; \vec{AC} = 4\hat{i} + 5\hat{j} + (\lambda - 10)\hat{k} ; \vec{AD} = 6\hat{i} + 2\hat{j} - 3\hat{k}$

$$\text{Volume} = \frac{1}{6} \begin{vmatrix} -2 & 3 & -3 \\ 4 & 5 & \lambda - 10 \\ 6 & 2 & -3 \end{vmatrix} = 11$$

$$\Rightarrow |-2(-15 - 2(\lambda - 10)) - 3[-12 - 6(\lambda - 10)] - 3(-22)| = 66 \Rightarrow \lambda = 7$$

3.(A) $[\lambda(\vec{a} + \vec{b}), \lambda^2 \vec{b}, \lambda \vec{c}] = [\lambda \vec{a} \lambda^2 \vec{b} \lambda \vec{c}] + 0 = -[\vec{a} \vec{b} \vec{c}] \Rightarrow \lambda^4 [\vec{a} \vec{b} \vec{c}] = -[\vec{a} \vec{b} \vec{c}] \Rightarrow \lambda^4 = -1$ (Not possible)

4.(C) $\vec{a} \times (\vec{b} \times \vec{c}) = (x^2 - 2x + 6)\vec{b} + \sin y \vec{c} \Rightarrow (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = (x^2 - 2x + 6)\vec{b} + \sin y \vec{c}$

As $\vec{b}$ and $\vec{c}$ are non-collinear $\Rightarrow \vec{a} \cdot \vec{c} = x^2 - 2x + 6$ and $\vec{a} \cdot \vec{b} = -\sin y$

Also, $\vec{a} \cdot (\vec{b} + \vec{c}) = 4 \Rightarrow x^2 - 2x + 6 - \sin y = 4 \Rightarrow (x - 1)^2 + (1 - \sin y) = 0$

Sum of two non-negative numbers is zero if $x = 1$ and $\sin y = 1$

$$\Rightarrow x = 1 \text{ and } y = (4n + 1)\pi/2 \Rightarrow (x, y) \text{ lies on line } x = 1$$

5.(A) $\vec{\alpha} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$ and $[\vec{a} \vec{b} \vec{c}] = \frac{1}{8}$;

$$\vec{\alpha} \cdot \vec{a} = y[\vec{a} \vec{b} \vec{c}] ; \vec{\alpha} \cdot \vec{b} = z[\vec{a} \vec{b} \vec{c}] ; \vec{\alpha} \cdot \vec{c} = x[\vec{a} \vec{b} \vec{c}]$$

$$\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c}) = (x + y + z)[\vec{a} \vec{b} \vec{c}] = \frac{1}{8}(x + y + z) \Rightarrow x + y + z = 8(\vec{\alpha} \cdot (\vec{a} + \vec{b} + \vec{c}))$$

6.(C) $|\vec{a} \times \vec{b}|^2 = 2$; $\vec{a} \cdot \vec{b} = 1$; As $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 \Rightarrow 2 + 1 = (3)|\vec{b}|^2 \Rightarrow |\vec{b}| = 1$

Only (C) has magnitude 1.

Another approach : $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$

$$\vec{a} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\hat{j} - \hat{k})$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix}$$

$$\Rightarrow \vec{a} - 3\vec{b} = -2\hat{i} + \hat{j} + \hat{k} \Rightarrow \vec{b} = \frac{1}{3}[\vec{a} + 2\hat{i} - \hat{j} - \hat{k}] = \hat{i}$$

7.(B) $A = |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$

$$= |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b} \cdot \vec{c} + |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

$$\Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq -3 \Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} \geq -\frac{3}{2}$$

Hence, max. value of A occurs if : $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2} \Rightarrow A|_{\max.} = 6 - 2\left(-\frac{3}{2}\right) = 9$

8.(D) $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = \frac{1}{3}|\vec{c}||\vec{b}||\vec{a}|$

$$\Rightarrow \vec{a} \cdot \vec{c} = 0 \text{ and } -(\vec{b} \cdot \vec{c}) = \frac{1}{3}|\vec{b}||\vec{c}| = -|\vec{b}||\vec{c}|\cos\theta \Rightarrow \cos\theta = -\frac{1}{3} \Rightarrow \sin\theta = \frac{2\sqrt{2}}{3}$$

9.(D) $\vec{u} \cdot \hat{n} = 0$; $\vec{v} \cdot \hat{n} = 0$

$$[\vec{u} \vec{v} \vec{w}] = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 2 & 3 \end{vmatrix} = -6 \neq 0$$

Hence, $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar

$$\hat{n} = l\vec{u} + m\vec{v} + p\vec{w} \Rightarrow \hat{n} \cdot \vec{u} = l(2) + m(\vec{u} \cdot \vec{v}) + p(\vec{u} \cdot \vec{w}) \Rightarrow 0 = 2l + 0 + p(3) \Rightarrow 2l + 3p = 0 \dots (i)$$

$$\text{and } \vec{v} \cdot \hat{n} = l(0) + m(2) + p(-1) = 0 \Rightarrow 2m = p \Rightarrow 2l = -3p = -6m$$

$$\hat{n} = -\frac{3}{2}p\vec{u} + \frac{p}{2}\vec{v} + p\vec{w} = p\left[-\frac{3}{2}(\hat{i} + \hat{j}) + \frac{1}{2}(\hat{i} - \hat{j}) + (\hat{i} + 2\hat{j} + 3\hat{k})\right] = p[3\hat{k}]$$

$$\text{As } \hat{n} \text{ is a unit vector, } p = \pm \frac{1}{3} \Rightarrow \hat{n} = \pm \hat{k} \Rightarrow |\vec{w} \cdot \hat{n}| = |\pm 3| = 3$$

$$10.(B) \quad [\vec{a} \vec{b} \vec{c}] \vec{r} - [\vec{a} \vec{b} \vec{r}] \vec{c} + [\vec{b} \vec{c} \vec{a}] \vec{r} - [\vec{b} \vec{c} \vec{r}] \vec{a} + [\vec{c} \vec{a} \vec{b}] \vec{r} - [\vec{c} \vec{a} \vec{r}] \vec{b} \\ = 3[\vec{a} \vec{b} \vec{c}] \vec{r} - ([\vec{a} \vec{b} \vec{r}] \vec{c} + [\vec{b} \vec{c} \vec{r}] \vec{a} + [\vec{c} \vec{a} \vec{r}] \vec{b}) \quad \dots (i)$$

$$\text{Consider } \vec{r} = \ell \vec{a} + m \vec{b} + n \vec{c}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{r} = n[\vec{a} \vec{b} \vec{c}]; (\vec{b} \times \vec{c}) \cdot \vec{r} = \ell[\vec{b} \vec{c} \vec{a}]; (\vec{c} \times \vec{a}) \cdot \vec{r} = m[\vec{c} \vec{a} \vec{b}]$$

$$\Rightarrow \quad \vec{r} = \frac{1}{[\vec{a} \vec{b} \vec{c}]} ([\vec{b} \vec{c} \vec{r}] \vec{a} + [\vec{c} \vec{a} \vec{r}] \vec{b} + [\vec{a} \vec{b} \vec{r}] \vec{c}) \quad \dots (ii)$$

$$\text{Using (ii) in (i), we have} \quad \Rightarrow \quad 3[\vec{a} \vec{b} \vec{c}] \vec{r} - [\vec{a} \vec{b} \vec{c}] \vec{r} = 2[\vec{a} \vec{b} \vec{c}] \vec{r}$$

$$11.(A) \quad \vec{\alpha} = (\vec{a} \cdot \vec{a}) \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b} + (\vec{a} \cdot \vec{c}) \vec{c}$$

$$\vec{\beta} = (\vec{a} \cdot \vec{b}) \vec{a} + (\vec{b} \cdot \vec{b}) \vec{b} + (\vec{a} \cdot \vec{c}) \vec{c}$$

$$\vec{\gamma} = (\vec{a} \cdot \vec{c}) \vec{a} + (\vec{b} \cdot \vec{c}) \vec{b} + (\vec{c} \cdot \vec{c}) \vec{c}$$

$$[\vec{\alpha} \vec{\beta} \vec{\gamma}] = \begin{bmatrix} (\vec{a} \cdot \vec{a}) & (\vec{a} \cdot \vec{b}) & (\vec{a} \cdot \vec{c}) \\ (\vec{a} \cdot \vec{b}) & (\vec{b} \cdot \vec{b}) & (\vec{b} \cdot \vec{c}) \\ (\vec{a} \cdot \vec{c}) & (\vec{c} \cdot \vec{b}) & (\vec{c} \cdot \vec{c}) \end{bmatrix} [\vec{a} \vec{b} \vec{c}]$$

[$\because$ using Ill - 53]

$$= [\vec{a} \vec{b} \vec{c}]^2 [\vec{a} \vec{b} \vec{c}] \quad [\because \text{using standard result 9}]$$

$$= [\vec{a} \vec{b} \vec{c}]^3 = V^3$$

$$12.(A) \quad V_1 = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$$

$$g_1 = \frac{1}{3}(\vec{0} + \vec{b} + \vec{c}) \quad ; \quad g_2 = \frac{1}{3}(\vec{0} + \vec{c} + \vec{a}) \quad ; \quad g_3 = \frac{1}{3}(\vec{0} + \vec{a} + \vec{b})$$

$$V = [\vec{g}_1 \vec{g}_2 \vec{g}_3] = \frac{1}{27} [\vec{b} + \vec{c} \quad \vec{c} + \vec{a} \quad \vec{a} + \vec{b}] = \frac{2}{27} [\vec{a} \vec{b} \vec{c}]$$

$$V_2 = \frac{2}{27} \times 6V_1 \Rightarrow 4V_1 = 9V_2$$

$$13.(D) \quad \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = (1+abc) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 0 \quad \dots \dots \dots (\text{By using the properties of determinants})$$

$$\Rightarrow \quad (1+abc)[\vec{a} \vec{b} \vec{c}] = 0$$

$$\Rightarrow \quad \text{But } [\vec{a} \vec{b} \vec{c}] \neq 0 \quad [\because \vec{a}, \vec{b}, \vec{c} \text{ are non-coplanar}]$$

$$\Rightarrow \quad abc = -1$$

14.(B) $|\vec{a}|^2 = 4p^2 + 1 = (p+1)^2 - 1 \dots (\because \text{length of vector is fixed irrespective of co-ordinate system})$

$$\Rightarrow 3p^2 - 2p - 1 = 0$$

$$\Rightarrow (p-1)(3p+1) = 0 \Rightarrow p = 1, -\frac{1}{3}$$

15.(C) $\vec{a}_1, \vec{b}_1$ and $\vec{c}_1$ form the reciprocal system of $\vec{a}, \vec{b}$ and $\vec{c}$

Hence, $\vec{a}_1 = \frac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}, \vec{b}_1 = \frac{\vec{c} \times \vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]}, \vec{c}_1 = \frac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$

$$\begin{aligned} \vec{a}_1 \times \vec{b}_1 &= \frac{(\vec{b} \times \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]} \times \frac{(\vec{c} \times \vec{a})}{[\vec{a} \ \vec{b} \ \vec{c}]} = \frac{[\vec{b} \ \vec{c} \ \vec{a}]\vec{c} - [\vec{b} \ \vec{c} \ \vec{c}]\vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]^2} \\ &= \frac{\vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]} \quad [\because [\vec{b} \ \vec{c} \ \vec{c}] = 0] \end{aligned}$$

Similarly, $\vec{b}_1 \times \vec{c}_1 = \frac{\vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]}$ and $\vec{c}_1 \times \vec{a}_1 = \frac{\vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$

Hence, $\vec{a}_1 \times \vec{b}_1 + \vec{b}_1 \times \vec{c}_1 + \vec{c}_1 \times \vec{a}_1 = \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]}(\vec{a} + \vec{b} + \vec{c}) = [\vec{a} \ \vec{b} \ \vec{c}]^{-1}(\vec{a} + \vec{b} + \vec{c})$

16.(B) $|\vec{AB} \times \vec{CD} + \vec{BC} \times \vec{AD} + \vec{CA} \times \vec{BD}|$

$$= |(\vec{b} - \vec{a}) \times (\vec{d} - \vec{c}) + (\vec{c} - \vec{b}) \times (\vec{d} - \vec{a}) + (\vec{a} - \vec{c}) \times (\vec{d} - \vec{b})|$$

$$= |\vec{b} \times \vec{d} + \vec{d} \times \vec{a} + \vec{c} \times \vec{b} + \vec{a} \times \vec{c} + \vec{c} \times \vec{d} + \vec{d} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{a} \times \vec{d} + \vec{d} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{b}|$$

$$\Rightarrow 2|\vec{b} \times \vec{a} + \vec{c} \times \vec{b} + \vec{a} \times \vec{c}| = 4 \text{ Area of } \Delta ABC$$

17.(B) $\vec{a} = \hat{i} - \hat{j} - 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{c} = -5\hat{i} + 2\hat{j} - 6\hat{k}$

$$|\vec{AB}| = \sqrt{(2-1)^2 + (1+1)^2 + (-2+3)^2} = \sqrt{6} \text{ units}$$

$$|\vec{AC}| = \sqrt{(-5-1)^2 + (2+1)^2 + (-6+3)^2} = 3\sqrt{6} \text{ units}$$

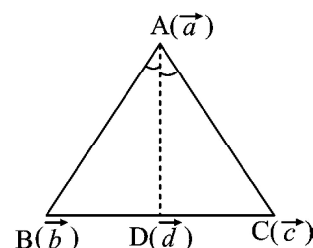
As AD is the angle bisector of angle A.

Hence D divides BC in the ratio of AB : AC

$$\Rightarrow \vec{d} = \frac{\vec{c} + 3\vec{b}}{1+3} = \frac{\hat{i} + 5\hat{j} - 12\hat{k}}{4}$$

$$\vec{AD} = \frac{\hat{i} + 5\hat{j} - 12\hat{k}}{4} - (\hat{i} - \hat{j} - 3\hat{k}) = \frac{-3}{4}\hat{i} + \frac{9}{4}\hat{j}$$

The length of angle bisector AD is given as : $|\vec{AD}| = \frac{3}{4}\sqrt{10} \text{ units}$



18.(A) We have, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 = [\vec{a} \vec{b} \vec{c}]^2$

But $\vec{c}$ is perpendicular to $\vec{a}$ and $\vec{b}$

$\Rightarrow \vec{c}$ is parallel to $\vec{a} \times \vec{b}$

$\Rightarrow \vec{c} = k(\vec{a} \times \vec{b})$ where $|\vec{c}| = 1$ (i)

$\Rightarrow |\vec{c}| = |k| |\vec{a} \times \vec{b}| \Rightarrow 1 = |k| |\vec{a}| |\vec{b}| \sin \frac{\pi}{4} \Rightarrow k = \pm \frac{\sqrt{2}}{|\vec{a}| |\vec{b}|}$

$[\vec{a} \vec{b} \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c} = k |\vec{a} \times \vec{b}|^2$ [$\because$ using (i)]

$= \pm \frac{\sqrt{2}}{|\vec{a}| |\vec{b}|} |\vec{a}|^2 |\vec{b}|^2 \times \frac{1}{2}$

$\Rightarrow [\vec{a} \vec{b} \vec{c}]^2 = \frac{1}{2} |\vec{a}|^2 |\vec{b}|^2 = \frac{1}{2} (a_1^2 + a_2^2 + a_3^2) (b_1^2 + b_2^2 + b_3^2)$

19.(A) $\vec{a} + \vec{b} + \vec{c} = \alpha \vec{d}$ (i)

and $-\beta \vec{a} + \vec{b} + \vec{c} = -\vec{d}$

$\Rightarrow \alpha \vec{a} - \alpha \vec{b} - \alpha \vec{c} = \alpha \vec{d}$ (ii)

Using (i) and (ii)

$\Rightarrow \alpha \vec{a} - \alpha \vec{b} - \alpha \vec{c} = \vec{a} + \vec{b} + \vec{c}$

Now as $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, comparing coefficients on both sides.

$\alpha \beta = 1, \alpha = -1 \Rightarrow \beta = -1$

Hence $\vec{a} + \vec{b} + \vec{c} = -\vec{d} \Rightarrow \vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$

20.(D) $\left. \begin{array}{l} \vec{a} \cdot (\vec{b} + \vec{c}) = 0 \\ \vec{b} \cdot (\vec{c} + \vec{a}) = 0 \\ \vec{c} \cdot (\vec{a} + \vec{b}) = 0 \end{array} \right\} \Rightarrow 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$

$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 16 + 16 + 25 + 0$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = \sqrt{57}$

21.(A) $|\vec{u} \times \vec{v}|^2 + (\vec{u} \cdot \vec{v})^2 = |\vec{u}|^2 |\vec{v}|^2$ and $\vec{u} \cdot \vec{v} = |\vec{a}|^2 - |\vec{b}|^2 = 0$

$\Rightarrow |\vec{u} \times \vec{v}|^2 = |\vec{u}|^2 |\vec{v}|^2$

$|\vec{u}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} = 8 - 2\vec{a} \cdot \vec{b} \Rightarrow |\vec{v}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = 8 + 2\vec{a} \cdot \vec{b}$

$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| = \sqrt{4 \times (4 - \vec{a} \cdot \vec{b})(4 + \vec{a} \cdot \vec{b})} = 2\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$

22.(A) $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $\vec{m} = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$

$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2m$; $0 - 2m = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 > 0 \Rightarrow m < 0$

$$23.(B) \quad \vec{A} = \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k} ; \vec{B} = \beta \hat{i} + \gamma \hat{j} + \alpha \hat{k} ; \vec{C} = \gamma \hat{i} + \alpha \hat{j} + \beta \hat{k}$$

$$|\vec{AB}| = |\vec{BC}| = |\vec{CA}| \Rightarrow \Delta \text{ is equilateral.}$$

$$24.(C) \quad s = |\vec{r} \times \hat{i}|^2 + |\vec{r} \times \hat{j}|^2 + |\vec{r} \times \hat{k}|^2$$

$$|\vec{r} \times \hat{i}|^2 = (\vec{r} \times \hat{i}) \cdot (\vec{r} \times \hat{i}) = (\vec{r} \cdot \vec{r})(\hat{i} \cdot \hat{i}) - (\vec{r} \cdot \hat{i})(\hat{i} \cdot \vec{r}) = |\vec{r}|^2 - (r_x)(r_x)$$

Similary, find for others. $s = 3|\vec{r}|^2 - (r_x^2 + r_y^2 + r_z^2) = 2|\vec{r}|^2$

$$25.(A) \quad \left| \frac{\vec{a} - \vec{b}}{2} \right|^2 = \frac{1}{4}(|\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b}) = \frac{1}{4}(1+1-2\cos\theta) = \sin^2 \frac{\theta}{2} \Rightarrow \left| \frac{\vec{a} - \vec{b}}{2} \right| = \sin \frac{\theta}{2}$$

$$26.(C) \quad w = \vec{F} \cdot \vec{S} = (2\hat{i} - \hat{j} - \hat{k}) \cdot (3\hat{i} + 2\hat{j} - 5\hat{k}) = 6 - 2 + 5 = 9 \text{ units}$$

$$27.(D) \quad \text{Moment} = \vec{r} \times \vec{F} = [(9\hat{i} - \hat{j} + 2\hat{k}) - (3\hat{i} - 2\hat{j} + \hat{k})] \times (5\hat{i} + \hat{k})$$

$$= [6\hat{i} + \hat{j} + \hat{k}] \times (5\hat{i} + \hat{k}) \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 1 & 1 \\ 5 & 0 & 1 \end{vmatrix} = \hat{i} - \hat{j} - 5\hat{k}$$

$$28.(B) \quad \text{Area} = \frac{1}{2} |\vec{AC} \times \vec{BD}| = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ -1 & 2 & -5 \end{vmatrix} = \frac{1}{2} |8\hat{i} + 4\hat{j}| = 2\sqrt{5} \text{ sq. units}$$

$$29.(D) \quad \vec{a} \cdot \vec{b} > 0 \Rightarrow 2x^2 - 3x + 1 > 0 \Rightarrow (2x-1)(x-1) > 0, \vec{b} \cdot \hat{j} < 0 \Rightarrow x < 0$$

$$\Rightarrow \text{Common solution : } x < 0$$

$$30.(B) \quad \begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

$$c^2 - ab = 0$$

$$\begin{aligned}
 31.(D) \quad (\bar{a} \times \hat{i})^2 + (\bar{a} \times \hat{j})^2 + (\bar{a} \times \hat{k})^2 &= \left[|\bar{a}|^2 |\hat{i}|^2 - (\bar{a} \cdot \hat{i})^2 \right] + \left[|\bar{a}|^2 |\hat{j}|^2 - (\bar{a} \cdot \hat{j})^2 \right] + \left[|\bar{a}|^2 |\hat{k}|^2 - (\bar{a} \cdot \hat{k})^2 \right] \\
 &= 3|\bar{a}|^2 - \left[(\bar{a} \cdot \hat{i})^2 + (\bar{a} \cdot \hat{j})^2 + (\bar{a} \cdot \hat{k})^2 \right] = 2|\bar{a}|^2
 \end{aligned}$$

$$32.(C) \quad \text{Consider } A \text{ to be origin, then p.v.s. of } B \text{ and } C \text{ are } \bar{a} \text{ and } \bar{b}, \text{ then } AG = \bar{g} - \bar{0} = \left(\frac{\bar{a} + \bar{b} + \bar{0}}{3} \right) = \frac{\bar{a} + \bar{b}}{3}$$

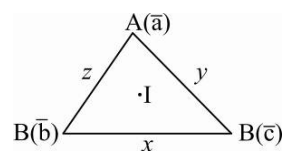
$$33.(C) \quad \bar{a} = \frac{2\bar{m} + 3(\bar{a} + 2\bar{b})}{5} \Rightarrow \bar{m} = \frac{2\bar{a} - 6\bar{b}}{2} = \bar{a} - 3\bar{b}$$

$$\begin{aligned}
 34.(B) \quad \text{Equation of line through } A(6\bar{a} - 4\bar{b} - 5\bar{c}), B(-4\bar{c}) \text{ is : } \bar{r} &= -4\bar{c} + k(6\bar{a} - 4\bar{b} - \bar{c}) \\
 \text{Equation of line through } C(-\bar{a} - 2\bar{b} - 3\bar{c}), D(\bar{a} + 2\bar{b} - 5\bar{c}) \text{ is : } \bar{r} &= (\bar{a} + 2\bar{b} - 5\bar{c}) + \lambda(2\bar{a} + 4\bar{b} - 2\bar{c}) \\
 \text{If lines intersect, then : } 6k &= 2\lambda + 1, -4k = 4\lambda + 2, -4 - k = -5 - 2\lambda \\
 \text{Solve to get } \lambda &= -\frac{1}{2}, k = 0 \\
 \text{Hence, point of intersection} &\equiv -4\bar{c}
 \end{aligned}$$

$$\begin{aligned}
 35.(D) \quad \text{Let } \bar{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\
 \text{As } \bar{r} \cdot \hat{i} &= \bar{r} \cdot \hat{j} = \bar{r} \cdot \hat{k}, \text{ hence } x = y = z \Rightarrow \bar{r} = x(\hat{i} + \hat{j} + \hat{k}) \\
 |\bar{r}| = 3 &\Rightarrow \bar{r} = \pm\sqrt{3}(\hat{i} + \hat{j} + \hat{k})
 \end{aligned}$$

$$36.(A) \quad \text{Consider } I \text{ to be origin} \Rightarrow I \equiv \frac{x\bar{a} + y\bar{b} + z\bar{c}}{x + y + z} = \bar{0}$$

$$\begin{aligned}
 |\overline{BC}| |\overline{IA}| + |\overline{CA}| |\overline{IB}| + |\overline{AB}| |\overline{IC}| \\
 = x\bar{a} + y\bar{b} + z\bar{c} = \bar{0}
 \end{aligned}$$



$$37.(D) \quad (\bar{\alpha} \times \bar{\beta}) \cdot (\bar{\alpha} \times \bar{\gamma}) = (\bar{\alpha} \cdot \bar{\alpha})(\bar{\beta} \cdot \bar{\gamma}) - (\bar{\alpha} \cdot \bar{\beta})(\bar{\alpha} \cdot \bar{\gamma}) = (14)(-3) - (8)(4) = -74$$

$$38.(A) \quad \begin{bmatrix} \bar{A} & \bar{B} + \bar{C} & \bar{A} + \bar{B} + \bar{C} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \bar{A} \bar{B} \bar{C} \end{bmatrix} = 0 \quad 39.(D) \quad \begin{bmatrix} \bar{a} - \bar{b} & \bar{b} + \bar{c} & \bar{c} + \bar{a} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \bar{a} \bar{b} \bar{c} \end{bmatrix} = 0$$

$$40.(A) \quad \text{Consider } A \text{ to be origin and } B(\bar{b}), C(\bar{c})$$

$$\begin{aligned}
 \text{Now, } \bar{d} &= \frac{\bar{b}}{2}, \bar{e} = \frac{\bar{c}}{2}, \bar{f} = \frac{\bar{c} + \bar{b}}{2} \\
 \overline{BE} + \overline{AF} &= \frac{\bar{c}}{2} - \bar{b} + \frac{\bar{c} + \bar{b}}{2} = \bar{c} - \frac{\bar{b}}{2} = \bar{c} - \bar{d} = \overline{DC}
 \end{aligned}$$

$$41.(B) \quad \overline{PA} + \overline{PB} = \bar{a} + \bar{b} - 2\bar{p} = 2\bar{c} - 2\bar{p} = 2\overline{PC}$$

$$42.(C) \quad \bar{a} \cdot \bar{a} = \bar{b} \cdot \bar{b} = \bar{c} \cdot \bar{c} = k^2, \bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{c} = \bar{c} \cdot \bar{a} = 0$$

$$\cos \theta = \frac{(\bar{a} + \bar{b} + \bar{c}) \cdot \bar{a}}{|\bar{a} + \bar{b} + \bar{c}| |\bar{a}|} = \frac{k^2}{k^2 |\bar{a} + \bar{b} + \bar{c}|}$$

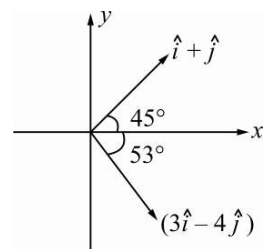
$$\text{where } |\bar{a} + \bar{b} + \bar{c}|^2 = 3k^2 + 0 \Rightarrow |\bar{a} + \bar{b} + \bar{c}| = \sqrt{3}k \quad \text{Hence } \cos \theta = 1/\sqrt{3}$$

43.(D) Let vector be $x\hat{i} + y\hat{j}$ where $x^2 + y^2 = 1$

$$\text{Such that } (x\hat{i} + y\hat{j}) \cdot (\hat{i} + \hat{j}) = (1)\sqrt{2} \cos \frac{\pi}{4} \Rightarrow x + y = 1$$

$$\text{and } (x\hat{i} + y\hat{j}) \cdot (3\hat{i} - 4\hat{j}) = (1)(5) \cos \frac{\pi}{3} \Rightarrow 3x - 4y = \frac{5}{2}$$

Solve to get : $x, y \in \{ \cdot \}$



44.(C) $\vec{b} \times \vec{c} = \vec{b} \times \vec{d} \Rightarrow \vec{b} \times (\vec{c} - \vec{d}) = 0 \Rightarrow \vec{b}$ is collinear with $\vec{c} - \vec{d}$

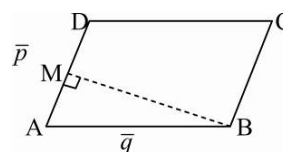
$$\vec{c} - \vec{d} = k\vec{b} \Rightarrow \vec{d} = \vec{c} - k\vec{b}$$

Take dot with $\vec{a}$ to get : $\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} = k$

$$\text{Hence } \vec{d} = \vec{c} - \left(\frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \right) \vec{b}$$

45.(B) $\vec{MB} = \vec{q} - (\vec{q} \cdot \hat{p}) \hat{p}$ [Resolution of vectors]

$$\Rightarrow \vec{BM} = -\vec{q} + \frac{(\vec{p} \cdot \vec{q}) \vec{p}}{(\vec{p} \cdot \vec{p})}$$



Integer type

46.(2) Let position vectors of orthocentre be r .

$$\begin{array}{ccc} \text{Orthocentre} & 2 & \text{Centroid} & 1 & \text{Circumcentre} \\ \vec{r} & & \frac{\vec{a} + \vec{b} + \vec{c}}{3} & & \frac{\vec{a} + \vec{b} + \vec{c}}{4} \end{array}$$

$$\therefore \frac{\vec{a} + \vec{b} + \vec{c}}{3} = \frac{2 \left(\frac{\vec{a} + \vec{b} + \vec{c}}{4} \right) + 1 \cdot \vec{r}}{2+1} \Rightarrow \vec{r} = \frac{\vec{a} + \vec{b} + \vec{c}}{2}$$

47.(5) Given $|\vec{a}| = |\vec{b}| = 1$

$$|\vec{a} + \vec{b}|^2 = 3 \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = 3$$

$$\Rightarrow 2\vec{a} \cdot \vec{b} = 1 \Rightarrow 2|\vec{a}||\vec{b}|\cos\theta = 1 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$|\vec{c}| = |\vec{a} + 2\vec{b} + 3(\vec{a} \times \vec{b})|^2 = |\vec{a}|^2 + 4|\vec{b}|^2 + 9|\vec{a} \times \vec{b}|^2 + 4\vec{a} \cdot \vec{b} + 6\vec{a} \cdot (\vec{a} \times \vec{b}) + 12\vec{b} \cdot (\vec{a} \times \vec{b})$$

$$= 1 + 4 + 9|\vec{a}|^2|\vec{b}|^2 \sin^2\theta + 2 + 0 + 0 = 2 + 9 \times \frac{3}{4} = \frac{55}{4} \therefore 4|\vec{c}|^2 = 55 \Rightarrow 2|\vec{c}| = \sqrt{55}$$

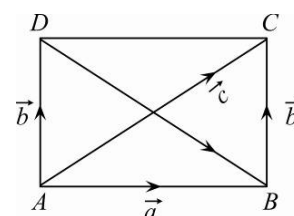
48.(3) ABCD is parallelogram

Let $\vec{AB} = \vec{a}$, $\vec{AD} = \vec{b}$ and $\vec{AC} = \vec{c}$ such that $|\vec{AB}| = |\vec{a}|$, $|\vec{AD}| = |\vec{b}|$, $|\vec{AC}| = |\vec{c}| = c$

Also, $\vec{BC} = \vec{AD} = \vec{b}$ $\{ \because 11gm \}$

In $\triangle ABC$, $\vec{c} = \vec{a} + \vec{b}$

$$|\vec{c}|^2 = |\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$$



$$c^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$$

$$\vec{a} \cdot \vec{b} = \frac{c^2 - a^2 - b^2}{2}$$

$$\text{In } \triangle ABD, \overrightarrow{BD} = \vec{a} - \vec{b}$$

$$\therefore \overrightarrow{DB} \cdot \overrightarrow{AB} = (\vec{a} - \vec{b}) \cdot \vec{a} = |\vec{a}|^2 - \vec{a} \cdot \vec{b} = a^2 - \frac{c^2 - a^2 - b^2}{2} = \frac{3a^2 + b^2 - c^2}{2}$$

49.(5)

$$|2\vec{a} - \vec{b}|^2 = 25 \Rightarrow 4|\vec{a}|^2 + |\vec{b}|^2 - 4\vec{a} \cdot \vec{b} = 25$$

$$\Rightarrow 4 \cdot 4 + 9 - 4\vec{a} \cdot \vec{b} = 25 \Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\therefore |2\vec{a} + \vec{b}|^2 = 4|\vec{a}|^2 + |\vec{b}|^2 + 4\vec{a} \cdot \vec{b} = 16 + 9 = 25 \Rightarrow |2\vec{a} + \vec{b}| = 5$$

50.(3) $\vec{c} \times (\hat{i} + 2\hat{j} + 5\hat{k}) = 0$

$$\Rightarrow \vec{c} \text{ is parallel to } \hat{i} + 2\hat{j} + 5\hat{k}$$

$$\therefore \vec{c} = \lambda(\hat{i} + 2\hat{j} + 5\hat{k})$$

$$\text{But } |\vec{c}| = 60 \Rightarrow \lambda^2(1 + 4 + 25) = 60 \Rightarrow \lambda = \pm\sqrt{2}$$

$$\therefore \vec{c} = \pm\sqrt{2}(\hat{i} + 2\hat{j} + 5\hat{k}) \text{ and } \vec{c} \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k}) = \pm 12\sqrt{2}$$

51.(3) Given $|\hat{x}| = |\hat{y}| = |\hat{z}| = 1$

$$\therefore |\hat{x} + \hat{y}| + |\hat{y} + \hat{z}| + |\hat{z} + \hat{x}|^2 = 6 + 2(\hat{x} \cdot \hat{y} + \hat{y} \cdot \hat{z} + \hat{z} \cdot \hat{x}) \quad \dots\dots (i)$$

$$\text{Now } |\hat{x} + \hat{y} + \hat{z}| \geq 0$$

$$\Rightarrow |\hat{x}|^2 + |\hat{y}|^2 + |\hat{z}|^2 + 2(\hat{x} \cdot \hat{y} + \hat{y} \cdot \hat{z} + \hat{z} \cdot \hat{x}) \geq 0$$

$$\Rightarrow 2(\hat{x} \cdot \hat{y} + \hat{y} \cdot \hat{z} + \hat{z} \cdot \hat{x}) \geq -3 \Rightarrow 6 + 2(\hat{x} \cdot \hat{y} + \hat{y} \cdot \hat{z} + \hat{z} \cdot \hat{x}) \geq 3 \quad (\text{adding } 6)$$

From (i), minimum value = 3

52.(7) $(\vec{x} \times \vec{y}) \cdot \vec{z} = \begin{vmatrix} 3 & -6 & -1 \\ 1 & 4 & -3 \\ 3 & -4 & -12 \end{vmatrix} = -182 \text{ and } |\vec{z}| = 13$

$$\text{Magnitude of projection of } \vec{x} \times \vec{y} \text{ on } \vec{z} = \left| \frac{(\vec{x} \times \vec{y}) \cdot \vec{z}}{|\vec{z}|} \right| = \left| \frac{-182}{13} \right| = 14$$

53.(3) $|\hat{a}| = |\hat{b}| = |\hat{c}| = 1$

$$\hat{a} + \hat{c} = \sqrt{3}\hat{b} \Rightarrow |\hat{a} + \hat{c}|^2 = 3|\hat{b}|^2 \Rightarrow |\hat{c}|^2 + |\hat{a}|^2 + 2\hat{a} \cdot \hat{c} = 3$$

$$\Rightarrow 1 + 1 + 2|\hat{a}||\hat{c}|\cos\theta = 3 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

54.(5) $\vec{u}$ and $\vec{v}$ are collinear.

$$\therefore \vec{v} = \lambda\vec{u} \Rightarrow (2 + 3\alpha)\vec{a} - 3\vec{b} = \lambda(\alpha - 2)\vec{a} + \lambda\vec{b}$$

As $\vec{a}$ and $\vec{b}$ are non-collinear, so equate on both sides

$$\lambda = -3 \text{ and } 2 + 3\alpha = \lambda(\alpha - 2)$$

$$\Rightarrow 2 + 3\alpha = -3(\alpha - 2) \Rightarrow \alpha = 2/3$$

$$55.(3) \quad 2\bar{a} - 3\bar{b} + 4\bar{c} = \lambda_1(\bar{a} - \bar{b} + \bar{c}) + \lambda_2(\bar{b} + \bar{c} - \bar{a}) + \lambda_3(\bar{c} + \bar{a} + \bar{b})$$

$$\Rightarrow \lambda_1 - \lambda_2 + \lambda_3 = 2, \quad -\lambda_1 + \lambda_2 + \lambda_3 = -3, \quad \lambda_1 + \lambda_2 + \lambda_3 = 4$$

$$\text{Solve to get : } \lambda_1 = \frac{7}{2}, \quad \lambda_2 = 1, \quad \lambda_3 = -\frac{1}{2} \quad \text{Hence, } \lambda_1 + \lambda_3 = 3 \quad [\text{or simply add (i) and (iii) eqn.}]$$

$$56.(5) \quad |\hat{a} + \hat{b} + \hat{c}|^2 = \sum |\hat{a}|^2 + 2 \sum \hat{a} \cdot \hat{b} \Rightarrow 0 = 3 + 2 \sum \hat{a} \cdot \hat{b}$$

$$57.(5) \quad [\bar{a} + 2\bar{b} \quad 2\bar{a} - \bar{b} \quad \bar{a} \times \bar{b}]$$

$$= -(\bar{a} \times \bar{b}) \cdot (\bar{a} \times \bar{b}) + 4(\bar{b} \times \bar{a}) \cdot (\bar{a} \times \bar{b}) = -5|\bar{a} \times \bar{b}|^2 = -5[|\bar{a}|^2 |\bar{b}|^2 - (\bar{a} \cdot \bar{b})^2] = -5[(1)(1) - (0)] = -5$$

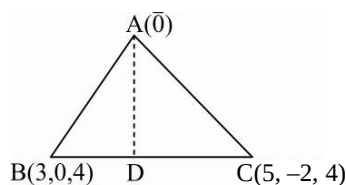
$$58.(3) \quad (\hat{a} + 2\hat{b}) \cdot (5\hat{a} - 4\hat{b}) = 0 \Rightarrow 5 - 8 + 6\hat{a} \cdot \hat{b} = 0 \Rightarrow \hat{a} \cdot \hat{b} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$59.(1) \quad [\bar{a} \times \bar{b} \quad \bar{b} \times \bar{c} \quad \bar{c} \times \bar{a}] = [\bar{a} \bar{b} \bar{c}]^2 \Rightarrow \lambda = 1 \quad [\text{Standard Result}]$$

60.(3) Consider A to be origin

$$D \equiv \left[\frac{3+5}{2}, \frac{0-2}{2}, \frac{4+4}{2} \right] \equiv (4, -1, 4)$$

$$|\overline{AD}| = \sqrt{4^2 + 1^2 + 4^2} = \sqrt{33}$$



Three Dimensional Geometry

$$1.(C) \quad \cos \alpha = \frac{1}{\sqrt{2}}; \cos \beta = \frac{1}{\sqrt{2}} \Rightarrow n^2 = 1 - \frac{1}{2} - \frac{1}{2} = 0 \Rightarrow \cos \gamma = 0 \Rightarrow \gamma = \frac{\pi}{2}$$

$$2.(D) \quad \vec{r} = |\vec{r}| \hat{r} : \hat{r} = \frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \text{ (positive octant)} \Rightarrow \vec{r} = 2\sqrt{3}(\hat{i} + \hat{j} + \hat{k})$$

$$3.(D) \quad \frac{1}{c^2} + \frac{1}{c^2} + \frac{1}{c^2} = 1 \Rightarrow c = \pm \sqrt{3}$$

$$4.(B) \quad (-3\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (3k\hat{i} + \hat{j} - 5\hat{k}) = 0 \Rightarrow -9k + 2k - 10 = 0 \Rightarrow k = -10/7$$

$$5.(D) \quad A(5, 2, 4) ; B(6, -1, 2) ; C(8, -7, k)$$

$$\overline{AB} = \hat{i} - 3\hat{j} - 2\hat{k} ; \overline{AC} = 3\hat{i} - 9\hat{j} + (k-4)\hat{k}$$

$$\overline{AB} \times \overline{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & -2 \\ 3 & -9 & k-4 \end{vmatrix} = \vec{0} \Rightarrow [-3(k-4) - 18]\hat{i} - \hat{j}(k+2) = \vec{0} \Rightarrow (-3k-6)\hat{i} - \hat{j}(k+2) = \vec{0} \Rightarrow k = -2$$

$$6.(C) \quad \frac{x-1}{0} = \frac{y-2}{0} = \frac{z-0}{1} = k ; \quad \frac{x-0}{1} = \frac{y+1}{0} = \frac{z-0}{0} = \lambda$$

$$\Rightarrow \text{Angle between the vectors : } (0\hat{i} + 0\hat{j} + \hat{k}) \cdot (\hat{i} + 0\hat{j} + 0\hat{k}) = 1.1 \cos \alpha \Rightarrow \alpha = \frac{\pi}{2}$$

$$7.(D) \quad P(x, y, z) \equiv \left(\frac{k(-a)+a}{k+1}, \frac{k(-c)+b}{k+1}, \frac{k(-b)+c}{k+1} \right)$$

$$\text{But } P \text{ lies on } XY \text{ Plane, so } z = 0 \Rightarrow \frac{k(-b)+c}{k+1} = 0 \Rightarrow k = \frac{c}{b}$$

$$8.(D) \quad \text{Eq. of line : } \frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1} = k$$

$$9.(B) \quad \text{Direction ratios of Line } \perp r \text{ to both lines is given as :}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ l_1 & m_2 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = (m_1 n_2 - m_2 n_1)\hat{i} + (n_1 l_2 - n_2 l_1)\hat{j} + (l_1 m_2 - l_2 m_1)\hat{k}$$

So, direction cosines will be proportional to above vector.

$$10.(C) \quad \text{Unit vector bisecting angle between two vectors is given as :}$$

$$\hat{c} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|} = \frac{(l_1 + l_2)\hat{i} + (m_1 + m_2)\hat{j} + (n_1 + n_2)\hat{k}}{\sqrt{(l_1 + l_2)^2 + (m_1 + m_2)^2 + (n_1 + n_2)^2}}$$

Direction ratios are : $l_1 + l_2, m_1 + m_2, n_1 + n_2$

$$11.(A) \quad S.D = \left| \frac{(\vec{a} - \vec{a}) \cdot (\vec{b} \times \vec{c})}{(\vec{b} \times \vec{c})} \right| \quad 12.(B) \quad \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 4 \\ 1 & 1 & 0 \end{vmatrix} = -4\hat{i} + 4\hat{j} + \hat{k}$$

13.(A) $(x-2)1 + (y-3)2 + (z-4)4 = 0 \Rightarrow x + 2y + 4z = 24$

14.(C) Mid-point of $BC = \left(\frac{-1+\lambda}{2}, \frac{3+5}{2}, \frac{\mu+2}{2} \right) \equiv D$

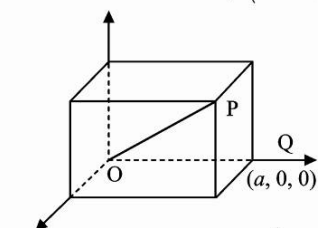
$$\overrightarrow{AD} = \left(\frac{\lambda-1}{2} - 2 \right) \hat{i} + (4-3) \hat{j} + \left(\frac{\mu+2}{2} - 5 \right) \hat{k} = \frac{\lambda-5}{2} \hat{i} + \hat{j} + \frac{\mu-8}{2} \hat{k}$$

As it is equally inclined to the axes ; $\frac{\lambda-5}{2} = 1, \frac{\mu-8}{2} = 1$; $\lambda = 7, \mu = 10$

15.(C) $\overrightarrow{OP} = a\hat{i} + a\hat{j} + a\hat{k}$

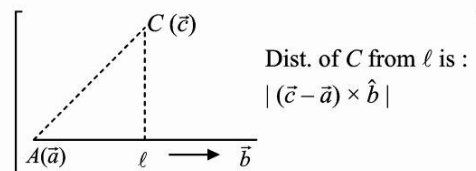
Equation of OP is $\vec{r} = (0\hat{i} + 0\hat{j} + 0\hat{k}) + \lambda(a\hat{i} + a\hat{j} + a\hat{k})$

$$\vec{r} = \mu(a\hat{i} + a\hat{j} + a\hat{k})$$



Distance of Q from OP is : $\left| (a\hat{i}) \times \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}} \right| = \sqrt{\frac{2}{3}} a$

For a unit cube : Distance = $\sqrt{\frac{2}{3}}$



16.(B) Line $L : \frac{x-1}{3} = \frac{y+1}{2} = \frac{z-3}{1}$ and plane $\pi : x - 2y + z = 0$ where normal to plane $\equiv \hat{i} - 2\hat{j} + \hat{k} = \vec{n}$ and parallel vector to line $\equiv \vec{b} = 3\hat{i} + 2\hat{j} + \hat{k}$

$\vec{b} \cdot \vec{n} = 3(1) + 2(-2) + 1(1) = 0$. Hence line is parallel to the plane. Also point $(1, -1, -3)$ of line satisfies the equation of plane $x - 2y + z = 0$. Hence line lies on it.

17.(C) $\vec{r} = \vec{a} + k\vec{b}$ and $\vec{r} \cdot \vec{n} = p$

$$(\vec{a} + k_1 \vec{b}) \cdot \vec{n} = p \Rightarrow k_1 = \frac{p - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}}$$

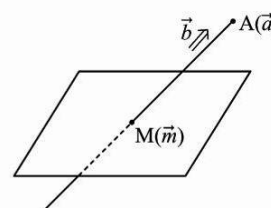
$$\Rightarrow \vec{m} = \vec{a} + \left(\frac{p - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}} \right) \vec{b} \text{ where } \vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} + 2\hat{k}, \vec{n} = 2\hat{i} - 6\hat{j} + 3\hat{k}$$

$$\Rightarrow \vec{m} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \left(\frac{-5 - (-1)}{4} \right) (2\hat{i} + \hat{j} + 2\hat{k}) = -\hat{i} + \hat{j} + \hat{k}$$

As $\vec{m}$ lies on $\vec{r} \cdot (\hat{i} + 75\hat{j} + 60\hat{k}) - \alpha = 0$

$$\Rightarrow (-\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + 75\hat{j} + 60\hat{k}) - \alpha = 0 \Rightarrow \alpha = 134$$

Hence $19\alpha + 17 = 2563$



18.(C) Equation of plane through p_1 and p_2 is given as :

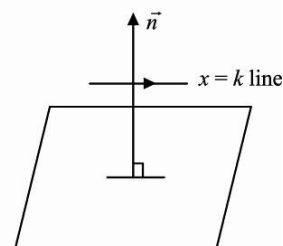
$$p_1 + kp_2 = 0 \quad \text{i.e. } (ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0$$

$$\Rightarrow (a + ka')x + (b + kb')y + (c + kc')z + (d + kd') = 0 \quad \dots (i)$$

$$\text{As it is parallel to line } \frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0} = k \quad (\text{obviously it should be } x = k \text{ line})$$

$$\Rightarrow (a + ka')(1) + (b + kb')(0) + (c + kc')(0) = 0 \quad \Rightarrow \quad k = \frac{-a}{a'}$$

$$\text{From (i) : } (a'b - ab')y + (a'c - ac')z + (a'd - ad') = 0$$



19.(B) Volume of tetrahedron = $\frac{1}{3} \times \text{area of base} \times \text{height}$

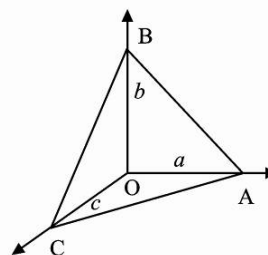
$$= \frac{1}{3} \times \left(\frac{1}{2} ab \right) \times c = \frac{1}{6} abc = 64k^3$$

$$\text{And equation of plane } ABC \text{ is } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\text{Centroid} \equiv (x_1, y_1, z_1) \equiv \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4} \right)$$

$$\Rightarrow (a, b, c) \equiv (4x_1, 4y_1, 4z_1)$$

$$\frac{1}{6} \times 4x_1 \times 4y_1 \times 4z_1 = 64k^3 \quad \Rightarrow \quad xyz = 6k^3$$

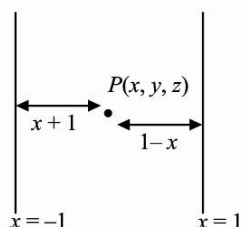


20.(C) $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1} = \lambda \quad \Rightarrow \quad x = 2 + 3\lambda, y = -1 + 2\lambda, 1 - \lambda = z$

$$xy = c^2 \text{ and } z = 0; \lambda - 1 = 0 \quad \Rightarrow \quad \lambda = 1 \text{ and } (2 + 3\lambda)(-1 + 2\lambda) = c^2 \quad \Rightarrow \quad (2 + 3)(-1 + 2) = c^2 \quad \Rightarrow \quad c = \pm \sqrt{5}$$

21.(B) $(x+1)^2 + (1-x)^2 + (y+1)^2 + (1-y)^2 + (z+1)^2 + (1-z)^2 = 10$

$$\Rightarrow 2x^2 + 2y^2 + 2z^2 - 4 = 0 \quad \Rightarrow \quad x^2 + y^2 + z^2 = 2$$



22.(A) $(\vec{r} - \vec{p}) \cdot \vec{b} = 0 \quad \Rightarrow \quad \vec{r} \cdot \vec{b} = \vec{p} \cdot \vec{b} \quad \Rightarrow \quad \vec{r} \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = -7$

23.(C) $(\vec{r} - \vec{b}) \cdot \vec{p} = 0 \quad \Rightarrow \quad \vec{r} \cdot \vec{p} = \vec{b} \cdot \vec{p} \quad \Rightarrow \quad \vec{r} \cdot (2\hat{i} + 3\hat{j} - 4\hat{k}) = -7$

$$\Rightarrow 2x + 3y - 4z + 7 = 0$$

24.(B) Equation of plane : $\frac{x}{\frac{-7}{2}} + \frac{y}{\frac{-7}{3}} + \frac{z}{\frac{7}{4}} = 1$

$$|X - \text{intl}| + |Y - \text{intl}| + |Z - \text{intl}| = \frac{7}{2} + \frac{7}{3} + \frac{7}{4} = \frac{91}{12}$$

$$25.(D) \sin \theta = \frac{(3\hat{i} + 2\hat{j} + 4\hat{k}) \cdot (2\hat{i} + \hat{j} - 3\hat{k})}{\sqrt{29}\sqrt{14}}$$

$$\Rightarrow \sqrt{406} \sin \theta = -4 \Rightarrow 1624 = 64 \operatorname{cosec}^2 \theta$$

$$26.(B) \overrightarrow{PQ} = 7\hat{i} + 8\hat{j} + 13\hat{k}, \overrightarrow{RS} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$s = \text{projection of } \overrightarrow{PQ} \text{ on } \overrightarrow{RS} = \overrightarrow{PQ} \cdot \widehat{RS} = (7\hat{i} + 8\hat{j} + 13\hat{k}) \cdot \frac{(2\hat{i} + 3\hat{j} + 4\hat{k})}{\sqrt{29}} = \frac{90}{\sqrt{29}}$$

$$\Rightarrow 29s^2 + 29 = 8100 + 29 = 8129$$

$$27.(B) \ell = \pm m = \pm n \Rightarrow \ell^2 + m^2 + n^2 = 1 \Rightarrow \ell = m = n = \pm \frac{1}{\sqrt{3}}$$

Hence 8 possible direction cosines. But 4 lines are possible

$$28.(A) x = 4 + \lambda, y = 2 + \lambda, z = k + 2\lambda \Rightarrow 2(4 + \lambda) - 4(2 + \lambda) + k + 2\lambda = 7 \Rightarrow k = 7$$

$$29.(B) 3\overrightarrow{p} + 8\overrightarrow{q} = 6\overrightarrow{r} + 5\overrightarrow{s} \Rightarrow \frac{3\overrightarrow{p} + 8\overrightarrow{q}}{11} = \frac{6\overrightarrow{r} + 5\overrightarrow{s}}{11} \Rightarrow \frac{2\overrightarrow{p} + 8\overrightarrow{q}}{3 + 8} = \frac{6\overrightarrow{r} + 5\overrightarrow{s}}{6 + 5}$$

Hence, lines PQ and RS intersect.

$$30.(C) x = \lambda, y = 2\lambda, z = 3\lambda$$

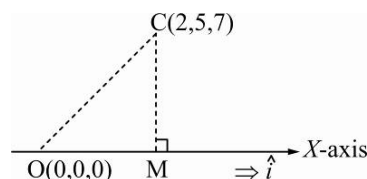
$$x - 2y + z = 0 \Rightarrow \lambda - 4\lambda + 3\lambda = 0. \text{ is true for all values of } \lambda$$

Note : Obviously line and plane passes through origin and $\vec{b}$ (along vector) and $\vec{n}$ (normal) are such that $\vec{b} \cdot \vec{n} = 0$
line lies on the plane

$\Rightarrow$ Hence infinite points of intersection.

$$31.(A) \vec{m} = \vec{0} + \{(\vec{c} - \vec{0}) \cdot \hat{i}\} \hat{i}$$

$$\vec{m} = \{2\hat{i} + 5\hat{j} + 7\hat{k}\} \cdot \hat{i} \hat{i} = 2\hat{i}$$



$$32.(D) \text{ Reflection of } (\alpha, \beta, \gamma) \text{ in } XY \text{ plane gives } (\alpha, \beta, -\gamma)$$

$$33.(C) \sin \alpha = \frac{(2\hat{i} - 3\hat{j} + 6\hat{k}) \cdot \hat{i}}{\sqrt{2^2 + 3^2 + 6^2} (1)} = \frac{2}{7}$$

$$34.(C) \text{ Vector along the line is } \ell\hat{i} + m\hat{j} + n\hat{k}. \text{ As line is parallel to } XY \text{ plane i.e. it is perpendicular to normal of the plane.}$$

$$\Rightarrow (\ell\hat{i} + m\hat{j} + n\hat{k}) \cdot \hat{k} = 0 \Rightarrow n = 0$$

$$35.(D) a_1a_2 + b_1b_2 + c_1c_2 = (2)(3) + (-1)(-2) + (2)(6) > 0$$

Hence, +ve sign gives obtuse angle bisector.

$$\frac{2x - y + 2z + 3}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{3x - 2y + 6z + 8}{\sqrt{3^2 + 2^2 + 6^2}}$$

$$\Rightarrow 7(2x - y + 2z + 3) = 3(3x - 2y + 6z + 8) \Rightarrow 5x - y - 4z + 3 = 0$$

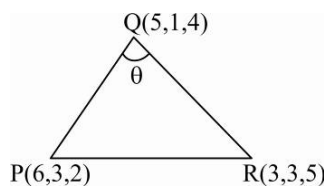
$$36.(B) \text{ Use } \frac{h - x_1}{A} = \frac{k - y_1}{B} = \frac{\ell - z_1}{C} = \frac{-2(Ax_1 + By_1 + Cz_1 + D)}{A^2 + B^2 + C^2}$$

$$\frac{h - 1}{2} = \frac{k - 3}{-1} = \frac{\ell - 4}{1} = \frac{-2(2 - 3 + 4 + 3)}{2^2 + 1^2 + 1^2} = -2 \Rightarrow (h, k, \ell) = (-3, 5, 2)$$

$$37.(D) \quad \overrightarrow{QP} = \hat{i} + 2\hat{j} - 2\hat{k}, \quad \overrightarrow{QR} = -2\hat{i} + 2\hat{j} + \hat{k}$$

$$\cos \theta = \frac{\overrightarrow{QP} \cdot \overrightarrow{QR}}{|\overrightarrow{QP}| |\overrightarrow{QR}|} = \frac{(-2) + (4) - 2}{\sqrt{9} \sqrt{9}} = 0$$

$$\Rightarrow \theta = \pi/2$$



$$38.(A) \quad \text{Point lying on the line} = \left(\frac{2+6}{2}, \frac{3+7}{2}, \frac{4+8}{2} \right) = (4, 5, 6)$$

$$\text{Vector along the normal} = 4\hat{i} + 4\hat{j} + 4\hat{k}$$

$$(\vec{r} - (4\hat{i} + 5\hat{j} + 6\hat{k})) \cdot (\hat{i} + \hat{j} + \hat{k}) = 0$$

$$(x-4) + (y-5) + (z-6) = 0 \Rightarrow x + y + z - 15 = 0$$

$$39.(A) \quad \text{Mid-point of } (1, 0, 7) \text{ and } (1, 6, 3) \text{ is } (1, 3, 5)$$

$$\text{Which satisfies } \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

$$\text{Also, line joining } (1, 0, 7) \text{ and } (1, 6, 3) \text{ has vector } 6\hat{j} - 4\hat{k} \perp r \text{ to } \hat{i} + 2\hat{j} + 3\hat{k} \text{ (vector along the line)}$$

$$40.(C) \quad \text{Equation of lines in vector form are}$$

$$\vec{r} = \lambda(2\hat{i} + 2\hat{j} + \hat{k}) \text{ and } \vec{r} = -2\hat{i} + 4\hat{j} + 5\hat{k} + \mu(-\hat{i} + 8\hat{j} + 4\hat{k})$$

$$\vec{a}_2 - \vec{a}_1 = -2\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ -1 & 8 & 4 \end{vmatrix} = -9\hat{j} + 18\hat{k}$$

$$\text{S.D.} = \frac{|-36+90|}{\sqrt{9^2+18^2}} = \frac{6}{\sqrt{5}} = 2.6 \in (2, 3]$$

$$41.(A) \quad \text{Points } (1, 1, \lambda) \text{ and } (-3, 0, 1) \text{ are equidistant from the plane } 3x + 4y - 12z + 13 = 0$$

$$\left| \frac{3 \cdot 1 + 4 \cdot 1 - 12\lambda + 13}{\sqrt{3^2 + 4^2 + (-12)^2}} \right| = \left| \frac{3(-3) + 4 \cdot 0 - 12 \cdot 1 + 13}{\sqrt{3^2 + 4^2 + (-12)^2}} \right|$$

$$\Rightarrow |3 + 4 - 12\lambda + 13| = |-9 - 12 + 13| \Rightarrow |20 - 12\lambda| = |-8| = 8$$

$$20 - 12\lambda = \pm 8 \Rightarrow \lambda = 1, \lambda = 7/3$$

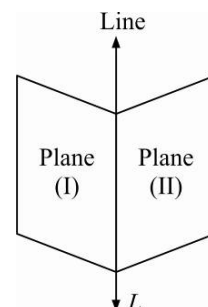
$$\text{Clearly, } 1 \text{ and } 7/3 \text{ satisfy } 3x^2 - 10x + 7 = 0$$

$$42.(C) \quad x + y + z + 1 = 0 = 2x - y + z + 3 \text{ is a line which passes through the intersection of the planes } x + y + z + 1 = 0 \text{ and } 2x - y + z + 3 = 0.$$

$$\text{Direction of this line } L \text{ is given by}$$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$\text{To find a point on the line } L, \text{ put } x = 0, \text{ we get } y + z + 1 = 0 \text{ and } -y + z + 3 = 0$$



Solving we get $y = 1, z = -2$

$\therefore$ A point on the line L is $(0, 1, -2)$

Thus, L is $\vec{r} = \vec{a} + \lambda \vec{b} = \hat{j} - 2\hat{k} + \lambda(2\hat{i} + \hat{j} - 3\hat{k})$

Given line is $\frac{x-1}{\alpha} = \frac{y+1}{-1} = \frac{z}{1}$

$\vec{r} = \hat{i} - \hat{j} + \mu(\alpha\hat{i} - \hat{j} + \hat{k})$

$$\text{S. D.} = \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \quad \dots\dots (i)$$

$$\vec{a}_2 - \vec{a}_1 = \hat{i} - 2\hat{j} + 2\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -3 \\ \alpha & -1 & 1 \end{vmatrix} = -2\hat{i} - (2+3\alpha)\hat{j} - (2+\alpha)\hat{k}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{4 + (2+3\alpha)^2 + (2+\alpha)^2}$$

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = -2 + 2(2+3\alpha) - 2(2+\alpha)$$

From (i)

$$\frac{1}{\sqrt{3}} = \frac{-2 + 2(2+3\alpha) - 2(2+\alpha)}{\sqrt{4 + (2+3\alpha)^2 + (2+\alpha)^2}} \Rightarrow \frac{1}{\sqrt{3}} = \frac{|4\alpha - 2|}{\sqrt{10\alpha^2 + 16\alpha + 12}} \Rightarrow 10\alpha^2 + 16\alpha + 12 = 3(4\alpha - 2)^2$$

$$\Rightarrow 38\alpha^2 - 64\alpha = 0 \Rightarrow 2\alpha(19\alpha - 32) = 0 \Rightarrow \alpha = 0, \alpha = 32/19$$

- 43.(B)** $x + y + 2z - 3 = 0$ and $2x + 3y + 4z - 4 = 0$ represents through a line L which passes through the intersection of the planes $x + y + 2z - 3 = 0$ and $2x + 3y + 4z - 4 = 0$.

Direction ratios of the line L are given by

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{vmatrix} = -2\hat{i} + \hat{k} = \vec{b}$$

A point on the line L is a common point on both the planes.

To find a common point on both planes, put $z = 0$.

$\therefore$ $x + y = 3$ and $2x + 3y = 4$

Solving them, we get $x = 5, y = -2$

$\therefore$ A point on the line L is $(5, -2, 0)$

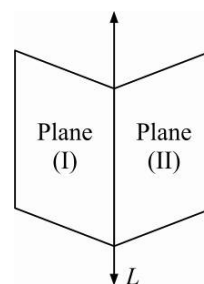
Equation of the line L is

$$\vec{r} = \vec{a} + \lambda \vec{b}$$

$$\vec{r} = 5\hat{i} - 2\hat{j} + \lambda(-2\hat{i} + \hat{k})$$

Direction ratios of z -axis are $0, 0, 1$ and it passes through the origin, so equation of z -axis is $\vec{r} = \mu \vec{k}$

Now, we have to find the shortest distance between the line L and z -axis.



$$\text{S.D.} = \frac{|\vec{a}_2 - \vec{a}_1 \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 2\hat{j}$$

$$\vec{a}_2 \times \vec{a}_1 = 5\hat{i} + 2\hat{j}$$

$$\text{S.D.} = \frac{|(-5\hat{i} + 2\hat{j}) \cdot (2\hat{j})|}{2} = \frac{|4|}{2} = 2$$

- 44.(B) Clearly, the line $\frac{x-1}{1} = \frac{y-2}{5} = \frac{z-3}{4}$ passes through the point (1, 2, 3) and its direction ratios are 1, 5, 4. Let equation of the plane passing through the point (3, 2, 0) be

$$a(x-3) + b(y-2) + c(z-0) = 0 \quad \dots\dots (i)$$

This plane contains the given line

$$\therefore a + 5b + 4c = 0 \quad \dots\dots (ii)$$

Plane contains the given line, so it passes through (1, 2, 3) also.

$$a(1-3) + b(2-2) + c(3) = 0$$

$$2a + 0.b + 3c = 0 \quad \dots\dots (iii)$$

Solving (ii) and (iii) by cross-multiplication

$$\frac{a}{15} = \frac{b}{-8-3} = \frac{c}{10} = \lambda \text{ (let)}$$

$$a = 15\lambda, b = -11\lambda, c = 10\lambda$$

$$\text{From (i), } 15\lambda(x-3) - 11\lambda(y-2) + 10\lambda z = 0$$

$$\lambda[15x - 45 - 11y + 22 + 10z] = 0$$

$$15x - 11y + 10z = 23$$

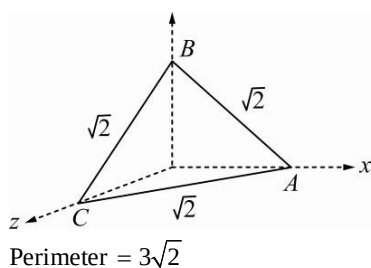
Obviously, this plane contains the point (0, 7, 10).

- 45.(C) $l = \cos \theta, m = \cos \theta, n = \cos \gamma$

$$\text{But } l^2 + m^2 + n^2 = 1 \Rightarrow 2\cos^2 \theta + \cos^2 \gamma = 1 \Rightarrow 2\cos^2 \theta = \sin^2 \gamma \leq 1$$

$$\Rightarrow \cos^2 \theta \leq \frac{1}{2} \Rightarrow 0 \leq \cos \theta \leq \frac{1}{\sqrt{2}} \therefore \theta \in \left(0, \frac{\pi}{2}\right] \Rightarrow \theta = \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$$

- 46.(3)



47.(4) Family of planes : $(x + 2y + 3z - 9) + k(2x - y + 2z - 11) = 0$

Normal to the plane is : $(1 + 2k)\hat{i} + (2 - k)\hat{j} + (3 + 2k)\hat{k}$

$$\Rightarrow 4(1 + 2k) + 3(2 - k) + 5(3 + 2k) = 0$$

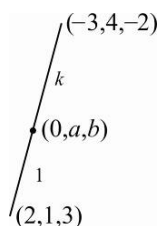
$$15k + 25 = 0 \Rightarrow k = -5/3$$

Equation of plane is : $5(2x - y + 2z - 11) - 3(x + 2y + 3z - 9) = 0 \Rightarrow 7x - 11y + z - 28 = 0$

48.(1) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$(1 + \cos 2\alpha) + (1 + \cos 2\beta) + (1 + \cos 2\gamma) = 2 \Rightarrow \cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$$

49.(3)



$$\Rightarrow \frac{2k - 3}{k + 1} = 0 \Rightarrow k = 3/2$$

50.(1) $\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and plane $x + 2y + 3z = 4$

$$\sin \theta = \frac{(1)(1) + (2)(2) + \lambda(3)}{\sqrt{1^2 + 2^2 + \lambda^2} \sqrt{1^2 + 2^2 + 3^2}} = \frac{3\lambda + 5}{\sqrt{\lambda^2 + 5} \sqrt{14}} = \sqrt{1 - \cos^2 \theta}$$

$$\Rightarrow \frac{3\lambda + 5}{\sqrt{\lambda^2 + 5}} = 3 \Rightarrow 9\lambda + 25 + 30\lambda = 9\lambda^2 + 45 \Rightarrow \lambda = 2/3$$

51.(3) Equation of a plane is : $x - 2y + 2z + k = 0$

Distance from origin = $\frac{|k|}{\sqrt{1^2 + 2^2 + 2^2}} = 1 \Rightarrow k = \pm 3$

$$x - 2y + 2z \pm 3 = 0$$

52.(9) $\begin{bmatrix} \bar{a}_2 - \bar{a}_1 & \bar{b}_1 & \bar{b}_2 \end{bmatrix} = 0$

$$\begin{vmatrix} (3-1) & (k+1) & (0-1) \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0 \Rightarrow 2[3-8] - (k+1)[2-4] - [4-3] = 0 \Rightarrow k = 9/2$$

53.(5) $\vec{b} \cdot \vec{n} = (3\hat{i} + \hat{j} - 5\hat{k}) \cdot (2\hat{i} - \hat{j} + \hat{k}) = 0$. Hence line is parallel to plane

Image of $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$ is : $\frac{h-1}{2} = \frac{k-3}{-1} = \frac{\ell-4}{1} = \frac{-2(2-3+4+3)}{2^2+1^2+1^2} = -2$

$$(h, k, \ell) = (-3, 5, 2)$$

Hence, equation of reflected line is : $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$

$$54.(6) \quad \ell^2 = m^2 + n^2 \text{ and } \ell^2 + m^2 + n^2 = 1 \Rightarrow \ell = \pm \frac{1}{\sqrt{2}}$$

$$\ell + m + n = 0 \Rightarrow \ell^2 = m^2 + n^2 + 2mn \Rightarrow mn = 0$$

$$m = 0 \Rightarrow \ell = -n \Rightarrow \left(\pm \frac{1}{\sqrt{2}}, 0, \mp \frac{1}{\sqrt{2}} \right)$$

$$n = 0 \Rightarrow \ell = -m \Rightarrow \left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{\sqrt{2}}, 0 \right)$$

$$\text{Hence, DCs can be } \left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}} \right), \left(\frac{-1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right), \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$\text{Angle between lines is : } \cos \theta = \frac{\left(\frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{k} \right) \cdot \left(\frac{1}{\sqrt{2}} \hat{i} - \frac{1}{\sqrt{2}} \hat{j} \right)}{(1)(1)}$$

$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \pi/3$$

$$55.(7) \quad 4x + 2y + 4z = 16 \text{ and } 4x + 2y + 4z = -5$$

$$\text{Distance between planes} = \frac{|16 + 5|}{\sqrt{4^2 + 2^2 + 4^2}} = \frac{7}{2}$$

$$56.(2) \quad \text{As lines are coplanar, } \begin{vmatrix} \bar{a}_2 - \bar{a}_1 & \bar{b}_1 & \bar{b}_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2-1 & 3-4 & 4-5 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix} = 0 \Rightarrow 1[1+2k] + (1+k^2) - (2-k) = 0 \Rightarrow k^2 + 3k = 0$$

$$57.(2) \quad \text{Equation of plane passing through } (1, 2, 2) \text{ and perpendicular to the planes } x - y + 2z = 3 \text{ and } 2x - 2y + z + 12 = 0$$

$$\text{is } \begin{vmatrix} x-1 & y-2 & z-2 \\ 1 & -1 & 2 \\ 2 & -2 & 1 \end{vmatrix} = 0 \Rightarrow x + y - 3 = 0 \quad \dots\dots (i)$$

$$\text{Distance of plane (i) from } (1, -2, 4) = \frac{|1-2-3|}{\sqrt{1^2+1^2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$58.(5) \quad \text{To find the point of intersection on lines.}$$

$$\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2} = \lambda$$

$$x = 3\lambda + 1, y = \lambda + 2, z = 2\lambda + 3$$

$$\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \mu$$

$$x = \mu + 3, y = 2\mu + 1, z = 3\mu + 2$$

Both the lines must have a common point.

$$\therefore 3\lambda + 1 = \mu + 3, \lambda + 2 = 2\mu + 1, 2\lambda + 3 = 3\mu + 2 \Rightarrow \lambda = 1, \mu = 1. \text{ So point of intersection is } A(4, 3, 5)$$

For the largest distance of plane from origin, the plane must pass through the point of intersection A and OA must be perpendicular to the plane.

$$\therefore \vec{a} = 4\hat{i} + 3\hat{j} + 5\hat{k} \text{ and } \vec{n} = OA = 4\hat{i} + 3\hat{j} + 5\hat{k}$$

$$\text{So, plane is } (\vec{r} - \vec{a}) \cdot \vec{n} = 0 \Rightarrow 4x + 3y + 5z = 50$$

$$59.(1) \quad x = ay + b \Rightarrow \frac{x-b}{a} = \frac{y}{1}$$

$$z = cy + d \Rightarrow \frac{z-d}{c} = \frac{y}{1}$$

$$\therefore \frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$$

So, direction ratios of line intersection of planes are $a, 1, c$.

$\therefore$ Option **(C)** and **(D)** are possible

But, only option **(D)** which passes through the point $(b+a, 1, d+c)$ satisfies both the given planes. Hence **(D)**.

$$60.(1) \quad L_1 : \frac{x}{1} = \frac{y}{-1} = \frac{z}{1} = \lambda$$

$$L_2 : \frac{x-1}{0} = \frac{y+1}{-2} = \frac{z}{1} = \mu$$

$$\therefore A(\lambda, -\lambda, \lambda), B(1, -2\mu-1, \mu).$$

Direction ratios of AB

$$1-\lambda, -2\mu+\lambda-1, \mu-\lambda$$

Let AB be the shortest distance, then $AB \perp L_1, AB \perp L_2$.

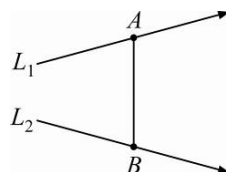
$$\therefore (1-\lambda) + (-2\mu-1+\lambda)(-1) + (\mu-\lambda)1 = 0 \Rightarrow 3\mu - 3\lambda + 2 = 0 \quad \dots\dots (i)$$

$$\text{and } (1-\lambda)0 + (-2\mu-1+\lambda)(-2) + (\mu-\lambda)1 = 0 \Rightarrow 5\mu - 3\lambda + 2 = 0 \quad \dots\dots (ii)$$

$$(i) - (ii) \Rightarrow \mu = 0 \text{ and } \lambda = 2/3.$$

$$\therefore A(2/3, -2/3, 2/3) \text{ and } B(1, -1, 0)$$

$$\therefore \text{Equation of } AB \text{ is } \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{-2}$$



Probability

1.(A) Let the first event be A and second event be B .
According to question we can assume,
 $P(A) = x^2$ and $P(B) = x$

$$\Rightarrow \text{Odds against } A = \frac{1-x^2}{x^2}$$

$$\text{Odds against } B = \frac{1-x}{x}$$

It is given in the question that odds against A is cube of odds against B .

$$\text{So we can take, } \left(\frac{1-x}{x}\right)^3 = \frac{1-x^2}{x^2}$$

$$\Rightarrow \frac{(1-x)^3}{x} = (1-x^2) \Rightarrow \frac{(1-x)^3}{x} = (1+x)(1-x)$$

$$\Rightarrow \text{Either } x = 1 \text{ or } \frac{(1-x)^2}{x} = 1+x$$

$$\Rightarrow x^2 - 2x + 1 = x^2 + x$$

$$\Rightarrow x = \frac{1}{3}$$

$$\Rightarrow P(A) = \frac{1}{9} \text{ and } P(B) = \frac{1}{3}$$

2.(B) Let the first event be E and Second Event be F .

Odds against E is $5 : 2 \Rightarrow P(E) = \frac{2}{7}$

Odds in favour of F is $6 : 5 \Rightarrow P(F) = \frac{6}{11}$

$$\Rightarrow P(\text{at least one of events})$$

$$= 1 - P(\text{none of the event happen})$$

$$= 1 - P(\overline{E} \cap \overline{F}) = 1 - P(\overline{E}) P(\overline{F}) \quad [\text{Use Section - 5}]$$

$$= 1 - [1 - P(E)] [1 - P(F)] = 1 - \frac{5}{7} \times \frac{5}{11} = \frac{52}{77}$$

3.(C) At least one : $P(A \cup B) = 0.6$
Simultaneous : $P(A \cap B) = 0.2$
We know $P(A \cup B) + P(A \cap B) = P(A) + P(B)$
 $\Rightarrow P(A) + P(B) = 0.8$
 $\Rightarrow 1 - P(\overline{A}) + 1 - P(\overline{B}) = 0.8$
 $\Rightarrow P(\overline{A}) + P(\overline{B}) = 1.2$
Hence (C).

4.(C) $P(A) = \frac{2}{3}$ [$\because$ odds in favour of $A = 2 : 1$]
 $P(A \cup B) = \frac{3}{4}$ [$\because$ odds in favour of $A \cup B = 3 : 1$]
We know,
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $\Rightarrow \frac{3}{4} = \frac{2}{3} + P(B) - P(A \cap B)$
 $\Rightarrow P(B) = \frac{1}{12} + P(A \cap B)$
 $\Rightarrow$ The smallest value of $P(B) = \frac{1}{12}$
which occurs when $P(A \cap B) = 0$
 $\Rightarrow P(B) \geq \frac{1}{12} \quad \dots(i)$
Also, $P(B) \leq P(A \cup B)$
 $\Rightarrow P(B) \leq \frac{3}{4} \quad \dots(ii)$
Combining (i) and (ii), we get $\frac{1}{12} \leq P(B) \leq \frac{3}{4}$

5.(B) Let E = The number is divisible by 6

F = The number is divisible by 8

Numbers divisible by 6 are : 6, 12, 18, 24,, 198 i.e..

There are 33 numbers in first 200 positive integers that are divisible by 6.

$$\Rightarrow P(E) = \frac{33}{200}$$

Numbers divisible by 8 are : 8, 16, 24, 32,, 200 i.e..

There are 25 numbers in first 200 positive integers that are divisible by 8.

$$\Rightarrow P(F) = \frac{25}{200}$$

Number divisible by 6 and 8 both are : 24, 48,, 192.

These are 8 numbers in first 200 positive integers that are divisible by both 6 and 8

$$\Rightarrow P(E \cap F) = \frac{8}{200}$$

$P(\text{integer is divisible by 6 or 8})$ is given as :

$$= P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$= \frac{33}{200} + \frac{25}{200} - \frac{8}{200} = \frac{1}{4}$$

$$8.(B) \quad P\left(\frac{A=7}{B=0}\right) = \frac{n(A=7 \cap B=0)}{n(B=0)} = \frac{n(\text{sum of the digits}=7 \text{ and their product is } 0)}{n(\text{product of digits is } 0)}$$

$$= \frac{n(07, 70)}{n(00, 01, \dots, 09, 10, 20, 30, \dots, 90)} = \frac{2}{19}$$

9.(D) Let $P\left(\frac{\text{Smaller number is } 5}{\text{Greater number is not more than } 10}\right)$

$$= \frac{n(\text{Smaller number is } 5 \cap \text{greatest number not more than } 10)}{n(\text{greatest number not more than } 10)}$$

$$= \frac{(\text{number of ways to choose } 5) \times (\text{number of ways to choose other number from } 6 \text{ to } 10)}{(\text{number of ways to choose } 2 \text{ numbers from } 1 \text{ to } 10)} = \frac{1 \times 5}{{}^{10}C_2} = \frac{1}{9}$$

10.(D) Check manually. Don't guess of two events are independent unless obvious.

$$P(E_1) = \frac{1}{2} \quad ; \quad P(E_1 / E_2) = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$P(E_2) = \frac{1}{2} \quad ; \quad P(E_2 / E_3) = \frac{1}{2}$$

$$P(E_3) = \frac{1}{2} \quad ; \quad P(E_3 / E_1) = \frac{1}{2}$$

6.(B) $P(\text{all gets one}) = P(\text{none of them is empty})$

$$= \frac{10!}{10^{10}}$$

$$\Rightarrow P(\text{atleast one doesn't get anything}) = 1 - P(\text{none of them is empty})$$

$$= 1 - \frac{10!}{10^{10}} = \frac{10^{10} - 10!}{10^{10}}$$

7.(B) $P(\bar{A}) = 0.3, \quad P(B) = 0.4 \quad \text{and} \quad P(A \cap \bar{B}) = 0.5$

$$\Rightarrow P(A) = 0.7, \quad P(\bar{B}) = 0.6$$

$$P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A \cap \bar{B})$$

$$\Rightarrow P(A \cup \bar{B}) = 0.7 + 0.6 - 0.5$$

$$\Rightarrow P(A \cup \bar{B}) = 0.8 \quad \dots (i)$$

$$P(B \cap (A \cup \bar{B})) = P(A \cap B)$$

$$= P(A) - P(A \cap \bar{B}) = 0.7 - 0.5 = 0.2 \quad \dots (ii)$$

$$\Rightarrow P(B / A \cup \bar{B}) = \frac{P(B \cap (A \cup \bar{B}))}{P(A \cup \bar{B})} = \frac{0.2}{0.8} = \frac{1}{4}$$

[using (i) and (ii)]

Thus E_1, E_2, E_3 are pairwise independent

$$P(E_1 \cap E_2 \cap E_3) = \frac{1}{4}$$

$$\text{But } P(E_1) \cdot P(E_2) \cdot P(E_3) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Hence $P(E_1 \cap E_2 \cap E_3) \neq P(E_1) \cdot P(E_2) \cdot P(E_3)$

$\Rightarrow E_1, E_2, E_3$ are not independent.

11.(D) Two throws are independent

So $P(\text{no. on } D_1 < \text{no. on } D_2) = P(\text{no. of } D_2 < \text{no. on } D_1)$

$\Rightarrow$ Let $E_1 = \text{Number on } D_1 < D_2$

$E_2 = \text{Number on } D_2 < D_1$

$F = \text{Number on } D_2 = D_1$

$\Rightarrow P(E_1) + P(E_2) + P(F) = 1$

$\Rightarrow 2P(E_1) + P(F) = 1 \quad \dots (i)$

$P(F) = P(1)P(1) + P(2)P(2) + \dots + P(6)P(6)$

$$= \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} + \dots + \frac{1}{6} \times \frac{1}{6} = \frac{1}{6}$$

$\Rightarrow P(E_1) = \frac{5}{12}$ (using (i))

13.(B) Let Event $I = \text{Drawing card } I$

Event $T = \text{Drawing card } T$

$$\begin{aligned} \Rightarrow P(I \cap I \cap T) &= P(I)P\left(\frac{I}{I}\right)P\left(\frac{T}{I \cap I}\right) \\ &= \frac{10}{20} \cdot \frac{9}{19} \cdot \frac{10}{18} = \frac{5}{38} \end{aligned}$$

14.(C) Let the event $A_i = A$ wins in i^{th} try

$B_i = B$ wins in i^{th} try

$\Rightarrow P(A \text{ wins}) = P(A \text{ wins in 1st attempt}) + P(A \text{ wins in 2nd attempt}) + \dots \infty$

$$= P(A_1) + P(\bar{A}_1 \cap \bar{B}_1 \cap A_2) + P(\bar{A}_1 \cap \bar{B}_1 \cap \bar{A}_2 \cap \bar{B}_2 \cap A_3) + \dots \infty$$

$$= P(A_1) + P(\bar{A}_1)P(\bar{B}_1 / \bar{A}_1)P(A_2 / \bar{A}_1 \cap \bar{B}_1) + \dots \infty$$

$$= \frac{m}{m+n} + \frac{n}{m+n} \frac{n}{m+n} \frac{m}{m+n} + \left(\frac{n}{m+n}\right)^4 \frac{m}{m+n} + \dots \infty$$

The above series is a G.P. with infinite terms, where common ratio $= \left(\frac{n}{m+n}\right)^2$

$$\Rightarrow P(A \text{ wins}) = \frac{m}{m+n} \frac{1}{1 - \left(\frac{n}{m+n}\right)^2} = \frac{m+n}{m+2n}$$

$\Rightarrow P(B \text{ wins}) = 1 - P(A \text{ wins}) \quad (\because \text{There is no possibility of draw})$

$$= 1 - \frac{m+n}{m+2n} = \frac{n}{m+2n}$$

It is given that $P(A) = 2P(B)$

$$\Rightarrow \frac{m+n}{m+2n} = 2 \frac{n}{m+2n} \quad \Rightarrow m = n \quad ; \quad \text{i.e. } m : n = 1 : 1$$

12.(AC)

$$P(\text{Both } A \text{ and } B \text{ occur}) = P(A \cap B) = \frac{1}{8}$$

$$P(\text{neither of } A \text{ and } B \text{ occur}) = P(\bar{A} \cap \bar{B}) = \frac{3}{8}$$

As A and B are independent events, we can take,

$$\Rightarrow P(A \cap B) = P(A)P(B) = \frac{1}{8}$$

$$\text{Similarly, } P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B}) = \frac{3}{8}$$

$$\Rightarrow [1 - P(A)][1 - P(B)] = \frac{3}{8}$$

$$\Rightarrow 1 - P(A) - P(B) + P(A)P(B) = \frac{3}{8}$$

$$\Rightarrow P(A) + P(B) = \frac{3}{4} \quad \dots (i)$$

$$\text{Also } P(A)P(B) = \frac{1}{8} \quad \dots (ii)$$

From (i) and (ii), we get

$$P(A) - P(B) = \pm \frac{1}{4} \quad \dots (iii)$$

Adding (i) and (iii), we get :

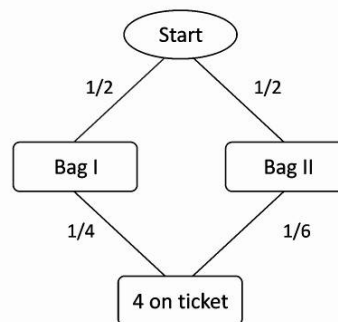
$$P(A) = \frac{1}{2} \text{ or } \frac{1}{4}$$

15.(C) Let E_1 be the event that bag I is chosen $P(E_1) = \frac{1}{2}$

Let E_2 be the event that bag II is chosen $P(E_2) = \frac{1}{2}$

Let A be the event that ticket drawn is 4

$$\begin{aligned} \Rightarrow P(A) &= P(E_1) P\left(\frac{A}{E_1}\right) + P(E_2) P\left(\frac{A}{E_2}\right) \\ &= \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{6} = \frac{5}{24} \end{aligned}$$



16.(B) Let G be the event that student guesses

$\bar{G}$ be the event that student knows

Let E be the event that answer is correct

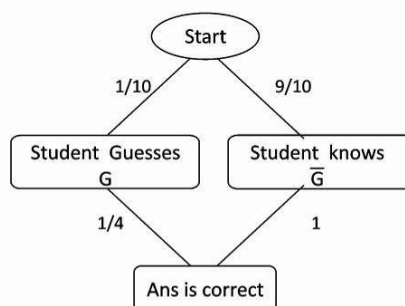
$$\Rightarrow P\left(\frac{E}{G}\right) = \frac{1}{4}, P\left(\frac{E}{\bar{G}}\right) = 1$$

$$P(\bar{G}) = \frac{9}{10}, P(G) = \frac{1}{10}$$

$\Rightarrow P(\text{Student guesses / Ans is correct})$

$$\begin{aligned} &= P(G/E) = \frac{P(G \cap E)}{P(E)} \\ &= \frac{P(G) P\left(\frac{E}{G}\right)}{P(G) P\left(\frac{E}{G}\right) + P(\bar{G}) P\left(\frac{E}{\bar{G}}\right)} \end{aligned}$$

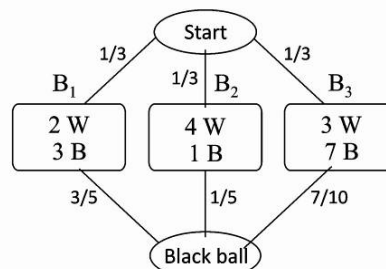
$$= \frac{\frac{1}{10} \cdot \frac{1}{4}}{\frac{1}{10} \cdot \frac{1}{4} + \frac{9}{10} \cdot 1} = \frac{1}{37}$$



17.(A) $P(B_i) = \frac{1}{3} \quad i = 1, 2, 3$

Let B = Black ball is drawn

$$\begin{aligned} \Rightarrow P\left(\frac{B_3}{B}\right) &= \frac{P(B \cap B_3)}{P(B)} \\ &= \frac{\frac{1}{3} \times \frac{7}{10}}{\frac{1}{3} \times \frac{3}{5} + \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{7}{10}} = \frac{7}{15} \end{aligned}$$



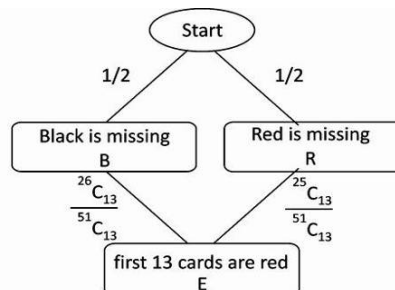
18.(D) Let B = Black card is missing

R = Red card is missing

$$\Rightarrow P(B) = P(R) = \frac{1}{2}$$

$\Rightarrow$ Let E = First 13 cards drawn are red

$$\Rightarrow P(E/B) = \frac{{}^{26}C_{13}}{{}^{51}C_{13}}; P(E/R) = \frac{{}^{25}C_{13}}{{}^{51}C_{13}}$$



19.(B) Place '2' ten paise coin at extreme points in a line. Now arrange rest $(m + n - 2)$ coins.

$$\Rightarrow \text{Total no. of favorable ways} = \frac{(m + n - 2)!}{m! \cdot (n - 2)!}; \text{Total ways} = \frac{m + n!}{m! \cdot n!}$$

$$\Rightarrow \text{Required probability} = \frac{(m + n - 2)!}{m! \cdot (n - 2)!} \cdot \frac{m! \cdot n!}{(m + n)!} = \frac{n(n - 1)}{(m + n)(m + n - 1)}$$

20.(A) Total arrangement = 3^{12} [As each ball have '3' choices]

Favorable ways = (Choose '3' balls out of 12 and place in 1st box) $\times$ (Remaining balls each has 2 choices)

$$= {}^{12}C_3 \times 2^9 \Rightarrow P(\text{required}) = \frac{{}^{12}C_3 \times 2^9}{3^{12}} = \frac{110 \times 2^{10}}{3^{12}}$$

21.(C) Favorable ways = (Particular couple in committee) + (Particular couple not in committee) = ${}^7C_3 + {}^7C_5$

$$\text{Total ways} = {}^9C_5 \Rightarrow \text{Probability (required)} = \frac{{}^7C_3 + {}^7C_5}{{}^9C_5} = \frac{4}{9}$$

$$\text{22.(D)} \quad P(\text{required}) = \frac{n(3 \text{ bowlers}) + n(4 \text{ bowlers}) + n(5 \text{ bowlers})}{\text{total no. of ways}} = \frac{{}^5C_3 \cdot {}^{10}C_8 + {}^5C_4 \cdot {}^{10}C_7 + {}^5C_5 \cdot {}^{10}C_6}{{}^{15}C_{11}} = \frac{12}{13}$$

23.(D) 9 can be obtained by permutation of $B B 3 3 3$ and $2 2 2 3 B$;

$$P(B) = \frac{4}{6} ; \quad P(2) = \frac{1}{6} ; \quad P(3) = \frac{1}{6}$$

$$\Rightarrow P(\text{obtaining 9}) = \left(\frac{4}{6} \cdot \frac{4}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} \right) \times \frac{5!}{2! 3!} + \left[\frac{4}{6} \cdot \left(\frac{1}{6} \right)^4 \right] \times \frac{5!}{3!} = \frac{5}{162}$$

24.(C) Given, $x_3 = 30 \Rightarrow$ Two tickets should be less than 30 and two should be more.

$$\Rightarrow P(x_3 = 30) = [2 \text{ tickets from } 1 - 29 \text{ and } 2 \text{ tickets from } 31 - 50] = \frac{{}^{29}C_2 \cdot {}^{20}C_2}{{}^{50}C_5}$$

25.(A) Let success : Hitting of target $\Rightarrow P(\text{atleast one hit}) = 1 - P(\text{none})$

$$\text{Failure : Non Hitting of target} \quad = 1 - {}^{10}C_0 \left(\frac{4}{5} \right)^{10} = 1 - \left(\frac{4}{5} \right)^{10}$$

$$\text{26.(A)} \quad P(\text{toss result is } H) = \left(\frac{n}{2n+1} \right) \cdot 1 + \left(\frac{n+1}{2n+1} \right) \cdot \frac{1}{2} = \frac{31}{42} \Rightarrow n = 10$$

27.(D) Let each of A and B has 'n' number of daughters.

$$\Rightarrow P(\text{all tickets go to daughter of A}) = \frac{{}^nC_3}{{}^{2n}C_3} = \frac{1}{20} \Rightarrow n = 3$$

28.(D) $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B) \Rightarrow P(A \cap B) = P(A) \cdot P(B) \Rightarrow$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

29.(A)

$$30. (C) \quad P(\text{draw ends at } 18^{\text{th}} \text{ draw}) = \left(\frac{{}^4C_1 \cdot {}^{48}C_{16}}{{}^{52}C_{17}} \times \frac{{}^3C_1}{{}^{35}C_1} \right) = \frac{561}{15925}$$

[First 17 draw : 1 Ace, 16 non - Ace , 18th draw : 2nd Ace]

$$31. (C) \quad \begin{array}{ll} \text{Let success : Draw of } W \text{ ball} & ; \quad \text{Ist 6 draw : } 3W, 3B \\ \text{failure : Draw of } B \text{ ball} & ; \quad \text{7th draw : 4th time } W \end{array}$$

$$\Rightarrow P(7^{\text{th}} \text{ draw is 4th time } W \text{ ball}) = P(3 \text{ success in Ist 6 draw}) \times P(7^{\text{th}} \text{ draw is success}) = {}^6C_3 \left(\frac{1}{2} \right)^3 \cdot \left(\frac{1}{2} \right)^3 \cdot \frac{1}{2} = \frac{5}{32}$$

$$32. (B) \quad \text{Choose 'm' floors out of 'n' floors} = {}^nC_m \text{ (one floor for each person)}$$

Arrange 'm' person on these chosen floors in $m!$ ways.

$$\Rightarrow \text{Favorable ways} = {}^nC_m \cdot m!, \text{ Total ways} = n^m [\text{as each person has } m \text{ choice}] \Rightarrow P(\text{required}) = \frac{{}^nC_m \cdot m!}{n^m} = \frac{{}^nP_m}{n^m}$$

$$33. (B) \quad \text{Total ways} = 8!$$

Favorable outcomes = (choose '3' places for ABC) $\times$ arrange rest 5 people = ${}^8C_3 \cdot 5!$

$$P(\text{req.}) = ({}^8C_3 \times 5!) / 8! = 1/6$$

First 9 throws	10 th throw
3 times '6'	4 th time 6
6 times (No '6')	

$$P(\text{required}) = {}^9C_3 \left(\frac{1}{6} \right)^3 \left(\frac{5}{6} \right)^6 \cdot \frac{1}{6} = \frac{84 \times 5^6}{6^{10}}$$

$$35. (A) \quad P(\text{Heads in even toss}) = P(\text{II}^{\text{nd}} \text{ toss}) + P(\text{IV}^{\text{th}} \text{ toss}) + \dots = P(TH) + P(TTTH) + \dots = (1-p) \cdot p + (1-p)^3 \cdot p + \dots$$

$$p + \dots = \frac{(1-p) \cdot p}{1 - (1-p)^2} = \frac{1-p}{2-p} = \frac{2}{5} \Rightarrow p = \frac{1}{3}$$

$$36. (B) \quad P(\text{teacher misses at least 1 test})$$

$$= 1 - P(\text{misses no test}) = 1 - P(\text{both days test was not given})$$

$$= 1 - \frac{4}{5} \times \frac{4}{5} = \frac{9}{25}$$

$$37. (C) \quad P(\text{none is duplicate}) = \frac{{}^{10}C_3}{{}^{15}C_3} = \frac{10 \times 9 \times 8}{15 \times 14 \times 13} = \frac{24}{91}$$

$$38. (A) \quad P(13^{\text{th}} \text{ day is second saturday}) = P(1^{\text{st}} \text{ day is a monday})$$

$$= \frac{1}{7} \text{ (as month is chosen randomly so each day has equal probability)}$$

39.(D) Probability (throws are unequal)

$$= 1 - \text{Probability (both are equal)}$$

$$= 1 - \sum_{i=1}^6 \text{Probability (both throws} = i)$$

$$P(\text{a throw results in 2}) = \frac{1}{36} = P(\text{a throw result in 12})$$

$$P(\text{a throw results in 3}) = \frac{2}{36} = P(\text{a throw result in 11})$$

$$P(\text{a throw results in 4}) = \frac{3}{36} = P(\text{a throw result in 10})$$

$$P(\text{a throw results in 5}) = \frac{4}{36} = P(\text{a throw result in 9})$$

$$P(\text{a throw results in 6}) = \frac{5}{36} = P(\text{a throw result in 8})$$

$$P(\text{a throw results in 7}) = \frac{6}{36}$$

$$\text{Probability} = 1 - \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2}{36^2} = 1 - \frac{73}{648} = \frac{575}{648}$$

$$40.(B) \quad E(X) = \sum x_i p_i = (30)\left(\frac{1}{5}\right) + (10)\left(\frac{3}{10}\right) + (-10)\left(\frac{1}{2}\right) = 4$$

$$41.(D) \quad P(B|A) + P(A|B) = \frac{P(A \cap B)}{P(A)} + \frac{P(A \cap B)}{P(B)} \quad P(A \cap B) = \frac{3}{10} + \frac{2}{5} - \frac{3}{5} = \frac{1}{10}$$

$$P(A|B) + P(B|A) = \frac{1}{10} \left[\frac{10}{3} + \frac{5}{2} \right] = \frac{7}{12}$$

$$42.(C) \quad P(A'|B') \cdot P(B'|A') = \frac{P(A' \cap B')}{P(B')} \cdot \frac{P(A' \cap B')}{P(A')} = \frac{[1 - P(A \cup B)]^2}{(1 - P(A))(1 - P(B))} = \frac{\left[1 - \left(\frac{2}{5} + \frac{3}{10} - \frac{1}{5}\right)\right]^2}{\left(\frac{3}{5}\right)\left(\frac{7}{10}\right)} = \frac{5 \times 10}{4 \times 3 \times 7} = \frac{25}{42}$$

$$43.(C) \quad P(A'|B') = \frac{P(A' \cap B')}{P(B')} = \frac{1 - P(A \cup B)}{P(B')}$$

44.(D) If two events are independent, then $P(A \cap B) = P(A)P(B)$

If two events are mutually exclusive, then $P(A \cap B) = 0$. Hence, none of the above is correct.

$$45.(B) \quad P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\ P(A)P(B)P(\bar{C}) + P(A)P(\bar{B})P(C) + P(\bar{A})P(B)P(C) \\ (0.4)(0.3)(0.8) + (0.4)(0.7)(0.2) + (0.6)(0.3)(0.2) = 0.188$$

$$46.(C) \quad P(\text{sum} = 3 | \text{sum} < 6) = \frac{P(\text{sum} = 3 \cap \text{sum} < 6)}{P(\text{sum} < 6)}$$

$$\text{sum} < 6 \Rightarrow \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (3,1), (3,2), (4,1)\}; \text{sum} = 3 \Rightarrow \{(1,2), (2,1)\}$$

$$P(\text{sum} = 3 | \text{sum} < 6) = \frac{2}{10} = \frac{1}{5}$$

47.(C) For binomial distribution outcomes may or may not be dependents on each other.

$$48.(A) \quad P(S) = 0.7 = p. \quad P(4 \text{ out of } 5 \text{ are swimmers}) = {}^5C_4 p^4 q = {}^5C_4 (0.7)^4 (0.3)$$

$$49.(D) \quad E(x^2) = \sum x_i^2 p_i = 1^2 \left(\frac{1}{10}\right) + 2^2 \left(\frac{1}{5}\right) + 3^2 \left(\frac{3}{10}\right) + 4^2 \left(\frac{2}{5}\right) = \frac{1}{10} [1 + 8 + 27 + 64] = 10$$

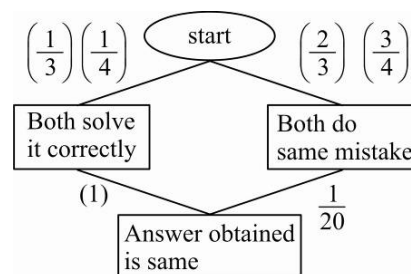
$$50.(A) \quad \frac{p(x=r)}{p(x=n-r)} = \frac{{}^nC_r p^r q^{n-r}}{{}^nC_{n-r} p^{n-r} q^r} = p^{2r-n} q^{n-2r} = \left(\frac{p}{1-p}\right)^{2r-n}$$

It becomes independent of n and r if $p = 1 - p \Rightarrow p = 1/2$

$$51.(D) \quad P(\text{answer is same}) = \left(\frac{1}{12}\right)(1) + \left(\frac{2}{3} \times \frac{3}{4}\right)\left(\frac{1}{20}\right)$$

$P(\text{both solve it correctly} | \text{Answer is same})$

$$= \frac{\left(\frac{1}{3}\right)\left(\frac{1}{4}\right)}{\frac{1}{12} + \left(\frac{2}{3} \times \frac{3}{4}\right)\left(\frac{1}{20}\right)} = \frac{1/12}{1/12 + 1/40} = \frac{10}{13}$$



52.(D) $P(\text{at most one defective}) = P(\text{no defective}) + P(\text{one defective})$

$$= {}^5C_5 \left(\frac{9}{10}\right)^5 + {}^5C_4 \left(\frac{9}{10}\right)^4 \left(\frac{1}{10}\right) = \left(\frac{9}{10}\right)^5 + \frac{1}{2} \left(\frac{9}{10}\right)^4$$

53.(B) $P(\text{at least one failure}) = 1 - P(\text{no failure}) \geq \frac{31}{32}$

$$\Rightarrow P(\text{no failure}) \leq \frac{1}{32} \Rightarrow {}^5C_5 p^5 \leq \frac{1}{32} \Rightarrow p \leq \frac{1}{2}$$

$$\text{Also, } P \geq 0 \Rightarrow p \in \left[0, \frac{1}{2}\right]$$

54.(A) $C \subset D \Rightarrow P(C) < P(D)$

$$P(C/D) = \frac{P(C \cap D)}{P(D)} \quad \text{where } P(C \cap D) = P(C)$$

$$= \frac{P(C)}{P(D)} \Rightarrow P(C/D) \geq P(C)$$

55.(B) $P(\text{min is 3} \cap \text{max is 6}) = \frac{P(\text{min is 3} \cap \text{max is 6})}{P(\text{max is 6})} = \frac{{}^2C_1}{{}^5C_2} = \frac{1}{5}$

56.(A) $P(\overline{A \cup B}) = \frac{1}{6} \Rightarrow P(A \cup B) = \frac{5}{6} = P(A) + P(B) - P(A \cap B)$

$$P(B) = \frac{5}{6} - \frac{3}{4} + \frac{1}{4} = \frac{1}{3}$$

$$P(A)P(B) = \left(\frac{3}{4}\right)\left(\frac{1}{3}\right) = \frac{1}{4} = P(A \cap B)$$

Hence A and B are independent but not equally likely.

57.(D) Numbers are 2, 4, 6, 8, 100 where mean $(\bar{x}) = \frac{2+4+\dots+100}{50} = 51$

$$Var(x) = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} + \frac{\sum \bar{x}^2}{n} - \frac{2\bar{x} \sum x_i}{n} = \frac{\sum x_i^2}{n} + \bar{x}^2 - 2\bar{x}^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{\sum_{r=1}^{50} (2r)^2}{50} - 51^2 = 833$$

58.(C) Required probability = ${}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right) + {}^5C_5 \left(\frac{1}{3}\right)^5 = \frac{10}{3^5} + \frac{1}{3^5} = \frac{11}{3^5}$

59.(B) $P(A \cap B') = P(A) - P(A \cap B)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cap A' = \phi$

$$P(A|(A' \cup B')) = \frac{P(A \cap (A' \cup B'))}{P(A' \cup B')} = \frac{P((A \cap A') \cup (A \cap B'))}{P((A \cap B)')}$$

$$= \frac{P(\phi \cup (A \cap B'))}{P((A \cap B)')} = \frac{P(A \cap B')}{P((A \cap B)')} = \frac{P(A) - P(A \cap B)}{1 - P(A \cap B)} = \frac{\frac{2}{5} - \frac{3}{20}}{1 - \frac{3}{20}} = \frac{5}{17}$$

60.(C) Let $p = P(\text{Success})$ and $q = P(\text{Failure})$

Given $p = 2q$. But $p + q = 1$

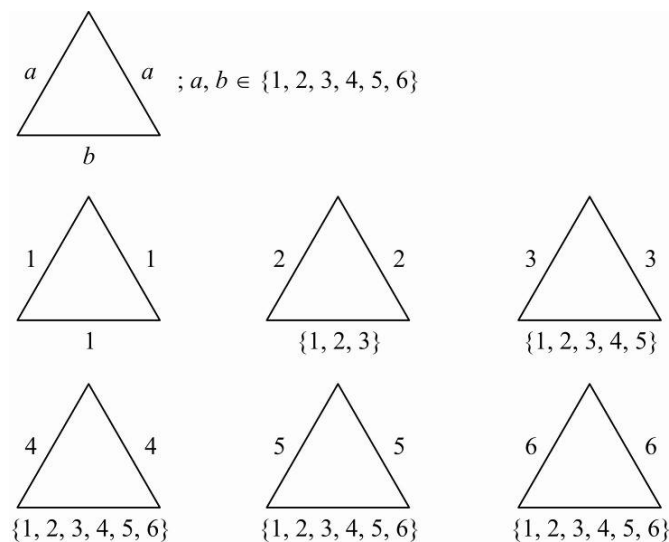
$$\therefore 2q + q = 1 \Rightarrow q = \frac{1}{3} \text{ and } p = \frac{2}{3}$$

Let X be number of success. Here $n = 6$

$$\text{Req. Probability } P = (X \geq 5) = P(X = 5) + P(X = 6) = {}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^1 + {}^6C_6 \left(\frac{2}{3}\right)^6 = \frac{256}{729}$$

61.(D)

62.(B) In the triangle, sum of two sides is greater than the third side. In isosceles triangle, two sides are equal. Also, equilateral triangle is also an isosceles triangle. For isosceles triangle.



$$\therefore \text{Number of isosceles triangles} = 1 + 3 + 5 + 6 + 6 + 6 = 27$$

But only one triangle has the greatest area.

$$\therefore \text{Required probability} = \frac{1}{27}$$

63.(B) S has 7 elements

$$\therefore \text{Total number of its subsets} = 2^7 = 128 \quad \therefore \text{Total number of non-empty subsets} = 128 - 1 = 127$$

$$\therefore n(A) = 127$$

$\therefore$ Now, we have to find the number of subsets containing x .

Therefore, x will be element, two elements, three elements,and 7 elements.

$$\therefore \text{Number of subsets contain } x = 1 + {}^6C_1 + {}^6C_2 + \dots + {}^6C_6 = 2^6$$

$$\therefore \text{Required probability} = \frac{2^6}{127} = \frac{64}{127}$$

64.(B) Mean = np

$$P(X = 2) = P(X = 3)$$

$$\Rightarrow {}^nC_2 p^2 q^{n-2} = {}^nC_3 p^3 q^{n-3} \Rightarrow \frac{q}{p} = {}^nC_3 / {}^nC_2 = \frac{n-2}{3} \Rightarrow \frac{1-p}{p} = \frac{n-2}{3} \Rightarrow np = 3 - p$$

65.(B)

$$A = \{10, 11, 12, \dots, 29, 50, 51, \dots, 100\} \Rightarrow n(A) = 71, n(S) = 100, P(A) = \frac{n(A)}{n(S)} = 0.71$$

66.(A) Statement I : $P(E/A) \geq P(A/E)P(E)$

$$\Rightarrow \frac{P(E \cap A)}{P(A)} \geq \frac{P(A \cap E)}{P(E)} P(E) \Rightarrow 1 \geq P(A) \text{ True}$$

Statement II : $P(A/E) \geq P(A \cap E)$

$$\Rightarrow \frac{P(A \cap E)}{P(E)} \geq P(A \cap E) \Rightarrow 1 \geq P(E) \text{ True}$$

67.(A) Let $p = \text{probability of hitting} = \frac{2}{5}, q = \frac{3}{5}$

$$P(x=r) = {}^k C_r p^r q^{k-r}$$

$$\text{Given } P(x \geq 1) > \frac{7}{10} \Rightarrow 1 - P(x=0) > \frac{7}{10}$$

$$\Rightarrow 1 - \left(\frac{3}{5}\right)^k > \frac{7}{10} \Rightarrow \frac{3}{10} > \left(\frac{3}{5}\right)^k \Rightarrow \frac{1}{2} > \left(\frac{3}{5}\right)^{k-1}$$

$$\left(\frac{5}{3}\right)^{k-1} > 2 \Rightarrow k=3$$

68.(A) Let A and B be two independent events.

$$P(A) = x, P(B) = y$$

$$P(\text{exactly one of them occurs}) = \frac{26}{49}$$

$$\Rightarrow P(A \cap B') + P(A' \cap B) = \frac{26}{49} \Rightarrow P(A)P(B') + P(A')P(B) = \frac{26}{49}$$

$$\Rightarrow x(1-y) + (1-x)y = \frac{26}{49} \Rightarrow x+y-2xy = \frac{26}{49} \dots\dots (i)$$

$$P(\text{none of them occurs}) = \frac{15}{49} \Rightarrow P(A' \cap B') \Rightarrow P(A') \cdot P(B') = \frac{15}{49}$$

$$\Rightarrow (1-x)(1-y) = \frac{15}{49} \dots\dots (ii)$$

$$\text{Solving (i) and (ii), we get : } x = \frac{2}{7}, y = \frac{4}{7} \text{ or } x = \frac{4}{7}, y = \frac{2}{7}$$

$\therefore$ Probability of more probable event is $4/7$.

69.(A) $P(A) = \frac{3}{4}, P(B) = \frac{1}{2}, P(C) = \frac{5}{8}$

Here events are independent.

$$\text{Req. probability} = P((A \cup B) \cap C') = P((A \cup B)P(C')) = [P(A) + P(B) - P(A)P(B)] \frac{3}{8} = \frac{21}{64}$$

70.(B) $3C_1 \times 4C_1 \times 5C_1 + 3C_2 \times 4C_1 + 3C_2 \times 5C_1 + 4C_2 \times 3C_1 + 4C_2 \times 5C_1 + 5C_2 \times 3C_1 + 5C_2 \times 4C_1$

71.(3) $P(\text{sum} = 5) = \frac{4}{36} = \frac{1}{9}$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

$P(7 \text{ comes before } 5)$

$$= P(7) + P(\sim 7 \sim 5) \cdot P(7) + \dots$$

$$= \frac{1}{6} + \frac{13}{8} \times \frac{1}{6} + \left(\frac{13}{18}\right)^2 \times \frac{1}{6} + \dots$$

$$= \frac{\frac{1}{6}}{1 - \frac{13}{18}} = \frac{1}{6} \times \frac{18}{5} = \frac{3}{5}$$

72.(5) $\begin{array}{|c|} \hline 5W \\ \hline 3B \\ \hline A \end{array} \quad \begin{array}{|c|} \hline 3W \\ \hline 1B \\ \hline B \end{array}$

$$P(B/W) = \frac{P(W/B) \cdot P(B)}{P(W/A) \cdot P(A) + P(W/B) \cdot P(B)} = \frac{\frac{3}{4}}{\frac{5}{8} + \frac{3}{4}} = \frac{6}{11}$$

73.(11)

$$\begin{array}{ll} 32 = 1 \times 32 & 58 = 1 \times 58 \\ = 2 \times 16 & = 2 \times 29 \\ = 4 \times 8 & = 29 \times 2 \\ = 8 \times 4 & = 58 \times 1 \\ = 16 \times 2 & \\ = 32 \times 1 & \end{array}$$

$$\text{Required probability} = \frac{2}{6 \times 4} = \frac{1}{12}$$

74.(40) $\bar{7} \times \bar{7} \times \bar{7} \times \bar{7}$ Total 7^4 ways

$\bar{3} \times \bar{7} \times \bar{7} \times \bar{3}$ Favorable $3^2 \cdot 7^2$ ways

75.(5) For divisible by 4 last 2 digits can be 12, 32, 52, 72, 24, 64, 16, 36, 56, 76, i.e. 10 ways

$$\text{Probability} = \frac{5 \times 4 \times 3 \times 10}{7 \times 6 \times 5 \times 4 \times 3} = \frac{5}{21}$$

76.(26) $P(A \cup B) = 0.42 + 0.48 - 0.16 = 0.74$

$$P((A \cup B)') = 0.26$$

77.(6) $p = \frac{1}{7}$

78.(11) $\frac{2/4/6}{I}, \frac{5/3/1}{II}$

i.e. (2, 5), (4, 3), (6, 1) only 3 favourable cases

$$P = \frac{3}{36} = \frac{1}{12}$$

$$79.(3) \quad P(B' / A') = \frac{P(B' \cap A')}{P(A')} = \frac{1 - P(A \cup B)}{1 - P(A)} = \frac{1 - \left(\frac{1}{3} + \frac{1}{4} - \frac{1}{5}\right)}{1 - \frac{1}{3}}$$

$$= \frac{60 - 20 - 15 + 12}{60} \times \frac{3}{2} = \frac{37}{60} \times \frac{3}{2} = \frac{37}{40}$$

$$80.(4) \quad \text{Probability of occurrence of event} = \frac{4}{9}$$

$$81.(4) \quad P(25p \text{ or } 50p) = P(25p) + P(50p) = \frac{3}{9} + \frac{2}{9} = \frac{5}{9}$$

$$82.(5) \quad \begin{array}{|c|} \hline 4W \\ \hline 5B \\ \hline \text{I} \end{array} \quad \begin{array}{|c|} \hline 5W \\ \hline 4B \\ \hline \text{II} \end{array}$$

$$P = (I / W) = \frac{P(W / I) \cdot P(I)}{P(W / I) \cdot P(I) + P(W / II) \cdot P(II)} = \frac{\frac{4}{9} \times \frac{1}{2}}{\frac{4}{9} \times \frac{1}{2} + \frac{5}{9} \times \frac{1}{2}} = \frac{4}{9}$$

$$83.(6) \quad \begin{array}{ccc} A & B & C \\ B & B & R \end{array} \text{ or } \begin{array}{ccc} A & B & C \\ B & B & B \end{array} \begin{array}{ccc} A & B & C \\ B & B & R \end{array} \text{ or } \dots$$

$$\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^9 + \dots = \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{1}{7}$$

$$84.(5) \quad \begin{array}{|c|} \hline 2W \\ \hline 3B \\ \hline \end{array} \quad \begin{array}{|c|} \hline 4W \\ \hline 2B \\ \hline \end{array} \quad \begin{array}{|c|} \hline 3W \\ \hline 2B \\ \hline \end{array}$$

$$P(B) = P(B / I) \cdot P(I) + P(B / II) \cdot P(II) + P(B / III) \cdot P(III)$$

$$= \frac{1}{3} \left(\frac{3}{5} + \frac{2}{6} + \frac{2}{5} \right) = \frac{4}{9}$$

$$85.(12) \quad P(G / R) = \frac{P(R / G) \cdot P(G)}{P(R / G) \cdot P(G) + P(R / K) \cdot P(K)} = \frac{\frac{1}{4} \times \frac{1}{4}}{\frac{1}{4} \times \frac{1}{4} + 1 \times \frac{3}{4}} = \frac{1}{13}$$